Singularity

Relational Heights and Arithmetic Visibility - Research Monograph

Singularity project contributors

18 October 2026

Contents

Reading guide **2**

1 What the population observes **4**
1.1 A relation before a prediction 4
1.2 Objects, assignments, and the meaning of a coefficient 4
1.3 Removing a repetition that carries no relational information 5
1.4 Labeled populations can define unlabeled observations 7
1.5 What a result in this book means 8

2 Graphs, integral cut lattices, and the height mark **10**
2.1 The quadratic form and its kernel 10
2.2 An orientation is a bookkeeping choice 10
2.3 A determinant that counts trees 11
2.4 Unmarked theta and the extra information in a mark 12
2.5 The height mark is an asymmetric gauge 13
2.6 Two endpoint calculations 14
2.7 Disjoint unions and recoverable component sizes 14

3 Nested heights and the interaction of boundaries **16**
3.1 A height profile is a stack of threshold sets 16
3.2 Squared differences do not add level by level 17
3.3 Two examples where repetition matters 17
3.4 An exact chain expansion of the marked series 18
3.5 The binary sector and its reciprocity 19
3.6 Minimum cuts are the first positive energy shell 20
3.7 Support restrictions useful for exact computation 21
3.8 Primitive height scale and exact dilation 21

4 Visibility depth and the loss of correspondence **23**
4.1 Increasing volume as an observation hierarchy 23
4.2 The first coefficients can be read by hand 23
4.3 What the second coefficient adds 25
4.4 A precise finite-depth comparison 26
4.5 What an occupancy coefficient can depend on 27
4.6 Why labeled probes make reconstruction easy 28
4.7 Marks, specializations, and the conclusions they permit 28

5 Prime arithmetic: assembling relational height populations **30**
5.1 The arithmetic weight and its exact assembly 30
5.2 Removing the common factor exactly 31
5.3 Synchronization and independence as exact endpoints 32
5.4 Minimum cuts are the first arithmetic correction 33

5.5 Two small models that expose the retained information 34

6 Euler analysis: formal coordinates and controlled analytic factors 36
6.1 Logarithmic coordinates from finite data 36
6.2 A different transform keeps integral bidegrees 38
6.3 Leading divergence and what it forgets 39
6.4 Finite-order zeta factorization 40
6.5 Why ratios are cleaner than individual continuations 40
6.6 A complete low-order transform example 41
6.7 Uniformity and the boundary of the weight parameter 42

7 Visibility singularities: the first distinction survives arithmetic assembly 43
7.1 A precise first-discrepancy theorem 43
7.2 A volume-two calculation and its actual meaning 45
7.3 A volume-three witness beyond the binary census 45
7.4 Effective control of the finite-prime approximation 46
7.5 The complete binary comparison and the remaining elementary distinction 51
7.6 A secondary boundary layer at an endpoint 52

8 Synchronization limits, uniform scaling, and exact contractions 54
8.1 An explicit estimate uniform in the energy weight 54
8.2 The double-scaling law 56
8.3 Optional extension: vertex masses 56
8.4 Optional extension: partial edge synchronization 57
8.5 Positive ordering and probabilistic derivatives 58
8.6 A worked contraction sequence 59

9 Exact examples: what the new population distinguishes 60
9.1 Two different comparisons 61
9.2 The exact Laplacian comparison 61
9.3 All binary cut-volume profiles 62
9.4 The first local coefficients and their span decomposition 63
9.5 What the analytic signatures see 66
9.6 A regular calibration: the cube and the Möbius ladder 67
9.7 Reading the examples as a controlled experiment 68

10 Inverse problems and the depth of a height population 70
10.1 Which equivalence is being studied? 70
10.2 Why retaining labels makes reconstruction easy 71
10.3 Universal relations within each depth 72
10.4 Connected components and exact recovery at zero parameter 73
10.5 Easy classes and the limits of degree information 74
10.6 A family with exactly controlled visibility depth 76
10.7 Terminal kernels and the Whitney twist 78
10.8 Restricted contexts and interface rank under a volume budget 82
10.9 Every late census collision is separated by triangles 92
10.10 Observation depth and connected supports 93
10.11 What should be searched next? 99

11 Archival bridges: what is inherited and what changes 101
11.1 From multiplicative integers to relational exponent geometry 101
11.2 From exact populations to navigable height assignments 102

11.3 From all-future behavioral equivalence to weighted transfer states . 103
11.4 The relation to classical cut lattices 104
11.5 Finite conductors and the part that is already old 105
11.6 What the new direction actually adds 106

12 Exact computation, certificates, and reproducible experiments 108
12.1 What a computation is intended to establish 108
12.2 A reference enumerator whose meaning is transparent 108
12.3 Three independent representations of the same coefficients 109
12.4 Exact polynomials before numerical evaluation 110
12.5 Ranking, unranking, and exact sampling 110
12.6 Certificates and the trusted boundary 111
12.7 Analytic experiments and graph boundaries 112

13 A mathematical research program with distinct proof routes 113
13.1 The question that organizes the program 113
13.2 Tree messages and separator geometry 113
13.3 What nested boundaries remember 116
13.4 Matroids, cut lattices, and the role of a positive chamber 117
13.5 Root-of-unity specializations as finite exact laboratories 118
13.6 Order chambers and weighted lattice-point geometry 120
13.7 Visibility strata and formal proof priorities 123

14 Literature, attribution, and publication claims 125
14.1 The contribution must be stated at the right scale 125
14.2 Binary heights and the established Ising polynomial 125
14.3 Cut lattices, spectra, and Gaussian comparisons 126
14.4 Orders, matroids, and Euler products 126
14.5 A claim register for papers and releases 127
14.6 Candidate paper abstracts with stated prerequisites 128
14.7 Verified references and their intended use 128

15 Context operators, cheapest experiments and periodic sectors 130
15.1 Background and what is new 130
15.2 Operator systems, experiments and word order 131
15.3 Closure, quotient and cheapest experiments 131
15.4 Certificates and their contracts 133
15.5 The graph context compiler 134
15.6 Exact robustness for finite candidate families 135
15.7 The periodic sector 136
15.8 Edge-factor families and compressed interfaces 137
15.9 Interfaces and runnable examples 137
15.10 Formal status 138
15.11 Open questions 138

16 Implementation and retained computational results 139
16.1 Installing and running the project 139
16.2 Implemented mathematical modules 139
16.3 Population navigation and exact bounds 142
16.4 Forest transfer experiment 143
16.5 Complete small-graph census 143
16.6 Reproduction and document generation 144

17 Thirty-six concrete research pushes 145

17.1 P01 — Freeze the coefficient semantics . . . 145
17.2 P02 — Add independent level-chain enumeration . . . 145
17.3 P03 — Publish portable coefficient certificates . . . 146
17.4 P04 — Give the spectral pair a short coefficient proof . . . 146
17.5 P05 — Explain the binary-census separation . . . 146
17.6 P06 — Derive universal low-volume formulas . . . 146
17.7 P07 — Formalize translation normalization . . . 146
17.8 P08 — Formalize the nested-level energy identity . . . 146
17.9 P09 — Implement bounded-height tree messages . . . 147
17.10 P10 — Prove safe volume truncation . . . 147
17.11 P11 — Recover span by bounded-box subtraction . . . 147
17.12 P12 — Extend messages to bounded separators . . . 147
17.13 P13 — Classify the first positive-energy layer . . . 147
17.14 P14 — Compute the nested-pair census . . . 147
17.15 P15 — Separate lattice information from chamber information . . . 148
17.16 P16 — Test Whitney switches systematically . . . 148
17.17 P17 — Add exact volume congruence filters . . . 148
17.18 P18 — Prove root-of-unity rationality in energy . . . 148
17.19 P19 — Analyze finite quadratic Gauss sums . . . 148
17.20 P20 — Build the disjoint order-chamber formula . . . 148
17.21 P21 — Relate constrained heights to order polytopes . . . 149
17.22 P22 — Establish one genuine reciprocity statement . . . 149
17.23 P23 — Implement formal logarithmic extraction . . . 149
17.24 P24 — Determine exceptional energy parameters . . . 149
17.25 P25 — Organize successive visibility levels . . . 149
17.26 P26 — Make analytic remainder bounds explicit . . . 149
17.27 P27 — Produce controlled real singularity experiments . . . 150
17.28 P28 — State the available complex domain exactly . . . 150
17.29 P29 — Specify a complete finite graph census . . . 150
17.30 P30 — Measure collision classes across invariants . . . 150
17.31 P31 — Resolve targeted collisions structurally . . . 150
17.32 P32 — Seek a separation bound for one graph family . . . 150
17.33 P33 — Verify concrete coefficient certificates in a kernel . . . 151
17.34 P34 — Formalize the visibility comparison boundary . . . 151
17.35 P35 — Release the first focused manuscript . . . 151
17.36 P36 — Choose the second paper from actual results . . . 151

Reading guide

This monograph develops a precise question: how much of a relation structure can be recovered from integer-height populations, and where does its first hidden difference appear in arithmetic? Chapters 1-4 establish the objects and finite observation hierarchy. Chapters 5-8 develop primewise assembly, local analytic comparison and singular synchronization. Chapters 9-11 give exact examples, inverse questions and the bridges to earlier work. Chapters 12-14 develop computational methodology, the research program and the literature. Chapter 15 adds context operators, certified cheapest experiments, exact robustness and the periodic residue sector. The implementation appendix records the software contracts and validated computational domains. The final roadmap preserves the original thirty-six development pushes; their completed acceptance records and subsequent research frontiers are maintained in the handoff ledger.

The central distinction is between an observation and the correspondence it forgets. A binary census can agree for every subset size and boundary size while a nested-height census differs. The arithmetic lift retains these local distinctions and turns the first discrepancy into a controlled ratio singularity. It adds analytic structure rather than new graph information. The full marked hierarchy explicitly contains the binary sector; complete graph reconstruction remains a research question.

Definitions and theorems are supplied with hypotheses and proofs. Finite graph calculations have explicit inputs and independently replayable outputs. Established antecedents are attributed in the literature chapter and bibliography. The book is a research foundation and executable companion, not a claim of a completed priority search or unrestricted reconstruction.

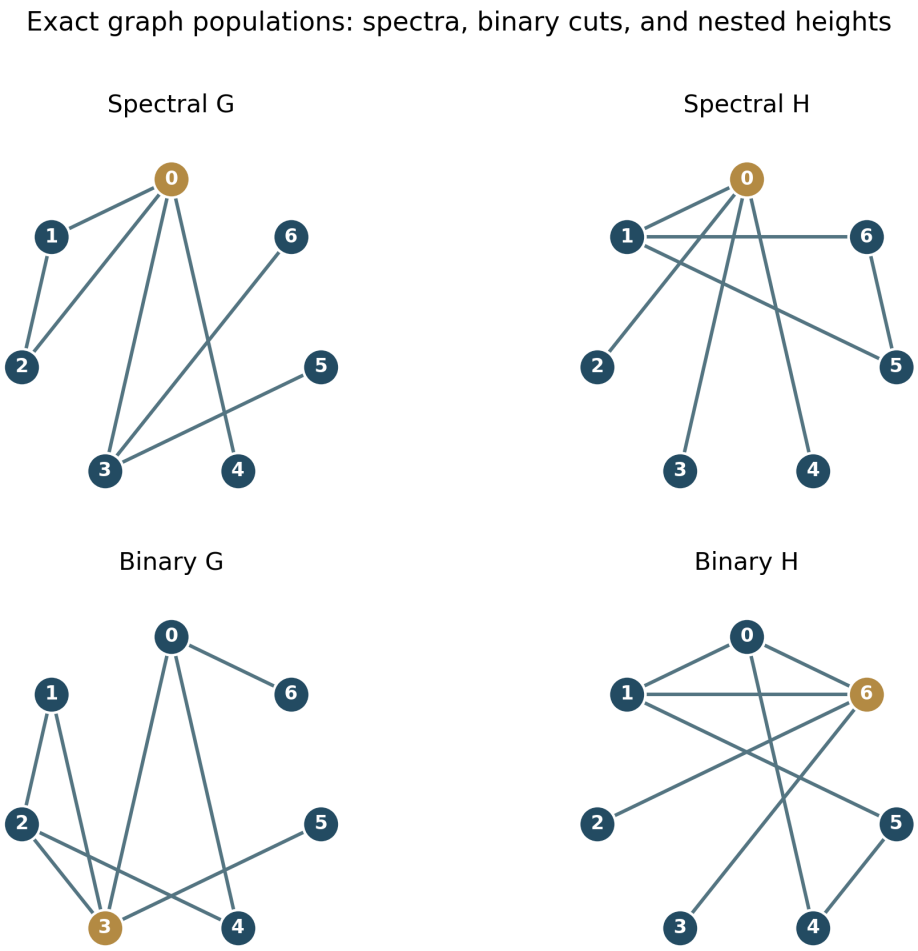


Figure 1: Exact reference graphs. Top: equal Laplacian spectra. Bottom: equal complete binary cut-volume census. Vertex labels remain original.

Chapter 1

What the population observes

1.1 A relation before a prediction

This book begins with a finite graph and a deliberately restricted experiment. Each vertex receives a nonnegative integer height. An edge pays the square of the difference between its endpoint heights. We count all height assignments having a specified total height and a specified energy. We then ask how much of the original relation between vertices survives in those counts. The question concerns the information contained in an exact mathematical observation. It does not require interpreting vertices as people, particles, assets, or physical locations.

The restriction is essential. If an experiment reports every vertex height and every labeled edge contribution, the graph can be read directly from the report. The interesting problem begins after that correspondence is removed. A histogram may tell us how many assignments have a certain energy without telling us which assignments they were. The resulting observation can preserve information that a familiar spectrum loses, while still losing information of its own. Establishing either possibility requires a proof or a concrete exact comparison.

There are consequently two distinct tasks. The forward task computes the population produced by a supplied graph. The inverse task decides what a population determines about its source. An efficient and correct forward computation does not settle the inverse question. Nor does a pair of distinguishable examples establish that every pair is distinguishable. Throughout the manuscript, reconstruction is a question to investigate, not an advertised theorem.

1.2 Objects, assignments, and the meaning of a coefficient

Let $G = (V, E)$ be a finite simple undirected graph with labeled vertex set

$$V = \{0, 1, \dots, k-1\}, \qquad k \geq 1.$$

An edge is an unordered pair of distinct vertices, and each unordered pair occurs at most once. Write d_i for the degree of vertex i, and $A_{ij} \in \{0, 1\}$ for the adjacency matrix. Connectedness will be required explicitly when needed. The basic formal counting series itself is defined for disconnected graphs as well.

A height profile is a vector $a = (a_0, \dots, a_{k-1}) \in \mathbb{Z}_{\geq 0}^k$. Its volume, energy, and span are

$$|a| = \sum_i a_i, \qquad E_G(a) = \sum_{\{i,j\} \in E} (a_i - a_j)^2, \qquad h(a) = \max_i a_i - \min_i a_i.$$

The word **volume** always means total integer height. It is not a Euclidean volume, a determinant, the number of occupied vertices, or the number of nonzero levels. A profile of volume four might put height four at one vertex, height two at two vertices, or height one at four vertices. These assignments generally have different energies.

Define the local series

$$L_G(x,q) = \sum_{a \in \mathbb{Z}^k_{\geq 0}} x^{|a|} q^{E_G(a)} = \sum_{m \geq 0} C_{G,m}(q) x^m.$$

The coefficient $[x^m q^e] L_G$ is the number of original labeled assignments of volume m and energy e. It does not count graph automorphism orbits. If a symmetry exchanges two assignments, both remain in the population. This convention makes coefficients directly comparable to enumeration and avoids inserting unannounced symmetry factors.

Theorem SNG-F01 (finite local coefficients). For every finite graph on $k \geq 1$ vertices and every $m \geq 0$, $C_{G,m}(q)$ is a polynomial in q with nonnegative integer coefficients. Its coefficient sum is

$$C_{G,m}(1) = \binom{m+k-1}{k-1}.$$

If $\Delta = \max_i d_i$, its degree is at most Δm^2.

Proof. The nonnegative integer solutions of $a_0 + \cdots + a_{k-1} = m$ are the weak compositions of m into k parts. The usual separator construction gives their number as the displayed binomial coefficient. In particular, there are finitely many terms in $C_{G,m}$. Every energy is a nonnegative integer. Moreover,

$$E_G(a) = \sum_i d_i a_i^2 - 2 \sum_{\{i,j\} \in E} a_i a_j \leq \Delta \sum_i a_i^2 \leq \Delta \Big(\sum_i a_i \Big)^2.$$

The first inequality uses nonnegativity of all heights; the second follows by expanding the square. This proves all assertions. □

Thus L_G is well defined as an element of $\mathbb{Z}[q][[x]]$. This assertion is formal: no choice of numerical x or q, and no convergence argument, is required. Later analytic interpretations must separately state the region in which a formal identity can be evaluated.

1.3 Removing a repetition that carries no relational information

Adding the same integer to every height leaves every edge difference unchanged:

$$E_G(a + t\mathbf{1}) = E_G(a), \qquad |a + t\mathbf{1}| = |a| + kt.$$

Every nonnegative profile has a unique decomposition

$$a = b + t\mathbf{1}, \qquad t = \min_i a_i, \qquad \min_i b_i = 0.$$

We therefore retain the normalized series

$$B_G(x,q) = \sum_{\substack{b \geq 0 \\ \min b = 0}} x^{|b|} q^{E_G(b)}.$$

The normalization removes the common-height tower, not an assignment-specific ambiguity. It retains one explicitly specified representative of each integer profile modulo common translation.

To distinguish assignments with one nonzero level from assignments with several levels, introduce the full marked series

$$\mathcal{B}_G(x,q,z) = \sum_{\substack{b\geq 0\\ \min b=0}} x^{|b|} q^{E_G(b)} z^{\max b}.$$

The symbol $B_G(x,q,z)$ may be used for this same three-variable object when its arguments are displayed. In these foundational chapters, $\mathcal{B}_G$ makes the extra mark visually explicit, while $B_G(x,q) = \mathcal{B}_G(x,q,1)$. On normalized profiles, maximum height equals span. Consequently z records a property that common translation does not change.

Theorem SNG-F02 (normalization and span marking). For every finite graph on $k \geq 1$ vertices,

$$B_G(x,q) = (1-x^k)L_G(x,q).$$

More generally, if

$$\widetilde{L}_G(x,q,z) = \sum_{a\geq 0} x^{|a|} q^{E_G(a)} z^{h(a)},$$

then

$$\mathcal{B}_G(x,q,z) = (1-x^k)\widetilde{L}_G(x,q,z).$$

Both normalized series have nonnegative integer coefficients despite the subtraction in these formulas.

Proof. Use the unique decomposition $a = b + t\mathbf{1}$. Energy and span are unchanged, whereas volume increases by kt. Hence $\widetilde{L}_G = \mathcal{B}_G \sum_{t\geq 0} x^{kt} = \mathcal{B}_G/(1-x^k)$. Specializing $z=1$ gives the first identity. Each coefficient of $\mathcal{B}_G$ counts an explicitly defined finite set, proving nonnegativity without relying on cancellation. □

Keeping k fixed, B_G and L_G contain exactly the same formal information: either is recovered from the other by multiplication or division by $1-x^k$. The span mark is a different matter. Specializing $z=1$ merges populations of different spans. We do not assume that this merging can be inverted.

Observation	Coefficient meaning	Correspondence deliberately removed
$L_G(x,q)$	All profiles by total height and energy	Identity of each profile
$B_G(x,q)$	Translation-normalized profiles by total height and energy	Identity of each normalized profile
$\mathcal{B}_G(x,q,z)$	Normalized profiles by total height, energy, and span	Identity of each profile within a marked cell
$[z]\mathcal{B}_G$	Binary profiles, excluding the two constant profiles	Identity of the vertex subset
$B_G(1,q)$, when defined	Translation classes by energy alone	Total height and span

1.4 Labeled populations can define unlabeled observations

Labels specify the population being counted, but the aggregate series does not privilege a particular labeling. Let $\sigma : V(G) \to V(H)$ be a graph isomorphism. Transport a profile by $a'_{\sigma(i)} = a_i$. This operation permutes the coordinates; it does not merge assignments.

Theorem SNG-F03 (isomorphism invariance with original multiplicity). Graph isomorphisms preserve L_G, B_G, and $\mathcal{B}_G$.

Proof. Coordinate permutation is a bijection on profiles and on normalized profiles. It preserves the sum, minimum, and maximum of their coordinates. Since it is a bijection on edges, it preserves the sum of squared differences. Each coefficient population is therefore carried bijectively to the corresponding population for H. □

The theorem proves one implication: isomorphic graphs have equal observations. It does not prove the converse. An invariant is allowed to identify nonisomorphic graphs. The inverse program investigates whether this particular invariant does so, and, if it does, which features remain recoverable despite that blindness.

For example, a single edge has energy $(a_0 - a_1)^2$. A normalized nonzero profile is either $(h, 0)$ or $(0, h)$, so

$$\mathcal{B}_{K_2}(x, q, z) = 1 + 2 \sum_{h \geq 1} x^h q^{h^2} z^h.$$

The coefficient two records the two original assignments. It is not replaced by one because the edge has an endpoint-exchanging symmetry. This elementary example is a useful check on every later convention.

For a slightly richer example, take the three-vertex path $0 - 1 - 2$. Its degrees are $(1, 2, 1)$, and direct enumeration gives

$$[x]B_G = 2q + q^2, \qquad [x^2]B_G = 2q + q^2 + 2q^4 + q^8.$$

At volume two, placing both units at an endpoint contributes q^4, placing both at the middle contributes q^8, and placing one unit at each of two vertices contributes the remaining terms. The full mark separates the two mechanisms:

$$[x^2]\mathcal{B}_G = z(2q + q^2) + z^2(2q^4 + q^8).$$

The span-one term consists of three binary subsets. The span-two term consists of three profiles concentrated at one vertex. Setting $z = 1$ adds these two populations without recording which mechanism produced each contribution.

At volume three, the unnormalized count contains the all-one profile, whose energy is zero. The normalized coefficient removes that one translation of the zero profile:

$$[x^3]B_G = 2q^2 + 4q^5 + 2q^9 + q^{18}.$$

Its coefficient sum is nine, compared with ten weak compositions of three into three parts. Thus normalization changes an original population by a precisely specified quotient rule; it does not arbitrarily discard an inconvenient solution. A useful calculation record should make this distinction visible whenever normalized and unnormalized outputs appear together.

For comparison, the triangle has

$$[x]B_{K_3} = 3q^2, \qquad [x^2]B_{K_3} = 3q^2 + 3q^8.$$

Already the first coefficient distinguishes the two graphs because their degree histograms differ. Such an example checks conventions but is not a demanding test of an inverse method. Later comparisons should deliberately match the simpler observations before attributing a discrepancy to a deeper mechanism.

1.5 What a result in this book means

Several mathematical activities appear together, and their conclusions should not be conflated. A proved identity between formal series applies coefficientwise to every allowed input. An exact finite computation establishes the cells actually enumerated. A comparison through volume M establishes equality or discrepancy through that volume. A numerical graph search may suggest a pattern without proving that no later counterexample exists. An analytic theorem requires its own convergence and continuation hypotheses even when its starting Euler factors have exact coefficients.

The book therefore separates mathematical propositions, software behavior, finite examples, and research targets. The numbered foundational statements below are proved from their displayed hypotheses. They are mathematical statements, not assertions that a proof assistant checked the implementation. Software validation supplies another kind of evidence: it tests whether a program realizes the stated mathematics on the cases examined.

Type of statement	Appropriate evidence	Scope of conclusion
Exact forward coefficient	Enumeration, recurrence, or an exact algebraic identity	The stated coefficient or family
Prefix agreement	Equal coefficient dictionaries through a declared volume	That finite prefix
Strict separation	One exact unequal coefficient	The two observations differ
Reconstruction theorem	A proof recovering isomorphism from equality	Every graph in the proved class
Reconstruction conjecture	A precise proposed implication	A question, not an established result
Analytic continuation	A convergent factorization on a stated domain	That domain and chosen branches

There is also a useful distinction between a new question for this project and a historically new mathematical construction. Integer graph lattices, Ising-type observations, graph partition functions, and inverse problems all have substantial literatures. The graph cut-lattice viewpoint belongs to established background, including [BHN1997]. Binary energy populations overlap directly with graph Ising polynomials with an external field; [AM2009] is relevant background. General graph reconstruction from families of observations must be compared carefully with the much richer families studied in [Lovasz2012]. No historical priority claim is needed to make the present calculations worthwhile.

The particular program here is to track exactly which relational information survives a succession of increasingly marked arithmetic height observations. Its central difficulty is **lost correspondence**. Counts can be exact while the identities responsible for them are absent. An inverse theorem must recover what that loss hides; an impossibility theorem must demonstrate that the lost information cannot be recovered from the declared observation. Both are substantive mathematical outcomes.

The next chapter puts the translation quotient on an integral lattice. The third chapter explains why multiple nested boundaries retain more information than a single binary cut and why repeated levels cannot be omitted. The fourth chapter turns coefficient order into a precise hierarchy of visibility, beginning with degrees and degree mixing and ending with the open reconstruction question.

Chapter 2

Graphs, integral cut lattices, and the height mark

2.1 The quadratic form and its kernel

Let G have adjacency matrix A, degree matrix $D = \mathrm{diag}(d_i)$, and combinatorial Laplacian $\Delta_G = D - A$. Expanding each square in the definition of energy gives

$$E_G(a) = a^\top \Delta_G a = \sum_i d_i a_i^2 - 2 \sum_{\{i,j\} \in E} a_i a_j.$$

The Laplacian is symmetric and positive semidefinite. This is stronger than a numerical spectral observation: the displayed edge-sum expression proves positivity for every real vector.

Theorem SNG-F04 (zero energy and component constants). For $a \in \mathbb{R}^k$, $E_G(a) = 0$ if and only if a is constant on each connected component. In particular, for connected G, $\ker \Delta_G = \mathbb{R}\mathbf{1}$.

Proof. A finite sum of nonnegative squares vanishes precisely when every summand vanishes. Thus adjacent vertices must have equal coordinates. Equality propagates along paths, proving constancy on every component. The converse follows because every edge lies within a component. Finally, $\Delta_G a = 0$ implies $a^\top \Delta_G a = 0$; component constancy also directly implies $\Delta_G a = 0$. □

For connected graphs, the zero direction is common translation. For disconnected graphs there are additional zero directions: different components can be shifted independently. This difference matters when a formal series is evaluated without its height mark. Quotienting only by global translation does not remove all zero modes of a disconnected graph.

2.2 An orientation is a bookkeeping choice

Choose an arbitrary orientation of every edge. Let ∂ be the $|E|$-by-k matrix whose row for an oriented edge $i \to j$ has -1 in column i, $+1$ in column j, and zero elsewhere. Then

$$(\partial a)_{i \to j} = a_j - a_i, \qquad \partial^\top \partial = \Delta_G.$$

The image

$$\mathcal{C}(G) = \partial \mathbb{Z}^k \subseteq \mathbb{Z}^E$$

is the integral cut, or tension, lattice in this convention. It consists of edge differences produced by integer vertex potentials. An arbitrary integer edge vector need not lie in this image: differences around every oriented cycle must sum to zero.

Reversing an edge orientation multiplies the corresponding coordinate by -1. This is an integral orthogonal transformation of $\mathbb{Z}^E$. It preserves inner products, squared lengths, and every energy count. An orientation is therefore needed to write vectors, but none of the observations depends on it.

Theorem SNG-F05 (integer quotient and cut-lattice coordinates). If G is connected, $a \mapsto \partial a$ induces an isomorphism of abelian groups

$$\mathbb{Z}^k/\mathbb{Z}\mathbf{1} \;\cong\; \mathcal{C}(G).$$

Under this isomorphism, squared lattice length is $E_G(a)$. If a root vertex r is fixed and $\Delta_G^{(r)}$ is obtained by deleting row and column r, the energy in rooted coordinates is the positive-definite integral form

$$u^\top \Delta_G^{(r)} u, \qquad u \in \mathbb{Z}^{k-1}.$$

Proof. The map is surjective onto its stated image. Its integer kernel consists of constant integer vectors by Theorem SNG-F04. The first isomorphism theorem gives the quotient description. The equality of squared length and energy follows from $\partial^\top \partial = \Delta_G$. Every integer translation class has a unique representative with coordinate $a_r = 0$, obtained by subtracting $a_r\mathbf{1}$. Substitution gives the reduced quadratic form. If that form vanishes, its full representative is constant and has root coordinate zero, so it is zero. The form is therefore positive definite. □

The root-zero representative is particularly convenient for linear algebra. It may have negative coordinates. The minimum-zero representative used by B_G is particularly convenient for a nonnegative height mark. These are two coordinate choices on the same quotient, not two different populations of translation classes.

2.3 A determinant that counts trees

The determinant of the reduced Laplacian gives a useful consistency check connecting the integer form with ordinary graph structure. We include a proof because it makes clear which combinatorial data enter the lattice and because no floating-point eigenvalue calculation is needed.

Theorem SNG-F06 (matrix-tree determinant). For connected G on $k \geq 2$ vertices and any root r,

$$\det \Delta_G^{(r)} = \tau(G),$$

where $\tau(G)$ is the number of spanning trees of G.

Proof. Delete column r of ∂, obtaining M. Then $\Delta_G^{(r)} = M^\top M$. The Cauchy–Binet formula gives

$$\det(M^\top M) = \sum_{F \subseteq E,\ |F|=k-1} \det(M_F)^2,$$

where M_F restricts M to the selected edge rows. If (V, F) is disconnected, choose a component not containing r. Its indicator vector, restricted to nonroot coordinates, is a nonzero vector killed by M_F, so the determinant is zero. A connected graph with

k vertices and $k-1$ edges is a tree. For a tree, the determinant is ± 1: repeatedly choose a nonroot leaf, expand in its column, and delete the leaf and its incident edge. Each expansion contributes a sign and an entry ± 1; the final empty determinant is one. Thus exactly the spanning-tree subsets contribute one to the sum. □

The theorem identifies a lattice determinant. It does not say that this determinant determines the lattice or the graph. Many nonisomorphic graphs have the same number of spanning trees. Even a complete list of real Laplacian eigenvalues describes the quadratic form only up to real orthogonal change of coordinates, whereas an integer height population depends on where the integer lattice sits inside that real space.

This distinction is elementary but decisive. A real orthogonal matrix preserves lengths of all real vectors. It need not map $\mathbb{Z}^{k-1}$ to itself. An integral change of lattice basis must lie in $\mathrm{GL}_{k-1}(\mathbb{Z})$, a much more restrictive group. Consequently equal real spectra alone do not supply a bijection between integer height profiles.

2.4 Unmarked theta and the extra information in a mark

For connected G, the unmarked cut-lattice theta series is

$$\Theta_G(q) = \sum_{u \in \mathbb{Z}^{k-1}} q^{u^\top \Delta_G^{(r)} u}.$$

Changing the root changes the integral coordinates but not the translation classes being counted. Likewise, an integral isometry of cut lattices preserves this series. Graph cut lattices and their relationship with graph structure have a substantial established theory; [BHN1997] provides relevant background. No converse classification from theta-series equality is assumed here.

Theorem SNG-F07 (finite energy shells and convergence). If G is connected and $k \geq 2$, every energy coefficient of Θ_G is finite, and the series converges absolutely for $|q| < 1$. Moreover,

$$\Theta_G(q) = B_G(1, q)$$

as a coefficientwise formal energy series and, for $|q| < 1$, as a convergent sum.

Proof. In root-zero coordinates, join r to each vertex i by a simple path of length at most $k-1$. The telescoping sum of edge differences and Cauchy–Schwarz give

$$u_i^2 \leq (k-1)E_G(u), \qquad \sum_{i \neq r} u_i^2 \leq (k-1)^2 E_G(u).$$

An energy shell $E_G(u) = e$ therefore lies in a finite integer box. If $0 < \rho < 1$, then

$$\rho^{E_G(u)} \leq \rho^{\|u\|_2^2/(k-1)^2},$$

and the sum of the right side is a product of convergent one-dimensional Gaussian sums. The case $q = 0$ is immediate. Finally, root-zero and minimum-zero profiles each select exactly one representative of every translation class and preserve energy. Their energy sums coincide. □

For a disconnected graph with at least two nonempty components, $B_G(1, q)$ generally diverges, even for $0 < q < 1$: set one component to height zero and another to arbitrary common height. These are distinct normalized zero-energy profiles. The true cut lattice quotients by componentwise constants, while our normalization quotients by one global constant. Those constructions must not be silently identified.

The total-height mark retains a feature that the lattice norm alone does not name. For an arbitrary integer representative a, define

$$\mu([a]) = \sum_i a_i - k \min_i a_i, \qquad \eta([a]) = \max_i a_i - \min_i a_i.$$

Both are well defined on translation classes. They equal volume and span on the minimum-zero representative. The marked theta population is therefore a lattice together with extra functions on its elements. An abstract norm-preserving lattice bijection need not preserve those functions.

2.5 The height mark is an asymmetric gauge

The function μ is not a linear functional on the quotient. A change in which vertex attains the minimum changes its formula. This nonlinearity is part of what the mark observes, rather than a defect in the normalization.

Theorem SNG-F08 (gauge identities and complement reciprocity). On $\mathbb{Z}^k/\mathbb{Z}\mathbf{1}$, the functions μ and η satisfy

$$\mu(\alpha+\beta) \le \mu(\alpha) + \mu(\beta), \qquad \mu(t\alpha) = t\mu(\alpha) \quad (t \in \mathbb{Z}_{\ge 0}),$$

$$\mu(\alpha) + \mu(-\alpha) = k\eta(\alpha).$$

For fixed span h, put $B_{G,h}(x,q) = [z^h]\mathcal{B}_G(x,q,z)$. This is a polynomial and obeys

$$B_{G,h}(x,q) = x^{kh} B_{G,h}(x^{-1},q).$$

Proof. For representatives a, b, $\min_i(a_i+b_i) \ge \min_i a_i + \min_i b_i$; subtracting k times this inequality from the coordinate sum proves subadditivity. Nonnegative integer homogeneity is immediate. Also $\min(-a) = -\max(a)$, so adding the formulas for $\mu([a])$ and $\mu([-a])$ gives $k(\max a - \min a)$. For reciprocity, every normalized span-h profile has coordinates in $\{0,\dots,h\}$, so the set is finite. The involution $a \mapsto h\mathbf{1} - a$ preserves energy, exchanges minimum and maximum, and changes volume from m to $kh - m$. Summing over the involution proves the identity. □

The reciprocity is spanwise. After setting $z = 1$, profiles with many different h are merged, and there is no single factor x^{kh} applying to their sum. This is a concrete example of a relation visible with the full mark but not immediately expressible after that mark is forgotten.

Structure retained	Natural coordinate change	Quantity necessarily preserved
Real quadratic space	Real orthogonal isometry	Real energy
Integral cut lattice	Integral lattice isometry	Integer energy shells
Height-marked lattice population	Bijection preserving energy and μ	Joint volume-energy counts
Fully marked population	Bijection preserving energy, μ, and η	Joint volume-energy-span counts

These rows are preservation requirements, not a claim that every bijection in a lower row arises from graph relabeling. The inverse problem asks precisely how much of the source graph is forced by increasingly restrictive observations.

2.6 Two endpoint calculations

At $q = 1$, energy is forgotten completely. Every coordinate contributes its own geometric series, so

$$L_G(x,1) = \frac{1}{(1-x)^k}, \qquad B_G(x,1) = \frac{1-x^k}{(1-x)^k}.$$

No edge appears in these expressions. At $q = 0$, only zero-energy assignments remain. If the connected components have sizes $k_1, \dots, k_c$, Theorem SNG-F04 gives

$$L_G(x,0) = \prod_{j=1}^{c} \frac{1}{1-x^{k_j}}, \qquad B_G(x,0) = (1-x^k)\prod_{j=1}^{c} \frac{1}{1-x^{k_j}}.$$

For connected graphs this reduces to $L_G(x,0) = 1/(1-x^k)$ and $B_G(x,0) = 1$. Thus both endpoints erase most graph structure: one ignores every edge energy, and the other retains only exact agreement within components. An inverse experiment must retain intermediate energy information rather than relying on these endpoint values alone.

2.7 Disjoint unions and recoverable component sizes

The unnormalized series factors naturally over independent components. Normalization by one global minimum does not factor in the same way, because choosing the minimum of the combined profile couples the choice of representatives in its components. This is a useful instance in which an exact quotient must be carried through an otherwise simple product rule.

Theorem SNG-F19 (component factorization and its endpoint inverse). For graphs G, H on $k, \ell \geq 1$ vertices,

$$L_{G \sqcup H}(x,q) = L_G(x,q) L_H(x,q),$$

$$B_{G \sqcup H}(x,q) = \frac{1-x^{k+\ell}}{(1-x^k)(1-x^\ell)} B_G(x,q) B_H(x,q).$$

Moreover, the zero-energy series $L_G(x,0)$ determines the multiset of connected-component sizes.

Proof. A profile on a disjoint union is uniquely a pair of profiles on its components. Both volume and energy add, proving the first identity. Substituting $L_G = B_G/(1-x^k)$, and the corresponding formulas for H and the union, proves the second identity in formal series. For the final assertion, let c_j be the number of components of size j. Then

$$L_G(x,0) = \prod_{j \geq 1} (1-x^j)^{-c_j}.$$

Formal logarithms are legitimate because the constant term is one. Expanding $-\log(1-x^j) = \sum_{r \geq 1} x^{jr}/r$, put $b_n = n[x^n] \log L_G(x,0)$. Then

$$b_n = \sum_{j \mid n} j c_j.$$

Recover $c_1 = b_1$ and recursively

$$c_n = \frac{1}{n}\left(b_n - \sum_{\substack{j \mid n \\ j<n}} j c_j\right).$$

This uniquely determines every component-size multiplicity. Only finitely many are nonzero. □

The inverse statement is deliberately modest. Zero energy knows component sizes but not the edges within a component. A path and a complete graph on the same connected vertex set have identical zero-energy series. The formal logarithm extracts exactly the available component information without pretending that it contains finer internal structure.

There is a related warning about the full span mark. The span of a concatenated pair of arbitrary profiles depends on their relative vertical placement, not just on the two individual spans. Consequently one should not multiply $\mathcal{B}_G$ and $\mathcal{B}_H$ and declare the result to be $\mathcal{B}_{G \sqcup H}$. A correct marked factorization would need enough additional data to retain minima, maxima, and their relative shifts. The scalar unnormalized product avoids that missing correspondence because it sums actual coordinates before applying a quotient.

The cut-lattice description clarifies why integer heights are a useful meeting point. They retain the arithmetic placement of a quadratic form, while their minimum-zero representatives supply a nonlinear order mark. The next chapter expresses that order mark through nested threshold sets. This reveals precisely what lies beyond a single binary cut.

Chapter 3

Nested heights and the interaction of boundaries

3.1 A height profile is a stack of threshold sets

A binary profile chooses a subset of vertices: height one on the subset and zero outside. A general integer profile chooses several nested subsets. This description is exact, but only if consecutive threshold sets are allowed to be equal. A jump of three at one vertex crosses three integer levels even if the set of vertices above those levels never changes.

For a normalized profile $a \geq 0$ with $\min a = 0$ and $\max a = h$, define

$$S_t(a) = \{i \in V : a_i \geq t\}, \qquad 1 \leq t \leq h.$$

Then

$$\emptyset \neq S_h \subseteq S_{h-1} \subseteq \cdots \subseteq S_1 \subsetneq V, \qquad a = \sum_{t=1}^{h} \mathbf{1}_{S_t}.$$

The inclusions are weak. Replacing them by strict inclusions changes the population by discarding legitimate profiles.

Theorem SNG-F09 (threshold-chain bijection). Normalized profiles of span $h \geq 1$ are in bijection with weakly nested chains of h nonempty proper vertex subsets. The inverse map is $a_i = \#\{t : i \in S_t\}$, and

$$|a| = \sum_{t=1}^{h} |S_t|.$$

Proof. Threshold sets of a profile have the stated nesting. Because some coordinate is zero, S_1 is proper; because some coordinate is h, S_h is nonempty. Conversely, nesting implies that, for each vertex, membership occurs at an initial segment of the thresholds. Thus the formula for a_i gives exactly the original membership sets when thresholds are taken again. A vertex outside S_1 has height zero and a vertex in S_h has height h, proving normalization and span. The volume identity follows by counting all incidences (i, t) with $i \in S_t$ in two orders. □

The chain is not an auxiliary encoding with multiple extensions. It is uniquely determined by the original profile, and it reconstructs that profile uniquely. Consequently a chain-based counting algorithm preserves original assignment multiplicity without an additional correction factor.

3.2 Squared differences do not add level by level

Write $c_G(S) = |\partial S|$ for the number of edges with one endpoint in S and one outside. For disjoint vertex sets U, W, write $e_G(U, W)$ for the number of edges joining them. The ordinary integer coarea identity says

$$\sum_{\{i,j\}\in E} |a_i - a_j| = \sum_{t=1}^{h} c_G(S_t).$$

Our energy uses squares, so this identity alone is insufficient. An edge crossing two levels contributes four, not two. Cross terms record which nested boundaries are crossed by the same edge.

Theorem SNG-F10 (nested-boundary energy formula). For the threshold chain of a normalized profile,

$$E_G(a) = \sum_{t=1}^{h} c_G(S_t) + 2 \sum_{1\le r<t\le h} e_G(S_t, V \setminus S_r).$$

Equivalently,

$$E_G(a) = \sum_{r,t=1}^{h} e_G(S_{\max(r,t)}, V \setminus S_{\min(r,t)}).$$

Proof. Use the oriented edge-difference matrix ∂ from Chapter 2 and substitute $a = \sum_t \mathbf{1}_{S_t}$ into $E_G(a) = \|\partial a\|_2^2$. Diagonal inner products equal $c_G(S_t)$. If $S_t \subseteq S_r$, an edge contributes to both gradients exactly when it joins S_t to $V \setminus S_r$. On such an edge the two gradient signs agree, irrespective of the chosen orientation; otherwise at least one gradient is zero. Their inner product is therefore $e_G(S_t, V \setminus S_r)$. Expanding the squared norm proves the first formula, and pairing the two orders of each off-diagonal term proves the second. □

An edgewise proof gives the same mechanism. If endpoint heights are $u < v$, the edge crosses exactly $v - u$ thresholds. The ordered pairs of crossed thresholds number $(v - u)^2$. The double sum counts those pairs edge by edge. This argument also explains why repeated threshold sets must remain: an edge can cross several numerically different thresholds whose vertex subsets happen to coincide.

The formula concerns pairwise edge energies throughout. The cross terms are interactions between observations of nested boundaries; they do not turn the graph Hamiltonian into a new three-body or higher-body edge law. Several boundaries can jointly reveal information that isolated boundary histograms lose, even though the underlying energy remains a sum over edges.

3.3 Two examples where repetition matters

Consider the path $0 - 1 - 2$. For $a = (0, 2, 2)$, the threshold chain is

$$S_1 = S_2 = \{1, 2\}.$$

Each set has cut size one. The sum of the two cut sizes is two, but the actual energy is $(0 - 2)^2 + (2 - 2)^2 = 4$. The cross term contributes the missing two, because one edge joins S_2 to $V \setminus S_1$. A strict-chain enumeration would omit this profile entirely.

For $a = (0, 1, 3)$, the same path gives

$$S_1 = \{1, 2\}, \qquad S_2 = S_3 = \{2\}.$$

The diagonal cut sum is three. The boundaries of S_1 and S_2 lie on different edges, so their overlap term is zero; the same holds for S_1, S_3. The repeated pair S_2, S_3 contributes two. The total is five, agreeing with $1^2 + 2^2$.

Path profile	Volume	Span	Diagonal cut sum	Cross-term contribution	Energy
$(0,1,1)$	2	1	1	0	1
$(0,2,2)$	4	2	2	2	4
$(0,1,3)$	4	3	3	2	5
$(0,1,2)$	3	2	2	0	2

The last row is also useful. Distinct boundaries can have no overlap, making the squared energy equal the coarea sum. There is no universal rule replacing the cross terms by a fixed function of the individual cut sizes. Their value depends on how the boundaries sit together in the same graph.

A four-cycle makes this failure especially explicit while holding both individual sizes and individual cut counts fixed. Let the cycle be $0-1-2-3-0$, and let $S = \{0, 1, 2\}$. Consider either $T = \{0\}$ or $T' = \{1\}$ as the inner threshold set. Both chains have outer size three, inner size one, and cut counts two and two. However,

$$e_G(T, V \setminus S) = 1, \qquad e_G(T', V \setminus S) = 0.$$

The two profiles are $(2, 1, 1, 0)$ and $(1, 2, 1, 0)$; both have volume four and span two. Their energies are respectively six and four. Thus even the ordered list of threshold sizes and cut counts does not determine the height energy. One must also retain which boundary edges are shared.

This example does not prove that two graphs with equal complete binary cut polynomials must have different height series. Both profiles occur in the same graph, and a global inverse implication could in principle use information unavailable in this two-boundary summary. What the example proves is the narrower and useful fact that the displayed chain energy cannot be computed from its separate boundary marginals alone.

3.4 An exact chain expansion of the marked series

Combining the bijection and energy formula gives

$$\mathcal{B}_G(x, q, z) = 1 + \sum_{h \geq 1} z^h \sum_{\emptyset \neq S_h \subseteq \cdots \subseteq S_1 \subsetneq V} x^{\sum_t |S_t|} q^{\sum_t c_G(S_t) + 2\sum_{r<t} e_G(S_t, V \setminus S_r)}.$$

This is an identity of formal series. At a specified total volume m, only chains of length at most m can contribute, so the coefficient remains a finite sum. The expression is also a useful specification for an independent implementation: direct weak-composition enumeration and threshold-chain enumeration must return the same marked cells.

Proposition SNG-F21 (volume-pruned chain enumeration). Fix $M \geq 0$. The following search visits every normalized profile of volume at most M exactly once:

emit the empty chain; for every nonempty proper $S_1 \subsetneq V$ with $|S_1| \leq M$, emit (S_1) and recursively extend a chain $(S_1, \dots, S_t)$ by every nonempty $S_{t+1} \subseteq S_t$, equal sets included, with $\sum_{r \leq t+1} |S_r| \leq M$. If each appended level is charged $c_G(S_{t+1}) + 2\sum_{r \leq t} e_G(S_{t+1}, V \setminus S_r)$, the accumulated charge is the energy of the profile, the accumulated size is its volume and the chain length is its span.

Proof. Every emitted sequence is a weakly nested chain of nonempty sets with proper first set, so SNG-F09 maps it to a normalized profile of span equal to its length and volume equal to its total size. Conversely the threshold chain of a normalized profile of volume at most M satisfies these constraints, and each of its prefixes has total size at most M, because sizes are positive; the search therefore reaches it through its prefixes. Distinct emitted chains differ in some level and hence give distinct profiles, since SNG-F09 is a bijection; and the search emits each chain once, because the chain determines the sequence of choices that produced it. Termination follows because each level has size at least one, so the length is at most M. The charge identity is SNG-F10 grouped by the later index t of each pair $r < t$: appending S_{t+1} adds its own cut and twice its overlap with every earlier level, and with no other pair. □

Repetitions are essential. Forbidding $S_{t+1} = S_t$ removes exactly the profiles in which some height between 0 and the span is unused, for instance $(0, 2, 2)$ and $(0, 1, 3)$ on the path. The search is implemented in `src/singularity/chains.py`. It shares neither its generator nor its energy evaluation with the weak-composition engine, so agreement of the two on complete marked histograms is a genuine cross-check. For every labeled simple graph on at most five vertices and every volume at most six, the two engines agree on all 142,416 marked cells and 479,963 profiles (`results/chain_agreement.json`). This is a finite computation on that declared population, not a proof beyond it.

The expansion shows exactly what a binary observation omits. The binary sector reports one subset at a time, through its size and cut. A chain term refers to several nested subsets in the same graph and additionally uses their boundary-overlap counts. Separate histograms of subset sizes and cuts do not retain the correspondence needed to form these overlaps.

This statement is directional. The full marked series certainly contains the binary histogram because its span-one coefficient extracts it. It does not follow that the two-variable specialization $B_G(x, q) = \mathcal{B}_G(x, q, 1)$ permits that same extraction. After span is forgotten, a cell of given volume and energy can receive both binary and nonbinary contributions. No inversion of that mixture is proved here.

3.5 The binary sector and its reciprocity

Define the binary cut polynomial

$$I_G(x, q) = \sum_{S \subseteq V} x^{|S|} q^{c_G(S)}.$$

Its height interpretation overlaps the external-field Ising polynomial studied in [AM2009], subject to the stated normalization of the energy and spin variables. The present definition is explicit, so no external convention is needed to compute it.

Theorem SNG-F11 (binary sector extraction). For $k \geq 2$,

$$[z]\mathcal{B}_G(x, q, z) = I_G(x, q) - 1 - x^k.$$

Also $I_G(x, q) = x^k I_G(x^{-1}, q)$.

Proof. A minimum-zero span-one profile takes values zero and one and contains both values. It is uniquely the indicator of a nonempty proper subset. Its volume is subset size and its energy is cut size. The omitted empty and full subsets give 1 and x^k. Complementation preserves cut size and sends subset size to $k - |S|$, proving reciprocity. □

The symmetry pairs the two sides of a cut. We count both sides as different profiles. A side of size r and its complement of size $k - r$ appear at their respective powers of x; when the sizes are equal they still contribute two unless other multiplicities happen to coincide.

3.6 Minimum cuts are the first positive energy shell

For a connected graph on $k \geq 2$ vertices, define edge connectivity by

$$\lambda(G) = \min_{\emptyset \neq S \subsetneq V} c_G(S) > 0.$$

This definition suffices here; no equivalence with a disjoint-path formulation is needed.

Theorem SNG-F12 (minimum-energy shell). Every nonzero normalized profile of span h obeys

$$E_G(a) \geq \lambda(G)h.$$

The smallest positive energy is $\lambda(G)$, and its full marked population is

$$[q^{\lambda(G)}]\mathcal{B}_G(x, q, z) = z \sum_{\substack{\emptyset \neq S \subsetneq V \\ c_G(S) = \lambda(G)}} x^{|S|}.$$

Proof. For every nonnegative integer d, $d^2 \geq d$. Hence the energy is at least the absolute-difference sum, which by coarea is $\sum_t c_G(S_t)$. Every threshold set is nonempty and proper, so each cut is at least $\lambda(G)$. This gives the bound. An indicator of a minimum cut has energy $\lambda(G)$, so the bound is attained. If a nonzero profile has that energy, the bound forces $h = 1$, and then its subset must be a minimum cut. □

For a path, $\lambda = 1$, so the first positive shell consists of the two orientations of every bridge cut. For a triangle, $\lambda = 2$, and its six nonempty proper subsets give

$$[q^2]\mathcal{B}_{K_3}(x, q, z) = 3z(x + x^2).$$

These formulas identify the very first nonconstant energy sector. They do not recover arbitrary boundary overlaps at higher energy.

Both hypotheses are needed. For a single vertex there is no nonempty proper subset, every normalized profile is zero, and $\mathcal{B}_{K_1} = 1$. For a disconnected graph the minimum cut is zero, and the indicator of a component is a nonzero normalized profile of energy zero. For $K_2 \sqcup K_1$, the profile $(1, 1, 0)$ has energy 0, and the smallest positive energy, 1, is attained by $(1, 0, 0)$, which is not a minimum-cut side. So the shell statement fails without connectedness. Both sides of every minimum cut are counted, at volumes $|S|$ and $k - |S|$, and the shell is symmetric under $r \mapsto k - r$ by SNG-F11. The theorem orders profiles by energy. It says nothing directly about the first discrepancy of two graphs ordered by volume, which can occur at energies far above λ. Its formal ingredient, the nested identity $E = \sum_r c(S_r) + 2\sum_{r<s}(\text{cross terms})$ with nonnegative cross terms, is kernel-checked as `energy_nested` in `formal/Singularity/Thresholds.lean` (P08).

3.7 Support restrictions useful for exact computation

Several finite bounds follow directly from the profile interpretation. They identify impossible marked cells before any enumeration and provide consistency checks afterward.

Theorem SNG-F13 (volume, span, and extreme energy). Let $k \geq 2$, and let a be a nonzero normalized profile of volume m and span h. Then

$$h \leq m \leq (k-1)h.$$

If G is connected,

$$E_G(a) \geq \max\left\{\lambda(G)h, \left\lceil \frac{h^2}{\operatorname{diam}(G)} \right\rceil\right\}.$$

For any graph with maximum degree $\Delta > 0$, the largest energy among normalized volume-m profiles is Δm^2, attained exactly by $a = me_i$ at maximum-degree vertices.

Proof. Some coordinate equals h, so the sum is at least h. At least one coordinate is zero and every other coordinate is at most h, giving the upper bound. The minimum-cut bound was proved above. Choose a shortest path between a minimum-height and a maximum-height vertex. If its length is ℓ, telescoping and Cauchy–Schwarz give $h^2 \leq \ell \sum_{e \text{ on path}} (\partial a)_e^2 \leq \operatorname{diam}(G) E_G(a)$. Energy is integral, giving the ceiling. Finally, the bound $E_G(a) \leq \Delta \sum_i a_i^2 \leq \Delta m^2$ was proved in Chapter 1. The profile me_i at a maximum-degree vertex attains it. If equality holds and $\Delta > 0$, then $\sum_i a_i^2 = m^2$. Expanding m^2 shows that all products $a_i a_j$, $i \neq j$, vanish; exactly one coordinate is nonzero. Its degree must be Δ. □

Thus only

$$\left\lceil \frac{m}{k-1} \right\rceil \leq h \leq m$$

can occur in a nonzero volume-m normalized population. For fixed h, complementation from Chapter 2 pairs volumes m and $kh - m$. A computation violating either restriction has lost a normalization condition or a multiplicity.

Nested thresholds provide the human explanation for the hierarchy studied next. One occupied unit reveals a degree. Two units reveal degree mixing. Several units can occupy different heights and thereby test several linked boundaries. The information comes from their shared placement in one relation, not from replacing the original edge law with an unrelated interaction.

3.8 Primitive height scale and exact dilation

Repeated sets also encode a common integer scale. If every nonzero coordinate of a normalized profile is divisible by d, division by d gives another normalized profile. Volumes and spans divide by d, while energy divides by d^2. This is an exact quadratic scaling law, not an approximation valid only at large height.

Call a nonzero normalized profile primitive when the greatest common divisor of its coordinates is one; zeros do not change that greatest common divisor. Let

$$\mathcal{P}_G(x, q, z) = \sum_{\substack{a \geq 0,\ \min a = 0 \\ a \neq 0,\ \gcd(a_0, \ldots, a_{k-1}) = 1}} x^{|a|} q^{E_G(a)} z^{\max a}.$$

The zero profile is excluded because its common divisor is not a positive scale.

Theorem SNG-F20 (primitive-height decomposition). For every graph on $k \geq 2$ vertices,

$$\mathcal{B}_G(x, q, z) - 1 = \sum_{d \geq 1} \mathcal{P}_G(x^d, q^{d^2}, z^d).$$

If μ is the arithmetic Möbius function, characterized by $\sum_{d|n} \mu(d) = 1$ for $n = 1$ and zero otherwise, then

$$\mathcal{P}_G(x, q, z) = \sum_{d \geq 1} \mu(d)(\mathcal{B}_G(x^d, q^{d^2}, z^d) - 1).$$

Both identities are coefficientwise finite in every positive volume.

Proof. Every nonzero integer profile has a unique positive greatest common divisor d, and a unique primitive quotient $b = a/d$. Normalization is preserved. Direct substitution gives $|a| = d|b|$, $E_G(a) = d^2 E_G(b)$, and $\max a = d \max b$, proving the first identity. For inversion, write $T_d F = F(x^d, q^{d^2}, z^d)$; then $T_d T_e = T_{de}$. Substitute the first identity in the proposed inverse. The coefficient multiplying $T_n \mathcal{P}_G$ is $\sum_{d|n} \mu(d)$, leaving only $n = 1$. At volume $m > 0$, only $d \mid m$ can contribute, so the rearrangement is finite coefficientwise. □

For the path example $(0, 2, 2)$, the primitive quotient is $(0, 1, 1)$: volume two becomes four, energy one becomes four, and span one becomes two. This decomposition distinguishes a genuinely new shape of height profile from an integral dilation of an earlier shape. It provides another exact observation hierarchy that can be implemented independently of threshold-chain enumeration.

Primitive height here refers only to common divisibility of integer coordinates. It does not mean a primitive graph, a primitive conductor phase, or an element primitive in a different lattice. Those notions may interact in later calculations, but their definitions must not be merged because they share a word. The decomposition itself uses no connectedness assumption, so it also applies when nonconstant componentwise profiles have zero energy.

Chapter 4

Visibility depth and the loss of correspondence

4.1 Increasing volume as an observation hierarchy

The coefficient $C_{G,m}(q) = [x^m]L_G(x, q)$ reports every energy attainable with exactly m units of nonnegative integer height, together with original assignment multiplicities. Increasing m enlarges the set of placements that can be tested. It does not simply increase numerical precision in an unchanged observation. New occupancy patterns become available: repeated occupation of one vertex, occupation of several vertices, and different heights on linked supports.

For equal-order graphs, normalization does not change the first place at which their scalar-height observations differ. If k is fixed, multiplication by $1 - x^k$ has constant term one. A first nonzero coefficient cannot be removed by that multiplication because no earlier nonzero coefficient exists to cancel it.

The hierarchy must be specified before it is interpreted. A volume prefix of the two-variable series is one hierarchy; a span prefix of the fully marked series is another. The binary sector is one span layer, not one volume layer: a binary subset of size ten has volume ten and span one. Conversely, a one-vertex profile of height ten has the same volume and span ten. Keeping those distinctions prevents an observation about a binary cut polynomial from being silently transferred to an unmarked height specialization.

4.2 The first coefficients can be read by hand

Write e_i for the unit profile at vertex i. At volume one the only profiles are e_i, and

$$E_G(e_i) = d_i.$$

At volume two the profiles are $2e_i$ and $e_i + e_j$, $i < j$. The latter subtracts two from the sum of endpoint degrees precisely when the endpoints are adjacent. This is the first layer that couples local degrees to a relation between vertices.

Theorem SNG-F14 (coefficients through volume three). For every simple graph,

$$C_{G,0}(q) = 1, \qquad C_{G,1}(q) = \sum_i q^{d_i},$$

$$C_{G,2}(q) = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j-2A_{ij}},$$

and

$$C_{G,3}(q) = \sum_i q^{9d_i} + \sum_{i\neq j} q^{4d_i+d_j-4A_{ij}} + \sum_{i<j<\ell} q^{d_i+d_j+d_\ell-2e_G(\{i,j,\ell\})}.$$

Here $e_G(S)$ denotes the number of edges induced by S. Empty index sums are zero. For normalized coefficients,

$$[x^m]B_G = C_{G,m} - C_{G,m-k},$$

where $C_{G,t} = 0$ for $t < 0$.

Proof. Partition the weak compositions of one, two, and three by their positive parts. The possibilities are respectively 1; 2 or $1+1$; and 3, $2+1$, or $1+1+1$. Substitute each profile in

$$E_G(a) = \sum_i d_i a_i^2 - 2 \sum_{\{i,j\}\in E} a_i a_j.$$

The pattern $2e_i + e_j$ is indexed by ordered distinct vertices because the heights distinguish the two positions. The three-unit pattern is indexed by unordered triples. These are disjoint exhaustive cases, giving the formulas without missing multiplicities. The normalized identity is coefficient extraction from Theorem SNG-F02. □

The classification can be read as a table. Write N_d for the number of vertices of degree d, $P^{\epsilon}_{a,b}$ for the number of **ordered** pairs (i,j), $i \neq j$, with $d_i = a$, $d_j = b$, $A_{ij} = \epsilon$, and $T_{s,t}$ for the number of unordered triples with degree sum s spanning t edges. These statistics are defined from degrees and adjacency alone.

Volume	Pattern	Index set	Support	Span	Energy
1	e_i	vertices	1	1	d_i
2	$2e_i$	vertices	1	2	$4d_i$
2	e_i+e_j	unordered pairs	2	1	$d_i+d_j-2A_{ij}$
3	$3e_i$	vertices	1	3	$9d_i$
3	$2e_i+e_j$	ordered pairs	2	2	$4d_i+d_j-4A_{ij}$
3	$e_i+e_j+e_k$	unordered triples	3	1	$d_i+d_j+d_k-2e_G(\{i,j,k\})$

The corresponding coefficient contributions are separated below so each formula remains readable.

Volume	Pattern	Contribution
1	e_i	$\sum_d N_d q^d$
2	$2e_i$	$\sum_d N_d q^{4d}$
2	e_i+e_j	$\frac{1}{2}\sum P^{\epsilon}_{a,b} q^{a+b-2\epsilon}$
3	$3e_i$	$\sum_d N_d q^{9d}$
3	$2e_i+e_j$	$\sum P^{\epsilon}_{a,b} q^{4a+b-4\epsilon}$
3	$e_i+e_j+e_k$	$\sum T_{s,t} q^{s-2t}$

A pattern lies in the normalized population exactly when its support is smaller than k; then its span is the displayed value. For $k \leq 3$ some patterns cover every vertex

and are translations instead: the profile $(1,1)$ on two vertices, $(2,1)$ and $(1,2)$ on two vertices, and $(1,1,1)$ on three. The normalized identity removes exactly these. The table is implemented without profile enumeration in `src/singularity/lowvolume.py`. `tests/test_low_volume.py` compares its local, normalized and marked outputs with both the weak-composition and the level-chain enumerators on all 75 labeled graphs with at most four vertices, on the fixed pairs, and on regular graphs including the cube, the Möbius ladder, the Petersen graph, K_4, $K_{3,3}$, C_5 and two disjoint triangles, where the closed form in terms of order, degree and triangle count (Proposition SNG-I008) is also checked.

For example, $C_{G,1}$ determines the degree histogram exactly: its coefficient of q^d is the number of vertices of degree d. It does not identify which vertex has each degree, because coordinate labels have been summed out. That missing assignment of degrees to positions is already a small instance of lost correspondence.

Normalization matters at low volume in small graphs. For $k=2$, $[x^2]B_G = C_{G,2} - 1$, removing the constant profile $(1,1)$. For $k=3$, the corresponding first subtraction occurs at volume three. The uniform formula is safer than assuming that normalized and unnormalized coefficients agree at every displayed low order.

4.3 What the second coefficient adds

Define the edge degree-sum polynomial

$$J_G(q) = \sum_{\{i,j\}\in E} q^{d_i+d_j}.$$

This is one measure of how degrees mix across edges. It is coarser than the full joint degree matrix because different degree pairs can have the same sum, but it is already more relational than the degree histogram alone.

Theorem SNG-F15 (second-layer degree mixing). For every graph,

$$C_{G,2}(q) = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j} + (q^{-2}-1)J_G(q).$$

The expression is a polynomial: every edge has $d_i + d_j \geq 2$. For graphs G, H with the same degree histogram,

$$C_{G,2} = C_{H,2} \quad\Longleftrightarrow\quad J_G = J_H.$$

Proof. Separate the pair sum in Theorem SNG-F14 into edges and nonedges. Start by assigning $q^{d_i+d_j}$ to every pair; on an edge replace it by $q^{d_i+d_j-2}$. The change is the displayed correction. If degree histograms agree, both of the first two sums agree, so the difference of second coefficients is $(q^{-2}-1)(J_G - J_H)$. A nonzero Laurent polynomial cannot annihilate another nonzero Laurent polynomial over the integers. Thus the difference vanishes exactly when $J_G - J_H$ does. □

This proposition prevents overinterpreting an attractive example. Suppose an exact comparison gives

$$C_{G,2} - C_{H,2} = 2q^2(1-q)^3(1+q).$$

If the graphs have the same degree histogram, then

$$J_G - J_H = 2q^4(1-q)^2.$$

The observation has detected a difference in degree mixing. That is a valid strict separation from any matched invariants, but it is not evidence that a genuinely high-order height interaction was needed. A stronger next example should first match J_G, then look for a later discrepancy.

Volume	Allowed positive height patterns	Direct information visible in the formula
0	Empty support	One zero profile
1	1	Degree histogram
2	2, 1+1	Degree data and edge degree-sum mixing
3	3, 2+1, 1+1+1	Ordered degree-adjacency weights and degree-decorated triple edge counts
m	Integer partitions of m placed on vertices	Aggregated degree-decorated supports of size at most m

The final row deliberately says "aggregated." It is not a claim that every decorated induced-subgraph count can be independently recovered from one coefficient polynomial. Different patterns can contribute the same power of q, and their multiplicities are then added.

4.4 A precise finite-depth comparison

For graphs G, H of the same order, define their scalar-height visibility depth by

$$r(G,H) = \min\{m \geq 0 : C_{G,m}(q) \neq C_{H,m}(q)\},$$

with $r = \infty$ when no discrepancy exists. The same definition using $[x^m]B_G$ gives the same value. A depth records how much total height is required to separate the two exact observations. It is not a claim about computation time, and it is not automatically a graph-theoretic reconstruction radius.

Theorem SNG-F16 (prefix equivalence and an ultrapseudometric). For each $M \geq 0$, agreement of $C_{G,m}$ for all $m \leq M$ is an equivalence relation on graphs of fixed order. Visibility depths satisfy

$$r(G,H) \geq \min\{r(G,K), r(K,H)\}.$$

Consequently $d(G,H) = 2^{-r(G,H)}$, with $2^{-\infty} = 0$, is an ultrapseudometric. It is a metric on the quotient obtained by identifying graphs with identical full scalar-height series.

Proof. Equality of finite coefficient lists is reflexive, symmetric, and transitive. If both pairs (G,K) and (K,H) agree below an index r_0, then G, H agree below that index as well. This gives the displayed inequality, including infinite depths. Exponentiating reverses the inequality and gives $d(G,H) \leq \max\{d(G,K), d(K,H)\}$. Zero distance means equality of the full series, which is exactly the relation used in the stated quotient. □

Proposition SNG-F33 (finite visibility filtration). Fix a domain $\mathcal{D}$: the symbolic coefficients in $\mathbb{Z}[q]$, the span-marked counts, or the values at a parameter q_0 that is rational or an isolated real algebraic number. For a finite family of series with common prefix through R, let $\sim_v$ be agreement in $\mathcal{D}$ of every coefficient at volumes $0, \ldots, v$. Then (i) each $\sim_v$ is an equivalence relation and $\sim_{v+1}$ refines $\sim_v$; (ii) if two

series are $\sim_{v-1}$-equivalent and differ at v, then v is their first difference in $\mathcal{D}$, and the signed difference at v is a witness; (iii) symbolic equivalence implies equivalence at every parameter, and marked equivalence implies symbolic equivalence, while the converses can fail; (iv) a class that survives to R records agreement through R only.

Proof. (i) Agreement of finitely many values in a set with equality is an equivalence relation, and agreeing through $v+1$ includes agreeing through v. (ii) follows from the definition of $\sim_{v-1}$. (iii) Evaluation at q_0 and forgetting the span mark are functions of the finer data, so they preserve equality. The converses fail: at $q = 1$ every series on n vertices has coefficients $\binom{m+n-1}{n-1}$, so the spectral and binary pairs merge there although they differ symbolically. (iv) The data beyond R were not examined. □

`src/singularity/visibility.py` computes these partitions with deterministic class identifiers, memberships at every volume, and a signed witness for every split. For an algebraic parameter it decides equality exactly through SNG-F31's gcd test. It refuses a bound beyond the common prefix, and a marked dictionary must be given the bound it was computed through, since it carries no length of its own. On the census of CENSUS-v1, the symbolic and marked class counts at each volume equal those of `results/small_census.json`. Every connected graph with at most six vertices is separated from every other by volume six (`results/visibility_filtration.json`). This is a finite statement about that population, not a separation bound for larger graphs.

Calling this a pseudometric is substantive caution. If nonisomorphic graphs share the entire series, their distance is zero. Proving that this cannot happen in a graph class would be a reconstruction theorem for that class. The present definitions do not assume it.

4.5 What an occupancy coefficient can depend on

A profile of volume m has at most m occupied vertices. Its energy depends on the induced adjacency among those vertices and on their ambient degrees. Edges from an occupied vertex to an unoccupied vertex contribute through its degree; no further structure among unoccupied vertices enters that profile's energy.

To state this precisely, regard the induced graph $G[S]$ as decorated by the ambient degree d_i at every vertex of S. Two decorated induced graphs are isomorphic when a graph isomorphism also matches those degree labels. Let the decorated t-deck record how many subsets of size t have each such type.

Theorem SNG-F17 (decorated-deck sufficiency for a prefix). If two graphs of the same order have equal decorated t-decks for every $t \le M$, then their scalar-height coefficients agree for every $m \le M$. Their fully marked coefficients agree at those volumes as well.

Proof. Fix a nonempty support S of size $t \le m$. Profiles with that exact support assign positive integers b_i to its vertices, summing to m. Their energies are

$$\sum_{i \in S} d_i b_i^2 - 2 \sum_{\{i,j\} \in E(G[S])} b_i b_j.$$

A degree-preserving induced-graph isomorphism gives a coordinate-permutation bijection on these positive assignments, preserving volume, energy, and span. For normalized profiles, exact supports are proper subsets; the full-support case is excluded on both graphs because their common order is the same. Summing over matching

decorated support types proves equality at each $m \le M$. The zero profile contributes equally. □

This is a one-way implication. The coefficient hierarchy can be coarser than the decorated decks because it aggregates different supports and positive assignments into the same energy cells. The proposition identifies a sufficient source of indistinguishability and explains the local nature of a bounded-volume experiment without asserting an unproved inversion.

It suggests several distinct research questions. Can nonisomorphic graph families have arbitrarily large finite visibility depth? Which natural graph classes admit a uniform depth bound? Can decorated-deck constructions produce exact equal-series examples, or do higher heights eventually distinguish them? These are precise targets. Finite searches can provide examples and exclusions within their declared range, but cannot settle the unrestricted questions merely by growing a table.

4.6 Why labeled probes make reconstruction easy

Consider instead the fully vertex-marked series

$$\mathcal{L}_G(z_0, \dots, z_{k-1}; q) = \sum_{a \ge 0} \Big(\prod_i z_i^{a_i}\Big) q^{E_G(a)}.$$

The variables now identify where each unit was placed. This is a much stronger observation than scalar total height.

Theorem SNG-F18 (labeled reconstruction at degree two). The coefficients of $\mathcal{L}_G$ through total z-degree two determine the labeled adjacency matrix of G.

Proof. The coefficient of z_i is the single monomial q^{d_i}, recovering d_i. For distinct i, j, the coefficient of $z_i z_j$ is $q^{d_i + d_j - 2A_{ij}}$. Its exponent, together with the two recovered degrees, gives

$$A_{ij} = \frac{d_i + d_j - \deg_q([z_i z_j]\mathcal{L}_G)}{2}.$$

Every coefficient used is a single monomial with coefficient one, so its exponent is unambiguous. □

Thus a theorem saying that arbitrary vertex-marked probes determine the graph would not resolve the difficult scalar-height inverse problem. The correspondence that makes reconstruction easy has been retained explicitly. The interesting restriction is the specialization $z_i = x$ for all i, which merges those individually identifiable monomials into common volume cells.

This is the same reason that general graph-observation frameworks must be compared with their actual permitted test families. Results for all graph homomorphism counts, as discussed in [Lovasz2012], concern an enormously rich set of probes. One particular partition function or one particular family of height weights does not acquire the same reconstruction power by resemblance.

4.7 Marks, specializations, and the conclusions they permit

For any observation, forgetting a mark is a deterministic map on its data. Equality before forgetting implies equality afterward. A discrepancy afterward also proves a

discrepancy beforehand. The reverse implications need not hold. In particular, full span-marked equality implies binary-profile equality by Theorem SNG-F11; scalar-height equality alone is not presently proved to imply it.

Numerical specialization of q can lose information as well. The endpoints are extreme examples: at $q = 1$ every graph of the same order has the same scalar-height series, while at $q = 0$ every connected graph has $B_G = 1$. More generally, a nonzero polynomial discrepancy can vanish at a selected algebraic value of q. Exact polynomial comparison is stronger than comparison at one such value. If q_0 is transcendental, no nonzero integer polynomial vanishes there, so an exact evaluation would preserve coefficient distinctions; that logical observation does not provide an effective exact arithmetic method for an arbitrary transcendental input.

Established implication	Reason	Converse available here?
Graph isomorphism $\Rightarrow$ full marked equality	Coordinate-permutation bijection	Not proved
Full marked equality $\Rightarrow$ scalar-height equality	Set $z{=}1$	Not proved
Full marked equality $\Rightarrow$ binary-profile equality	Extract $[z]$	Not proved
Connected scalar-height equality $\Rightarrow$ unmarked theta equality	Sum finite energy shells	Not proved
Same decorated decks through size $M \Rightarrow$ height-prefix equality through volume M	Support decomposition	Not proved
Exact coefficient discrepancy $\Rightarrow$ different graph observations	Definition of equality	Yes, at some coefficient if the formal series differ

Finally, a later primewise Euler lift of L_G should be understood as an analytic reorganization. Its coefficient at a prime power recovers the corresponding local coefficient, and the local coefficients specify every Euler factor. That lift can expose useful singularities and arithmetic asymptotics, but it does not restore correspondence already discarded by the local histogram. The forward computation, the observation hierarchy, and the inverse reconstruction problem remain distinct throughout.

The resulting program is deliberately open in the right place. We can prove what the first layers observe, identify exact sources of finite-depth agreement, and formulate complete-series reconstruction as a concrete question. We can also search for persistent blindness and treat an exact counterexample as a mathematical result. What we cannot do is promote a successful prefix calculation into a conclusion about every unseen layer. The hierarchy provides the language for making the next calculation precise.

Chapter 5

Prime arithmetic: assembling relational height populations

The height population becomes an arithmetic object when one integer height is assigned to every prime valuation. This step has a precise meaning. A vertex is no longer assigned one nonnegative integer; it is assigned a positive integer, whose exponent at each prime supplies a local height. The graph compares those exponents independently at each prime. Unique factorization then assembles the comparisons into an Euler product. No relation of the form $n_u + n_v = n_w$ is imposed, and numerical closeness of two ordinary integers is not the quantity being measured. The construction records multiplicative disagreement, prime by prime.

Throughout this chapter, $G = (V, E)$ is a finite loopless simple graph with $k \geq 2$ vertices. Connectedness is stated where needed. For a nonnegative height vector a, write

$$|a| = \sum_{v \in V} a_v, \qquad E_G(a) = \sum_{\{u,v\} \in E} (a_u - a_v)^2, \qquad L_G(x, q) = \sum_{a \geq 0} x^{|a|} q^{E_G(a)}.$$

The coefficient of every fixed power of x is a polynomial in q with nonnegative integer coefficients. Thus this definition is meaningful formally before convergence is discussed. For numerical analysis we take $0 \leq q \leq 1$, use $0^0 = 1$, and initially require $|x| < 1$. At fixed volume n, there are exactly $\binom{n+k-1}{k-1}$ labeled profiles. Consequently

$$|L_G(x, q)| \leq (1 - |x|)^{-k} \qquad (|q| \leq 1).$$

This elementary majorant will support every exchange of sums made below. The number of vertices, rather than the number of edges, controls this convenient uniform bound. Graph structure remains in the distribution of energy among profiles of the same volume.

5.1 The arithmetic weight and its exact assembly

For a positive integer tuple $n = (n_v)_{v \in V}$, define its relational valuation energy by

$$\mathcal{E}_G(n) = \sum_{p \ mprime} E_G\big((v_p(n_v))_{v \in V}\big).$$

Only finitely many terms are nonzero. The definition is invariant under simultaneously multiplying all coordinates by the same positive integer: this adds the same prime valuation to every local height and does not change any difference. Such multiplication does change the product $\prod_v n_v$, so energy and size remain separate statistics. Define

$$Z_G(s,q) = \sum_{n_v \geq 1} \frac{q^{\mathcal{E}_G(n)}}{(\prod_v n_v)^s}.$$

Theorem SNG-A01 — valuation assembly. For $0 \leq q \leq 1$ and $\Re s > 1$, the preceding Dirichlet series is absolutely convergent and

$$Z_G(s,q) = \prod_p L_G(p^{-s}, q).$$

Proof. Absolute values of the summands are bounded by $\prod_v n_v^{-\sigma}$, where $\sigma = \Re s$. Their sum is $\zeta(\sigma)^k < \infty$. Restrict first to tuples supported on a finite set of primes. Unique factorization identifies such a tuple with one height vector for each of those primes. Its denominator factors as $\prod_p p^{s|a(p)|}$, and its weight factors as $\prod_p q^{E_G(a(p))}$. Multiplication of the finitely many absolutely convergent local series gives the finite Euler product. Increasing the finite prime set exhausts all tuples. Dominated convergence, or monotone convergence for real positive s, proves the identity. The local estimate $L_G(p^{-s}, q) = 1 + O_G(p^{-\sigma})$ also establishes absolute convergence directly. □

The theorem does not claim that arithmetic assembly creates extra information about the graph. The complete local series determines the Euler product. Its value is that it transfers a local relational statistic into an analytic language in which the first surviving distinction has a sharply identifiable singularity. Subsequent chapters prove that transfer rather than treating the Euler product as decoration.

Consider a single edge with heights a, b. Its energy is $(a-b)^2$. At a prime dividing only one coordinate to exponent one, the local penalty is q. A common factor at that prime contributes no penalty. A quotient with valuation difference two contributes q^4, not q^2. Thus the energy is more sensitive to repeated unequal valuation than a simple count of unmatched prime factors. The quadratic choice is part of the definition, not a claim that it is uniquely natural. Other edge costs can be studied, but their theorems require separate hypotheses.

5.2 Removing the common factor exactly

Translation invariance of energy has a local and a global consequence. Locally it removes the common minimum height. Globally it removes the greatest common divisor of all coordinates. These are the same operation viewed through prime valuations.

Theorem SNG-A02 — common-shift factorization. Put

$$B_G(x,q) = \sum_{\substack{a \geq 0 \\ \min_v a_v = 0}} x^{|a|} q^{E_G(a)}.$$

Then, as a formal power series and for $|x| < 1$,

$$L_G(x,q) = \frac{B_G(x,q)}{1-x^k}.$$

Equivalently, on the quotient lattice $\mathbb{Z}^k/\mathbb{Z}\mathbf{1}$, define

$$\nu([a]) = \sum_v a_v - k \min_v a_v.$$

This function is well defined, and

$$B_G(x,q) = \sum_{[a]\in\mathbb{Z}^k/\mathbb{Z}\mathbf{1}} x^{\nu([a])} q^{E_G(a)}.$$

Proof. Every nonnegative profile has a unique decomposition $a = b + m\mathbf{1}$, with $m = \min a$ and $\min b = 0$. Its energy equals $E_G(b)$, and its volume is $|b| + km$. Summing x^{km} over $m \geq 0$ gives the geometric factor. Every integer profile has a unique translate with minimum zero, which proves the quotient-lattice formula. □

The quotient formula explains what the volume mark retains. The quadratic form E_G measures distances transverse to the constant direction. The function ν additionally remembers the distinguished positive cone supplied by vertex coordinates. An abstract isometry of quadratic lattices need not preserve that function. Removing common shifts does not discard the order information; it records it on a canonical representative. This distinction is central when two unmarked energy distributions agree but their volume-marked distributions differ.

Corollary SNG-A03 — primitive tuple factorization. For $\Re s > 1$,

$$\frac{Z_G(s,q)}{\zeta(ks)} = \sum_{\substack{n_v\geq 1 \\ \gcd(n_v : v\in V)=1}} \frac{q^{\mathcal{E}_G(n)}}{(\prod_v n_v)^s} = \prod_p B_G(p^{-s}, q).$$

Proof. Write every integer tuple uniquely as $n_v = d\, m_v$, where the coordinates of m have common gcd one. Its energy is unchanged, and its denominator gains d^{ks}. Alternatively, common gcd one says that at every prime the minimum coordinate valuation is zero, exactly the defining local condition for B_G. Absolute convergence justifies both factorizations. □

Here “primitive” means a common gcd of one. It does not mean pairwise coprime coordinates. A tuple can share a factor between two vertices while having no factor common to every vertex. Confusing these conditions would change the local population and invalidate the shift factor.

5.3 Synchronization and independence as exact endpoints

At $q = 1$, all energy penalties disappear. At $q = 0$, only zero-energy profiles survive. For a connected graph these are precisely the constant profiles. Thus the same family interpolates between independent coordinates and complete synchronization, in an exact combinatorial sense.

Theorem SNG-A04 — endpoint identities. If G is connected, then

q	$L_G(x,q)$	$B_G(x,q)$	$Z_G(s,q)$
1	$(1-x)^{-k}$	$(1-x^k)(1-x)^{-k}$	$\zeta(s)^k$
0	$(1-x^k)^{-1}$	1	$\zeta(ks)$.

The Euler products converge respectively for $\Re s > 1$ and $\Re s > 1/k$. If the graph has connected component sizes $k_1, \dots, k_c$, the synchronized endpoint is instead $\prod_j \zeta(k_j s)$.

Proof. Independence gives a geometric series for each vertex. At zero energy every edge has equal endpoint heights, and therefore heights are constant on each connected component. The arithmetic tuple is correspondingly constant on each component. The indicated geometric and zeta factors follow. □

These endpoint identities require no limiting argument. A limit involving both q and s is a different problem because the singularity at $s = 1$ can amplify a small nonzero energy weight. Chapter 8 proves a uniform estimate that makes this distinction quantitative.

5.4 Minimum cuts are the first arithmetic correction

Assume now that the graph is connected, and let $\lambda(G)$ be its edge connectivity. A minimum cut is a nonempty proper vertex set whose edge boundary has size λ. We count both sides: S and $V \setminus S$ are separate height profiles and generally have different volumes.

Theorem SNG-A05 — first positive energy. Among nonconstant integer profiles, the least energy is λ. Among minimum-zero profiles, the profiles of energy λ are exactly $\mathbf{1}_S$ for minimum cuts S. Consequently

$$[q^\lambda] B_G(x,q) = \sum_{\substack{\emptyset \neq S \subsetneq V \\ |\partial S| = \lambda}} x^{|S|}, \qquad [q^e] B_G = 0 \quad (0 < e < \lambda).$$

Proof. For an integer difference d, $d^2 \geq |d|$. If a has minimum zero and span h, then the discrete coarea identity gives

$$\sum_{\{u,v\}} |a_u - a_v| = \sum_{t=0}^{h-1} |\partial\{v : a_v > t\}|.$$

Every level set in that sum is nonempty and proper, so $E_G(a) \geq \lambda h$. A nonzero minimum-zero profile has $h \geq 1$. A binary minimum-cut profile attains energy λ. If energy equals λ, the bound forces $h = 1$, and the profile is the indicator of a minimum cut. □

Write $P(z) = \sum_p p^{-z}$ for the prime zeta function on its initial half-plane. The first positive energy coefficient now has an exact analytic description.

Theorem SNG-A06 — minimum-cut prime-zeta coefficient. Initially for $\Re s > 1$,

$$[q^\lambda] \log \frac{Z_G(s,q)}{\zeta(ks)} = \sum_{\substack{\emptyset \neq S \subsetneq V \\ |\partial S| = \lambda}} P(|S| s).$$

Proof. The constant coefficient in q of B_G is one, and there are no positive coefficients below degree λ. Hence its first positive coefficient is unchanged by the formal logarithm. Apply Theorem SNG-A05 to $\log B_G(p^{-s}, q)$, and sum over primes. Absolute convergence on $\Re s > 1$, uniformly for q in a sufficiently small complex disk, justifies coefficient extraction. This uniformity also follows from the local majorant and the fact that each normalized factor equals one at $q = 0$. □

A cut with a large side contributes at a different scaled argument than a cut with a small side. The coefficient therefore contains the sizes of oriented minimum-cut sides, not merely their number. It also identifies the scope of this first correction: it sees minimum cuts only. Higher energy coefficients mix larger cuts with profiles having several height levels. Extracting those sectors requires retained marks or additional identities; the first coefficient alone does not reconstruct the graph.

5.5 Two small models that expose the retained information

For the graph consisting of a single edge, a minimum-zero profile is either $(0, 0)$, $(h, 0)$, or $(0, h)$, with $h \geq 1$. Thus there is a closed local formula

$$B_{K_2}(x, q) = 1 + 2\sum_{h\geq 1} q^{h^2} x^h, \qquad L_{K_2}(x, q) = \frac{1 + 2\sum_{h\geq 1} q^{h^2} x^h}{1 - x^2}. \tag{5.1}$$

The exponent h^2 records a genuinely unequal-height sector. Replacing it by h would give a different model even though the first positive-energy coefficient would agree. The first minimum-cut correction is $2P(s)$: either coordinate can be the raised endpoint. At volume two the three original profiles are $(2, 0), (1, 1), (0, 2)$, with energy histogram $1 + 2q^4$. Normalization removes the common-shift profile $(1, 1)$, leaving $2q^4$. This small calculation separates labeled population counting, common-factor removal, and energy weighting without relying on a large graph example.

For the path on three vertices, choose its middle vertex as a root and write the edge differences as $u = a_1 - a_0$ and $v = a_2 - a_0$. There is one integer profile modulo common translation for every pair $(u, v) \in \mathbb{Z}^2$. Its minimum-zero volume is

$$\nu = u + v - 3\min(0, u, v), \qquad E = u^2 + v^2.$$

Consequently

$$B_{P_3}(x, q) = \sum_{u,v\in\mathbb{Z}} q^{u^2+v^2} x^{u+v-3\min(0,u,v)}. \tag{5.2}$$

The energy has independent coordinates, but the volume does not. The minimum couples them through which vertex is lowest. This is a direct elementary instance of metric information being simpler than order information. It also illustrates why a factorization of an unmarked theta series does not automatically factor its volume-marked refinement.

Theorem SNG-A19 — all trees share the unmarked energy theta. Let T be any tree with k vertices, and let

$$\vartheta(q) = \sum_{h \in \mathbb{Z}} q^{h^2}.$$

For $0 \le q < 1$,

$$B_T(1, q) = \vartheta(q)^{k-1}. \tag{5.3}$$

Proof. Choose a root and orient every edge away from it. Sending a vertex profile modulo common integer translation to its $k-1$ oriented edge differences is a bijection with $\mathbb{Z}^{k-1}$. Injectivity follows by propagating equal differences along the unique root-to-vertex paths. Surjectivity follows by starting the root at zero and integrating the prescribed integer differences along those paths. Energy becomes the sum of squares of the independent edge differences. At $x = 1$, the minimum-zero representative introduces no extra weight, so the sum factors into $k-1$ copies of ϑ. Absolute convergence follows from the convergent one-dimensional Gaussian series. □

This is a useful family of exact collisions, not merely a single searched pair. Every tree of a fixed order has the same unmarked energy distribution. The marked series can already distinguish a four-vertex path from a four-vertex star at volume one: the first has degree polynomial $2q+2q^2$, the second $3q+q^3$. Their unmarked quadratic lattices agree, yet the coordinate cone places the lattice vectors into different volume layers. More demanding matched examples must remove this elementary degree explanation, but the family makes the logical source of the restored information unmistakable.

Connectedness matters to an unmarked specialization. If the graph has several components, removing one global common shift leaves independent relative component shifts of zero energy. The sum $B_G(1, q)$ then diverges, even when $q < 1$. The local series at $|x| < 1$ remains perfectly well defined. On disconnected graphs it is preferable to normalize one common shift per component before setting the volume variable to one. Ordinary component factorization gives

$$L_G(x, q) = \prod_{j=1}^{c} L_{G_j}(x, q),$$

while the single-shift normalized series is

$$B_G(x, q) = \frac{1 - x^k}{\prod_j (1 - x^{k_j})} \prod_j B_{G_j}(x, q).$$

The remaining denominator is the exact record of those unconstrained relative shifts. This is an example where checking the meaning of a normalization prevents a misleading finite theta interpretation.

Chapter 6

Euler analysis: formal coordinates and controlled analytic factors

An Euler product has two distinct kinds of structure in this project. Its coefficients are exact finite polynomials, so logarithms and Euler transforms can be manipulated formally. Its values are infinite products, so analytic conclusions additionally require convergence and choices of branches. The distinction is useful rather than restrictive. Formal calculations reveal which factor ought to be removed; a remainder estimate proves where that removal has analytic meaning. The results below keep those two steps explicit.

Fix a graph with k vertices, write

$$L(x,q) = 1 + \sum_{n\geq 1} \ell_n(q) x^n, \qquad \ell_n(q) = \sum_{|a|=n} q^{E(a)},$$

and initially regard q as an indeterminate. Every ℓ_n belongs to $\mathbb{Z}_{\geq 0}[q]$. Formal logarithms belong to $\mathbb{Q}[q][[x]]$, because division by positive integers enters the logarithm expansion. No numerical approximation to q, the coefficients, or the graph enters this part of the theory.

6.1 Logarithmic coordinates from finite data

Put $\log L = \sum_{n\geq 1} c_n(q) x^n$. The identity $L' = L(\log L)'$ gives a triangular recurrence that computes each logarithmic coefficient using only the local coefficients through the same volume:

$$c_n = \ell_n - \frac{1}{n} \sum_{j=1}^{n-1} j\, c_j \ell_{n-j}. \tag{6.1}$$

In particular,

$$c_1 = \ell_1, \quad c_2 = \ell_2 - \ell_1^2/2, \quad c_3 = \ell_3 - \ell_1\ell_2 + \ell_1^3/3.$$

The recurrence is a coefficient identity, not a fitted model. If two graphs agree through volume $r-1$, they also agree in logarithmic coordinates through that volume, and their difference at volume r is unchanged. This triangular property is the reason first-discrepancy singularities can be read directly from the original local coefficients.

Theorem SNG-A07 — rational polynomial Euler transform. There is a unique sequence $b_m(q) \in \mathbb{Q}[q]$ such that, formally in x,

$$L(x,q) = \prod_{m\geq 1} (1-x^m)^{-b_m(q)}. \tag{6.2}$$

It is given by

$$b_m(q) = \sum_{d|m} \frac{\mu(d)}{d}\, c_{m/d}(q). \tag{6.3}$$

Only factors with $m \leq N$ affect coefficients through volume N.

Proof. Taking formal logarithms of the proposed product gives

$$c_n = \frac{1}{n}\sum_{m|n} m b_m.$$

Ordinary Möbius inversion gives (6.3). Conversely those coefficients satisfy the displayed logarithmic identity, and formal exponential recovers L. Uniqueness follows from the same triangular relation. Rational polynomial exponents are meaningful through the formal exponential definition of $(1-x^m)^{-b_m}$. □

The first exponents are

$$b_1 = \ell_1, \qquad b_2 = \ell_2 - \frac{\ell_1^2+\ell_1}{2},$$

$$b_3 = \ell_3 - \ell_1\ell_2 + \frac{\ell_1^3-\ell_1}{3}.$$

The implementation follows these identities literally and keeps them independent. `series_log` uses the recurrence (6.1); `euler_exponents` solves $nc_n = \sum_{d|n} db_d$ upward; `euler_exponents_mobius` applies (6.3); and `recompose_euler` rebuilds L by multiplying the generalized binomial series $(1-x^m)^{-b_m} = \sum_j \binom{b_m+j-1}{j} x^{mj}$, without any logarithm. All arithmetic is in $\mathbb{Q}[q]$ with reduced fractions, and a prefix through volume N determines exactly the exponents $b_1, \ldots, b_N$. Tests check the triangular identity on a seven-vertex graph series, the single-edge values (6.10), recomposition of every retained certificate's exponents, nonconstant rational exponents and a symbolic discrepancy that vanishes at $q=1$, so that the specialized first difference moves to a later volume.

These formulas are economical coordinates for local information. They do not make every exponent nonnegative or integral as a polynomial in q. Such claims would be false for general integer-coefficient series. The existence theorem is over $\mathbb{Q}[q]$, where the necessary divisions are legitimate.

Common-shift removal has an especially simple Euler-coordinate description. If $B = (1 - x^k)L$, then

$$b_m(L) = b_m(B) + \mathbf{1}_{m=k}.$$

At the connected synchronized endpoint the only nonzero exponent is $b_k = 1$. At the independent endpoint the only nonzero exponent is $b_1 = k$. These are useful exact consistency checks, but they do not determine intermediate exponents from the endpoints.

6.2 A different transform keeps integral bidegrees

There is a second, genuinely different product decomposition. Instead of putting a polynomial exponent on $1 - x^m$, retain energy inside the individual factors:

$$L(x, q) = \prod_{m \geq 1} \prod_{e \geq 0} (1 - x^m q^e)^{-N_{m,e}}. \tag{6.4}$$

At each fixed m, only finitely many $N_{m,e}$ are nonzero. The exponents can be negative. This is a formal decomposition and does not identify an actual population of positive "atoms" without a separate positivity theorem.

Theorem SNG-A08 — integral bidegree transform. For every series $L \in 1 + x\mathbb{Z}[q][[x]]$, there are unique integers $N_{m,e}$ satisfying (6.4). Their generating series is the plethystic logarithm

$$\sum_{m,e} N_{m,e} x^m q^e = \sum_{d \geq 1} \frac{\mu(d)}{d} \log L(x^d, q^d). \tag{6.5}$$

Proof. Suppose all factors of volume below m have been removed. The remaining series has the form $1 + x^m P_m(q) + O(x^{m+1})$, with integral polynomial P_m. For every coefficient $N_{m,e} = [q^e]P_m$, multiply by $(1 - x^m q^e)^{N_{m,e}}$. This removes volume m and preserves integrality. Induction gives existence and uniqueness. Taking logarithms of (6.4), and applying Möbius inversion to the simultaneous substitutions $(x, q) \mapsto (x^d, q^d)$, gives (6.5). Thus the apparently rational expression in (6.5) has integral coefficients. □

The substitution $q \mapsto q^d$ is essential. Omitting it computes the transform in Theorem SNG-A07 rather than the bidegree transform. The two descriptions organize the same formal series differently. For example, a factor $(1 - xq)^{-1}$ has one bidegree atom, yet its ordinary exponents $b_m(q)$ are generally nontrivial polynomials. Coefficient-level equality permits conversion; the interpretations should remain distinct.

The corresponding weighted arithmetic factor is

$$\mathcal{Z}_{m,e}(s, q) = \prod_p (1 - q^e p^{-ms})^{-1},$$

with logarithm, on its absolutely convergent domain,

$$\log \mathcal{Z}_{m,e}(s,q) = \sum_{j\geq 1} \frac{q^{ej}}{j} P(mjs).$$

It is not equal to $\zeta(ms)^{q^e}$. Replacing one expression by the other discards all the terms involving q^{ej} for $j > 1$. The ordinary rational-exponent transform and the integral bidegree transform therefore lead to different useful analytic factorizations.

6.3 Leading divergence and what it forgets

The first local coefficient has a simple graph formula:

$$\alpha_G(q) = \ell_1(q) = \sum_{v\in V} q^{d_v}.$$

Only one vertex has height one in a volume-one profile. Its energy is the number of incident edges. Thus the dominant singularity records the degree multiset and no finer graph information.

Theorem SNG-A09 — leading real singularity. For fixed $0 \leq q \leq 1$, define

$$H_G(s,q) = \prod_p (1-p^{-s})^{\alpha_G(q)} L_G(p^{-s},q). \tag{6.6}$$

The product is holomorphic for $\Re s > 1/2$, using the analytic logarithm of $1-p^{-s}$ defined by its convergent power series on $\Re s > 0$. It is positive at the real point $s = 1$ and holomorphic and nonzero in a complex neighborhood of that point. Initially for $\Re s > 1$,

$$Z_G(s,q) = \zeta(s)^{\alpha_G(q)} H_G(s,q).$$

Along the positive real axis,

$$Z_G(1+\varepsilon,q) \sim H_G(1,q)\varepsilon^{-\alpha_G(q)} \qquad (\varepsilon \downarrow 0). \tag{6.7}$$

Proof. Locally $L = 1 + \alpha x + O(x^2)$, uniformly for $q \in [0,1]$. Multiplication by $(1-x)^\alpha$ cancels the linear coefficient, leaving $1 + O(x^2)$. The prime sum of the remainder converges normally on compact subsets of $\Re s > 1/2$. Each individual factor is holomorphic there; L is analytic on $|x| < 1$, and the chosen power is holomorphic on $\Re s > 0$. For real positive s, all factors are positive. The infinite tail is nonzero by absolute convergence of its logarithm; finitely many remaining factors are nonzero in a neighborhood of $s = 1$. The factorization follows on the original domain from the usual zeta Euler product. Finally $\zeta(1+\varepsilon) \sim 1/\varepsilon$. □

The word "holomorphic" in the theorem does not assert that H_G is zero-free throughout its half-plane. A finite local factor can have complex zeros. Nonvanishing is asserted near the positive real point where it is proved. Similarly, fractional powers of zeta are unambiguous on the original half-plane but need branch choices if continued around zeros or poles elsewhere.

6.4 Finite-order zeta factorization

Theorem SNG-A10 — controlled finite Euler extraction. Fix an integer $M \geq 1$ and a numerical $q \in [0,1]$. Put

$$H_{G,M}(s,q) = \prod_p L_G(p^{-s},q) \prod_{m=1}^{M} (1-p^{-ms})^{b_m(q)}. \tag{6.8}$$

This product is holomorphic on $\Re s > 1/(M+1)$. It is positive for real $s > 1/(M+1)$, and nonzero in a neighborhood of every point on that positive real interval. On the initial half-plane,

$$Z_G(s,q) = \left(\prod_{m=1}^{M} \zeta(ms)^{b_m(q)} \right) H_{G,M}(s,q). \tag{6.9}$$

Proof. Theorem SNG-A07 implies

$$L_G(x,q) \prod_{m=1}^{M} (1-x^m)^{b_m(q)} = 1 + O(x^{M+1})$$

near zero. For primes sufficiently large, this estimate is uniform on any compact subset of the indicated half-plane, and its prime sum converges. The finitely many small-prime factors are holomorphic on $\Re s > 0$; the powers use the logarithms specified above. Positivity on the real axis follows directly from (6.8), and absolute logarithmic convergence proves local nonvanishing there. Equation (6.9) follows by finite multiplication of zeta Euler products on $\Re s > 1$. □

This theorem removes any need to assume convergence of the infinite product over all m in an analytic setting. Formal identity (6.2) does not by itself license an unrestricted exchange of its two infinite products, over primes and over volumes. Small-prime local zeros can limit the radius of the logarithmic power series, and the exponents can grow. Keeping M finite is both sufficient for a prescribed visibility calculation and analytically safer.

Equation (6.9) supplies local continuations after branches have been chosen in regions avoiding relevant zeta zeros and poles. It is not a blanket meromorphic-continuation theorem. Nonintegral exponents generally produce branched functions. Even on the positive real axis below one, separate zeta factors can be negative, so assigning real fractional powers independently does not reproduce an automatically positive partition function. The positive quantity directly defined there is the remainder product (6.8); its relation to a continued Z_G must respect the chosen branches.

6.5 Why ratios are cleaner than individual continuations

If two graphs have identical low-volume coefficients, their low-order zeta exponents cancel before any continuation is attempted. The quotient Euler product can consequently converge farther left as a positive real product. Its first remaining zeta factor has argument approaching one from above. No lower-argument zeta powers are needed to state the resulting real asymptotic. This is the analytic advantage of

comparing matched structures rather than interpreting every formal singularity of an individual object as an independently observable feature.

The hierarchy is exact but limited. It proves where a known local distinction appears arithmetically. It does not prove that a distinction always exists between nonisomorphic graphs, nor that all formal exponent locations are actual singularities after every cancellation. If the local series agree, every arithmetic lift in this chapter agrees. If the first discrepancy vanishes at a special numerical q, the visibility order at that parameter must be recomputed. These qualifications identify the genuine inverse problem rather than weakening a theorem that never claimed more.

6.6 A complete low-order transform example

The single-edge model makes the difference between polynomial exponents and positive constituent populations concrete. Its first coefficients are

$$\ell_1 = 2q, \qquad \ell_2 = 1 + 2q^4, \qquad \ell_3 = 2q + 2q^9.$$

Substitution into the rational Euler transform gives

$$\begin{aligned} b_1(q) &= 2q, \\ b_2(q) &= 1 + 2q^4 - 2q^2 - q, \\ b_3(q) &= 2q^9 - 4q^5 + (8q^3 - 2q)/3. \end{aligned} \tag{6.10}$$

At $q = 1/2$, these are $1, 1/8, -31/256$. Thus a valid finite extraction reads

$$Z_{K_2}(s, 1/2) = \zeta(s)\zeta(2s)^{1/8}\zeta(3s)^{-31/256}H_{K_2,3}(s, 1/2) \qquad (\Re s > 1). \tag{6.11}$$

The negative third exponent is not a negative number of arithmetic objects. It is a subtraction introduced by logarithmic and Möbius coordinates. Likewise, the exponent one eighth does not describe one eighth of an independent integer coordinate. Those are exact algebraic coordinates on a positive local series. They are useful precisely because they can cancel lower-order contributions, not because each factor has a separate population interpretation.

At the endpoints the same formulas reduce correctly: at $q = 0$, the second exponent is one and the other displayed exponents are zero; at $q = 1$, the first exponent is two and the next two are zero. Agreement at both endpoints therefore places only weak restrictions on the intermediate polynomial behavior. The coefficient recurrence supplies the actual intermediate values. A numerical plot interpolating only the endpoints would have no authority to invent them.

The bidegree decomposition has a different bookkeeping purpose. Its first volume layer contains the atom count $N_{1,1} = 2$. Removing $(1 - xq)^{-2}$ from the single-edge series leaves, at volume two, the polynomial $1 - 3q^2 + 2q^4$. Hence the second bidegree layer already contains a negative exponent, $N_{2,2} = -3$. Even integral bidegree exponents are therefore not automatically positive atoms. The notation "plethystic" supplies a formal inversion rule; it does not establish a free-generation theorem for an underlying combinatorial class.

6.7 Uniformity and the boundary of the weight parameter

For each fixed extraction order M, the remainder product in Theorem SNG-A10 is jointly continuous in real $q \in [0,1]$ and real $s > 1/(M+1)$. On compact subsets of that parameter region its large-prime logarithmic tail converges uniformly. To see this, the finitely many polynomials $b_m(q)$ are bounded on the compact weight interval, and the coefficient majorant bounds the Taylor remainders uniformly for x in a small fixed disk. The canceled coefficients through order M are identically zero as polynomials, so the remaining estimate is uniformly $O(x^{M+1})$. A convergent prime-power sum dominates the tail. Finitely many positive real local factors then preserve continuity and local nonvanishing.

This continuity statement is often enough to pass a weight limit through a regularized product. It does not mean that the unregularized Euler product converges uniformly as $s \downarrow 1$. The leading zeta factor carries the obstruction to that stronger statement. Chapter 8 isolates it in a quantitative estimate rather than suppressing it through a formal exchange of limits.

The chosen weight range also has a mathematical boundary. Suppose the graph has an edge, choose a vertex of positive degree d_v, and fix $0 < x < 1$. The terms coming from profiles ne_v are $q^{d_v n^2} x^n$. For every real $q > 1$, these terms fail even to tend to zero as $n \to \infty$. Thus the unbounded local height series diverges. The restriction $q \leq 1$ cannot simply be removed by calling the series a partition function. A finite height cutoff would define a different object and would require its own limiting argument.

For $|q| < 1$, the local series is analytic in the complex weight parameter. At a boundary value such as $q = 1$, derivatives from the allowed real side can still exist to every finite order, because each derivative introduces only polynomial energy factors and the geometric volume decay dominates them. Such smooth one-sided behavior should not be mistaken for an unrestricted analytic extension beyond the unit disk. These domain distinctions matter especially when symbolic exponents are evaluated numerically: exact polynomial coefficients can be evaluated anywhere, while the infinite series from which they were extracted has the domain proved for it.

Chapter 7

Visibility singularities: the first distinction survives arithmetic assembly

A comparison is strongest when simpler explanations have been removed explicitly. Two graphs can have the same degree multiset and therefore the same leading divergence. They can agree on additional finite invariants and still differ in a later local height coefficient. The arithmetic question is then not whether an Euler product can be attached to each graph: that is already automatic. The useful question is where the first unmatched local observation becomes analytically visible.

For $0 \le q \le 1$, put $\ell_{G,n}(q) = [x^n]L_G(x,q)$. In this chapter, q is fixed before an analytic comparison is made. Polynomial equality in q is stronger than numerical equality at a selected parameter. A polynomial discrepancy can vanish at an endpoint or an interior root, in which case the first visible volume at that numerical parameter can change.

7.1 A precise first-discrepancy theorem

Theorem SNG-A11 — first-discrepancy ratio law. Let G, H have the same finite number k of vertices. Fix $q \in [0,1]$. Suppose

$$\ell_{G,n}(q) = \ell_{H,n}(q) \quad (0 \le n < r), \qquad \delta = \ell_{G,r}(q) - \ell_{H,r}(q) \neq 0. \tag{7.1}$$

For real $s > 1/r$, the ratio product

$$R_{G,H}(s,q) = \prod_p \frac{L_G(p^{-s},q)}{L_H(p^{-s},q)} \tag{7.2}$$

converges absolutely to a positive number. It agrees with Z_G/Z_H on the original domain $s > 1$. Define

$$K_{G,H}(s,q) = \prod_p (1-p^{-rs})^{\delta} \frac{L_G(p^{-s},q)}{L_H(p^{-s},q)}. \tag{7.3}$$

The latter product converges absolutely and is positive for real $s > 1/(r+1)$. It is holomorphic and nonzero in a complex neighborhood of $s_0 = 1/r$. For real $s > s_0$,

$$R_{G,H}(s,q) = \zeta(rs)^{\delta} K_{G,H}(s,q), \qquad R_{G,H}(s,q) \sim C(q)(rs-1)^{-\delta}, \tag{7.4}$$

where $C(q) = K_{G,H}(1/r, q) > 0$.

Proof. Condition (7.1) gives

$$\frac{L_G(x,q)}{L_H(x,q)} = 1 + \delta x^r + O(x^{r+1})$$

near zero, because the denominator has constant term one. The prime sum of the deviation converges for $rs > 1$. No local denominator vanishes on the positive real axis: both local series have nonnegative coefficients and constant term one. Thus the product is positive, and its logarithm converges absolutely. Multiplication by $(1-x^r)^{\delta}$ cancels the first discrepancy, leaving $1 + O(x^{r+1})$. This proves convergence of (7.3) on the stated real interval.

To obtain a complex neighborhood, separate finitely many small primes from the normally convergent tail. Every small-prime denominator is positive at s_0, hence nonzero in a sufficiently small disk about that point. The tail is analytic and nonzero there by its convergent logarithm. The factors $(1-p^{-rs})^{\delta}$ use the analytic logarithm defined on $\Re s > 0$. This gives the asserted local holomorphic nonzero function. On $s > 1/r$, the ordinary zeta Euler product yields the factorization. Finally $\zeta(1+t) \sim 1/t$ with $t = rs - 1$. □

A positive discrepancy makes the ratio diverge at its visibility point; a negative discrepancy makes it tend to zero. Interchanging the graphs reverses that interpretation. If δ is nonintegral, the complex singularity is a fractional-power branch singularity. It should not be called a pole. The theorem supplies a local branch and a positive-real asymptotic, not a globally single-valued meromorphic function.

There is also no requirement that the original partition functions converge separately at s_0. Their common low-order divergences have been canceled in the ratio product before the limit is taken. This cancellation is the main analytic mechanism. It makes a comparison meaningful farther left without pretending that either original positive Dirichlet series has acquired an absolutely convergent definition there.

The positive-real comparison now has a complete Lean proof over actual Mathlib real numbers and the Riemann zeta function. In `formal-analysis/`, `SingularityAnalysis.visibility_of_coefficients` proves the theorem for nonnegative polynomial-growth coefficient sequences, and `SingularityAnalysis.graph_visibility` applies it to graph height coefficients for $q \in [0,1]$. The chain proves the local logarithmic expansion, prime-product convergence and boundary continuity, the zeta Euler product and its residue limit. It identifies the continued ratio with the quotient of the original prime products for $s > 1$. The concrete statements `SingularityAnalysis.spectral_visibility_half` and `SingularityAnalysis.binary_visibility_half` import P33 and prove the exact exponents at $q = 1/2$. No unverified analytic premise structure remains. The local-complex holomorphy and branch-continuation argument in this chapter, and the later interval endpoints, remain ordinary or computer-assisted mathematics rather than part of that real-analytic Lean theorem.

7.2 A volume-two calculation and its actual meaning

At volume two there are only two kinds of profile: $2e_v$, and $e_u + e_v$ with $u < v$. Their energies give the identity

$$\ell_{G,2}(q) = \sum_v q^{4d_v} + \sum_{u<v} q^{d_u+d_v-2A_{uv}}. \tag{7.5}$$

For graphs with the same degree multiset, define the edge degree-sum polynomial

$$M_G(q) = \sum_{\{u,v\}\in E} q^{d_u+d_v}.$$

Then

$$\ell_{G,2} - \ell_{H,2} = (q^{-2} - 1)(M_G - M_H). \tag{7.6}$$

Although written with q^{-2}, the right side is a polynomial: every edge has degree sum at least two. Formula (7.6) follows by starting with the sum over all unordered vertex pairs and correcting the exponent by minus two precisely on edges.

Calculation SNG-A12 — first reference pair. On vertices $0, \dots, 6$, take

$$E(G) = \{01, 02, 03, 04, 12, 35, 36\},$$
$$E(H) = \{01, 02, 03, 04, 15, 16, 56\}.$$

Both have degree multiset $\{4, 3, 2, 2, 1, 1, 1\}$. Exact profile enumeration gives equality at volume one and

$$\delta_2(q) = \ell_{G,2} - \ell_{H,2} = 2q^2 - 4q^3 + 4q^5 - 2q^6 = 2q^2(1-q)^3(1+q). \tag{7.7}$$

It is positive on $0 < q < 1$, so Theorem SNG-A11 applies with $r = 2$. At $q = 1/2$,

$$R_{G,H}(s, 1/2) \sim C_2(2s-1)^{-3/32}.$$

The arithmetic conclusion is exact. Its local source is also elementary: $M_G - M_H = 2q^4(1-q)^2$. Thus this example illustrates arithmetic visibility of degree-assortativity information. It does not demonstrate that volume two captures a fundamentally higher-order graph relation. Matching more demanding invariants is a separate task, and the interpretation of a collision should state precisely which invariants were matched.

7.3 A volume-three witness beyond the binary census

At volume three, the profile types are $3e_v$, $2e_u + e_v$ with $u \neq v$, and indicators of three-element subsets. Hence

$$\ell_{G,3}(q) = \sum_v q^{9d_v} + \sum_{u\neq v} q^{4d_u+d_v-4A_{uv}} + \sum_{|S|=3} q^{|\partial S|}. \tag{7.8}$$

This identity keeps the explanation transparent. The last term is binary. The middle term tests unequal heights at two vertices. It can distinguish a joint degree-mixing pattern that the degree-sum polynomial merges. Every energy remains a sum of pairwise edge terms; a later observation order does not introduce a new three-body interaction.

Calculation SNG-A13 — binary-census reference pair. Take

$$E(G) = \{03, 04, 06, 12, 13, 23, 24, 35\},$$
$$E(H) = \{01, 04, 06, 15, 16, 26, 36, 45\}.$$

Both have degree multiset $\{4, 3, 3, 2, 2, 1, 1\}$. The exact finite binary census agrees for every subset size and cut size. The local coefficients agree through volume two, but their volume-three difference is

$$\begin{aligned}\delta_3(q) &= q^3 - q^4 - 2q^6 + 2q^8 + q^9 + 3q^{10} - 3q^{11} - q^{12} - 2q^{13} \\ &\quad + 2q^{15} + q^{17} - q^{18} \\ &= q^3(1-q)^5(1+q)^2(1 + 2q + 5q^2 + 6q^3 + 8q^4 \\ &\qquad + 6q^5 + 5q^6 + 2q^7 + q^8).\end{aligned} \tag{7.9}$$

The factorization proves positivity throughout $0 < q < 1$; it is not a numerical sign test. At $q = 1/2$, the discrepancy is $11025/262144$. Therefore

$$R_{G,H}(s, 1/2) \sim C_3(3s - 1)^{-11025/262144} \qquad (s \downarrow 1/3). \tag{7.10}$$

The finite calculation can be reproduced without a symbolic algebra system. Generate every weak composition of volumes zero through three into seven labeled coordinates, evaluate the integer energy from the supplied edges, and accumulate multiplicities by energy. There are only $1, 7, 28, 84$ profiles at those volumes. Subtract the two energy histograms. Multiplication of the explicit factors in (7.9) independently checks the polynomial identity. The full binary claim requires the separate exhaustive census of 2^7 subsets; it is not inferred from the first four height coefficients.

This pair does not match the full joint degree table on edges. Its degree-sum polynomial agrees, but the more refined table does not. Consequently the witness proves that the unequal-height sector carries information beyond the aggregated binary census. It does not prove survival after every degree-assortativity statistic has been matched. A stronger search should match that full table as well and seek discrepancies at volume four or later. The first-discrepancy theorem will still apply; the difficult new work is finding and interpreting the local witness.

7.4 Effective control of the finite-prime approximation

A singularity law is more useful when its regular factor can be bounded rather than merely named. The following bounds require only local coefficient estimates and ordinary convergent sums. They do not require assumptions about prime gaps or an unproved asymptotic formula for primes.

For an integer cutoff $P \geq 2$, define $Z_{\leq P}$ by retaining primes at most P. For real $s > 1$, put $\rho = P^{-s} < 1$. Since

$$0 \le \log L(t,q) \le L(t,q) - 1 \le \frac{kt}{(1-\rho)^{k+1}} \qquad (0 \le t \le \rho),$$

we obtain the explicit interval

$$0 \le \log Z_G(s,q) - \log Z_{\le P}(s,q) \le \frac{k}{(1-\rho)^{k+1}} \frac{P^{1-s}}{s-1}. \tag{7.11}$$

Here the tail over primes was bounded by the tail over all integers, followed by the integral test. Exponentiating gives a multiplicative enclosure. The bound is deliberately conservative near $s = 1$, where raw Euler products converge slowly.

For the regularized ratio factor a faster bound is available. Set $A_n = \binom{n+k-1}{k-1}$, choose $0 < \rho < 1$, and define the explicit nonnegative constants

$$D = \frac{(1-\rho)^{-k} - 1}{\rho}, \qquad T = \frac{2}{\rho^{r+1}} \left((1-\rho)^{-k} - \sum_{n=0}^{r} A_n \rho^n \right),$$

$$C = T + |\delta| D + \frac{|\delta| \rho^{r-1}}{2(1-\rho^r)}. \tag{7.12}$$

Theorem SNG-A14 — computable regularized tail. Under the assumptions of Theorem SNG-A11, if real $s > 1/(r+1)$ and $P^{-s} \le \rho$, then

$$\left| \log K_{G,H}(s,q) - \log K_{G,H,\le P}(s,q) \right| \le C \frac{P^{1-(r+1)s}}{(r+1)s - 1}. \tag{7.13}$$

Proof. Write $A = L_G(t,q)$, $B = L_H(t,q)$, and $D_t = A - B$. Matching the coefficients below r gives

$$|D_t - \delta t^r| \le T t^{r+1}, \qquad \max(A-1, B-1) \le Dt.$$

Because $A, B \ge 1$,

$$\log A - \log B = \int_0^1 \frac{D_t}{B + u D_t} \, du.$$

The denominator is at least one. Subtracting δt^r therefore gives an error at most $(T + |\delta| D) t^{r+1}$. Finally

$$|\log(1 - t^r) + t^r| \le \frac{t^{2r}}{2(1-\rho^r)} \le \frac{\rho^{r-1} t^{r+1}}{2(1-\rho^r)}.$$

Adding these estimates bounds the absolute logarithm of one regularized factor by $C t^{r+1}$. Sum over primes beyond P, enlarge to all integers, and apply the integral test. □

The constant (7.12) is implemented exactly as `regularized_tail_constant` in `src/singularity/arithmetic.py`. Its proof uses only $0 \le \ell_n(q) \le A_n$, valid for $q \in [0,1]$, and $t \le \rho < 1$. The older routine `normalized_ratio_tail_constant` bounds the same quantity through the coefficient ℓ^1 norm of the logarithm. It requires $U = (1-\rho)^{-k} - 1 < 1$ and is far weaker: for the binary pair at $q = 1/2$ and $\rho = 1/32$ it gives about 6.9×10^5, against about 451 from (7.12). Both are retained, and (7.12) is preferred. For rational $s = a/b$ the cutoff is taken as $P = N^b$, so the tail bound $CP^{1-(r+1)s}/((r+1)s-1)$ and the radius condition $P^{-s} = N^{-a} \le \rho$ are exact rational statements. Primes $p \le P$ are kept exactly in the finite product, and those beyond it satisfy $p^{-s} < \rho$. At $s = 1/r$ the bound is Cr/N. The function `certified_tail_record` recomputes r and δ from the two supplied coefficient prefixes at the chosen q before it produces a bound. It records the remaining hypotheses explicitly, refuses a prefix that ends before the discrepancy, and refuses a parameter such as $q = 1$ at which the discrepancy vanishes. The retained records for the three reference pairs at $q = 1/2$ are in `results/tail_records.json`. A test also checks the local inequality for the spectral pair at three rational points $t \le \rho$, using rigorous rational enclosures of both local series by (7.14) and of $(1-t^r)^\delta$ by Bernoulli's inequalities, with no floating point.

Theorem SNG-F35 (an explicit complex disc around the visibility point). Keep the hypotheses of SNG-A11 with $q \in [0,1]$ fixed, and let $s_0 = 1/r$, $D_\eta = \{s : |s - s_0| \le \eta\}$ and $\sigma = s_0 - \eta$. Suppose

$$\text{(a) } (r+1)\sigma > 1, \qquad \text{(b) } r\eta \le \tfrac{1}{4}, \qquad \text{(c) } \frac{\eta\, k}{e\,\sigma}\,(1 - 2^{-\sigma})^{-(k+1)} < 1.$$

Then for every prime p and every $s \in D_\eta$, the local factors $L_G(p^{-s}, q)$ and $L_H(p^{-s}, q)$ have positive real part, and so does $1 - p^{-rs}$. With principal logarithms, the function

$$K(s) = \exp \sum_p \big(\log L_G(p^{-s}) - \log L_H(p^{-s}) + \delta \log(1 - p^{-rs})\big)$$

is holomorphic and nonzero on D_η, the series converges uniformly there, and K equals the positive product (7.3) on $D_\eta \cap \mathbb{R}$. On $D_\eta \setminus [s_0 - \eta, s_0]$,

$$R(s) = (rs-1)^{-\delta}\big((rs-1)\zeta(rs)\big)^{\delta} K(s),$$

with principal powers, continues the Euler-product ratio from $s \in (s_0, s_0 + \eta]$. For nonintegral δ, s_0 is a branch point, not a pole. For integral δ, R is meromorphic on D_η with a pole of order δ (or a zero of order $-\delta$) at s_0.

Proof. Write $t = p^{-s}$ and $t_0 = p^{-s_0}$. Then $|t| = p^{-\Re s} \le p^{-\sigma}$, and $|t - t_0| \le |s - s_0| \log p \max_{w \in [s_0, s]} |p^{-w}| \le \eta \log p\, p^{-\sigma}$. For a local series $L = \sum \ell_n t^n$ with $0 \le \ell_n \le A_n$ (true for $q \in [0,1]$), $|L(t) - L(t_0)| \le |t - t_0| \sup_{|u| \le p^{-\sigma}} |L'(u)| \le |t - t_0|\, k(1 - p^{-\sigma})^{-k-1}$. Using $\max_{u>1} u^{-\sigma} \log u = 1/(e\sigma)$ and $p \ge 2$, condition (c) makes this less than $1 \le L(t_0)$, so $\Re L(t) > 0$, and the principal logarithm is holomorphic in s. Also $|p^{-rs}| < 1$, so $\Re(1 - p^{-rs}) > 0$. Let $h_p(s)$ be the summand. As a function of t, $h(t) = \log L_G(t) - \log L_H(t) + \delta \log(1 - t^r)$ is holomorphic for $|t| \le \rho$ when $U(\rho) = (1-\rho)^{-k} - 1 < 1$. By (7.1) it vanishes to order $r+1$ at $t = 0$, and $|h| \le M = 2U/(1-U) + |\delta|\rho^r/(1-\rho^r)$ there, using $|\log(1+z)| \le |z|/(1-|z|)$. Schwarz's lemma applied to $h(t)/M$ gives $|h(t)| \le M(|t|/\rho)^{r+1}$. This is the constant of `normalized_ratio_tail_constant`, which is therefore valid for complex t, unlike (7.12), whose proof

used positivity. For $p > P$ with $P^{-\sigma} \le \rho$ we get $|h_p(s)| \le Mp^{-(r+1)\sigma}/\rho^{r+1}$. By (a) the tail converges uniformly on D_η, and K is holomorphic and nonzero as the exponential of a uniform limit of holomorphic functions. For real s the principal logarithms are real, so K agrees with (7.3). Finally, with $w = rs$ and $|w-1| \le r\eta \le \frac{1}{4}$, the representation $(w-1)\zeta(w) = w - w(w-1)\int_1^\infty \{u\}u^{-w-1}\,du$ for $\Re w > 0$ gives $|(w-1)\zeta(w) - 1| \le |w-1|(1 + |w|/\Re w) \le \frac{1}{4}(1 + \frac{5/4}{3/4}) = \frac{2}{3}$. The principal power $((w-1)\zeta(w))^\delta$ is therefore holomorphic on D_η. For real $s > s_0$, the identity $R = \zeta(rs)^\delta K$ of SNG-A11 holds with positive real powers. The displayed expression is that identity rewritten, and $(rs-1)^{-\delta}$ with the principal branch is holomorphic off the cut. □

The disc is explicit and very small. `complex_disc_certificate` in `src/singularity/arithmetic.py` checks (a)–(c) exactly, using the rational bounds $2^{-\sigma} \le 2^{-1/r}(1+\eta)$ and $e > 27/10$. It admits $\eta = 10^{-5}$ for the spectral pair ($k = 7$, $r = 2$), $\eta = 10^{-7}$ for the binary pair ($r = 3$) and $\eta = 10^{-9}$ for the cube and Möbius ladder ($k = 8$, $r = 4$), and it refuses $\eta = 10^{-4}$, 10^{-6} and 10^{-8} respectively. The radius is limited by the crude derivative bound at $p = 2$. It is not a claim that larger discs contain zeros. Nothing here concerns complex energy parameters q, whose local factors need their own zero analysis, or a global continuation.

Theorem SNG-F39 (normalized local factors are entire). Let G be connected on $n \ge 2$ vertices with diameter D, and fix $0 < q < 1$. Then $B_G(t,q) = (1-t^n)L_G(t,q)$ is an entire function of t, and $L_G(t,q)$ is meromorphic in t, with poles at most at the n-th roots of unity. For every $H \ge 0$ and complex t,

$$\big|B_G(t,q) - B_{G,\le H}(t,q)\big| \le \sum_{h>H} N_n(h)\max(|t|^h, |t|^{(n-1)h})q^{\lceil h^2/D\rceil},$$
$$N_n(h) = (h+1)^n - 2h^n + (h-1)^n.$$

where $B_{G,\le H}$ sums over minimum-zero profiles of maximum at most H.

Proof. A minimum-zero profile of maximum h has a vertex at height 0 and one at h, joined by a path of at most D edges. Cauchy–Schwarz along that path gives $h^2 \le D\,E_G(a)$. Its volume lies in $[h, (n-1)h]$, and $N_n(h)$ counts the profiles in $\{0,\dots,h\}^n$ with minimum 0 and maximum h, by inclusion–exclusion. This proves the tail bound. A profile of volume m has $h \ge m/(n-1)$, so $|[t^m]B_G| \le \binom{m+n-1}{n-1} q^{m^2/((n-1)^2 D)}$. Hence $|[t^m]B_G|^{1/m} \to 0$, and the radius of convergence is infinite. □

For disconnected graphs the zero-energy profiles that are constant on components give $B_G(t,0)$-type poles at roots of unity of the component sizes (SNG-F19), so connectedness is necessary. The theorem concerns one local factor at fixed q. It says nothing about continuation of the product over primes.

Theorem SNG-F38 (certified enclosure of the regularized product). Under SNG-A11 with G, H connected and $q \in (0,1)$ rational, fix rational $s = a/b$ with $(r+1)s > 1$, an integer N, the cutoff $P = N^b$ and $\rho = N^{-a} = P^{-s}$. Let ℓ_n, m_n be the local coefficients of G and H at q for $n \le M$, $A_n = \binom{n+k-1}{k-1}$, and $S_M =$

$(1-\rho)^{-k} - \sum_{n\le M} A_n\rho^n$. Put

$$T' = \sum_{r<n\le M} |\ell_n - m_n|\rho^{n-r-1} + \frac{2S_M}{\rho^{r+1}},$$
$$D' = \sum_{1\le n\le M} \max(\ell_n, m_n)\rho^{n-1} + \frac{S_M}{\rho},$$
$$C' = T' + |\delta|D' + \frac{|\delta|\rho^{r-1}}{2(1-\rho^r)}.$$

For the finite summands, abbreviate

$$\theta_p(s) = \delta \log(1-p^{-rs}) + \log \frac{B_G(p^{-s})}{B_H(p^{-s})}.$$

Then

$$\left|\log K_{G,H}(s,q) - \sum_{p\le P} \theta_p(s)\right| \le C' \frac{P^{1-(r+1)s}}{(r+1)s-1},$$

and each finite summand is enclosed by rational intervals computed from: integer-root enclosures of $p^{-a/b}$; monotonicity of B in $t \ge 0$; the height tail of SNG-F39; outward-rounded sums over a checked tree decomposition; and series enclosures of logarithms.

Proof. $L_G/L_H = B_G/B_H$ because the orders agree. The bound for omitted primes repeats the proof of (7.12) with sharper majorants. For $0 \le t \le \rho$, $|L_G(t) - L_H(t) - \delta t^r| \le \sum_{n>r} |\ell_n - m_n| t^n \le T' t^{r+1}$, using $|\ell_n - m_n| \le 2A_n$ beyond M. Likewise $\max(L_G, L_H) - 1 \le \sum_{n\ge 1} \max(\ell_n, m_n) t^n \le D' t$, using $0 \le \ell_n, m_n \le A_n$. Hence one regularized logarithm is at most $C' t^{r+1}$. Primes $p > P$ satisfy $p^{-s} < \rho$. Summing $C' p^{-(r+1)s}$ over all integers $p > P$ and applying the integral test gives the displayed bound. The finite enclosures were justified above: B_G has nonnegative coefficients, so it is increasing on $t \ge 0$, and rounding every product of nonnegative fixed-point numbers down (respectively up) gives a lower (respectively upper) bound. □

`src/singularity/certified.py` implements the theorem with 256 fractional bits. The minimum-zero condition is carried as a flag in the bag messages, so no subtraction of enclosures occurs. For $t = p^{-a/b}$ the factor $(1-t^r)$ is exact when $b \mid ra$, in particular at $s = 1/r$. The retained results, with exact dyadic endpoints, are in `results/certified_k.json`. All are at $q = 1/2$. The spectral pair has $r = 2$, $\delta = 3/32$; the binary pair has $r = 3$, $\delta = 11025/262144$:

Pair	s	Primes kept	Omitted log bound	K lower	K upper
spectral	1/2	1229 ($\le 10^4$)	4.6×10^{-3}	0.84973	0.85750
spectral	3/5	985 ($\le 6^5$)	2.2×10^{-4}	0.89931	0.89971
spectral	1	168 ($\le 10^3$)	1.1×10^{-7}	0.9678188	0.9678191
binary	1/3	2961 ($\le 30^3$)	2.2×10^{-2}	0.90084	0.94230

The local height tails total below 4×10^{-16} in every run, so the width comes almost entirely from the omitted primes; arithmetic rounding is below 10^{-60}. With $R = \zeta(rs)^\delta K$, the tables exhibit the proved branch singularity $(rs-1)^{-\delta}$ with an explicitly

enclosed positive constant $K(1/r, 1/2)$. The singularity exponent is proved by SNG-A11, not fitted. Narrowing the binary interval needs either more primes or a smaller constant C'; the bound decays only like $P^{-1/3}$ at $s = 1/3$. A separate test checks the following. A run at a smaller cutoff with 160-bit arithmetic overlaps the $s = 1/2$ enclosure. Reversing the pair negates the finite log enclosure. Identical graphs give a log ratio enclosing zero.

Local factors themselves can also be enclosed by finitely many height coefficients. If $L_N(t, q) = \sum_{n=0}^{N} \ell_n(q) t^n$, choose any $t < R < 1$. Then

$$0 \le L(t, q) - L_N(t, q) \le (t/R)^{N+1}(1 - R)^{-k}. \tag{7.14}$$

Indeed each omitted term gains at most $(t/R)^{N+1}$, and the complete series at R is bounded by $(1 - R)^{-k}$. Since $L_N \ge 1$, its logarithmic error is at most the same bound. Finite-prime and finite-height errors can thus be propagated separately and added in logarithmic coordinates. A numerical implementation claiming a rigorous enclosure must also bound its arithmetic rounding; the inequalities alone do not certify an unchecked floating-point product.

These results establish controllable local witnesses, arithmetic visibility, and finite approximation. They leave a genuine open inverse question: which graph distinctions are invisible to the entire univariate volume-marked local series? The examples show strict separations from specified coarser observations. They do not answer that complete question or establish historical priority for the construction.

7.5 The complete binary comparison and the remaining elementary distinction

For the second reference pair, the common binary polynomial at each side size is small enough to display completely. Let $C_j(q) = \sum_{|S|=j} q^{|\partial S|}$. Complementation gives $C_j = C_{7-j}$, and the common values are

Side sizes	Common cut-energy polynomial
0,7	1
1,6	$2q+2q^2+2q^3+q^4$
2,5	$2q^2+8q^3+5q^4+4q^5+2q^6$
3,4	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$

Evaluating a row at $q = 1$ gives respectively $1, 7, 21, 35$, the appropriate binomial subset counts. This check does not prove equality of the polynomials, but it catches a missing subset in a proposed census. The actual equality is coefficientwise, obtained by evaluating each labeled subset against the original edges.

The edge joint-degree table identifies information the common binary census has merged. Entries not displayed are zero in both graphs.

Unordered endpoint degrees	Edges in G	Edges in H
(1,3)	1	0
(1,4)	1	2
(2,2)	0	1

Unordered endpoint degrees	Edges in G	Edges in H
$(2,3)$	3	2
$(2,4)$	1	0
$(3,3)$	0	1
$(3,4)$	2	2

Both columns total eight edges. Collapsing endpoint degrees to their sum makes the tables agree: sums four, five, six, seven occur with multiplicities one, four, one, two. This explains agreement at volume two. Unequal heights two and one weight the endpoint degrees differently, through $4d_u + d_v$, so the collapse no longer hides the difference. The height witness can therefore be understood without asserting an unexplained new source of graph information.

The distinction also lies in a specific span sector. Volume-three profiles of the form $3e_v$ have span three and agree because degrees agree. Binary profiles have span one and agree because the full binary census agrees. The entire discrepancy (7.9) is consequently in span two. Retaining a span mark would isolate that sector directly. After setting the span mark to one, the same discrepancy survives, but the proof that it came from that sector uses the separate matched observations.

7.6 A secondary boundary layer at an endpoint

The first-discrepancy exponent itself can become small near an endpoint. This produces a second controlled scaling law, now for graph ratios rather than the original synchronized partition function.

Theorem SNG-A20 — endpoint scaling of visibility. Suppose the polynomial local coefficients of two graphs agree below volume r for every $q \in [0,1]$. Let $q_* \in [0,1]$ be a value where the entire local series agree, and write $\delta(q) = \ell_{G,r}(q) - \ell_{H,r}(q)$. Let $q_j \to q_*$, $\varepsilon_j \downarrow 0$, with

$$\delta(q_j)\log(1/\varepsilon_j) \longrightarrow \tau \in \mathbb{R}.$$

Then

$$R_{G,H}((1+\varepsilon_j)/r, q_j) \longrightarrow e^{\tau}. \tag{7.15}$$

Proof. The normalized local ratio in (7.3) has its first r nonconstant coefficients canceled identically in q. Its $O(x^{r+1})$ estimate can be chosen uniformly on the compact weight interval. Therefore the regularized prime product is jointly continuous near $(1/r, q_*)$, by uniform convergence of the large-prime logarithmic tail and positivity of the finitely many remaining factors. At q_*, every local ratio is one and $\delta(q_*) = 0$, so $K(1/r, q_*) = 1$. Finally

$$\log R = \delta(q_j)\log\zeta(1+\varepsilon_j) + \log K.$$

The second term tends to zero, while $\log\zeta(1+\varepsilon) - \log(1/\varepsilon) \to 0$ and δ stays bounded. This proves the limit. □

For the volume-two pair, $\delta_2(q) \sim 4(1-q)^3$ as $q \uparrow 1$. For the volume-three pair, $\delta_3(q) \sim 144(1-q)^5$, because the final polynomial factor in (7.9) has value 36 at

one. Thus $(1-q)^5 \log(1/\varepsilon) \to t$ yields the ratio limit e^{144t} for the second pair at $s = (1+\varepsilon)/3$. Near zero the corresponding leading behaviors are $\delta_2 \sim 2q^2$ and $\delta_3 \sim q^3$. These are proved parameter scales for particular exact witnesses. They do not establish a universal exponent for all graph pairs.

Chapter 8

Synchronization limits, uniform scaling, and exact contractions

Setting the energy weight to zero and letting the arithmetic size parameter approach its divergence point are different operations. At zero weight a connected graph forces every coordinate integer to agree, producing $\zeta(ks)$. At any fixed positive weight the system has rare unequal valuations at every prime, and their combined contribution diverges at $s = 1$. There are infinitely many opportunities for a rare local event. A small penalty need not remain negligible when those opportunities accumulate. This chapter turns that intuition into a uniform theorem and then develops optional extensions that retain the same exact arithmetic meaning.

Assume first that G is connected with $k \geq 2$, minimum degree $d \geq 1$, and edge connectivity $\lambda \geq 1$. Write m_d for the number of vertices of degree d, and $\alpha(q) = \sum_v q^{d_v}$. All limits below use real positive parameters. No complex continuation is needed for the double-scaling statements.

8.1 An explicit estimate uniform in the energy weight

We begin with constants that are intentionally simple rather than optimized. Define

$$D_k = 2(2^k - 1), \qquad C_k = 4(2^k - 1 - k/2), \qquad A_k = C_k + D_k^2/2 + k. \tag{8.1}$$

Let

$$J_k = \sum_{n \geq 2} \frac{\log n}{n^k} < \infty.$$

This last constant is computable with an elementary integral remainder. For a cutoff $N \geq 2$, the omitted tail is at most

$$\int_N^\infty \frac{\log t}{t^k}\, dt = N^{1-k} \left(\frac{\log N}{k-1} + \frac{1}{(k-1)^2} \right). \tag{8.2}$$

The summand is decreasing on $[2, \infty)$ for $k \geq 2$, so the integral bound applies directly.

Theorem SNG-A15 — uniform synchronization estimate. For $0 \le q \le 1$ and $\varepsilon > 0$,

$$\begin{aligned} &|\log Z_G(1+\varepsilon, q) - \log \zeta(k) - \alpha(q)\log(1/\varepsilon)| \\ &\quad \le A_k q^\lambda(\zeta(2)-1) + (kJ_k + k)\varepsilon. \end{aligned} \tag{8.3}$$

Proof. Write $B_G(x,q) = 1 + u(x,q)$. Every nonzero minimum-zero profile has energy at least λ. Therefore, for real $0 \le x \le 1/2$,

$$0 \le u(x,q) \le q^\lambda D_k x,$$

and, since the volume-one coefficient is $\alpha(q)$,

$$0 \le u(x,q) - \alpha(q)x \le q^\lambda C_k x^2.$$

To verify the constants, bound each minimum-zero coefficient by the number of all weak compositions. At $x \le 1/2$, dividing the first bound by x and the second by x^2 leaves geometric-composition sums bounded respectively by D_k and C_k. These bounds discard graph information but retain the crucial factor q^λ.

For $u \ge 0$, integration of $1 - 1/(1+u)$ gives

$$0 \le u - \log(1+u) \le u^2/2.$$

Also $|\log(1-x) + x| \le x^2/[2(1-x)]$, and $\alpha(q) \le kq^\lambda$. Combining the estimates gives

$$|\log B_G(x,q) + \alpha(q)\log(1-x)| \le A_k q^\lambda x^2. \tag{8.4}$$

Here $q^{2\lambda} \le q^\lambda$ was used in the quadratic term. Substitute $x = p^{-s}$, with real $s \ge 1$, and sum over primes. By common-shift factorization,

$$\log H_G(s,q) - \log \zeta(ks) = \sum_p \{\log B_G(p^{-s}, q) + \alpha(q)\log(1-p^{-s})\}.$$

Thus its absolute value is at most $A_k q^\lambda(\zeta(2)-1)$. The derivative bound $|(\log\zeta)'(t)| \le \sum_{n\ge 2}(\log n)n^{-k} = J_k$ for $t \ge k$ gives

$$|\log \zeta(k(1+\varepsilon)) - \log\zeta(k)| \le kJ_k\varepsilon.$$

Finally the integral test yields $1/\varepsilon \le \zeta(1+\varepsilon) \le 1 + 1/\varepsilon$. Hence

$$0 \le \log\zeta(1+\varepsilon) - \log(1/\varepsilon) \le \varepsilon.$$

Multiply by $\alpha(q) \le k$, use the leading factorization, and add the three error bounds. □

This proof provides more than continuity of a regular factor. It shows exactly which uniform estimate permits the two parameters to move together. The graph-dependent

minimum-cut exponent controls the error in replacing the regular factor by its synchronized value. The minimum-degree exponent will control the rare unequal valuations that are amplified by the singular factor. Those exponents need not be equal, and they play different mathematical roles.

8.2 The double-scaling law

Theorem SNG-A16 — synchronization boundary layer. Let $q_j \to 0$, $\varepsilon_j \downarrow 0$. If

$$\alpha(q_j)\log(1/\varepsilon_j) \longrightarrow \tau \quad \text{in } [0,\infty],$$

then

$$Z_G(1+\varepsilon_j, q_j) \longrightarrow \zeta(k)e^{\tau}, \tag{8.5}$$

where $e^{+\infty} = +\infty$. In particular, if

$$q_j^d \log(1/\varepsilon_j) \longrightarrow t < \infty,$$

then the limit is $\zeta(k)e^{m_d t}$. If that product tends to infinity, the partition function tends to infinity.

Proof. The error in (8.3) tends to zero. Exponentiating gives the first statement, including the infinite case. For the second, use

$$\alpha(q) = m_d q^d + R(q), \qquad 0 \le R(q) \le (k-m_d)q^{d+1}.$$

When $q^d \log(1/\varepsilon)$ has a finite limit, the extra contribution $R(q)\log(1/\varepsilon)$ tends to zero. In the infinite case, $\alpha(q) \ge m_d q^d$ is enough. □

The scaling variable is logarithmic in the distance to the arithmetic singularity. For a graph with minimum degree one, maintaining a nontrivial finite limit requires q of order $1/\log(1/\varepsilon)$. With minimum degree two it requires order $1/\sqrt{\log(1/\varepsilon)}$. This is a conclusion of the exact estimate, not a power-law ansatz chosen from numerical plots. The coefficient m_d counts the cheapest single-vertex disagreements.

The theorem also explains the noncommuting iterated limits. At fixed $\varepsilon > 0$, sending $q \to 0$ gives $\zeta(k(1+\varepsilon))$, and then sending $\varepsilon \to 0$ gives $\zeta(k)$. At fixed $q > 0$, sending $\varepsilon \to 0$ gives infinity, so subsequently shrinking q does not recover that finite iterated limit. A family can therefore look synchronized throughout a large finite range and still have a different limiting divergence. The double-scaling variable states how close to one the size parameter must be before that discrepancy matters.

8.3 Optional extension: vertex masses

Positive integer vertex masses allow exact contraction without pretending that a merged vertex represents only one original coordinate. Assign a mass $m_v \ge 1$ to each vertex and let $M = \sum_v m_v$. Define

$$L_{G,m}(x,q)=\sum_{a\geq 0} q^{E_G(a)}x^{\sum_v m_v a_v}.$$

Its arithmetic counterpart weights an integer tuple by $\prod_v n_v^{-m_v s}$. All coefficients remain finite, and a common shift increases weighted volume by M.

Theorem SNG-A17 — weighted endpoints and first divergence. For connected G,

$$L_{G,m}=\frac{B_{G,m}}{1-x^M},\quad Z_{G,m}(s,0)=\zeta(Ms),\quad Z_{G,m}(s,1)=\prod_v \zeta(m_v s).$$

Let $r=\min_v m_v$ and fix $0<q\leq 1$. Put $\gamma(q)=\sum_{m_v=r} q^{d_v}$. Then the first nonconstant local term is $\gamma(q)x^r$, the Euler product converges for real $s>1/r$, and

$$Z_{G,m}(s,q)=\zeta(rs)^{\gamma(q)}H_{G,m}(s,q),$$

with a positive nonzero regular factor near $s=1/r$. Therefore its leading real singular exponent there is $\gamma(q)$.

Proof. The shift and endpoint arguments are unchanged after adding masses. The least positive weighted volume is r, attained exactly by the profiles e_v with $m_v=r$. Their energies are d_v. Every remaining weighted volume is an integer at least $r+1$. Multiplication by $(1-x^r)^{\gamma(q)}$ leaves $1+O(x^{r+1})$, and the same proof as Theorem SNG-A11 gives the regular factor. At least one vertex has minimum mass and $q>0$, so $\gamma(q)>0$; logarithmic divergence at $s=1/r$ also proves the sharp real convergence threshold. □

This extension changes the arithmetic size observable while retaining the same graph energy. It need not be interpreted as adding physical masses. Its exact purpose is to remember how many original coordinates have been identified, or how strongly their integer sizes are weighted. Ignoring those weights after contraction would move the synchronized zeta factor to the wrong argument.

8.4 Optional extension: partial edge synchronization

Allow one energy weight $q_e\in[0,1]$ for every edge. The local weight is $\prod_e q_e^{(a_u-a_v)^2}$. Setting a selected edge weight to zero imposes equality of its endpoint heights. Equality propagates through the connected components of the selected-edge subgraph.

Theorem SNG-A18 — exact contraction under zero edge weights. Let $F\subseteq E$, and set $q_e=0$ for $e\in F$. Let $C_1,\dots,C_t$ be the components of (V,F). Replace each component by one vertex of mass $|C_i|$. An unselected edge between C_i and C_j contributes its original factor $q_e^{(b_i-b_j)^2}$; an internal unselected edge contributes one. Then the resulting weighted, possibly multiple-edge local series equals the original specialized local series exactly. The equality also holds for the Euler products on every common domain of absolute convergence.

Proof. A profile survives the zero weights exactly when it is constant on each C_i. Such profiles correspond bijectively to $b\in\mathbb{Z}^t_{\geq 0}$. Its original volume is $\sum_i |C_i|b_i$. Internal

edges have zero difference, and every remaining edge retains its stated difference and weight. This is equality term by term. Euler assembly follows on an absolutely convergent domain. □

Parallel surviving edges can be combined: their weights multiply, since $q_1^{d^2} q_2^{d^2} = (q_1 q_2)^{d^2}$. Their presence is therefore supported mathematically even if the core graph class remains simple. The contraction description is an optional extended representation, not a request to accept duplicate edges silently in the ordinary constructor.

Contraction order does not matter. Applying zero constraints in stages produces the equivalence relation generated by all selected edges. Vertex masses add associatively, internal edges disappear, and external weights multiply associatively. This is a direct consequence of the termwise bijection, not an appeal to an unspecified categorical equivalence. If the selected edges connect the entire graph, the result is one vertex of mass k, recovering $\zeta(ks)$.

One can retain separate complex size parameters s_v as well. A synchronized component then has parameter $s_{C_i} = \sum_{v \in C_i} s_v$. A sufficient domain before contraction is $\Re s_v > 1$ for every vertex; after contraction the less restrictive sufficient condition is $\Re s_{C_i} > 1$ for every component. The exact local identity remains valid wherever the relevant sums converge. This multivariate refinement records how contraction redistributes size exponents without claiming a complete several-variable continuation theory.

8.5 Positive ordering and probabilistic derivatives

There is a further useful consequence of retaining positive weights. If $G \subseteq H$ on the same vertex set, then $E_H(a) \geq E_G(a)$. Hence, for $q \in [0,1]$, every local coefficient and every real convergent arithmetic value satisfies $L_H \leq L_G$ and $Z_H \leq Z_G$. Adding constraints reduces the positive population weight. The corresponding ratio need not be monotone in s, and no such stronger ordering is asserted.

For real $s > 1$ and $q = e^{-\beta}$, normalize the integer-tuple weights by Z_G. This defines an ordinary probability distribution. Its sufficient statistics are $U(n) = \log \prod_v n_v$ and $V(n) = \mathcal{E}_G(n)$. On compact parameter sets with $s > 1$ and $\beta \geq 0$, all their polynomial moments are finite. Differentiating gives

$$\partial_s \log Z = -\mathbb{E}[U], \qquad \partial_\beta \log Z = -\mathbb{E}[V],$$

$$\nabla^2_{(s,\beta)} \log Z = \begin{pmatrix} \operatorname{Var}(U) & \operatorname{Cov}(U,V) \\ \operatorname{Cov}(U,V) & \operatorname{Var}(V) \end{pmatrix} \succeq 0. \tag{8.6}$$

To justify finiteness, note that local energy is at most a graph-dependent constant times squared local volume, while the local geometric coefficient majorant has every polynomial volume moment at p^{-s}. Summing their prime contributions and logarithmic size factors is uniformly convergent when s stays above one. Alternatively, global energy is bounded by a constant times $(\log \prod_v n_v)^2$, and products of powers of logarithms remain summable against $\prod_v n_v^{-s}$. Dominated differentiation follows. The covariance identity is a general property of a normalized exponential family, used here with an explicitly defined arithmetic population.

These derivative identities supply rigorous sensitivity observations. They do not add a physical interpretation, empirical model, or claim of a thermodynamic phase transition. The proved limiting phenomenon is the uniform arithmetic boundary layer in Theorem SNG-A16. Its higher-order corrections, sharp graph-dependent constants, and possible multivariate visibility geometries remain separate analytic research questions.

8.6 A worked contraction sequence

Take the four-vertex path $0-1-2-3$, initially with mass one at every vertex and a common positive edge weight q. Its leading exponent at $s=1$ is $2q+2q^2$, because its endpoint degrees are one and its internal degrees are two. Now set the weight on edge 01 to zero while retaining weight q on the other edges. The surviving local profiles are exactly (a,a,b,c), and their local series is

$$\sum_{a,b,c\geq 0} x^{2a+b+c} q^{(a-b)^2+(b-c)^2}. \tag{8.7}$$

The contracted path has masses $2,1,1$. Its light vertices have degrees two and one, so Theorem SNG-A17 gives leading exponent q^2+q at $s=1$. The heavy vertex is still part of the graph energy, but it cannot occur in the lowest-volume profile. This is why simply counting contracted vertices would give the wrong answer.

Next synchronize edge 23 as well. The surviving profiles are (a,a,b,b). There are two contracted vertices, each of mass two, joined by the original middle edge. The local factor is exactly $L_{K_2}(x^2,q)$, and its arithmetic product is $Z_{K_2}(2s,q)$. The first divergence moves to $s=1/2$, with exponent $2q$. Finally synchronize the middle edge; all four heights become equal and the product becomes $\zeta(4s)$, whose first divergence is at $s=1/4$.

The changes in threshold have a direct combinatorial explanation. A minimal positive shift of a synchronized block raises every original coordinate in that block. Its arithmetic denominator is therefore the corresponding power of an integer, rather than the first power associated with a single original vertex. The contraction operation preserves that information through its masses. The example also shows why a statement about an exactly zero coupling should not be inferred from small positive couplings without a uniform estimate: rare violations of the equality constraint can restore a divergence at a different location.

For the derivative identities (8.6), values at $\beta=0$ mean derivatives from the allowed side $\beta\geq 0$. At an interior value $\beta>0$, ordinary derivatives are available in a neighborhood. The one-sided convention is necessary because rewarding unbounded quadratic energy with $\beta<0$ makes even a local height series diverge. This is a domain issue in the mathematical definition, not a numerical instability that can be fixed by increasing a cutoff.

Chapter 9

Exact examples: what the new population distinguishes

This chapter records the finite evidence on which the first release rests. The examples are deliberately small enough that their defining populations can be examined without numerical approximation. They serve a precise purpose: to show that integer heights retain relational information that certain familiar summaries discard. They do not establish that the resulting invariant determines every graph, that the Euler product adds information to its local factor, or that the construction has historical priority. The claims below distinguish direct arguments from exact finite computations. The computations can be repeated through the standard-library package accompanying this manuscript.

Throughout the chapter a graph is simple, undirected, and has the labeled vertex set $V = \{0, 1, \dots, 6\}$. Labels specify the input and the assignments being counted. The resulting polynomials are unchanged by relabeling the graph, because a permutation of vertices induces a bijection on all height assignments. An edge written 03 means the unordered pair $\{0, 3\}$. The height energy is

$$E_G(a) = \sum_{\{i,j\} \in E(G)} (a_i - a_j)^2, \qquad |a| = \sum_i a_i.$$

The unnormalized local population and the normalized marked population are

$$P_G(x, q) = \sum_{a \in \mathbb{N}^7} x^{|a|} q^{E_G(a)},$$

$$B_G(x, q, z) = \sum_{\substack{a \in \mathbb{N}^7 \\ \min_i a_i = 0}} x^{|a|} q^{E_G(a)} z^{\max_i a_i}.$$

Volume means total integer height. The variable z records height span after the common minimum has been removed. Consequently $B_G(x, q, 1) = (1 - x^7) P_G(x, q)$. For volumes below seven, the coefficients of the unmarked normalized and unnormalized series agree. Every assignment of such a volume already has a zero coordinate. This observation lets us discuss the first local coefficients and their span decomposition without changing normalization halfway through an argument.

9.1 Two different comparisons

The first pair has equal Laplacian spectra. The second pair has equal complete binary cut-volume populations. These are separate examples, and keeping them separate prevents an unsupported inference about a single pair defeating every existing invariant.

Name in this chapter	Package name	Edge set	Degrees in vertex order
G_{sp}	`spectral_g`	01,02,03,04,12,35,36	(4,2,2,3,1,1,1)
H_{sp}	`spectral_h`	01,02,03,04,15,16,56	(4,3,1,1,1,2,2)
G_{bin}	`binary_g`	03,04,06,12,13,23,24,35	(3,2,3,4,2,1,1)
H_{bin}	`binary_h`	01,04,06,15,16,26,36,45	(3,3,1,1,2,2,4)

All four graphs are connected. Each spectral graph has one triangle and four bridges. Each binary graph has the same two-connected block, a theta graph with three internally disjoint paths of lengths one, two, and three, together with two attached bridges. The attachments differ. This distinction between a common block structure and different vertex attachments will matter when we compare the marked population with an ordinary cut-lattice theta series.

Proposition SNG-E001: the spectral graphs are nonisomorphic. Each graph has a unique vertex of degree four, namely vertex zero. In G_{sp}, that vertex lies in the triangle $0, 1, 2$. In H_{sp}, the only triangle is $1, 5, 6$, which excludes vertex zero. An isomorphism preserves vertex degrees and triangle membership. It would therefore have to send the degree-four vertex to the degree-four vertex while preserving a property that differs, which is impossible.

Proposition SNG-E002: the binary graphs are nonisomorphic. The unique degree-four vertex of G_{bin} is vertex three, and its neighbors are $0, 1, 2, 5$. Exactly one of these neighbors, vertex five, is a leaf. The unique degree-four vertex of H_{bin} is vertex six, with neighbors $0, 1, 2, 3$. Exactly two of those neighbors are leaves. The number of leaf neighbors of a specified degree class is invariant under isomorphism. The graphs are therefore nonisomorphic.

These short arguments are logically independent of the population computation. A differing graph invariant would also prove nonisomorphism, but an elementary structural witness makes the examples easier to inspect. It also guards against a mistaken interpretation in which a relabeling of one graph has merely been counted in a different order.

9.2 The exact Laplacian comparison

The graph Laplacian has entries $L_{ii} = d_i$ and $L_{ij} = -\epsilon_{ij}$, where d_i is vertex degree and ϵ_{ij} is the adjacency indicator. We use the characteristic-polynomial convention $\chi_G(t) = \det(tI - L_G)$. The following values are exact finite computational results, obtained using integer arithmetic. They are not floating-point eigenvalue comparisons and are not described as proof-assistant-checked theorems.

Graphs	Characteristic polynomial
Both spectral graphs	$t^7-14t^6+73t^5-178t^4+209t^3-112t^2+21t$
G_{bin}	$t^7-16t^6+98t^5-290t^4+430t^3-300t^2+77t$
H_{bin}	$t^7-16t^6+98t^5-290t^4+432t^3-302t^2+77t$

The spectral pair is therefore Laplacian-cospectral. The binary pair is not. In particular, the second example must not be advertised as a simultaneous collision of the Laplacian spectrum and the binary cut polynomial. Its role is to isolate a different information gap. Both members of a pair have the same number of spanning trees: three for the spectral pair and eleven for the binary pair. These counts follow independently from the matrix-tree theorem or from the coefficient of t, whose absolute value is seven times the spanning-tree count in these connected seven-vertex graphs.

An integer implementation can compute the characteristic polynomial through traces of powers and Newton identities. If $p_j = \operatorname{tr}(L^j)$ and $c_0 = 1$, the descending coefficients satisfy

$$c_r = -\frac{1}{r}\sum_{j=1}^{r} c_{r-j}p_j.$$

Every division is exact for a matrix characteristic polynomial. This supplies one transparent computational route, although a determinant calculation is equally legitimate. The manuscript's conclusion depends on the exact resulting polynomial, not on a choice of numerical eigensolver or an eigenvalue tolerance.

9.3 All binary cut-volume profiles

For a subset $S \subseteq V$, let ∂S be its edge boundary and define

$$I_{G,r}(q) = \sum_{\substack{S\subseteq V\\ |S|=r}} q^{|\partial S|}.$$

The full binary polynomial is $C_G(x,q) = \sum_{r=0}^{7} x^r I_{G,r}(q)$. Its coefficients count every original subset. A graph automorphism does not reduce a coefficient by identifying subsets in the same orbit. Complementation gives $I_{G,r} = I_{G,7-r}$, because the two sides of a cut have the same boundary. At $q = 1$, the row sum is $\binom{7}{r}$. Both relations provide simple consistency checks on the tables.

r	$I_{G_{\mathrm{sp}},r}(q)$	$I_{H_{\mathrm{sp}},r}(q)$
0	1	1
1	$3q+2q^2+q^3+q^4$	$3q+2q^2+q^3+q^4$
2	$6q^2+7q^3+3q^4+5q^5$	$4q^2+11q^3+3q^4+q^5+2q^6$
3	$q+q^2+8q^3+15q^4+7q^5+3q^6$	$q+3q^2+4q^3+15q^4+11q^5+q^6$
4	$q+q^2+8q^3+15q^4+7q^5+3q^6$	$q+3q^2+4q^3+15q^4+11q^5+q^6$
5	$6q^2+7q^3+3q^4+5q^5$	$4q^2+11q^3+3q^4+q^5+2q^6$
6	$3q+2q^2+q^3+q^4$	$3q+2q^2+q^3+q^4$
7	1	1

The spectral graphs already differ in the binary sector. Their local height distinction does not require heights above one. This is useful evidence beyond the spectrum, but it does not yet demonstrate a benefit from nested thresholds. The binary pair was selected to make that second issue explicit.

r	Common value $I_{G_{\mathrm{bin}},r}=I_{H_{\mathrm{bin}},r}$	Row sum at $q=1$
0	1	1
1	$2q+2q^2+2q^3+q^4$	7
2	$2q^2+8q^3+5q^4+4q^5+2q^6$	21
3	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$	35
4	$q^2+4q^3+13q^4+10q^5+5q^6+2q^7$	35
5	$2q^2+8q^3+5q^4+4q^5+2q^6$	21
6	$2q+2q^2+2q^3+q^4$	7
7	1	1

Exact computation SNG-E003. Enumeration of all $2^7 = 128$ binary assignments gives the second table identically for both binary graphs. This is equality of the complete two-variable binary polynomial, not merely equality at a chosen value of q, equality of a few moments, or equality of one minimum-cut statistic. Under the usual changes of variables it is the same binary information encoded by an Ising polynomial with a field. The connection to that established literature does not make the next height sector redundant: it identifies exactly what the example must improve upon.

9.4 The first local coefficients and their span decomposition

The coefficients of P_G are easy to derive at small volume. A volume-two vector either puts height two at one vertex or height one at each of two vertices. A volume-three vector has one of the patterns 3, $2+1$, or $1+1+1$. Evaluating the edge energy on each pattern gives

$$A_1(q) := [x]P_G = \sum_i q^{d_i},$$

$$A_2(q) := [x^2]P_G = \sum_i q^{4d_i} + \sum_{i<j} q^{d_i+d_j-2\epsilon_{ij}},$$

$$A_3(q) := [x^3]P_G = \sum_i q^{9d_i} + \sum_{i\neq j} q^{4d_i+d_j-4\epsilon_{ij}} + \sum_{i<j<\ell} q^{d_i+d_j+d_\ell-2(\epsilon_{ij}+\epsilon_{i\ell}+\epsilon_{j\ell})}.$$

These identities are **Proposition SNG-E004**, valid for every finite simple graph. For the interpretation as separate normalized span sectors at these volumes, there must be a zero coordinate; seven vertices certainly suffice. In the marked population, the three summands of A_3 belong respectively to z^3, z^2, z^1. The middle sum is over ordered pairs: the vertex receiving height two is distinguished from the vertex receiving height one. Its population size at $q = 1$ is $7 \cdot 6 = 42$. The other two populations have sizes seven and thirty-five, adding to the eighty-four weak compositions of three into seven parts.

For the spectral pair the common linear coefficient is

$$A_1 = 3q + 2q^2 + q^3 + q^4.$$

The two quadratic coefficients are

$$A_{2,G_{\mathrm{sp}}} = 6q^2 + 7q^3 + 6q^4 + 5q^5 + 2q^8 + q^{12} + q^{16},$$

$$A_{2,H_{\mathrm{sp}}} = 4q^2 + 11q^3 + 6q^4 + q^5 + 2q^6 + 2q^8 + q^{12} + q^{16}.$$

Subtracting and factoring gives the exact identity

$$\boxed{[x^2](P_{G_{\mathrm{sp}}} - P_{H_{\mathrm{sp}}}) = 2q^2(1-q)^3(1+q).} \qquad \text{(SNG-E005)}$$

The height-two-at-one-vertex terms cancel because the degree multisets coincide. The difference is entirely in the binary span-one sector. It is positive for every real $0 < q < 1$, with exact value $3/32$ at $q = 1/2$. Thus the distinction is not a numerical coincidence at one parameter or a sign-sensitive coefficient observed only after rounding.

Proof of SNG-E005 from degree pairs. Volume zero contributes the zero profile to both graphs. Volume one contributes $\sum_i q^{d_i}$, and volume two contributes $\sum_i q^{4d_i}$ from the profiles $2e_i$; these agree because both degree sequences are $\{4, 3, 2, 2, 1, 1, 1\}$. The only remaining volume-two profiles are $e_i + e_j$, $i < j$, with energy $d_i + d_j - 2A_{ij}$. Summing $q^{d_i+d_j}$ over all pairs again depends only on the degree sequence, so by SNG-F15 the difference is $(q^{-2} - 1)(J_{G_{\mathrm{sp}}} - J_{H_{\mathrm{sp}}})$, where J sums $q^{d_i+d_j}$ over edges. The edges are distributed over unordered endpoint-degree pairs as follows.

Endpoint degrees	{1,3}	{1,4}	{2,2}	{2,3}	{2,4}	{3,4}
degree sum	4	5	4	5	6	7
G_{sp} edges	2	1	1	0	2	1
H_{sp} edges	0	3	1	2	0	1

Hence $J_{G_{\mathrm{sp}}} = 3q^4 + q^5 + 2q^6 + q^7$, $J_{H_{\mathrm{sp}}} = q^4 + 5q^5 + q^7$, their difference is $2q^4(1-q)^2$, and

$$[x^2](P_{G_{\mathrm{sp}}} - P_{H_{\mathrm{sp}}}) = q^{-2}(1-q)(1+q) \cdot 2q^4(1-q)^2 = 2q^2(1-q)^3(1+q).$$

Reversing the order of the pair reverses the sign. The degree-four vertex of G_{sp} is joined to the two degree-two vertices, while in H_{sp} it is joined to three leaves; this difference in degree mixing (degree assortativity) is the entire volume-two discrepancy. No higher-order phenomenon is involved: the pair is a cospectral example whose separation is already explained by adjacent-degree statistics. The closed formulas are implemented independently of enumeration in `src/singularity/lowvolume.py`, and `results/pair_acceptance.json` records their agreement with the weak-composition and level-chain enumerators and with the replayed certificate. □

The binary pair has common first and second coefficients

$$A_1 = 2q + 2q^2 + 2q^3 + q^4,$$

$$A_2 = 2q^2 + 8q^3 + 7q^4 + 4q^5 + 2q^6 + 2q^8 + 2q^{12} + q^{16}.$$

The span-one part of its cubic coefficient agrees because the binary table agrees. Its span-three part also agrees, since it only depends on degree:

$$[x^3z^3]B_{G_{\mathrm{bin}}} = [x^3z^3]B_{H_{\mathrm{bin}}} = 2q^9 + 2q^{18} + 2q^{27} + q^{36}.$$

It remains to inspect exactly forty-two ordered height patterns in the span-two part. The complete nonzero energy profiles are as follows; omitted energies have multiplicity zero in both graphs.

Energy	G_{bin} multiplicity	H_{bin} multiplicity	Difference
3	1	0	1
4	1	2	-1
5	2	2	0
6	4	6	-2
7	6	6	0
8	2	0	2
9	5	4	1
10	5	2	3
11	1	4	-3
12	3	4	-1
13	4	6	-2
14	2	2	0
15	4	2	2
17	1	0	1
18	1	2	-1
Total	42	42	0

Exact computation SNG-E006. The difference polynomial obtained from this profile is

$$\begin{aligned}\Delta(q) &= q^3 - q^4 - 2q^6 + 2q^8 + q^9 + 3q^{10} - 3q^{11} \\ &\quad - q^{12} - 2q^{13} + 2q^{15} + q^{17} - q^{18} \\ &= \boxed{q^3(1-q)(1-q^3)^2(1-q^4)^2}.\end{aligned}$$

The factorization is an exact polynomial identity, checkable by multiplication. Enumeration over all ternary assignments in $\{0,1,2\}^7$ independently reproduces the same x^3 span-two difference. Therefore

$$[x^3z^2](B_{G_{\mathrm{bin}}} - B_{H_{\mathrm{bin}}}) = [x^3](P_{G_{\mathrm{bin}}} - P_{H_{\mathrm{bin}}}) = \Delta(q).$$

Accounting for every lower coefficient. Volumes zero, one and two depend only on the degree sequence and on J_G (SNG-F14, SNG-F15). Both binary graphs have degree sequence $\{4,3,3,2,2,1,1\}$ and $J = q^4+4q^5+q^6+2q^7$, so these coefficients agree. At volume three, the $3e_i$ sector is again a function of degrees and the $e_i+e_j+e_k$ sector is the size-three binary row, equal by SNG-E003. Only the forty-two ordered $2e_i+e_j$ profiles remain; their energy is $4d_i+d_j-4A_{ij}$, so they record the ordered degree pair and the adjacency of the two positions together. The two graphs do not have the same

joint degree matrix: G_{bin} has edges of degree types $\{1,3\},\{1,4\},3\times\{2,3\},\{2,4\},2\times\{3,4\}$, and H_{bin} has $2\times\{1,4\},\{2,2\},2\times\{2,3\},\{3,3\},2\times\{3,4\}$, with equal degree sums edge by edge in aggregate. Nor are they cospectral. The separation claimed here is specifically beyond the complete binary cut-volume census, through the nested chain $S_1=\{i,j\}\supseteq S_2=\{i\}$, whose energy is $c(S_1)+c(S_2)+2e(\{i\},V\setminus\{i,j\})$ by SNG-F10.

For every $0<q<1$, this is strictly positive. At one half it is exactly $11025/262144$. The factors at zero and one have straightforward meanings: zero eliminates all non-constant heights on a connected graph, while one forgets energy and leaves the same number of assignments on seven coordinates. The additional factors at cube and fourth roots are an algebraic feature of this finite difference; no universal cyclotomic cancellation rule is inferred from them.

9.5 What the analytic signatures see

For the primewise Euler lift, the singularity exponent at $s=1$ is the common degree polynomial $c_G(q)=\sum_i q^{d_i}$. Accordingly neither pair is distinguished by that leading exponent. The spectral pair shares

$$c_{\text{sp}}(q)=3q+2q^2+q^3+q^4,$$

and the binary pair shares

$$c_{\text{bin}}(q)=2q+2q^2+2q^3+q^4.$$

Both zero-parameter Euler products are $\zeta(7s)$, because all four graphs are connected. The first minimum-cut correction also agrees within each pair. For the spectral graphs, the oriented minimum-cut side sizes occur with multiplicities three, one, one, three at sizes one, three, four, six. For the binary graphs only sizes one and six occur, each twice. If $\mathcal{P}(s)=\sum_p p^{-s}$, the respective first corrections to $\log(Z_G/\zeta(7s))$ are

$$q\big(3\mathcal{P}(s)+\mathcal{P}(3s)+\mathcal{P}(4s)+3\mathcal{P}(6s)\big)$$

and

$$2q\big(\mathcal{P}(s)+\mathcal{P}(6s)\big).$$

Those common signatures illustrate why a hierarchy is necessary. A graph can share degree data, a leading singular exponent, and its entire minimum-cut volume census with another graph while differing at a later local coefficient. After cancellation of the lower equal local coefficients, the first ratio singularity occurs at $s=1/2$ for the spectral pair and $s=1/3$ for the binary pair. Its exponent is respectively $2q^2(1-q)^3(1+q)$ or $\Delta(q)$. This is a local continuation statement on the positive-real branch for the ratio after cancellation; it is not a blanket meromorphic continuation of either original product.

The finite evidence supports a sharply bounded conclusion. The height-span marked invariant is strictly finer than its binary sector, and its unmarked specialization can distinguish a displayed Laplacian-cospectral pair. Both examples also distinguish vertex realizations with isometric ordinary cut lattices. The evidence does not prove

completeness, does not show that the unmarked series always recovers the binary sector, and does not establish an efficient algorithm for general graph isomorphism. What has been established is a concrete, exact relational distinction and an analytic language that records where that distinction first appears.

9.6 A regular calibration: the cube and the Möbius ladder

A third comparison tests the low-volume blindness of regular graphs proved in the inverse-problems chapter. Let Q_3 be the cube graph on binary vertex labels $0, \dots, 7$, with an edge whenever two labels differ in one binary digit. Let M_8 be the eight-cycle together with the four opposite-vertex chords. Their edge sets are fixed here to make the computation reproducible.

Graph	Edge set
Q_3	01,02,04,13,15,23,26,37,45,46,57,67
M_8	01,04,07,12,15,23,26,34,37,45,56,67

Both graphs have eight vertices, twelve edges, degree three at every vertex, and no triangles. They are nonisomorphic for a simple reason. The cube is bipartite: the parity of the number of ones in the binary label changes across every edge. The ladder contains the five-cycle $0, 1, 2, 3, 4, 0$, using its opposite chord 04, and therefore is not bipartite. No spectral calculation is needed for that nonisomorphism witness.

The common local coefficients through volume three are

$$A_0 = 1, \qquad A_1 = 8q^3,$$

$$A_2 = 12q^4 + 16q^6 + 8q^{12},$$

$$A_3 = 24q^5 + 24q^7 + 8q^9 + 24q^{11} + 32q^{15} + 8q^{27}.$$

Their span decompositions also agree. This agreement is predicted by the theorem that order, regular degree and triangle count determine the marked coefficients through volume three. It is therefore a mathematically controlled delay, rather than an unexplained failure of the population calculation. At volume four the binary subsets already distinguish the graphs:

Graph	Binary four-vertex cut polynomial	Number of subsets
Q_3	$6q^4+32q^6+30q^8+2q^{12}$	70
M_8	$4q^4+40q^6+18q^8+8q^{10}$	70

The six cube subsets of boundary size four are its six square faces. Its two four-vertex subsets of boundary size twelve are its bipartition classes, each containing no internal edges. The ladder has no such boundary-twelve four-subset, consistently with the absence of a bipartition into independent four-sets. The other coefficients count the remaining labeled subsets, rather than quotienting them by graph symmetries.

Exact computation SNG-E007. The first full local difference is

$$[x^4](P_{Q_3} - P_{M_8}) = [x^4 z](B_{Q_3} - B_{M_8}) = \boxed{2q^4(1-q^2)^4}.$$

All other span sectors at volume four agree. The value at $q = 1/2$ is $81/2048$. At the next volume the difference moves to span two:

$$[x^5](P_{Q_3} - P_{M_8}) = [x^5 z^2](B_{Q_3} - B_{M_8}) = \boxed{8q^9(1-q^2)^2(1-q^4)^2}.$$

Its value at one half is $2025/262144$. These identities were replayed using exact coefficient enumeration with the displayed edges. Their status is exact finite computation, followed by exact polynomial factorization. The first ratio singularity of this pair is at $s = 1/4$, with exponent $2q^4(1-q^2)^4$, after the equal lower-volume terms are cancelled. The common leading exponent at $s = 1$ is $8q^3$.

This pair is not another collision of the complete binary polynomial. It agrees only in the early coefficients and separates in a binary sector at volume four. The nonbinary discrepancy at volume five adds an example of how span information is distributed, but is not needed to distinguish the graphs once the binary four-subset census is available. Its role is calibration: it demonstrates a predicted limitation of very early coefficients and provides a positive check that the next level responds to a concrete structural difference.

9.7 Reading the examples as a controlled experiment

The two pairs answer related but different objections. One might first suspect that the construction is simply a complicated way to reproduce the eigenvalues of a graph Laplacian. The spectral pair rules out that interpretation: the entire Laplacian characteristic polynomial agrees, while a positive difference appears at volume two for every interior real parameter. One might then suspect that the extra information is entirely the familiar information obtained by considering all binary cuts and remembering their side sizes. The binary pair rules out that second interpretation: every binary coefficient agrees, while a positive difference appears in the next height span. The experimental sequence is therefore a sequence of exact controls, not a ranking based on a few numerical scores.

The controls are intentionally strong within their stated scope. Both pairs have the same degree multiset, so a singleton degree statistic cannot explain either separation. Both also share their number of components, spanning-tree count, edge connectivity and minimum-cut volume census. Those equalities do not make the examples universally difficult. A simple neighborhood statistic separates each pair, and the second pair has different Laplacian spectra. The point is to identify a definite piece of information that the height population retains, not to pretend that no other mathematical description can retain it.

For the binary pair, the first additional statistic is particularly transparent. An ordered pair (i, j) receives heights two and one. Its energy is $4d_i + d_j - 4\epsilon_{ij}$. A binary pair only sees $d_i + d_j - 2\epsilon_{ij}$, without distinguishing which endpoint is emphasized. The complete binary triple census adds many constraints, but it still does not force the displayed ordered-pair energy distribution. The table supplies a direct counterexample. The new information is a degree and adjacency mixing statistic that is visible through an asymmetric height pattern and is then aggregated without labels.

The analytic ratio does not create that statistic. It transports an already established local discrepancy into a location on the real singularity ladder. The spectral pair first differs at an exponent sum of two and therefore has its first ratio singularity at one half. The binary pair first differs at an exponent sum of three and therefore has its first ratio singularity at one third. This interpretation is helpful because it connects a combinatorial order of distinction to an analytic scale, while keeping the logical direction clear: the exact coefficient calculation comes first, and the analytic statement follows from it.

Chapter 10

Inverse problems and the depth of a height population

An invariant is a way of asking questions about an object while discarding its presentation. The inverse problem asks how much of the object can be recovered from the answers. For relational height populations, this question must be posed at several levels. A finite depth, the full marked series, an unmarked specialization, and a primewise Euler product are different data structures. Their relationships can be proved in some directions and remain open in others. Confusing them would replace a promising research question with an unsupported completeness claim.

The purpose of this chapter is to define those relationships and identify useful targets. The small examples show that the construction preserves some information absent from binary cut populations and the Laplacian spectrum. They do not tell us how many nonisomorphic graphs remain indistinguishable. In particular, a hierarchy can improve at one level and still stabilize long before it becomes complete. Determining whether that happens is a central mathematical task, rather than a detail to be deferred until after a universal reconstruction claim.

10.1 Which equivalence is being studied?

Fix the number of vertices n. Graphs are considered up to ordinary graph isomorphism, but every coefficient counts assignments on the original vertex set before relabeling. Let

$$B_G(x,q,z) = \sum_{\min a = 0} x^{|a|} q^{E_G(a)} z^{\max a}$$

be the normalized marked series. We say that $G \equiv_{\mathrm{mark}} H$ when these formal series agree. We say that $G \equiv_{\mathrm{unmark}} H$ when their specializations at $z = 1$ agree. The latter is equivalent to equality of the unnormalized local series P_G for graphs of the same order, because $B_G(x,q,1) = (1-x^n)P_G(x,q)$. These are definitions, not claims that either relation is graph isomorphism.

For an integer $D \geq 0$, define the depth-D population by

$$B_G^{\leq D}(x,q,z) = \sum_{h=0}^{D} z^h [z^h] B_G(x,q,z).$$

Equality of these finite polynomials is written $G \equiv_D H$. The depth is the largest height span retained; it is not the total volume and is not an iteration count in a numerical algorithm. For fixed D, every coordinate lies between zero and D, at least one coordinate is zero, and the number of assignments is exactly $(D+1)^n - D^n$. Thus each depth comparison is a finite exact question.

Proposition SNG-I001: hierarchy implications. Graph isomorphism implies marked equality. Marked equality implies equality at every finite depth and implies unmarked equality. Equality at depth $D+1$ implies equality at depth D. Conversely, equality at every finite depth implies equality of the full marked series, because every individual coefficient has a finite span. These statements follow directly from coefficient extraction and specialization. No converse from unmarked equality to marked equality is asserted.

The first depth is precisely the binary cut sector, after its two constant assignments are restored. If

$$C_G(x,q) = \sum_{S \subseteq V} x^{|S|} q^{|\partial S|},$$

then

$$C_G(x,q) = 1 + x^n + [z]B_G(x,q,z). \tag{SNG-I002}$$

The term one is the empty subset and the term x^n is the full subset. Both have zero boundary. The normalized span-one population excludes them because a representative must attain both zero and one. The binary pair in the preceding chapter agrees at depth one and differs at depth two, proving that this particular hierarchy step is strict. This is a theorem about the defined hierarchy, supported by an exact counterexample to equality of its first two levels.

For two graphs define their first marked distinguishing depth by

$$d_{\text{span}}(G,H) = \min\{D : G \not\equiv_D H\},$$

with value infinity when no finite depth distinguishes them. One may separately define a first distinguishing volume by the smallest r at which a coefficient of x^r differs. The two measures answer different questions. A large span can be concentrated on one vertex, while a low span can involve many vertices. The binary pair has distinguishing span two and distinguishing volume three. Its difference at volume three belongs specifically to span two; it is not evidence that every depth-two distinction must first occur at volume three.

10.2 Why retaining labels makes reconstruction easy

The intended inverse problem is difficult because vertex labels have been summed out. It is important to explain why a seemingly stronger multivariate formulation would trivialize part of that question. Define the fully labeled local series

$$\mathcal{P}_G(x_1, \dots, x_n; q) = \sum_{a \in \mathbb{N}^n} q^{E_G(a)} \prod_i x_i^{a_i}.$$

Its singleton coefficient is $[x_i]\mathcal{P}_G = q^{d_i}$, so it determines the degree of every named vertex. Its coefficient at two distinct variables is

$$[x_i x_j]\mathcal{P}_G = q^{d_i + d_j - 2\epsilon_{ij}}.$$

The exponent therefore reveals whether the named pair is an edge. This proves **Proposition SNG-I003**: the labeled formal series determines the entire labeled graph using coefficients of total degree at most two. A formal parameter avoids questions of numerical recognition. At a fixed exact real $q \in (0,1)$, the distinct integer powers are also distinct, but a floating-point implementation should not be substituted for that exact statement without an error analysis.

This easy reconstruction is not the problem Singularity is claiming to solve. The specialization $x_1 = \cdots = x_n = x$ combines many assignments into one volume class. Individual pair identities disappear, and their relational distribution remains. The challenge is to recover a graph from that distribution without quietly reintroducing one coordinate variable per vertex. This distinction also explains why a large formal generating function need not supply an efficient isomorphism algorithm. One can encode the adjacency matrix in a generating function almost immediately; the mathematical question is what survives a specified loss of labels.

10.3 Universal relations within each depth

Several identities are available before any difficult inverse problem is solved. They constrain every valid output and reveal how the hierarchy is organized. They also provide exact checks that are different from merely running the same enumeration twice.

Proposition SNG-I004: span reflection. Put $B_{G,h}(x,q) = [z^h]B_G(x,q,z)$. Then

$$B_{G,h}(x,q) = x^{nh} B_{G,h}(x^{-1},q).$$

For a normalized assignment of span h, the reflected assignment $a_i' = h - a_i$ again has minimum zero and maximum h. Every edge difference changes sign, so energy is unchanged, while volume becomes $nh - |a|$. Reflection is an involution, proving the identity coefficient by coefficient. For $h = 1$, this is the usual complement symmetry of binary cuts. For the seven-vertex binary pair, its proved x^3z^2 difference therefore also occurs at $x^{11}z^2$, since $14 - 3 = 11$. This second location follows from a bijection, not from a new empirical claim.

For $h \geq 1$, an assignment of exact span h has at least one coordinate zero and at least one equal to h. Consequently its volume lies between h and $(n-1)h$. Its energy is an integer between zero and $|E|h^2$. If the graph is connected and $h > 0$, the energy is positive. These bounds make $B_{G,h}$ a finite polynomial with explicitly bounded support. The reflection center $nh/2$ is consistent with the two volume bounds.

There is an inclusion-exclusion construction that does not require enforcing minimum and maximum conditions separately in a transfer algorithm. Define

$$T_{G,h}(x,q) = \sum_{a \in \{0,\ldots,h\}^n} x^{|a|} q^{E_G(a)}, \qquad T_{G,-1} = 0.$$

For $h \geq 1$, shift invariance of energy gives

$$B_{G,h} = T_{G,h} - (1 + x^n)T_{G,h-1} + x^n T_{G,h-2}. \tag{SNG-I005}$$

Assignments missing the top height contribute $T_{G,h-1}$. Assignments missing zero can be shifted down by one, contributing $x^n T_{G,h-1}$. Assignments missing both endpoints have been subtracted twice and contribute $x^n T_{G,h-2}$. The formula holds for every graph because only common-shift invariance was used. At $q = 1$, it reduces to a universal volume polynomial independent of graph edges. In particular,

$$B_{G,h}(1, 1) = (h + 1)^n - 2h^n + (h - 1)^n.$$

Summing through depth D gives $(D + 1)^n - D^n$, as required. At unmarked $q = 1$, one obtains the further check

$$B_G(x, 1, 1) = \frac{1 - x^n}{(1 - x)^n}.$$

These identities explain why all examples coincide when energy is forgotten. A successful distinguishing polynomial must vanish at that specialization. They also clarify why a zero at $q = 1$ is a necessary population constraint, rather than evidence by itself of a special physical or cyclotomic mechanism.

10.4 Connected components and exact recovery at zero parameter

Disconnected graphs require a different normalization argument from connected graphs. Energy zero means that height is constant on each connected component, not that all vertices share a common height. If the component sizes are $n_1, \ldots, n_c$, then

$$P_G(x, 0) = \prod_{j=1}^{c} \frac{1}{1 - x^{n_j}}, \qquad Z_G(s, 0) = \prod_{j=1}^{c} \zeta(n_j s). \tag{SNG-I006}$$

This statement follows by assigning an arbitrary nonnegative common height to each component. It includes isolated vertices, whose factors are $(1 - x)^{-1}$ and $\zeta(s)$. It shows why the connected-graph claim that the zero-parameter value is finite at $s = 1$ cannot be copied unchanged to a graph with isolated vertices.

Proposition SNG-I007: the zero-parameter local series determines the multiset of component sizes. Let m_d be the number of components of size d, and put $b_r = r[x^r] \log P_G(x, 0)$. Expanding the logarithm gives

$$b_r = \sum_{d \mid r} d\, m_d.$$

Möbius inversion yields

$$m_n = \frac{1}{n}\sum_{d|n}\mu(n/d)b_d.$$

All operations here are formal power-series operations; no analytic continuation is needed. This recovers component sizes, not the internal graph on each component. Within a fixed connected component size, all connected graphs have the same zero-energy population.

For unmarked populations, ordinary disjoint union gives $P_{G\sqcup H} = P_G P_H$. The globally normalized marked population does not simply factor as $B_G B_H$: the two components may be shifted by different heights before their common global minimum is removed, and the global span depends on those relative shifts. A depth algorithm must retain this information or work first with the bounded boxes $T_{G,h}$, which do multiply over disjoint components, and then use the inclusion-exclusion formula. Treating marked disjoint union as an automatic product would erase legitimate assignments.

10.5 Easy classes and the limits of degree information

The first local coefficient is the degree polynomial $c_G(q) = \sum_v q^{d_v}$. It determines the degree multiset, because its coefficient of q^d is the number of degree-d vertices. This immediately resolves several inverse questions under explicit class assumptions. Such small results are useful calibration points; they must not be expanded into a claim about arbitrary graphs with the same degree sequence.

For a star $K_{1,n-1}$,

$$c_G(q) = (n-1)q + q^{n-1}.$$

For a path P_n with $n \geq 3$,

$$c_G(q) = 2q + (n-2)q^2.$$

The two polynomials differ for $n \geq 4$, while the three-vertex star and path are the same graph. A degree multiset consisting of one degree-$n-1$ vertex and $n-1$ leaves forces the star even without a prior star-family promise. A connected graph whose vertices all have degree two is a cycle, so the degree polynomial determines it within that connected class. If connectedness is omitted, a disjoint union of cycles has the same degree polynomial for every partition of the total order into allowed cycle lengths. The zero-parameter component-size series then resolves those different cycle decompositions.

A connected tree of maximum degree at most two is a path: there can be no branch and no cycle. Complete graphs and edgeless graphs are also forced by their degree multisets. Within the family of complete bipartite graphs, the polynomial $aq^b + bq^a$ determines the unordered pair of part sizes $\{a, b\}$. This last statement has a family restriction. Other graph families may share that degree sequence, so the polynomial by itself does not certify complete bipartiteness.

The early coefficients become less informative on regular graphs. If G is d-regular on n vertices with $m = nd/2$, then

$$A_2 = nq^{4d} + mq^{2d-2} + \left(\binom{n}{2} - m\right) q^{2d}.$$

Thus no two d-regular graphs of the same order can be distinguished at volume two. At volume three, the ordered $2+1$ sector is

$$2mq^{5d-4} + (n(n-1) - 2m)q^{5d}.$$

The remaining binary triple sector depends on the numbers N_j of three-vertex subsets spanning exactly j edges. If $T = N_3$ is the triangle count, then

$$N_2 + 3T = n\binom{d}{2}, \qquad N_1 + 2N_2 + 3T = m(n-2),$$

$$N_0 + N_1 + N_2 + T = \binom{n}{3}.$$

These identities count wedges, edge-containing triples, and all triples. They show that n, d, T determine every N_j. Therefore **Proposition SNG-I008** states that regular graphs with the same order, degree and triangle count have identical marked coefficients through total volume three. This result supplies a useful negative prediction for searches. The heterogeneous degrees of the displayed binary pair are part of why an ordered height pattern can distinguish it so early.

There are also useful support bounds on the coefficients at arbitrary volume. Suppose the graph is connected on $n \geq 2$ vertices, with maximum degree $d_{\max} > 0$ and edge connectivity λ. For a nonnegative assignment of volume r,

$$E_G(a) = \sum_i d_i a_i^2 - 2\sum_{ij \in E} a_i a_j \leq d_{\max} \sum_i a_i^2 \leq d_{\max} r^2.$$

The upper bound is attained by placing all height r at a maximum-degree vertex. For $r \geq 1$, equality in $\sum_i a_i^2 \leq r^2$ requires that only one coordinate be positive. Therefore the coefficient of the highest possible energy $q^{d_{\max} r^2}$ in $[x^r]P_G$ is exactly the number of maximum-degree vertices. This is **Proposition SNG-I009**, an exact boundary constraint on every local population. It should not be mistaken for a new distinguishing statistic: it is already determined by the first coefficient's degree data.

For a normalized nonzero assignment, let $h = \max a$. The threshold-cut identity and integrality give $E_G(a) \geq \lambda h$. At least one coordinate is zero, so $r \leq (n-1)h$. Consequently every occupied coefficient of the normalized population at positive volume satisfies

$$E_G(a) \geq \lambda \left\lceil \frac{r}{n-1} \right\rceil.$$

This lower bound is easy to evaluate and does not require solving an optimization problem over all assignments. It is not asserted to be sharp at every volume. A spectral bound can be stronger at large volume: if λ_2 is the first nonzero Laplacian eigenvalue, subtract the mean of a to obtain

$$E_G(a) \geq \lambda_2 \left(\sum_i a_i^2 - \frac{r^2}{n} \right) \geq \frac{\lambda_2 r^2}{n(n-1)}.$$

The last step follows because a zero coordinate leaves at most $n-1$ nonzero coordinates and Cauchy's inequality gives $\sum_i a_i^2 \geq r^2/(n-1)$. These inequalities connect familiar spectral control to the support of the new population without equating the two objects. They help delimit an exact search region, while the examples show that knowledge of the spectrum does not determine the multiplicities within that region.

10.6 A family with exactly controlled visibility depth

The decorated-deck criterion SNG-F17 asks whether nonisomorphic graphs can agree through arbitrarily large volume. They can, with an explicit discrepancy. For $\ell \geq 2$ and $1 \leq j \leq \ell$, let $G_{\ell,j}$ be the cycle $C_{2\ell}$ on $0, \dots, 2\ell-1$ with a leaf u attached at vertex 0 and a leaf w attached at vertex j. The graphs have $n = 2\ell + 2$ vertices, the same degree sequence and the same multiset of endpoint-degree types on edges when $j \geq 2$. Their cut lattices are isometric: each is the cut lattice of $C_{2\ell}$ orthogonally summed with $\mathbb{Z}^2$ for the two pendant edges, whose differences are free. So the energy theta series that forgets volume is identical. They are not cospectral in general.

Theorem SNG-F36 (exact visibility depth for pendant cycles). For $2 \leq p < \ell$, the graphs $G_{\ell,p}$ and $G_{\ell,\ell}$ have equal local, normalized and span-marked coefficients at every volume $m \leq p$, and

$$[x^{p+1}](L_{G_{\ell,p}} - L_{G_{\ell,\ell}}) = q^2(1-q)^2(1-q^2)^p.$$

The discrepancy at volume $p+1$ lies entirely in span one. More precisely, the complete binary-sector difference is

$$\sum_{S \subseteq V} z^{|S|} \left(q^{c_{G_{\ell,p}}(S)} - q^{c_{G_{\ell,\ell}}(S)} \right) = q^2(1-q)^2(1-q^2)^p \, z^{p+1}(1-z)^2 \, U_{\ell-p}(z,q)^2,$$

where $U_0 = 0$, $U_1 = 1$ and $U_k = (1+z)U_{k-1} - z(1-q^2)U_{k-2}$. The binary formula holds for $1 \leq p < \ell$.

Proof. **Binary sector.** Summing over the leaves first: if its attachment vertex is outside S, a leaf contributes $1 + zq$; if inside, $q + z = (1+zq) - (1-q)(1-z)$. Write $\phi(s) = \phi_0 - (1-q)(1-z)s$ for $s \in \{0,1\}$. With $N = 2\ell$, $T = \begin{pmatrix} 1 & q\sqrt{z} \\ q\sqrt{z} & z \end{pmatrix}$ and $\Pi = \mathrm{diag}(0,1)$, the cycle part of the binary polynomial with both vertices 0 and j in S is $\Gamma(j) = \mathrm{tr}(\Pi T^j \Pi T^{N-j})$. Terms with fewer marked vertices in S do not depend on j, by rotation and reflection of the cycle. Hence the difference is $(1-q)^2(1-z)^2(\Gamma(p) - \Gamma(\ell))$. With orthonormal eigenvectors $v_\pm$ and eigenvalues $\lambda_\pm$ of T,

$$\Gamma(j) = \sum_{a,b \in \{\pm\}} (v_a)_2^2 (v_b)_2^2 \lambda_a^j \lambda_b^{N-j}.$$

The diagonal terms do not depend on j, and $(v_+)_2^2 (v_-)_2^2 = \beta^2/(\lambda_+ - \lambda_-)^2$ for the off-diagonal entry $\beta = q\sqrt{z}$. So

$$\Gamma(p) - \Gamma(\ell) = \frac{q^2 z}{(\lambda_+ - \lambda_-)^2} \left(\lambda_+^p \lambda_-^{N-p} + \lambda_-^p \lambda_+^{N-p} - 2(\lambda_+\lambda_-)^\ell \right) = q^2 z \, (\lambda_+\lambda_-)^p U_{\ell-p}^2,$$

with $\lambda_+\lambda_- = \det T = z(1-q^2)$ and $U_k = (\lambda_+^k - \lambda_-^k)/(\lambda_+ - \lambda_-)$. This satisfies the stated recurrence, since $\lambda_+ + \lambda_- = 1 + z$. Since $U_k = 1 + O(z)$, the lowest term is $q^2(1-q)^2(1-q^2)^p z^{p+1}$.

Decorated decks of size at most p. Let S have $t \le p$ vertices. Each component of $G_{\ell,j}[S]$ is either an isolated leaf whose attachment vertex is outside S, or an arc of the cycle with the leaves of its marked vertices that lie in S. An arc containing both marked vertices would need at least $p+1$ cycle vertices for $j = p$ and $\ell + 1$ for $j = \ell$, so each arc contains at most one. The degree-decorated type of S is therefore a multiset of components of three kinds: free arcs, which contain no marked vertex and are recorded by their lengths; marked arcs, recorded by length, the offset of the degree-three vertex and whether its leaf is present; and isolated leaves. The two marked vertices are interchangeable, each having degree three and one leaf. Fix a type τ and an assignment of its marked components to the two marked vertices; then the marked arcs are placed. The free arcs, of total length F in k blocks, must be placed pairwise non-adjacent, avoiding the marked vertices and non-adjacent to the marked arcs, in the two segments between the marked regions. Let α_1, α_2 be the segment lengths after removing one obligatory gap next to each marked arc. Placing labeled blocks in one segment of effective length α has $k!\binom{\alpha - F + 1}{k}$ solutions up to the labeling convention, so the two-segment count is a sum over splits of the blocks of products of binomials.

Placement lemma. This sum is a function of $\alpha_1 + \alpha_2$ alone whenever $\alpha_1, \alpha_2 \ge F - 1$. Gluing the segments with one separator vertex identifies the count with placements in a line of length $\alpha_1 + \alpha_2 + 1$, which depends only on the sum, minus placements in which some block covers the separator. The latter decompose, block by block and offset by offset, into the same two-segment problem with one block fewer. By induction on k, the count is a function of the sum once both lengths are at least F. As a polynomial identity in (α_1, α_2) it then holds everywhere. For $\alpha_i \ge F - 1$ every binomial $\binom{\alpha_i - F_I + 1}{|I|}$ has a nonnegative upper argument, so it equals the true count.

For $G_{\ell,\ell}$ both segments come from arcs with $\ell - 1 \ge p$ interior vertices. For $G_{\ell,p}$ the short arc has $p - 1$ interior vertices, and the marked arcs remove at most $(t - F)$ of them together with their gaps, so $\alpha_1 \ge p - 1 - (t - F) \ge F - 1$; the long arc is larger. The same counting shows that the marked arcs fit in both graphs whenever $t \le p$. The sum $\alpha_1 + \alpha_2$ depends only on τ and the assignment, so both graphs have the same number of subsets of each decorated type and size $t \le p$.

Conclusion. SNG-F17 gives equality of local and marked coefficients through volume p. At volume $p+1$, its proof matches every profile whose support has at most p vertices. The remaining profiles have $p+1$ support vertices and volume $p+1$, so they are binary. Their difference is the z^{p+1} coefficient above, and these subsets are proper because $p + 1 < n$. □

Taking $\ell = p + 1$ gives $n = 2p + 4$ and visibility depth $p + 1 = (n - 2)/2$, linear in the order. The theorem is unconditional and explicit, and it answers the first question raised after SNG-F17. It is not a uniform statement about all graph pairs of a given order.

Theorem SNG-F37 (sign and monotonicity of the pendant discrepancy). For real $0 < x < 1$ and $0 < q < 1$, and $1 \le j \le j' \le \ell$,

$$L_{G_{\ell,j}}(x, q) \ge L_{G_{\ell,j'}}(x, q).$$

Proof. Let $f(h) = \sum_{b \ge 0} x^b q^{(b-h)^2}$ be the leaf factor, bounded by $1/(1-x)$. Let T

be the operator on $\ell^2(\mathbb{Z}_{\geq 0})$ with kernel $x^{(h+h')/2}q^{(h-h')^2}$. It is Hilbert–Schmidt, and it is positive semidefinite because $q^{(h-h')^2} = e^{-c(h-h')^2}$ with $c = -\log q > 0$ is a positive-definite Gaussian kernel, and conjugating by the diagonal $x^{h/2}$ preserves this. Summing over leaf heights and then over cycle heights gives $L_{G_{\ell,j}} = \mathrm{tr}(FT^jFT^{N-j})$ with $F = \mathrm{diag}(f)$. In an orthonormal eigenbasis, with $\lambda_a \geq 0$ and $F_{ab} = \langle v_a, Fv_b\rangle$, the absolutely convergent expansion is

$$L_{G_{\ell,j}} = \tfrac{1}{2}\sum_{a,b} F_{ab}^2(\lambda_a^j\lambda_b^{N-j} + \lambda_b^j\lambda_a^{N-j}).$$

For $\lambda_a, \lambda_b \geq 0$ the bracket $\varphi(j)$ satisfies $\varphi(j) - \varphi(j+1) = (\lambda_a\lambda_b)^j(\lambda_b^{N-2j-1} - \lambda_a^{N-2j-1})(\lambda_b - \lambda_a) \geq 0$ for $j < \ell$. □

In particular $L_{G_{\ell,p}} - L_{G_{\ell,\ell}} = \frac{1}{2}\sum F_{ab}^2(\lambda_a\lambda_b)^p(\lambda_a^{\ell-p} - \lambda_b^{\ell-p})^2$ is a sum of squares, consistent with the positive leading coefficient. The arithmetic ratio of SNG-A11 therefore has exponent $\delta = q^2(1-q)^2(1-q^2)^p > 0$ at $s = 1/(p+1)$ for every $q \in (0,1)$. This exponent is largest at the root $q_p^* = 2/(1+\sqrt{4p+9})$ of $1 - q - (p+2)q^2$, where $q_p^* \sim p^{-1/2}$ and $\delta_p(q_p^*) \sim 1/(ep)$, whereas $\delta_p(1/2) = 3^p/4^{p+2}$ decays exponentially. Choosing the probe parameter thus turns an exponentially weakening signal into one of inverse-depth size.

The implementation is `src/singularity/families.py`. `tests/test_families.py` and `results/pendant_family.json` check the following: the first difference by weak-composition enumeration for all $2 \leq p < \ell \leq 7$; the closed binary form against complete subset enumeration for $1 \leq p < \ell \leq 7$; and equality of decorated decks for every size up to p, with inequality at $p+1$, on six pairs. They also check the optimal-probe equation exactly.

10.7 Terminal kernels and the Whitney twist

A **block** is a graph G with an ordered tuple $\mathbf{s} = (s_1, \dots, s_k)$ of distinct terminal vertices. Write $I = V \setminus \mathbf{s}$. Its **terminal kernel** is the family of series

$$K_G(\mathbf{b}; x, q) = \sum_{a_I \geq 0} x^{\sum_{i\in I} a_i}\, q^{E_G(a_I, \mathbf{b})}, \qquad \mathbf{b} \in \mathbb{Z}_{\geq 0}^k,$$

where $E_G(a_I, \mathbf{b})$ is the energy of the profile equal to $\mathbf{b}$ on the terminals and to a_I inside. Kernels retain terminal heights and interior volume separately, so they compose.

Theorem SNG-F41 (gluing and the Whitney twist). Let $(G_1, \mathbf{s})$ and $(G_2, \mathbf{t})$ be blocks with k terminals that share no edge, and let $G_1 \cup_\pi G_2$ identify s_i with $t_{\pi(i)}$ for a permutation π. Then

$$L_{G_1\cup_\pi G_2}(x,q) = \sum_{\mathbf{b}} x^{|\mathbf{b}|} K_{G_1}(\mathbf{b})\, K_{G_2}(\pi\cdot\mathbf{b}).$$

For $k = 2$, put $A_j(b_1,b_2) = \frac{1}{2}(K_{G_j}(b_1,b_2) - K_{G_j}(b_2,b_1))$. Then the straight and twisted gluings satisfy

$$L_{G_1\cup G_2} - L_{G_1\cup_{\mathrm{tw}} G_2} = 2\sum_{\mathbf{b}} x^{|\mathbf{b}|} A_1(\mathbf{b})A_2(\mathbf{b}).$$

In particular the twist is invisible at every volume if either kernel is symmetric, for instance if a block has an automorphism exchanging its terminals.

Proof. A profile on the glued graph is a choice of terminal heights $\mathbf{b}$ and independent interior profiles of the two blocks. Its volume is $|\mathbf{b}|$ plus the two interior volumes. Each edge lies in exactly one block, so its energy is the sum of the two block energies. Summing gives the gluing formula. For $k = 2$ write $K_j = S_j + A_j$ with S_j symmetric under exchanging the coordinates of $\mathbf{b}$. The weight $x^{|\mathbf{b}|}$ is symmetric, so $\sum_{\mathbf{b}} x^{|\mathbf{b}|} F(\mathbf{b}) = 0$ for every antisymmetric F. Expanding $(S_1 + A_1)(S_2 + A_2) - (S_1 + A_1)(S_2 - A_2) = 2S_1A_2 + 2A_1A_2$ and dropping the antisymmetric S_1A_2 proves the identity, coefficient by coefficient in x. □

Theorem SNG-F42 (twisting identical blocks). Let $G_1 = G_2 = G$ with terminals (s, t). Write $2A(\mathbf{b}) = \sum_v x^v P_{\mathbf{b},v}(q)$ with $P_{\mathbf{b},v} \in \mathbb{Z}[q]$, and let $v_{\mathbf{b}}$ be the least v with $P_{\mathbf{b},v} \neq 0$. If $A \not\equiv 0$, the straight and twisted self-gluings first differ at volume

$$\mu = \min_{\mathbf{b}}(|\mathbf{b}| + 2v_{\mathbf{b}}), \qquad [x^\mu](L_{G\cup G} - L_{G\cup_{\mathrm{tw}} G}) = \tfrac{1}{2} \sum_{|\mathbf{b}|+2v_{\mathbf{b}}=\mu} P^2_{\mathbf{b},v_{\mathbf{b}}} \neq 0.$$

For real $0 < x < 1$ and $0 < q < 1$, $L_{G\cup G} \geq L_{G\cup_{\mathrm{tw}} G}$.

Proof. By SNG-F41 the difference is $\frac{1}{2}\sum_{\mathbf{b}} x^{|\mathbf{b}|}(2A(\mathbf{b}))^2$. A term $x^{|\mathbf{b}|+v+v'} P_{\mathbf{b},v} P_{\mathbf{b},v'}$ is nonzero only if $v, v' \geq v_{\mathbf{b}}$, so it has volume at least $|\mathbf{b}| + 2v_{\mathbf{b}} \geq \mu$. At volume μ only $v = v' = v_{\mathbf{b}}$ with $|\mathbf{b}| + 2v_{\mathbf{b}} = \mu$ survives. A finite sum of squares of nonzero real polynomials is not the zero polynomial. For real parameters in the unit square every kernel converges absolutely, being dominated by L_G, so the difference is a convergent sum of squares of real numbers. □

The pendant-cycle family is an instance. Let $B_{m,k}$ be the path of length $2m$ from s to t with a leaf at distance $k < m$ from s. Gluing it to itself gives C_{4m} with leaves $2k$ apart, and the twist gives leaves $2m$ apart. These are $G_{2m,2k}$ and $G_{2m,2m}$ of SNG-F36. SNG-F42 then gives depth $2k + 1$ and leading coefficient $q^2(1-q)^2(1-q^2)^{2k}$, in agreement with SNG-F36, and its positivity is the even case of SNG-F37. The twist representation exists only for even p: with leaves at distances a, c from s on the two arcs, the straight gluing gives distance $a + c$ and the twist $a + \ell - c$, and matching (p, ℓ) forces $p = 2a$. The kernel point of view separates the mechanism from the family. Visibility of a twist is decided by the asymmetry order of the block, the least volume at which the two terminals' decorated neighbourhoods differ, and the sign is always that of a sum of squares.

Proposition SNG-F43 (marked composition). For blocks as in SNG-F41, let $K^{(H)}$ be the box kernel, with every height, terminals included, in $\{0, \ldots, H\}$. The glued box histogram is $C_H = \sum_{\mathbf{b}} x^{|\mathbf{b}|} K^{(H)}_{G_1}(\mathbf{b}) K^{(H)}_{G_2}(\pi \cdot \mathbf{b})$, and the span-marked series of the glued graph is recovered from these by $D_H = C_H - x^n C_{H-1}$ and exact span $D_H - D_{H-1}$.

Proof. The gluing argument of SNG-F41 applies verbatim inside the box. SNG-F25 converts box histograms into minimum-zero and exact-span counts, using one global translation of the glued graph. □

Global marks are not local: the minimum and maximum of a glued profile may sit in either block, so the marked series cannot be read from the unbounded kernels alone. The box kernels supply exactly the missing information. `marked_glue` in `src/singu`

`larity/terminals.py` implements this, and the tests compare it with direct marked enumeration for straight and twisted gluings and for a disconnected block.

Proposition SNG-F44 (star probes determine a block). Let G be a block with k terminals and $R \geq 0$. For $\mathbf{r} \in \{0, \dots, R\}^k$, let $S_{\mathbf{r}}$ be the context with r_i pendant leaves on terminal i. The series $L_{G \cup S_{\mathbf{r}}}$ modulo x^{R+1}, over all $(R+1)^k$ such $\mathbf{r}$, determine $x^{|\mathbf{b}|} K_G(\mathbf{b})$ modulo x^{R+1} for every $\mathbf{b}$, and this holds already after specializing q to any $q_0 \in (0,1)$. Consequently two k-terminal blocks satisfy $L_{G \cup C} = L_{G' \cup C}$ for every edge-disjoint context C if and only if $K_G = K_{G'}$.

Proof. A leaf on a terminal of height h contributes $f(h) = \sum_{c \geq 0} x^c q^{(c-h)^2}$, so $L_{G \cup S_{\mathbf{r}}} = \sum_{\mathbf{b}} y_{\mathbf{b}} \prod_i f(b_i)^{r_i}$ with $y_{\mathbf{b}} = x^{|\mathbf{b}|} K_G(\mathbf{b})$. Modulo x^{R+1} only $|\mathbf{b}| \leq R$ contributes, and in particular only $b_i \leq R$. In the ring $\mathbb{Q}(q)[x]/(x^{R+1})$, or $\mathbb{Q}[x]/(x^{R+1})$ after specializing q, the differences $f(h) - f(h') \equiv q^{h^2} - q^{h'^2} \pmod{x}$ are units for $h \neq h'$, because their constant terms are nonzero (for $0 < q_0 < 1$ the numbers $q_0^{h^2}$ are distinct). The tensor Vandermonde matrix $\left(\prod_i f(b_i)^{r_i}\right)_{\mathbf{r},\mathbf{b}}$, with $\mathbf{b}, \mathbf{r} \in \{0, \dots, R\}^k$, is therefore invertible: its determinant is a product of powers of these differences. Solving recovers every $y_{\mathbf{b}}$, including those with $|\mathbf{b}| > R$, which vanish. The 'if' part of the consequence is the gluing formula; the 'only if' part follows from the star contexts. □

So every distinction a block can make in any context is already made by star contexts. The kernel is the complete interface object for edge-disjoint gluing at its terminals. The count can be reduced to exactly $\binom{R+k}{k}$, the number of unknowns $y_{\mathbf{b}}$ with $|\mathbf{b}| \leq R$. The star probes with $|\mathbf{r}| \leq R$ suffice. Modulo x, their matrix is monomial interpolation on the lower set $\{|\mathbf{r}| \leq R\}$ at the grid points $(q^{b_1^2}, \dots, q^{b_k^2})$, $|\mathbf{b}| \leq R$, and interpolation on a lower set with tensor-grid nodes of distinct coordinates is unisolvent, as the Newton form on lower sets shows. The matrix is therefore invertible over $\mathbb{Q}[x]/(x^{R+1})$. The count is optimal in the following sense. The star blocks are themselves realizable k-terminal blocks, and by the same invertibility their kernels span the whole $\binom{R+k}{k}$-dimensional space of truncated $(y_{\mathbf{b}})$. A family of fewer contexts gives a linear map from that span with a nonzero kernel, so it cannot determine every element of the span. Swap-symmetric contexts are treated below (SNG-F47, SNG-F48); compression with proved savings remains open (P54). `recover_kernel_from_probes` in `src/singularity/terminals.py` carries out the inversion at a rational q_0. The test recovers a two-terminal kernel through volume three from sixteen glued graphs counted by plain enumeration.

Proposition SNG-F45 (rooted decks bound the asymmetry order). For a block G with terminals (s,t) and $S \subseteq I$, regard $G[S \cup \{s,t\}]$ as decorated by ambient degrees and by the roles of the two terminals. If for every $j \leq m$ the multiset of these types over $|S| = j$ is unchanged when the roles of s and t are exchanged, then $K_G(\beta, \gamma)$ and $K_G(\gamma, \beta)$ agree in every cell of interior volume at most m. Consequently twisting two copies of G is invisible below volume $2m+3$, and more generally twisting blocks with rooted-deck symmetry through sizes m_1 and m_2 is invisible below volume $m_1 + m_2 + 3$.

Proof. As in SNG-F17, interior profiles of volume v with exact support S have energy determined by the decorated type of $S \cup \{s,t\}$ and the terminal heights. The edges inside that set are read from the induced graph, and every other edge at a vertex u of the set contributes a_u^2 times its number, which is fixed by the degree label. A role-respecting isomorphism between a type and the role-exchanged type of another support gives a bijection of positive interior assignments preserving volume and energy, carrying terminal heights (β, γ) to (γ, β). Summing over supports of size at

most $v \le m$ gives the equality of cells. In the pairing of SNG-F41 a term at terminal heights $\mathbf{b}$ needs nonzero cells of both $A_1(\mathbf{b})$ and $A_2(\mathbf{b})$. These require $\beta \neq \gamma$, so $|\mathbf{b}| \ge 1$, and interior volumes at least $m_1 + 1$ and $m_2 + 1$. Its volume is therefore at least $m_1 + m_2 + 3$. □

In the exhaustive pairing of all 54 distinct asymmetric kernels of connected blocks with at most five vertices, no pairing vanished through volume six, and every twist was visible by volume three. Among blocks with at most six vertices and terminals of equal degree there are 104 distinct asymmetric kernels. None of their 5,460 pairings vanishes through volume six; the first visible volumes are 3, 5 and 6, the last attained by different blocks with $m_1 = 1$, $m_2 = 2$ as the bound allows. A cancelling pair of asymmetric kernels, which would make a twist invisible without a symmetry, has not been found. Its existence is open.

On every connected block with at most six vertices, with terminals 0 and 1 (27,474 blocks), the bound is sharp. Whenever the kernel is asymmetric, its least asymmetric interior volume equals the first size $m + 1$ at which the rooted decks lose their swap symmetry; this happens for 24,066 blocks. In the remaining blocks both are symmetric at every size. Whether this equality holds in general is open; it is a finite observation, not a theorem. The path blocks $B_{m,k}$ attain it with first asymmetric size k.

A systematic census of switches is retained in `results/whitney_switches.json`, produced by `scripts/whitney_switches.py` and checked in CI. Family A glues every connected block on three or four vertices to every such block without the terminal edge. Family B uses a first block on at most five vertices with terminals of equal degree, so that the twist preserves the degree sequence. For each of the 124 pairs kept after removing isomorphic duplicates, the receipt records simplicity, connectedness and cycle-matroid preservation, checked by comparing all circuits under the edge correspondence. It also records isomorphism by canonical code and the first local and marked differences through volume six, by weak compositions and level chains, together with the SNG-F41 prediction. In family A, 68 switches give isomorphic graphs and 28 separate at volume one, where the degree sequence changes. All 28 degree-preserving switches of family B separate at volume two: by SNG-F15 the difference is $(q^{-2} - 1)(J_G - J_{G'})$, and the twist changes the edge degree-sum polynomial J. No pair is unresolved at the bound. Deeper twists require blocks whose terminals agree in rooted decks to larger size (SNG-F45), and the path blocks show that the depth can be made arbitrarily large.

A twist changes the graph but preserves its cycle matroid, so the cut lattice and its energy theta series are unchanged. A block whose kernel is symmetric without an exchanging automorphism would therefore give two graphs with identical scalar height series. Among all connected blocks with at most six vertices and balanced terminals (7,456 at six vertices), the kernel through volume six is symmetric exactly when such an automorphism exists. This finite search does not decide the question in general. `src/singularity/terminals.py` computes kernels on a checked decomposition, with the terminals in every bag, and checks them against enumeration. `tests/test_terminals.py` verifies the gluing formulas for two and three terminals, the twist identity, invisibility for symmetric blocks, and the depth formula and identification above for $B_{m,k}$ with $m \le 4$.

10.8 Restricted contexts and interface rank under a volume budget

SNG-F44 identifies the kernel as the complete interface for unrestricted contexts. A natural restricted class consists of the contexts that cannot tell the two terminals apart. Call a two-terminal context C *swap-symmetric* if it has an automorphism exchanging its terminals. For terminal heights write an unordered orbit $o = \{s,t\}$, $|o| = s+t$, and put $K_G^+(o) = K_G(s,t) + K_G(t,s)$ for $s \neq t$ and $K_G^+(o) = K_G(s,s)$.

Proposition SNG-F47 (symmetric contexts see the symmetrized kernel). If C is swap-symmetric and edge-disjoint from G, then $L_{G\cup C} = \sum_o x^{|o|} K_G^+(o) K_C(o)$. Hence blocks with $K_G^+ = K_{G'}^+$ are indistinguishable by every swap-symmetric context, in particular G with terminals (s,t) and with terminals (t,s).

Proof. The automorphism carries interior assignments with terminal heights (s,t) to assignments with (t,s), preserving volume and energy, so $K_C(s,t) = K_C(t,s)$. Group the terms **b** and its reversal in SNG-F41. □

Whether swap-symmetric contexts see all of K^+ is a question of rank. Fix a budget R and $q_0 \in (0,1)$, and let the unknowns be the coefficients $Y_o[v]$ of $Y_o = x^{|o|} K_G^+(o)(x, q_0)$, $|o| \le v \le R$. The glued coefficient at volume w is $\sum_o \sum_{j \le w} Y_o[j]\, K_C(o)[w-j]$, so the probe map is block lower triangular in the degree, and its diagonal blocks are the interior-volume-zero values $K_C(o)[0]$.

Theorem SNG-F48 (interface rank of symmetric contexts). Let $m_o = s^2 + t^2$ and $d_o = (s-t)^2$.

(a) If C is swap-symmetric and has no edge between its terminals, then $K_C(o)[0] = q^{\delta m_o}$, where δ is the common degree of the terminals in C. Consequently every vector supported on the top degree, $\sum_o c_o Y_o[R]$ with $\sum_{m_o = m} c_o = 0$ for each m, is invisible to all such contexts at budget R. The rank defect at budget R is at least the number of orbits with $|o| \le R$ minus the number of distinct values m_o among them. It is positive from $R = 7$ on, where $\{0,5\}$ and $\{3,4\}$ share $m = 25$.

(b) Let $C_{\delta,\varepsilon}$ have δ pendant leaves on each terminal and $\varepsilon \in \{0,1\}$ terminal edges, and suppose G has no terminal edge. If no three orbits with $|o| \le R$ share a value of m_o, then the $2M$ series $L_{G\cup C_{\delta,\varepsilon}} \bmod x^{R+1}$, $\delta < M$, where M is the number of distinct m_o, determine every Y_o modulo x^{R+1}. The hypothesis holds for every $R \le 24$; the first triple is $325 = 1^2 + 18^2 = 6^2 + 17^2 = 10^2 + 15^2$.

Proof. (a) At interior volume zero all interior heights vanish. With no terminal edge the energy is $\deg_C(s) s^2 + \deg_C(t) t^2$, and the automorphism forces the two degrees to be equal. A coordinate $Y_o[R]$ enters only the glued coefficient at volume R, with coefficient $K_C(o)[0]$, so the stated vectors give zero in every coordinate.

(b) Induct on v. Once the $Y_o[j]$, $j < v$, are known, subtracting their contribution from the volume-v coefficients leaves $\sum_{|o| \le v} Y_o[v] q_0^{\delta m_o + \varepsilon d_o}$ for all probes. Suppose a combination $\sum_o c_o q_0^{\delta m_o + \varepsilon d_o}$ vanishes for $\delta < M$ and both ε. The numbers q_0^m are distinct, so the Vandermonde system in δ gives $\sum_{m_o = m} c_o q_0^{\varepsilon d_o} = 0$ for each m and ε. Two orbits with the same m and the same d have the same st and $s+t$, so they coincide. A group of at most two orbits therefore has distinct values of

d, and the 2×2 system forces $c_o = 0$. Hence the residual system is injective at every degree, and Y is determined. □

Part (b) is constructive: `recover_symmetrized_kernel` in `src/singularity/contexts.py` solves the triangular system exactly. `scripts/symmetric_contexts.py`, checked in CI, replays the recovery at $R = 3$, $q_0 = 1/2$, on all 15 connected blocks with at most four vertices and no terminal edge, against the symmetrized kernel computed by bag messages. For the family with δ leaves per terminal, $c \le R$ shared neighbours and an optional terminal edge, it also computes the exact rank of the probe map for $R \le 7$. With the edge the rank is full. Without it the rank equals the number of unknowns minus the top-degree defect of (a): full through $R = 6$, and 69 of 70 at $R = 7$. The same pattern holds at $R = 8$ (94 of 95, full with the edge) and $R = 9$ (124 of 125, full with the edge), computed once in the session that introduced it. Lower-degree collisions are thus resolved by the next volume, and the loss is confined to the top degree in the computed range. The fixed-budget exactness question outside the proved ranges remains open. Theorem SNG-F51 below proves that the edge-free class determines every finite prefix of K^+ when additional observation volume is allowed. For comparison, unrestricted contexts have full rank $\binom{R+3}{3}$ on the ordered unknowns by SNG-F44. With the edge, symmetric contexts reach $\sum_{v \le R} \#\{o : |o| \le v\}$, and the difference is exactly the information carried by the order of the terminals.

On the realizable side the quotient is no coarser than forced. Among the 12,753 connected blocks with at most six vertices and no terminal edge, grouped by K^+ through volume four, no class contains two kernels unrelated by the terminal swap (451 classes; `--census` mode of the same script). On that population, swap-symmetric contexts lose exactly the order of the terminals. This is a finite statement about one population and one budget.

10.8.1 The observable interface

Let V_R be the space of coefficients $Y_o[v]$, $|o| \le v \le R$, over $\mathbb{Q}(q)$. For a class $\mathcal{C}$ of contexts let $I_{\mathcal{C},R} = \bigcap_{C \in \mathcal{C}} \ker L_C$, where L_C sends the coefficients to the glued series modulo x^{R+1}. The *observable interface* is $\mathcal{O}_{\mathcal{C},R} = V_R / I_{\mathcal{C},R}$: two blocks have the same image exactly when every context in the class gives the same answer. Write $\kappa = 1$ for the simple swap-symmetric contexts without a terminal edge. Write $\kappa = 2$ for those that may use it, in which case the block has no terminal edge. Group the orbits of each degree by m_o, and put $y_o = q^{d_o}$ and

$$\tau_o = q^{1-2s} + q^{1-2t} - q^{2-2(s+t)}.$$

Theorem SNG-F49 (the observable interface of simple symmetric contexts).

(a) *Top degree.* Let N_R be the space of vectors supported on degree R whose restriction to each group is annihilated by y^e, $e < \kappa$. Then $N_R \subseteq I_{\mathcal{C},R}$, N_R is exactly the kernel of the degree-R diagonal block of the whole class, and $\dim N_R = \sum_g \max(0, |g| - \kappa)$. For $\kappa = 2$ the first nonzero N_R occurs at $R = 25$. It is spanned by

$$v = \left(q^{25} - q^{121},\ q^{289} - q^{25},\ q^{121} - q^{289}\right)$$

on the orbits $\{1, 18\}, \{6, 17\}, \{10, 15\}$ of 325, whose d_o are $289, 121, 25$.

(b) *Exactness.* Suppose that for every $v < R$ and every group at degree v the functionals $y^e \tau^i$, $e < \kappa$, $i \in \{0, 1\}$, are linearly independent on the group. Then

$I_{\mathcal{C},R} = N_R$. The interface has the explicit coordinates

$$\mathcal{O}_{\mathcal{C},R} \cong \big(Y_o[v]\big)_{v<R} \oplus \big(\textstyle\sum_{o\in g} y_o^e Y_o[R]\big)_{g,\, e<\kappa}.$$

The finitely many contexts with d pendant leaves per terminal, c shared neighbours and, when $\kappa = 2$, an optional terminal edge span every response of the class. The hypothesis holds for $\kappa = 1$ exactly when $R \le 25$, and for $\kappa = 2$ for every $R \le 91$.

(c) *Realizability.* Let a block have no terminal edge, and let its terminals have internal degrees a, b. Then $Y_o[|o|] = q^{as^2+bt^2} + q^{bs^2+at^2}$, or $q^{(a+b)s^2}$ for $s = t$, and $Y_{\{0,1\}}[1] = q^a + q^b$ determines $\{a, b\}$. Hence the difference of two blocks that no context in the class distinguishes vanishes in every coordinate with $|o| = v$. Parts (a) and (b) hold for such differences with each group restricted to $|o| < v$, and the exactness hypothesis then holds through $R = 26$ for $\kappa = 1$ and $R = 92$ for $\kappa = 2$. In particular:

- For $\kappa = 2$, simple swap-symmetric contexts separate the truncated symmetrized kernels of any two blocks for every $R \le 25$, and the blind direction v of (a) is not a difference of two blocks.
- At $R = 26$ the only possible invisible difference allowed by (c) alone is a multiple of v placed at degree 26. SNG-F52 below excludes it as well.
- For $\kappa = 1$, blocks are separated for $R \le 7$. At $R = 8$ the only possible invisible difference allowed by (c) alone is a multiple of $e_{\{0,5\}}[8] - e_{\{3,4\}}[8]$, which SNG-F52 also excludes.

Proof. (a) At interior volume zero a context in the class has kernel $q^{\delta m_o + e d_o}$, with δ the common terminal degree and $e \in \{0, 1\}$ ($e = 0$ if $\kappa = 1$), as in SNG-F48(a). A coordinate $Y_o[R]$ enters only the volume-R coefficient, multiplied by this value. A top-degree vector c is therefore invisible exactly when $\sum_o c_o q^{\delta m_o + e d_o} = 0$ for every realized pair (δ, e). Leaves realize every $\delta \ge 0$, and the numbers q^m are distinct, so a Vandermonde system in δ reduces this to $\sum_{o\in g} c_o y_o^e = 0$ for each group and $e < \kappa$. Within a group the y_o are distinct (SNG-F48(b)). The $\kappa \times |g|$ Vandermonde matrix therefore has rank $\min(\kappa, |g|)$, which gives the dimension, and Cramer's rule gives the basis. For the triple with nodes q^{289}, q^{121}, q^{25} the cofactors are v.

(b) Induct on $v < R$, assuming every coordinate of degree below v is determined. A leaf on a terminal of height h contributes $f(h) = q^{h^2}(1 + xq^{1-2h} + O(x^2))$, and a shared neighbour contributes $q^m(1 + xq^{2-2(s+t)} + O(x^2))$. The context with d leaves per terminal, c shared neighbours and e terminal edges therefore has kernel

$$q^{ed_o + (d+c)m_o}\big(1 + x(d\sigma_o + c\pi_o) + O(x^2)\big), \qquad \sigma_o = q^{1-2s} + q^{1-2t},\ \pi_o = q^{2-2(s+t)}.$$

After the known terms are removed, the volume-v coefficients give

$$\sum_o q^{\delta m_o} y_o^e Y_o[v].$$

In the volume-$v+1$ coefficient, $Y[v+1]$ appears with the split-independent factor $q^{ed_o + (d+c)m_o}$, and $Y[v]$ with the factor $q^{ed_o + (d+c)m_o}(d\sigma_o + c\pi_o)$. Subtracting the context $(d-1, c+1)$ from (d, c) removes $Y[v+1]$ and leaves $\sum_o q^{\delta m_o} y_o^e \tau_o Y_o[v]$, with $\delta = d + c \ge 1$. The Vandermonde system in δ separates the groups. What

remains on each group is the functionals $y^e\tau^i$, which are independent by hypothesis, so degree v is determined. The induction reaches $R-1$. At degree R, (a) shows that exactly the group sums are determined, so $I_{\mathcal{C},R} = N_R$. The probes used and the whole class have the same annihilator N_R, so the probes span every response.

For $\kappa = 1$ the hypothesis asks that $(1, \tau_o)$ separate each group. Two orbits of a group have different $s+t$. Since $1-2s$ is odd, the only even exponent of τ_o is $2-2(s+t)$, so the two τ_o differ and groups of two pass. A group of three cannot be separated by two functionals; the first one is complete at degree 25. For $\kappa = 2$, after scaling column o by $q^{2(s+t)}$ the four functionals are integer polynomials. Their matrix has full column rank modulo $2^{61}-1$ at a fixed residue of q for every prefix, ordered by $|o|$, of every group with $s+t \le 100$. A nonzero minor modulo p is nonzero over $\mathbb{Z}[q]$. The first failure is the five-orbit group $4225 = 0^2+65^2 = 16^2+63^2 = 25^2+60^2 = 33^2+56^2 = 39^2+52^2$ at degree 91.

(c) Interior volume zero is the single profile with every interior height 0, and its energy is as^2+bt^2. By (b) at degree 1, where $m = 0, 1$ are distinct, two indistinguishable blocks have the same q^a+q^b, hence the same $\{a, b\}$, and so the same coordinates $Y_o[|o|]$. Repeating (a) and (b) on the remaining coordinates gives the restricted statements. □

So the interface has a precise shape. Without the terminal edge, each budget loses the top-degree collision information and nothing else, through $R = 25$. With the edge, the formal interface is complete through $R = 24$ and loses one direction at $R = 25$, and this direction is not realized by blocks. Independently of the proof, the exact rank of the leaf-and-shared-neighbour probe map modulo $2^{61}-1$ equals $\dim V_R - \dim N_R$ for every $R \le 18$ without the edge. The values for $R = 8, \dots, 12$ are rechecked in CI. Beyond the stated budgets, whether the higher terms of the x-expansion keep resolving lower-degree collisions is open.

Proposition SNG-F50 (weighted terminal coupling). Let μ_R be the largest number of orbits with $|o| \le R$ that share m_o. Allow contexts to carry a terminal edge of integer weight $e \in \{0, \dots, W\}$ with $W+1 \ge \mu_R$. Then the $M(W+1)$ probes with $\delta < M$ leaves per terminal and terminal weight e determine every Y_o modulo x^{R+1}, at every budget R.

Proof. The induction of SNG-F48(b) applies with the Vandermonde system $(q^{ed_o})_{e \le W,\, o \in g}$, whose nodes are distinct within a group. □

The multiplicity grows slowly: $\mu_R = 2$ for $7 \le R < 25$, 3 from $R = 25$, 4 from $R = 47$, where 1105 has four orbits, and 5 from $R = 91$. Weighted coupling therefore enlarges the experimental language by a small and explicit amount. It is the extra capability that removes the obstruction of SNG-F49(a). The script replays weighted recovery on real weighted gluings, counted by weak compositions, for all fifteen blocks with at most four vertices ($R = 3$, weights $0, 1, 2$). It also checks the degree-25 diagonal ranks modulo p: 181 of 182 with simple contexts, 169 of 169 on the realizable coordinates, and 182 of 182 with weights up to 2.

Reuse. Rank reduction is not the only computational gain. A kernel is compiled once and then answers every attachment query by the contraction of SNG-F41. For a 2×6 ladder with terminals at opposite corners and $R = 6$, compiling the kernel takes 4.6 s. Each of 24 symmetric-context queries then takes 7.6 ms by contraction, against 1.33 s for a fresh decomposition of the glued graph, with identical exact output. The compilation pays for itself after about 3.5 queries (`--benchmark`, wall-clock on the

session machine, retained). Theorem SNG-F53 below supplies a compression family whose savings grow with the volume budget (P54).

Targeted realizability searches. The following observation gives a route to a symmetric kernel without a swap automorphism. If two blocks A, B have $K_A = K_B$, then joining A in series with the reversal of B, at an internal middle vertex, gives a block with $K(s,t) = \sum_w x^w K_A(s,w)K_A(t,w)$, which is symmetric. It has no swap automorphism unless such a map carries A onto the reversal of B. The search for kernel-equal blocks found nothing. The 777 terminal-isomorphism classes of connected blocks with at most six vertices and no terminal edge have 777 distinct kernels through volume four. Among the 7,682 two-terminal trees with at most ten vertices, up to isomorphism, no kernel is symmetric without a swap automorphism. Five pairs of non-isomorphic trees on ten vertices agree through volume four, and all five separate at volume five. For cancelling twists, the pairing of two asymmetric kernels vanishes only if its leading coefficient at volume $\mu = \min_{\mathbf{b}}(|\mathbf{b}| + v_{1,\mathbf{b}} + v_{2,\mathbf{b}})$ cancels, and this requires at least two orbits to reach μ. We tested this on the 1,522 distinct antisymmetric parts, through volume five, of balanced blocks: connected graphs with at most six vertices and two-terminal trees with at most ten vertices. Of the 1,159,003 pairs, self-pairings included, 12,112 have two or more leading orbits, and none has a vanishing leading coefficient. A cancelling pair, if one exists, must therefore come from a richer family or a deeper leading level.

10.8.2 Recovery with extra volume, and rigidity of realizable interfaces

The obstruction of SNG-F49(a) is tied to a fixed budget. Two questions remain: how much additional observation recovers an interface, and which formal ambiguities actual blocks can realize. The relevant object is $\ker \mathcal{O}_R \cap (\mathcal{B}_R - \mathcal{B}_R)$, where $\mathcal{B}_R$ is the set of truncated interfaces of actual blocks. SNG-F49 describes the first factor; the slice constraints below begin to describe the second.

Theorem SNG-F51 (recovery with extra volume). Let m_R be the largest number of orbits with $|o| \le R$ sharing a value of $s^2 + t^2$, and let G be the number of distinct values. Let $C_{d,c}$ be the context with d pendant leaves on each terminal, c shared neighbours and no terminal edge. Then the glued series of a block with the contexts $C_{d,c}$, $d + c < G + m_R - 1$, through volume $R + m_R - 1$ determine Y_o modulo x^{R+1} for every orbit. This holds over $\mathbb{Q}(q)$ and at every $q_0 \in (0,1)$. In particular, edge-free symmetric contexts determine every finite prefix of K^+: the obstruction of SNG-F49(a) delays observation by $m_R - 1$ units of volume, and loses nothing.

Proof. With $L_h(x) = \sum_u x^u q^{(u-h)^2}$, a pair of leaves contributes $P = L_s L_t$ and a shared neighbour contributes $S = \sum_u x^u q^{(u-s)^2+(u-t)^2}$. Both have constant term q^E, $E = s^2 + t^2$, and $S - P = -q^E \tau_o x + O(x^2)$. The context $C_{k-c,c}$ contributes $P^{k-c}S^c$, so for $k \ge j$

$$\sum_{c=0}^{j} (-1)^{j-c} \binom{j}{c} P^{k-c} S^c = P^{k-j}(S-P)^j = x^j q^{kE} (-\tau_o)^j + O(x^{j+1}).$$

This is an exact linear combination of responses of actual contexts, and it vanishes through degree $j - 1$.

Suppose every coordinate of degree below n is known, and subtract its contribution. At output volume $n + j$ the differenced measurement is $\sum_{|o| \le n} q^{kE_o} (-\tau_o)^j Y_o[n]$. Coor-

dinates of degree above n cannot enter, because the context factor supplies at least j units of volume. For fixed j, varying $k = j, \dots, j+G-1$ gives a Vandermonde system in the distinct numbers q^E, which separates the groups. Inside a group, $j = 0, \dots, |g|-1$ gives a Vandermonde system in the τ_o.

These are distinct. Write $\tau_o = 1 - (q^{1-2s}-1)(q^{1-2t}-1)$. For $t \geq 1$ the pair $\{0, t\}$ has $\tau > 1$, and pairs with $s, t \geq 1$ have $\tau < 1$. For the latter put $r = 1/q > 1$ and $g(y) = \log(r^{2\sqrt{y}-1} - 1)$ for $y \geq 1$. With $x = \sqrt{y}$, $h(x) = \log(r^{2x-1} - 1)$ and $L = \log r$, one has $h' = 2Lr^{2x-1}/(r^{2x-1}-1) > 0$ and $h'' = -4L^2 r^{2x-1}/(r^{2x-1}-1)^2 < 0$. Hence $g'' = (h''-h'/x)/(4x^2) < 0$, and on the circle $s^2+t^2 = E$ the sum $g(s^2)+g(t^2)$ strictly increases toward balance. So τ_o strictly decreases toward balance, and distinct orbits of a group have distinct τ_o. A group at degree n has at most m_R orbits, so the output volumes used are at most $R + m_R - 1$. □

If only connected contexts are allowed, keep one shared neighbour in every probe. The same argument goes through with the extra factor S, whose constant term is a unit.

The three mechanisms compare as follows. Ordinary symmetric contexts at the original budget give the quotient of SNG-F49. Finite differences, with $m_R - 1$ extra units of volume, produce the powers $1, \tau, \tau^2, \dots$ and recover everything (SNG-F51). A weighted terminal edge with weights $0, \dots, m_R-1$ produces the powers $1, y, y^2, \dots$ and recovers everything at the original budget (SNG-F50). Here $m_R = 2$ for $7 \leq R < 25$, 3 from 25, 4 from 47 and 5 from 91. `recover_by_finite_differences` in `src/singularity/contexts.py` carries out the reconstruction. `scripts/symmetric_contexts.py` replays it at $R = 7$, where the budget-7 quotient has a defect, with 39 probes through volume 8. Their glued series are computed by tree decomposition of each actual glued graph, independently of the contraction formula and of the closed-form context factors.

Proposition SNG-F52 (interior slices of a block kernel). Let a block have terminal degrees a, b, let $e \in \{0, 1\}$ count its terminal edge, and put $Q_0(s,t) = as^2 + bt^2 - 2est$, $U = q^{-2s}$, $V = q^{-2t}$. For every j,

$$[x^j]K(s,t) = q^{Q_0(s,t)} P_j(U, V),$$

where P_j has coefficients in $\mathbb{Z}_{\geq 0}[q]$ and degree at most j in each variable. In particular $P_1 = \sum_v q^{\deg v} U^{A_{sv}} V^{A_{tv}}$, summed over interior vertices, has at most four monomials whatever the size of the block.

(a) Suppose two blocks have the same $\{a, b\}$ and e. If their symmetrized kernels agree in interior volume one at the states $(0,0), (1,1), (2,2)$ and $(0,1)$, of total volume at most 5, then their interior-volume-one slices of K^+ agree at every state.

(b) If moreover $a = b$, agreement of slice j of K^+ at the states with $s, t \leq j$, of total volume at most $3j$, implies agreement of slice j at every state.

Proof. For interior heights h of total j, the energy is

$$Q_0 + [\sum_{\text{interior edges}} (h_u - h_v)^2 + \sum_{v \sim s} h_v^2 + \sum_{v \sim t} h_v^2] - 2s \sum_{v \sim s} h_v - 2t \sum_{v \sim t} h_v.$$

The bracket does not depend on (s, t), and each of the last two sums is at most j. For $j = 1$ the single raised vertex contributes $q^{\deg v} U^{A_{sv}} V^{A_{tv}}$.

(a) When $a \neq b$, order the terminals of both blocks so that s has degree a; this does not change K^+. Let $P = P_1 - P_1'$. At a diagonal state $K^+(s,s) = K(s,s)$, so agreement at $s = 0, 1, 2$ says that $P(z,z)$, of degree at most 2, vanishes at the three distinct nodes $z = 1, q^{-2}, q^{-4}$. Hence $P = c(U - V)$. At $s \neq t$ the symmetrized difference is $c(U-V)(q^{Q_0(s,t)} - q^{Q_0(t,s)})$. This vanishes identically when $a = b$. When $a \neq b$, at $(0,1)$ it equals $c(1-q^{-2})(q^b - q^a)$, forcing $c = 0$.

(b) With $a = b$, Q_0 is symmetric, and the symmetrized slice is $q^{Q_0}\Sigma(U,V)$ with $\Sigma(U,V) = P_j(U,V) + P_j(V,U)$ off the diagonal, and $q^{Q_0}\Sigma(z,z)/2$ on it. The symmetric polynomial Σ, of degree at most j in each variable, vanishes on the tensor grid of the $j+1$ distinct nodes q^{-2s}, $s \leq j$, and is therefore zero. □

Together with SNG-F49(c), which fixes interior volume zero, this sharpens the realizable statements. Two indistinguishable blocks agree below degree v by induction, and the degree data fix $\{a, b\}$ at degree 1 and e at degree 2. So their difference vanishes in slice one at every degree $v \geq 6$, and, when the terminal degrees are equal, in every slice j with $3j < v$. Rerunning SNG-F49 on the remaining coordinates (`realizable="slice"` or `"balanced"` in `interface_status`) gives:

- **With the terminal edge ($\kappa = 2$).** Simple symmetric contexts separate the truncated symmetrized kernels of any two blocks through $R = 26$. This removes the $R = 26$ candidate: its coordinate on $\{10, 15\}$ has interior volume one. At $R = 27$ the only possible invisible difference is v placed at degree 27, at interior volumes $8, 4, 2$. For equal terminal degrees, separation holds through $R = 37$.
- **Without the edge ($\kappa = 1$).** Separation holds through $R = 8$. At $R = 9$ the only possible invisible difference is a multiple of $e_{\{0,5\}}[9] - e_{\{3,4\}}[9]$. For equal terminal degrees, separation holds through $R = 10$.

On these coordinates the exactness hypothesis of SNG-F49(b) holds for $R \leq 27$ ($\kappa = 1$) and $R \leq 93$ ($\kappa = 2$) in general. For equal terminal degrees it holds for $R \leq 38$ and $R \leq 120$, the latter being the range checked. All statements are recorded in `results/symmetric_contexts.json`. For unequal terminal degrees, the analogue of (b) is the following interpolation theorem.

Theorem SNG-F59 (unequal-degree slice interpolation). Fix integers $j \geq 0$, $d \geq 1$, $\ell \geq 0$, and let P have degree at most j in each variable. Put

$$F_P(s,t) = q^{ds^2+\ell s}P(q^{-2s}, q^{-2t}) + q^{dt^2+\ell t}P(q^{-2t}, q^{-2s}).$$

If $F_P(s,t) = 0$ for all $s + t \leq 2j$, then $P = 0$. This holds over $\mathbb{Q}(q)$, at every transcendental q, and at every rational $q \in (0,1)$; more generally at every algebraic q that is not an algebraic unit. The unordered sample set has $\sum_{s \leq j}(2j - 2s + 1) = (j+1)^2$ states, the dimension of the space of such P.

Proof. For $j = 0$, $F_P(0,0) = 2P$. For $j \geq 1$, the state $(0,0)$ gives $P(1,1) = 0$, so $P(1,V) = (V-1)A(V)$ and $P(U,1) = (U-1)B(U)$ with $\deg A, \deg B \leq j-1$. The states $(0,t)$, $1 \leq t \leq 2j$, give $A(q^{-2t}) + q^{dt^2+\ell t}B(q^{-2t}) = 0$. With $z = q^{-1}$, the matrix of this system has rows $\left[z^{2kt} \mid z^{2kt-dt^2-\ell t}\right]_{0 \leq k < j}$.

Its determinant $D_{j,d,\ell}(z)$ is a Laurent polynomial whose extreme coefficients are ± 1. Once the rows assigned to the two column families are fixed, the rearrangement inequality uniquely maximizes the exponent by assigning increasing k, and uniquely minimizes it by assigning decreasing k. Consider adjacent rows $t, t+1$ holding a B-entry and an A-entry of family ranks b and a, so that $t = a + b - 1$. Moving the B-entry

one row later changes the optimized exponent by $2(b-a)-d(2t+1)-\ell \le -3-\ell$ for the maximum, and by $2(a-b)-d(2t+1)-\ell \le -3-\ell$ for the minimum. So the maximum is attained uniquely with all B-rows first, and the minimum uniquely with all B-rows last.

After clearing the lowest power, D is an integer polynomial with leading and constant coefficients ± 1. It is not zero, and by the rational-root theorem it has no rational root $z > 1$; every algebraic root is a unit. Hence $A = B = 0$, and $P = (U-1)(V-1)Q$ with Q of bidegree at most $(j-1, j-1)$. At states with $s, t \ge 1$, divide out the nonzero symmetric factor and write $s = u+1$, $t = v+1$. Then $q^{d(u+1)^2+\ell(u+1)} = q^{d+\ell}q^{du^2+(2d+\ell)u}$, and $Q(q^{-2s}, q^{-2t}) = \widetilde{Q}(q^{-2u}, q^{-2v})$ with $\widetilde{Q}(U,V) = Q(q^{-2}U, q^{-2}V)$. The states $u+v \le 2(j-1)$ give the same problem at bidegree $j-1$ with $\ell \mapsto \ell+2d$, and induction completes the proof. □

For a block with $a \ne b$, the symmetrized slice is $q^{b(s^2+t^2)-2est}F_P(s,t)$ with $d = |a-b|$ and $\ell = 0$. So slice j is fixed by its values at $s+t \le 2j$, of total volume at most $3j$, exactly as in the balanced case. The rule "slice j is rigid at degree v when $3j < v$" therefore holds for every block (`realizable="interpolated"`). Rerunning SNG-F49 on the remaining coordinates gives two bounds. With the terminal edge, simple symmetric contexts separate the truncated symmetrized kernels of any two blocks through $R = 37$, and the first possible ambiguity is at $R = 38$. Without the edge, separation holds through $R = 10$, and the first possible ambiguity is at $R = 11$.

The arithmetic hypothesis is necessary for the unrestricted statement. For $j = 3$, $d = 1$, $\ell = 0$,

$$D(z) = z^{-43}(z-1)^{11}H(z), \quad \deg H = 68,$$
$$H(1) = 3{,}870{,}720, \qquad H(0) = \operatorname{lead} H = -1.$$

H changes sign between $z = 1.1213855316120952$ and 1.1213855316120954, so the boundary system degenerates at $q_* \approx 0.8917539702535734$. This obstructs only unrestricted polynomials. Whether a graph slice can realize the degeneracy is open.

`boundary_determinant` and `interpolation_rank` in `src/singularity/contexts.py` back these statements. `scripts/overlap_hierarchy.py` checks the unit extreme coefficients for $j \le 3$, injectivity at $q = 1/2$ for $j \le 4$, $d \le 3$, and the exceptional factorization with its root bracket. Whether the remaining candidates are realized by actual blocks is open, but a realization is tightly constrained. Without the terminal edge at $R = 11$, the candidate is a multiple of $e_{\{0,5\}}[11] - e_{\{3,4\}}[11]$, at interior volumes 6 and 4. Two blocks realizing it must have equal symmetrized kernels at every coordinate of degree at most ten. In particular their interiors, pinned at terminal height zero, have equal series through interior volume eleven. Their slice-four difference P must satisfy $F_P(s,t) = 0$ at every state with $s+t \le 7$ except $\{3,4\}$. By SNG-F59 this leaves a five-dimensional affine family of slice-four polynomials, fixed by the values at $s+t = 8$. Dominating and other single-charge blocks cannot realize it: their slice differences are $q^{m(s^2+t^2)}\Delta A_j q^{-2j(s+t)}$, nonzero at the state $(0,0)$ as soon as they are nonzero anywhere. Blocks on at most six vertices are already separated by K^+ at volume four. A realization therefore needs larger blocks whose pinned interiors agree through volume eleven. By SNG-F55, equality of their decorated connected-set statistics through size eleven is sufficient for this agreement; the converse is not asserted, since different statistics can have cancelling contributions to the same series prefix. With the terminal edge, the first candidate is at $R = 38$: the vector v of SNG-F49(a) on $\{1,18\}, \{6,17\}, \{10,15\}$, placed at degree 38, at interior volumes 19, 15 and 13. The same constraints apply, with agreement required through degree 37.

10.8.3 Boundary charges and a growing compression family

The rank results above concern what restricted contexts can see. The other half of P54 asks how small an exact interface can be for all future contexts. The answer depends on how the interior meets the terminals. For a block with terminals $b_1, \dots, b_k$, give each interior vertex v its neighbourhood pattern $\eta_v \in \{0,1\}^k$, and call $Q(h) = \sum_v h_v \eta_v$ the *boundary charge* of interior heights h.

Theorem SNG-F53 (charge interfaces).

(a) Let d_i be the number of interior neighbours of terminal i, and let E_T be the energy of the edges among the terminals. For terminal heights $\mathbf{b}$ and interior heights h of total j, the energy is

$$E = [E_{\text{int}}(h) + \sum_v |\eta_v| h_v^2] + \sum_i d_i b_i^2 + E_T(\mathbf{b}) - 2\langle \mathbf{b}, Q(h)\rangle.$$

Hence $K(\mathbf{b}) = q^{\sum_i d_i b_i^2 + E_T(\mathbf{b})} \sum_{j,Q} x^j q^{-2\langle \mathbf{b}, Q\rangle} A_{j,Q}$, with

$$A_{j,Q} = \sum_{|h|=j,\ Q(h)=Q} q^{E_{\text{int}}(h) + \sum_v |\eta_v| h_v^2}.$$

The charges of volume j lie in $j\,\text{conv}\{\eta_v\}$. If the patterns have affine dimension d, the table through volume R has $O(R^{d+1})$ entries, against $\binom{R+k+1}{k+1}$ for the raw kernel.

(b) *Dominating terminals.* Let G_H consist of an arbitrary simple graph H on m vertices and two nonadjacent terminals, each joined to every vertex of H. Then every pattern is $(1,1)$, and

$$K_{G_H}(s,t) = q^{m(s^2+t^2)} \sum_{j \ge 0} A_{H,j} x^j q^{-2j(s+t)},$$
$$A_{H,j} = \sum_{|h|=j} q^{E_H(h) + 2\sum_i h_i^2}.$$

Through volume R the interface is the $R+1$ polynomials $A_{H,0}, \dots, A_{H,R}$. The ordered kernel table has $\binom{R+3}{3}$ entries and the symmetrized interface of SNG-F49 has 1,547 at $R = 24$ and 3,417 at $R = 32$, so the saving grows quadratically.

(c) *One closure is a complete interface.* Let C be any known context, and put $W_{C,j} = \sum_{s,t} x^{s+t} q^{m(s^2+t^2) - 2j(s+t)} K_C(s,t)$. Then the closed series is $Z_{H,C} = \sum_j A_{H,j} x^j W_{C,j}$, and $W_{C,j}$ has constant term 1. Consequently $A_{H,v} = [x^v] Z_{H,C} - \sum_{j<v} A_{H,j} [x^{v-j}] W_{C,j}$. Within the class with given m, for any fixed context C_0,

$$\begin{aligned} A_{H,\le R} = A_{H',\le R} \\ \iff \quad Z_{H,C_0,\le R} = Z_{H',C_0,\le R} \\ \iff \quad Z_{H,C,\le R} = Z_{H',C,\le R} \quad \text{for every context } C. \end{aligned}$$

So the sequence $A_{H,\le R}$ is the coarsest exact interface for future gluing at the original budget: any representation that determines every future closure determines the C_0-closure, hence A.

Proof. (a) Expand $(h_v - b_i)^2 = h_v^2 - 2b_i h_v + b_i^2$ over the edges between the interior and the terminals; the interior and terminal–terminal edges are unchanged. (b) This is (a) with $\eta_v = (1,1)$, $d_1 = d_2 = m$ and no terminal edge. (c) Gluing gives $Z = \sum_{\mathbf{b}} x^{|\mathbf{b}|} K_{G_H}(\mathbf{b}) K_C(\mathbf{b})$ (SNG-F41). The constant term of $W_{C,j}$ comes only from $s = t = 0$ with the whole context at height zero, which has energy zero. The recursion follows. The first equivalence follows from the recursion, the second from $Z_{H,C} = \sum_j A_{H,j} x^j W_{C,j}$, and the coarsest-interface statement from the first. □

The closing context may be connected and have no terminal edge: a single shared neighbour suffices.

Proposition SNG-F54 (context transport). Let A and Z_C be the coefficient vectors through R, and put

$$(T_C)_{v,j} = \sum_{s+t+u=v-j} q^{m(s^2+t^2)-2j(s+t)} [x^u] K_C(s,t).$$

Then $Z_C = T_C A$, T_C is lower triangular with unit diagonal, and it depends on m, q, C, R but not on H. Hence $Z_D = \mathcal{U}_{D\leftarrow C} Z_C$ with $\mathcal{U}_{D\leftarrow C} = T_D T_C^{-1}$, and $\mathcal{U}_{E\leftarrow D}\mathcal{U}_{D\leftarrow C} = \mathcal{U}_{E\leftarrow C}$. A compiled operator stores $O(R^2)$ scalars and applies in $O(R^2)$ operations. Building it from a context kernel takes $O(R^3)$ operations directly.

Proof. Take the coefficient of x^v in SNG-F53(c). □

The entries of T_C can carry negative powers of q. `closure_operator` in `src/singularity/charges.py` therefore computes them exactly at a rational q_0, while $A_{H,j}$ is kept as a polynomial. `charge_interface` and `kernel_from_charges` implement (a) for any number of terminals and rebuild the kernel exactly. `scripts/charge_interfaces.py`, checked in CI, verifies the following:

- all eight labelled interiors H on three vertices;
- three contexts: a shared neighbour, a leaf on each terminal, and an asymmetric context with a terminal edge;
- $q \in \{1/2, 2/3\}$: all 48 closures $T_C A$ equal the series of the actual glued graphs, counted by weak compositions;
- all 16 single shared-neighbour closures recover A through volume five;
- the composition law holds exactly.

For blocks whose patterns span two dimensions the charge table is not smaller than the raw table: 84 or 140 entries against 84 at $R = 6$. The compression is a property of attachment structure. False twins, with patterns $(0,0)$ and $(1,1)$, give $d = 1$ and a quadratic-size table.

Exact bits. The compression survives exact rational accounting. For $K_{2,3}$, that is H three isolated vertices, at $q = 1/2$, the value $A_j = \sum_{|h|=j} 2^{-2\sum h_i^2}$ has denominator exactly 2^{2j^2}. After clearing it, the three profiles concentrated at one vertex contribute 1 each and every other profile an even power of 2, so the numerator is odd. The interface $A_0, \dots, A_R$ therefore has $\Theta(R^3)$ bits. An ordered kernel entry is $2^{-(3(s^2+t^2)-2j(s+t))} A_j$, with 2-adic exponent $3(s^2+t^2) - 2j(s+t) + 2j^2$, a positive-definite quadratic form. Summing over $s + t + j \le R$ gives $\Theta(R^5)$ bits. At $R = 10$ the counts are $1{,}303$ bits against $26{,}066$.

Reuse. For the dominating block over C_6 at $R = 8$, $q = 1/2$:

- computing $A_0, \dots, A_8$ takes 0.014 s;

- each context operator takes 0.026 s, and is reusable for every interior with $m = 6$;
- each closure then takes 0.18 ms;
- a fresh tree decomposition of each glued graph takes 19.6 s, with identical exact output (`--benchmark`, wall-clock on the session machine, retained).

Future classes. For every isomorphism class of interiors on at most six vertices ($1, 2, 4, 11, 34, 156$ classes), the sequences $A_{H,0..5}$ are pairwise distinct. On that population, future equivalence of dominating blocks at budget five is isomorphism of the interiors. Whether non-isomorphic interiors can share the whole sequence, which by SNG-F53(c) would make their dominating blocks indistinguishable in every context, is open. Computing $A_{H,j}$ for large H is handled by the connected-support compiler of SNG-F55 below.

Charge dimension. If the terminal patterns present have affine dimension a, the cells (j, Q) with $j \le R$ number $\Theta(R^{a+1})$. With two distinct patterns $\chi_1 \neq \chi_2$, the charge $Q = J_1\chi_1 + (j - J_1)\chi_2$ determines J_1, so there are exactly $\binom{R+2}{2}$ cells, even when χ_1, χ_2 span two linear dimensions. For ordered terminal observations there is a reconstruction chart without exceptional parameters. Project the distinct charges of a slice injectively to a coordinates, and sample the terminal states at the projected charges. The matrix $M_{v,w} = q^{-2v\cdot w}$ is strictly positive definite, because $\sum c_v c_w M_{v,w} = \sum_\alpha \frac{(-2\log q)^{|\alpha|}}{\alpha!}\big(\sum_v c_v v^\alpha\big)^2$ and polynomial interpolation on a finite set makes some square nonzero. The total budget is at most $(a + 1)R$. With several charges per slice the single-closure inverse of SNG-F53(c) is replaced by such a chart J, and transport becomes $J'J^{-1}$.

Proposition SNG-F62 (ordered charge charts). Let the patterns of a block have affine dimension a. Choose a terminal coordinates on which the differences of patterns have full rank. Projection to those coordinates is then injective on the charges of every slice, because $Q - j\chi_0$ lies in the span of the differences. The ordered kernel at the embedded projected charges determines every $A_{j,Q}$, with total volume at most $(a + 1)R$. Two such charts are related by $\mathcal{U} = J'J^{-1}$, slice by slice.

Proof. At $\mathbf{b}$ supported on the chosen coordinates,

$$[x^j]K(\mathbf{b}) = q^{\sum d_i b_i^2 + E_T(\mathbf{b})} \sum_Q A_{j,Q} q^{-2\langle \pi\mathbf{b}, \pi Q\rangle}.$$

The matrix is the positive-definite kernel above, so each slice is a nonsingular square system. Each sampled coordinate is a charge coordinate, hence at most j. □

`charge_chart`, `recover_from_chart` and `chart_transport` in `src/singularity/charges.py` implement this. The tests recover the charge tables exactly from ordered kernels computed by bag messages: a false-twin block ($a = 1$), a block with four patterns ($a = 2$), and a three-terminal block, where transport between two different charts reproduces the second chart's observations.

10.9 Every late census collision is separated by triangles

For connected graphs with at most six vertices, the symbolic coefficients separate every pair of nonisomorphic graphs by volume three (`results/visibility_filtration.json`). The pairs that survive volume two are fourteen pairs on six vertices; they are retained with their graphs in `results/late_collisions.json`. In each of them the ordered adjacent-degree sector $2e_i + e_j$ of SNG-F14 also agrees, so the two graphs

have the same ordered degree-and-adjacency statistics. The separation comes entirely from the induced-triple sector $\sum T_{s,t} q^{s-2t}$, and in every pair the triangle counts differ, for example $K_{3,3}$ against the prism. None of the fourteen pairs is related by a single Whitney twist at a two-vertex separator. Their separation is therefore not the twist mechanism of SNG-F41; it is the classical one of SNG-I008, a different number of triangles under the same degree data. This is a complete finite classification of the late collisions in that population, with an explicit separating pattern for each pair. It is not a statement about larger graphs, where the pendant-cycle and twist families show that collisions can persist to arbitrarily large volume.

The same population is measured across invariants in `results/collision_census.json`, produced by `scripts/collision_census.py` and checked in CI. Counts are of connected isomorphism classes, one representative each. On six vertices, 112 classes give 68 degree sequences and 110 Laplacian spectra (two cospectral pairs). Volume one of the height series also gives 68 classes and volume two gives 100. The binary cut profile, the nested-pair census, and the height prefix through volume three each separate all 112, both symbolically in q and at $q = 1/2$. Both cospectral pairs are separated by the volume-three height prefix. All ten nontrivial classes of the volume-two prefix are split by the binary profile and also by the spectrum. The report lists every nontrivial class with its graph codes, and the number of classes of each invariant split by every other. It gives a signed difference witness for each pair first separated at volume two or three, and a chain-recomputed sample. There are no unresolved classes at volume three in this population.

10.10 Observation depth and connected supports

Which parts of a graph does a volume budget see, and what is the cheapest exact computation of exactly those parts? For a graph H and an integer $\lambda \geq 0$, put $w_v = \deg_H(v) + \lambda$ and

$$F_\lambda(H) = \sum_{h:V\to\mathbb{N}} x^{|h|} q^{\sum_v w_v h_v^2 - 2\sum_{uv\in E} h_u h_v}.$$

Here $\lambda = 0$ gives the local series L_H, and $\lambda = 2$ gives the interior sequence A_H of the dominating blocks of SNG-F53(b). For $T \subseteq V$ let Z_T be the same sum over heights supported in T, keeping the ambient degrees, and let $\Phi_S = \sum_{T\subseteq S} (-1)^{|S|-|T|} \log Z_T$.

Theorem SNG-F55 (connected-support locality). $\Phi_S = 0$ unless $H[S]$ is connected, $\Phi_S = O(x^{|S|})$, and Φ_S depends only on the isomorphism type of $H[S]$ decorated by the weights w_v. Hence

$$\log F_\lambda(H) = \sum_{H[S] \text{ connected}} \Phi_S,$$

and $\log F_\lambda(H) \bmod x^{R+1}$ is determined by the decorated connected induced subgraphs with at most R vertices.

Proof. Give each vertex its own variable x_v. Then Z_T is F with $x_v = 0$ for $v \notin T$, and since substitution commutes with the logarithm, $\log Z_T$ consists of the monomials of $\log F$ supported in T. Inclusion–exclusion leaves exactly the monomials whose support is S, and these have degree at least $|S|$. If $H[T]$ has no edge between T_1 and $T_2 = T \setminus T_1$, the exponent separates, so $Z_T = Z_{T_1} Z_{T_2}$. Then $\log F$ has no monomial meeting both parts, and $\Phi_S = 0$ for disconnected S. The value of Φ_S is computed from the weights and the edges inside S. □

`pinned_log` in `src/singularity/locality.py` compiles $\log F$ modulo x^{R+1}. It enumerates connected sets, memoizes Φ by an exact decorated-type key (an AHU encoding for trees, a class-respecting canonical form for small non-trees, and otherwise the set itself), and uses the additivity of $\log Z_T$ over the components of $H[T]$. The series is recovered by the exact recursion $nF_n = \sum_{r \le n} B_r F_{n-r}$, $B_r = r[x^r] \log F$. On graphs of maximum degree Δ, the number of connected sets of size at most R is $O(n(e\Delta)^R)$, so for fixed R and Δ the cost is linear in n, and treewidth does not enter.

On seeded random connected cubic graphs, `scripts/locality_family.py --benchmark` gives the following wall-clock times on the session machine (retained):

n	Compiler, $R{=}4$	Tree decomposition, $R{=}4$	Min-degree width	Compiler, $R{=}6$
16	0.03 s	0.24 s	4	0.90 s
32	0.03 s	52.6 s	7	0.82 s
64	0.04 s	about $8{\times}10^9$ bag assignments	13	0.90 s
128	0.04 s	about $5{\times}10^{19}$ bag assignments	27	0.80 s

Where both methods run, the output is identical. The two methods are complementary: separators suit large budgets on narrow graphs, and connected supports suit small budgets on large sparse graphs.

Theorem SNG-F56 (the first cycle a budget sees). Let H, H' be d-regular graphs on the same number of vertices, both of girth at least g, and let N_g count g-cycles. Then $F_\lambda(H) \equiv F_\lambda(H') \pmod{x^g}$, and

$$[x^g]\big(F_\lambda(H) - F_\lambda(H')\big) = \big(N_g(H) - N_g(H')\big)\, q^{(d+\lambda-2)g}(1-q^2)^g.$$

Proof. Every connected set with fewer than g vertices induces a tree. The number of sets inducing a given tree with $k < g$ vertices is the number of injective homomorphisms of the tree, divided by its automorphisms. All homomorphisms of a k-vertex tree into a d-regular graph number nd^{k-1}. A non-injective one has an image on fewer than k vertices, which is a tree because the girth exceeds k, so it is counted in the same way, and induction on k shows that the counts depend only on n and d. This gives agreement modulo x^g.

At degree g, only sets with exactly g vertices contribute something new, through the multilinear coefficient. By the cumulant expansion of $\log F$, the coefficient of $\prod_{v \in S} x_v$ is $q^{(d+\lambda)g} \sum_{E'} \prod_{e \in E'} f$, with $f = q^{-2} - 1$, summed over the connected spanning edge sets E' of $H[S]$. A set with g vertices that is not a tree induces a g-cycle, since a chord would make a shorter cycle. That cycle contributes $q^{(d+\lambda)g} f^{g-1}(f+g)$. Its vertex set is also one of the vertex sets that would otherwise induce a path, worth $q^{(d+\lambda)g} f^{g-1}$, and each g-cycle removes exactly g of them. Paths are the only trees affected. The net contribution per cycle is $q^{(d+\lambda)g} f^g = q^{(d+\lambda-2)g}(1-q^2)^g$. □

This explains the retained cube–Möbius comparison. Both graphs are cubic of girth four, with six and four squares, so the theorem predicts $[x^4](L_{\text{cube}} - L_{\text{Möbius}}) = 2q^4(1-q^2)^4$, exactly the fixture's first difference. In A the same pair gives $2q^{12}(1-q^2)^4$. The prism against $K_{3,3}$ (two triangles against none) gives $2q^3(1-q^2)^3$ at volume three, the triangle mechanism of the late census collisions below.

Cycles. For $H_1 = C_{2g}$ and $H_2 = C_g \sqcup C_g$, the transfer matrix $T_{ab} = x^b q^{(a-b)^2+2b^2}$ gives $A_{C_n} = \operatorname{tr} T^n$. Hence

$$A_{H_2} - A_{H_1} = (\operatorname{tr} T^g)^2 - \operatorname{tr} T^{2g} = 2\operatorname{tr}((\wedge^2 T)^g).$$

Column (c,d) of $\wedge^2 T$ carries x^{c+d}. The only column of minimal degree is $(0,1)$, with diagonal entry $xq^2(1-q^2)$. So the difference is $2[q^2(1-q^2)]^g x^g + O(x^{g+1})$, nonzero at every $q \in (0,1)$; this is SNG-F56 with $d = 2$, $\lambda = 2$, $\Delta N_g = 2$. Attach the two dominating terminals of SNG-F53(b). Both blocks have $2g$ interior vertices, so every context acts through the same unit-triangular operator T_C (SNG-F54). Every context therefore sees the same first discrepancy, at volume g, and no context makes it appear earlier. For $g = 6$ the blocks have fourteen vertices. Glued to one shared neighbour and counted directly by weak compositions on fifteen vertices, they agree through volume five and differ at six by $729/8{,}388{,}608$ at $q = 1/2$. These are finite-prefix collisions in an infinite family. Whether two non-isomorphic interiors share the whole sequence A remains open.

Theorem SNG-F57 (reconstruction of 2-regular interiors). Let $\kappa = q^2(1-q^2)$. Let $b = \sum_{k\ge 1} \varphi_k$, where φ_k is Φ of a k-vertex path whose vertices have ambient degree 2, and put $D_\ell = \log A_{C_\ell} - \ell b$. Then $D_\ell = \kappa^\ell x^\ell + O(x^{\ell+1})$. For a 2-regular graph on n vertices with c_ℓ cycles of length ℓ,

$$\log A_H - nb = \sum_{\ell \ge 3} c_\ell D_\ell, \qquad c_r = \kappa^{-r}[x^r](\log A_H - nb - \sum_{\ell < r} c_\ell D_\ell).$$

The budget $R = \lfloor n/2 \rfloor$ determines H, and for $n \ge 6$ no smaller budget does.

Proof. By SNG-F55, $\log A_{C_\ell} = \ell \sum_{k<\ell} \varphi_k + \Phi_{C_\ell}$, so $D_\ell = \Phi_{C_\ell} - \ell \sum_{k \ge \ell} \varphi_k$. Its degree-$\ell$ coefficient is the cycle-minus-paths count of SNG-F56, $q^{4\ell} f^\ell = \kappa^\ell$. Logarithms add over components, which gives the triangular system. After the cycles of length at most R are found, the remaining vertices can form at most one cycle, since two cycles longer than $n/2$ need more than n vertices. Sharpness: C_{2g} against $C_g \sqcup C_g$, and C_{2g+1} against $C_g \sqcup C_{g+1}$, agree below volume g, because the difference of logarithms is $D_g + D_{g+1} - D_{2g+1}$ or $2D_g - D_{2g}$. □

`reconstruct_cycle_type` implements the recursion at a rational q_0. The receipt reconstructs all 57 two-regular graphs on at most fourteen vertices at budget $\lfloor n/2 \rfloor$ and records the sharpness pairs. It checks the compiler against weak-composition enumeration on ten graphs for both λ, and the cycle identity for $g = 3, \dots, 8$, with an enumeration oracle for $g \le 6$; the session also confirmed $g = 9, 10$. This gives an infinite graph class with a proved reconstruction algorithm, a sharp observation threshold, and explicit pairs attaining it.

10.10.1 Squared height: a complete enrichment

Fixed pinning merges profiles with the same volume and energy but different squared height. Keeping the squared height as a second marker gives

$$\mathcal{P}_H(x,u,q) = \sum_{h: V \to \mathbb{N}} x^{J(h)} u^{S(h)} q^{E_H(h)}, \qquad J = \sum_v h_v, \quad S = \sum_v h_v^2,$$

so that $F_\lambda(H) = \mathcal{P}_H(x, q^\lambda, q)$.

Theorem SNG-F58 (the moment series).

(a) $[x^j u^j]\mathcal{P}_H = \sum_{|T|=j} q^{|\partial T|}$, the binary cut profile by size, and $[x^j u^{j^2}]\mathcal{P}_H = \sum_v q^{\deg(v) j^2}$ for $j \geq 1$, the degree distribution.

(b) Write $C_{j,S} = [x^j u^S]\mathcal{P}_H$. Since $h^2 \equiv h \pmod 2$, we have $S = j + 2k$ with $0 \leq k \leq \binom{j}{2}$, and $q^{-\lambda j}[x^j]F_\lambda = \sum_k C_{j,j+2k}(q^{2\lambda})^k$. Hence any $\binom{R}{2} + 1$ distinct pinnings λ determine $\mathcal{P}_H$ mod x^{R+1}, at every $q \in (0,1)$.

(c) For every $q_0 \in (0,1)$, $\mathcal{P}_H(x,u,q_0) = \mathcal{P}_{H'}(x,u,q_0)$ as formal series in x,u if and only if $H \cong H'$.

Proof. (a) $S = J$ forces heights in $\{0,1\}$, and $S = J^2$ forces the whole volume onto one vertex. (b) The system is a Vandermonde system in the distinct numbers $q^{2\lambda}$.

(c) By (a), the series determines n at q_0: the degree multiplicities follow by successive exponential dominance. Choose an integer $B > 2n$ with $\eta = q_0^{B^2} < 1/(n!+1)$, put $M_i = B^{3^i}$ for $0 \leq i < n$, $J_* = \sum M_i$ and $S_* = \sum M_i^2$.

First, these moments force the multiset of heights to be $\{M_i\}$. Let M be the largest and r the sum of the others, so $(r+1)^2 < M$. A competing n-tuple with maximum at most $M-1$ has square sum at most $(M-1)^2 + (r+1)^2 < M^2$. One with maximum at least $M+1$ has square sum above $M^2 + r^2 \geq S_*$. So its maximum is M; remove it and repeat. Hence

$$[x^{J_*} u^{S_*}]\mathcal{P}_H = \sum_\pi q_0^{E_\pi}, \qquad E_\pi = \sum_i d_\pi(i) B^{2 \cdot 3^i} - 2 \sum_{i<j} A_\pi(i,j) B^{3^i + 3^j},$$

summed over the $n!$ bijections of heights to vertices. The positions $2 \cdot 3^i$ and $3^i + 3^j$ are distinct, and the digits lie in $[-2, n-1]$, so with $B > 2n$ each E_π, read in balanced base B, returns the degrees and adjacencies of a labelling. Each E_π is a multiple of B^2, so the coefficient is $\sum_k c_k \eta^k$ with $0 \leq c_k \leq n!$.

Second, the specialization keeps this information. If the histograms of H and H' differ, factor out the first differing power of η. The remaining constant is a nonzero integer, while the tail has absolute value at most $n!\eta/(1-\eta) < 1$. So equal coefficients force equal histograms, which contain a common labelled adjacency matrix. □

The separating volumes in (c) are enormous. This is a completeness statement, not an isomorphism algorithm, and completeness of the fixed-pinning series $A_H = F_2(H)$ remains open. On small populations the fixed-pinning series is already complete. All 1,044 graphs on seven vertices, disconnected ones included, have distinct A_H and distinct L_H through volume five; a separate session run confirmed volume six. Among the 12,346 graphs on eight vertices, exactly twenty pairs agree through volume five, the same twenty for A_H, for L_H and for the full moment series. All of them are connected with equal degree sequences, and all separate at volume six (`collision_census` in `results/loop_series.json`). This is a finite statement about these populations.

At budgets the compiler reaches, (b) gives an inexpensive enrichment with an independent check. In moment mode, `moment_series` in `src/singularity/locality.py` carries S through the connected-support compiler, packed into the exponent key $E(R^2+1)+S$, which stays additive and decodable through degree R. `moments_from_pinnings` recovers $C_{j,S}(q_0)$ from $\binom{R}{2}+1$ ordinary pinned series. The receipt checks both against enumeration, as well as the strata of (a).

10.10.2 Beyond the first cycle: overlaps

SNG-F56 resolves cycles. The next information a budget reaches is how cycles overlap. Three exact computations make this concrete, and SNG-F60 and SNG-F61 below prove the pattern.

Rook against Shrikhande. The 4×4 rook graph and the Shrikhande graph are 6-regular on sixteen vertices, with the same spectrum, thirty-two triangles, sixty squares and forty-eight diamond subgraphs. They differ in K_4 subgraphs: eight against none. Their series agree below volume four, and

$$[x^4](F_\lambda(\text{rook}) - F_\lambda(\text{Shrikhande})) = 8q^{12+4\lambda}(1-q^2)^6$$

for $\lambda = 0, 2$, confirmed by enumeration. With $a = q^{d+\lambda}$, $f = q^{-2} - 1$ and $\kappa = af$, this is $\kappa^4 f^2 \Delta N_{K_4}$. It matches the normalized girth-three term $N_4 + fN_{\text{diamond}} + f^2 N_{K_4}$, with diamonds counted as subgraphs: the induced counts are 0 and 48, and each K_4 contains six diamonds.

The loop decomposition. For a regular graph the whole series below the first overlap is determined by cycle counts. Let $\mathbb{T}_d$ be the d-regular tree, and let $\mathcal{U}_\ell$ be the cycle C_ℓ with $d-2$ infinite $(d-1)$-ary branches at each vertex. Put $b_g = \lceil 3g/2 \rceil - 1$. Define the universal series

$$b_{d,\lambda} = \sum_{o \in S \subseteq \mathbb{T}_d} \frac{\Phi_S}{|S|}, \qquad W_{\ell,d,\lambda} = \sum_{C \subseteq S \subseteq \mathcal{U}_\ell} \Phi_S - \ell \sum_{S'} \Phi_{S'},$$

over finite connected sets. In the last sum S' ranges over the subtrees of $\mathbb{T}_d$ that meet a fixed bi-infinite line L exactly in a path $[0, m-1]$ with $m \geq \ell$. These are the tree unfoldings of the cycle, counted modulo translation along L, which lifts C.

Theorem SNG-F60 (loop decomposition below the first overlap). Let H be d-regular on n vertices with girth at least g, and let $R < b_g$. Then

$$\log F_\lambda(H) \equiv n\, b_{d,\lambda} + \sum_{\ell=g}^{R} N_\ell(H)\, W_{\ell,d,\lambda} \pmod{x^{R+1}},$$

and $W_\ell = \kappa^\ell x^\ell + O(x^{\ell+1})$, with $\kappa = af$, $a = q^{d+\lambda}$, $f = q^{-2} - 1$.

Proof. A connected graph with two independent cycles contains a theta, a figure-eight or a dumbbell. Under girth $\geq g$, a theta with path lengths α, β, γ needs $\alpha + \beta, \beta + \gamma, \gamma + \alpha \geq g$, hence at least b_g vertices; the other two need at least $2g - 1$. So for $|S| < b_g$, $H[S]$ is a tree or has exactly one cycle, which is chordless.

Let $p : \mathbb{T}_d \to H$ be the universal covering, and fix a lift $\tilde{v}$ of each vertex. Summing $\Phi_{S'}/|S'|$ over subtrees $S' \ni \tilde{v}$ with $|S'| \leq R$, and then over v, gives nb modulo x^{R+1}, since $\Phi_{S'}$ is computed with all weights equal to $d + \lambda$. Regroup by the image $p(S')$:

- If p is not injective on S', the geodesic between two points with the same image projects to a closed non-backtracking walk, so the image contains a cycle and hence exactly one.
- An image S inducing a tree arises exactly once through each of its $|S|$ vertices, and contributes Φ_S.
- An image S with one cycle contributes an amount $\Psi(H[S])$ that depends only on the type of $H[S]$. The relevant component of $p^{-1}(S)$ is the universal cover of $H[S]$.

For an ℓ-cycle C of H, the covering $q_C : \mathcal{U}_\ell \to H$ with $q_C(\tilde{C}) = C$ induces a type-preserving bijection. On one side are the finite connected $\tilde{S} \supseteq \tilde{C}$ with $|\tilde{S}| \le R$; on the other, the sets S of H with one cycle, namely C, and $|S| \le R$. By the lifting criterion every such S lifts uniquely, since its fundamental group is generated by C. Conversely, a collision or an extra edge in $q_C(\tilde{S})$ would give a loop not in $\langle C \rangle$, hence two independent cycles on fewer than b_g vertices.

Therefore $\log F = \sum_S \Phi_S = nb + \sum_{S \text{ unicyclic}} (\Phi_S - \Psi(H[S]))$, and the last sum equals $\sum_\ell N_\ell W_\ell$. For the cover $\mathbb{T}_d \to \mathcal{U}_\ell$, the preimage of C is the single line L, and the tree unfoldings of $\tilde{S}$ are the subtrees whose intersection with L covers every residue modulo ℓ. Those are the paths of length at least ℓ, counted modulo translation, which is the factor ℓ. Every set in W_ℓ has at least ℓ vertices. At degree ℓ only C and the ℓ-vertex paths enter, and SNG-F56 gives κ^ℓ. □

For $d = 2$, $\mathcal{U}_\ell = C_\ell$ and $W_\ell = D_\ell$ of SNG-F57.

Theorem SNG-F61 (the first overlap). In SNG-F60 take $R = b_g$. Then

$$\log F_\lambda(H) \equiv nb + \sum_{\ell=g}^{R} N_\ell W_\ell + x^R \sum_S a^R \sum_{E'} f^{|E'|} \pmod{x^{R+1}},$$

where S runs over the connected R-sets whose induced graph has cyclomatic number at least 2, and E' over the connected spanning edge sets of $H[S]$ with cyclomatic number at least 2.

Proof. Below degree R nothing changes. At degree R the only new sets are the multicyclic R-sets, and only their multilinear coefficient enters, which is $a^R \sum f^{|E'|}$ over all connected spanning E' (SNG-F56). The background assigns to S the lifts through its spanning trees, worth $\tau(H[S]) a^R f^{R-1}$. A cycle term W_ℓ assigns to each connected spanning subgraph U of $H[S]$ with one cycle the lifted set $\tilde{S}$, which induces exactly U. That contributes $a^R(f^R + \ell f^{R-1})$, minus its ℓ tree unfoldings, net $a^R f^R$. Subtracting the tree and unicyclic parts from the full sum leaves the stated correction. The covering counts break only at these sets, because two independent cycles need at least R vertices. □

For $g \ge 4$ the multicyclic b_g-sets are exactly the balanced thetas Θ_g, with path lengths $(g/2, g/2, g/2)$ or $((g-1)/2, (g+1)/2, (g+1)/2)$. Each contributes $a^R f^{R+1} = \kappa^R f$. For $g = 3$, $R = 4$, a diamond subgraph contributes $\kappa^4 f$ and a K_4 contributes $\kappa^4 f^2$ beyond the six diamonds it contains. This proves the normalized forms $N_{b_g} + f N_{\Theta_g}$ and $N_4 + f N_{\text{diamond}} + f^2 N_{K_4}$, and with them the rook–Shrikhande coefficient above and the two-temperature recovery below, in absolute form.

One volume past the first cycle. Put $\beta = a((d-2)f - 2) + a^3(q^{-2} + 1)^2$. Then $[x^{g+1}] W_{g,d,\lambda} = g\beta\kappa^g$ for every $g \ge 3$, $d \ge 2$ and $\lambda \ge 0$. For $g \ge 5$ we have $g + 1 < b_g$, so SNG-F60 and $[x] \log F = na$ give, for d-regular graphs of equal order and girth at least g,

$$\Delta[x^{g+1}] F_\lambda = \Delta N_{g+1} \kappa^{g+1} + \Delta N_g \kappa^g (g\beta + na).$$

Proof of the coefficient. At degree $g+1$, W_g receives contributions from the following sets:

- the cycle with one vertex at height two;

- the cycle with a pendant vertex ($g(d-2)$ choices);
- with sign $-g$: the path P_g with one vertex at height two, the path P_{g+1}, and P_g with a pendant.

The pendant terms combine to $g(d-2)a^{g+1}f^g\big((f+g)-g\big) = g\kappa^g \cdot a(d-2)f$.

For a vertex v at height two, the coefficient of $x_v^2\prod_{u\neq v}x_u$ in $\log Z$ is given by a *twin lemma*:

$$\tfrac{1}{2}a^{k+1}\sum_{E'\subseteq G^*}\prod f_e + \tfrac{1}{2}a^{k+3}\sum_{E'\subseteq G}\prod f'_e.$$

Here G^* adds a twin v' of v, joined to the neighbours of v with weight f, and to v with weight a^2-1. The weights f' equal $q^{-4}-1$ on the edges at v and f elsewhere. Both sums run over connected spanning edge sets. The lemma comes from writing the two units at v as labelled copies: multiset partitions are half the set partitions of $V\cup\{v'\}$, plus half of those keeping the copies together.

Enumerating the connected spanning edge sets of the cycle, the path and their twins gives a closed form in a, f and ℓ. After division by $a^{\ell+1}f^{\ell-3}$, the required identity is polynomial in a and f and linear in ℓ. `scripts/loop_series.py` checks it symbolically, with a and f independent, for $\ell = 3,\dots,8$, which is enough. It also checks each component against compiler coefficients. □

The first bicycle. For girth $g = 4$, the first footprint with two independent cycles is $K_{2,3}$, on $b_4 = 5$ vertices. Take three cubic graphs on twelve vertices, each with six squares, whose $(N_5, N_{K_{2,3}})$ counts are $(0,0)$, $(0,1)$ and $(4,2)$. For every pair and $\lambda = 0, 2$, the series agree below volume five, and

$$\Delta[x^5]F_\lambda = \kappa^5(\Delta N_5 + f\,\Delta N_{K_{2,3}}).$$

Since $f(1/2) = 3$ and $f(1/3) = 8$, the normalized differences $r(q) = \Delta[x^5]F/\kappa^5$ at two temperatures recover both counts: $\Delta N_{K_{2,3}} = (r(1/3) - r(1/2))/5$ and $\Delta N_5 = (8r(1/2) - 3r(1/3))/5$. The receipt recovers all three pairs exactly. By SNG-F61 the absolute version holds too: after subtracting nb and the cycle terms, the normalized degree-five coefficient is $N_5 + fN_{K_{2,3}}$.

So a budget resolves cycles first and then their overlaps, and two temperatures already separate a cycle count from a bicycle count. The computations are checked in `results/overlap_hierarchy.json`, produced by `scripts/overlap_hierarchy.py`. The decomposition is checked in `results/loop_series.json`, produced by `scripts/loop_series.py`; both scripts run in CI. That receipt compares SNG-F60 and SNG-F61 with the compiler, for $\lambda = 0, 2$, on the cube, $K_{3,3}$, the Petersen graph, the rook and Shrikhande graphs, K_5, C_5 and the three twelve-vertex cubic graphs, below and at b_g. The cycle-only prediction fails at b_g exactly for the graphs that contain a multicyclic b_g-set. A retained sweep over all twenty-five generalized Petersen graphs $GP(m,k)$ with $m \le 14$ and girth $4 \le g \le 6$ found every prediction exact at $R = b_g$. It also confirms $[x^{g+1}]W_g = g\beta\kappa^g$ on a grid of $d \le 4$, $\lambda \in \{0,2\}$, $g \le 6$.

10.11 What should be searched next?

The first inverse question is whether full marked equality implies graph isomorphism on any substantial class. Trees are an attractive target because the ordinary cut lattice of every tree is a standard integer lattice and therefore discards almost all attachment geometry. The marked volume population could preserve some of that geometry.

Completeness on trees is a question, not a conclusion. A single pair of nonisomorphic trees with equal full marked series would refute it; a finite census without a collision would only establish a bounded result.

A second question is whether the full hierarchy eventually stops improving on a fixed graph order. Since there are finitely many isomorphism classes on n vertices, if the full marked invariant is complete on that order then some finite maximum distinguishing depth exists. This existence is purely finite and does not supply a useful bound. If the invariant has collisions, the same argument instead gives a finite depth at which its equivalence partition stabilizes to its final, possibly nontrivial partition. A search can record that stabilization for a finite census, while a theorem must control all higher depths.

A third question concerns the loss of the span marker. It is possible that the unmarked series is complete on a class where no simple proof recovers the binary sector. It is also possible that different marked populations collapse to the same unmarked series. Neither possibility is settled by the current examples. The appropriate experiment compares marked equality and unmarked equality separately, rather than assuming that agreement or disagreement of one automatically decides the other.

A useful search ledger should retain graph order, component sizes, degree polynomial, binary profile, first different volume, first different span, and exact difference polynomial. It should separate pairs selected to collide under a background invariant from arbitrary pairs that happen to be easy to distinguish. Regular graphs with equal triangle count offer controlled difficult cases after the easy volume-three obstructions have been removed. Trees offer a different challenge, in which lattice isometry is automatic but attachment structure varies. Graphs related by changes of attachment within a common block structure test whether the volume marker preserves the intended vertex realization.

These are falsifiable targets. The hypothesis that depth two distinguishes all binary-profile collisions can be rejected by one depth-two collision. The hypothesis that a finite volume cutoff suffices for all graphs of a stated order needs a census whose completeness is itself justified. The hypothesis that a coefficient depends only on a known motif census can be tested by deriving its assignment patterns before scanning data. Every negative result would locate an exact limit of the representation. Such a limit would still be useful mathematics, because it would say what relational information the population has actually retained.

Chapter 11

Archival bridges: what is inherited and what changes

Singularity did not arise from an empty mathematical landscape. PerfectPower, PerfectJet, the SOE program, and older conductor and lattice investigations already contain several of its ingredients. The appropriate historical claim is therefore narrow: the present project selects a different object and a different inverse question from a substantial existing toolkit. It does not claim to invent prime valuations, graph Laplacians, discrete Gaussian heights, cut lattices, binary Ising polynomials, or Euler products. Those established subjects are part of the mathematical background [BHN1997; AM2009; BPR2024]. The value of the new direction must be demonstrated through its precise marking, its exact examples, and the questions those examples make available.

This chapter explains the bridges in mathematical terms. A shared vocabulary such as population, shadow, quotient or phase is not enough to establish a bridge. There must be a defined map, a preserved quantity, an exact identity, or a specifically stated proposal. The distinction matters because the archives contain both successful computations and ambitious descriptions whose full objects were not implemented. Reusing a phrase should not silently upgrade the status of either the old claim or the new one.

11.1 From multiplicative integers to relational exponent geometry

The oldest relevant arithmetic step is the valuation embedding

$$n \longmapsto v(n) = (v_2(n), v_3(n), v_5(n), ...).$$

Every positive integer gives a finitely supported nonnegative integer vector, and multiplication becomes addition: $v(mn) = v(m) + v(n)$. The June collision work used this geometry to define

$$\kappa(m, n) = \sum_p v_p(m) v_p(n).$$

It related shared prime support to a dot product and represented divisors of an integer as a product of finite chains. Its analytic deformation rewarded or penalized shared

exponents with $e^{t\kappa(m,n)}$. The exact Euler factor was $\sum_{a,b\geq 0} e^{tab}p^{-s(a+b)}$. A positive real reward is unstable because the terms with $a = b = N$ contain quadratic exponential growth against only a linear exponential penalty. This is a specific mathematical limitation, not a general failure of the valuation viewpoint [PJ2026].

The present construction uses exponent differences instead of that attractive dot product. For a graph G, put

$$\mathcal{E}_G(n_1, \dots, n_k) = \sum_p \sum_{ij\in E(G)} (v_p(n_i) - v_p(n_j))^2.$$

The sum is finite for every tuple because only finitely many primes divide the product. For $0 \leq q \leq 1$, the weight $q^{\mathcal{E}_G}$ is bounded by one. The resulting series is therefore dominated by the independent tuple series in its elementary convergence domain. This change does more than repair a divergent sign. It creates a zero-energy relation: two adjacent integer positions must have equal valuations at every prime, hence equal integers. On a connected graph, zero energy means the entire tuple is constant.

Proposition SNG-I020: exact synchronization at zero parameter. For a connected graph on k vertices,

$$\mathcal{E}_G(n_1, \dots, n_k) = 0 \quad \Longleftrightarrow \quad n_1 = \cdots = n_k.$$

The forward direction follows because each squared difference must vanish and equality propagates along graph paths. The reverse direction is immediate. For disconnected graphs, equality is required separately on each component. Thus the earlier multiplicative linearization now supports a relational arithmetic population. The graph specifies which exponent coordinates must agree; the primewise product turns that specification into a Dirichlet series.

There is no information gain in the Euler lift itself. Once $P_G(x, q)$ is known, $\prod_p P_G(p^{-s}, q)$ is fixed in its convergence domain. Conversely, the coefficient of a prime power p^r in the Dirichlet series recovers $[x^r]P_G$, because a product equal to p^r forces every tuple entry to be a power of that single prime. The arithmetic series is an exact re-expression with useful analytic operations, not an additional sensor that distinguishes graphs whose full local factors agree.

11.2 From exact populations to navigable height assignments

PerfectPower repeatedly studies a large finite population through counts, conditional counts, rankings and samples instead of explicitly listing every member. That approach transfers directly to bounded height populations [PP2026]. At depth D, the original objects are vectors $a \in \{0, \dots, D\}^k$ with minimum zero. Their number is $(D+1)^k - D^k$. If energy, volume and span constraints are imposed, the retained population is smaller but remains a population of original assignments. A conditional count can determine how many completions remain after the first few coordinates have been fixed.

The released `HeightPopulation` supplies exact completion counts; lexicographic ranking and unranking follow from the usual branching identity. At a prefix u, the total completion count is the sum of completion counts after each allowed next height. A

rank selects the first branch whose cumulative count exceeds it. An original assignment receives a rank by adding all earlier branch populations. This supplies an exact navigation mechanism for height configurations without identifying assignments that merely share a profile. It also makes uniform sampling possible by drawing a uniform integer rank in the feasible population.

The distinction between counting assignments and counting profile types is essential. The energy-volume-span histogram may have relatively few occupied cells, but a cell's coefficient is its multiplicity. Replacing that coefficient by one changes the invariant. The seven-vertex span-two cubic sector has forty-two ordered assignments even when only a modest set of energies occurs. Its successful distinction depends on the distribution of those multiplicities. The population machinery is therefore relevant as an exact computational method, while its prior application domains need not be imported into the present mathematical target.

An exact residual decomposition also becomes possible. A partial assignment contributes the energy of edges whose endpoints are both processed. Edges crossing to unprocessed vertices remain symbolic functions of their boundary heights. Their contributions should not be guessed from the already accumulated scalar energy. Keeping the boundary state allows future assignments to complete the squared differences exactly. This is the natural finite-state interface for dynamic programming over a path or tree decomposition. For a fixed cutoff D, a boundary of size b has at most $(D+1)^b$ raw height states before any further reduction. This is a conditional complexity observation about a chosen decomposition, not a universal small-state theorem.

11.3 From all-future behavioral equivalence to weighted transfer states

The SOE program formalizes the principle that two partial states should be identified exactly when every allowed future continuation gives the same observable behavior [SOE2026]. A direct analog exists for a fixed finite height-transfer computation. The observation is now a polynomial population, and a continuation adds vertices, edges, weights, or terminal conditions according to a declared processing scheme. Two boundary states can be merged only when all continuations permitted by that scheme produce the same remaining generating polynomial.

That sentence is a proposed computational use of the SOE idea, not a claim that the previous Lean theorem has already verified this graph algorithm. Several details must be fixed before a formal transport is legitimate. The state must include any boundary heights required by future edges, the accumulated or deferred volume information, and sufficient data to enforce the global span convention. Polynomial weights must be handled over a specified coefficient structure. A quotient of a two-state partial machine does not automatically verify a family of graph transfers with arbitrarily many boundary states.

The proposal nevertheless has a concrete exact form. Let a processing alphabet encode the legal remaining transfer steps for a fixed decomposition. To a partial state s, associate the function $f_s(w)$ giving the remaining polynomial after continuation word w. Define $s \sim t$ when $f_s(w) = f_t(w)$ for every permitted w. This equivalence is correct by definition and is stable under a common legal next step. The research problem is to compute it efficiently and provide a finite certificate that the retained tests suffice. On a fixed finite continuation space, exhaustive comparison is available. Across unbounded decompositions, the finite characterization requires an additional argument.

The distinction connects usefully to PerfectJet's exact representation work. A vector of sufficient boundary quantities should remain closed under every declared transfer. If it does not, the missing interaction is part of the returned mathematical diagnosis. In the present examples the binary population is insufficient for a specific higher-height question, and the span-two population supplies an explicit witness. This is an actual object-level result, stronger than merely observing that a chosen summary might omit information.

11.4 The relation to classical cut lattices

For a connected graph choose an orientation of its edges and write d for the vertex-to-edge incidence map. Then $E_G(a) = \|da\|^2$, and $\ker d$ consists of common constant shifts. The image of integer vertex heights under d is the integral cut lattice. Every integer height class modulo common shifts has a unique representative of minimum zero. Consequently

$$B_G(1, q, 1) = \sum_{a \in \mathbb{Z}^k / \mathbb{Z}\mathbf{1}} q^{\|da\|^2}$$

is its ordinary theta series for $0 < q < 1$. This is classical lattice territory. The relation between a cut lattice's minimum norm and graph edge connectivity is established background [BHN1997]. Singularity's elementary minimum-cut proof should be read as a proof in its chosen coordinates, not as a historical priority claim about that relationship.

The added volume and span markers retain the position of a height class relative to the vertex-height coordinate system. They are not invariants of an abstract Euclidean lattice by itself. An isometry of cut lattices need not preserve the minimum-normalized sum of vertex heights or their maximum. This is the precise sense in which the new object can distinguish vertex realizations that an ordinary theta series identifies.

The examples give a concrete instance. Gluing two graph blocks at a single vertex makes their cut lattices an orthogonal direct sum. To see this, independently choose integer heights on each block modulo constants, and shift their representatives so the common vertex agrees. The edge-energy sum splits into the two blocks, and the correspondence is bijective. A bridge contributes a one-dimensional standard integer lattice. Each spectral graph therefore has a cut lattice isometric to the triangle cut lattice plus four such bridge factors. Each binary graph has the cut lattice of its common theta block plus two bridge factors. The placement of those bridge attachments does not change this abstract lattice decomposition, but it does change the marked populations established in the preceding chapter.

This distinction also cautions against interpreting the construction as a new kind of Euclidean spectral statistic. The Laplacian spectrum, the abstract cut lattice, and the marked height population forget different information. Their comparison is valuable precisely because they are not interchangeable. The examples show a difference after some conventional data have been fixed; they do not prove a total ordering among every graph invariant.

11.5 Finite conductors and the part that is already old

The conductor archive studied rational phases and exact residue closure. The same algebraic mechanism works for the present unmarked local factor. Let $q = \zeta_L$ be an L-th root of unity. Changing a single coordinate by L changes energy by

$$E_G(a + Le_i) - E_G(a) = 2L \sum_{j \sim i} (a_i - a_j) + L^2 d_i,$$

which is divisible by L. The phase is therefore periodic in each height coordinate modulo L. Splitting $a_i = r_i + Lb_i$ gives the exact identity

$$P_G(x, \zeta_L) = \frac{N_{G,L}(x)}{(1 - x^L)^k}, \qquad N_{G,L}(x) = \sum_{r \in \{0,\dots,L-1\}^k} \zeta_L^{E_G(r)} x^{|r|}. \tag{SNG-I021}$$

The equality holds as a formal series and analytically for $|x| < 1$. It is a finite cyclotomic calculation. The archive already used this kind of residue decomposition for collision and Cartan quadratic forms [PJ2026]. Its presence here is an inherited bridge, not the new claim. The difference is that a graph Laplacian has a common-shift null direction and the project retains coordinate-sensitive marks around that direction.

The marked span variable prevents one from copying the same numerator formula unchanged. Adding L to one coordinate can alter the common minimum and maximum, so its effect on the span marker is not determined by that coordinate's residue alone. One may specialize to $z = 1$, keep a finite height cutoff, or develop an additional boundary description. The unmarked rational formula does not establish a correspondingly simple rational formula for the full marked series.

There is a further exact bridge to spanning trees. For an odd prime p, let

$$\Gamma_G(p) = \sum_{a \in \mathbb{F}_p^k} \exp(2\pi i E_G(a)/p).$$

Fix one vertex height to zero and separate the free common shift. The remaining quadratic form has a reduced Laplacian matrix with determinant $\tau(G)$, the spanning-tree count. If $p \nmid \tau(G)$, the reduced form is nondegenerate. Standard finite-field diagonalization then gives

$$\Gamma_G(p) = p \left(\frac{\tau(G)}{p} \right) g_p^{k-1}, \qquad g_p = \sum_{u \in \mathbb{F}_p} e^{2\pi i u^2/p}. \tag{SNG-I022}$$

In particular its absolute value is $p^{(k+1)/2}$. When p divides the spanning-tree count, the reduced quadratic form is singular and this nondegenerate formula must be replaced by its rank-sensitive version, which holds for every graph, connected or not.

Theorem SNG-F30 (rank-sensitive Gauss sum). Let p be an odd prime, $\omega = e^{2\pi i/p}$, and $g_p = \sum_{u \in \mathbb{F}_p} \omega^{u^2}$. Let $P \in GL_k(\mathbb{F}_p)$ satisfy $P^\top \Delta_G P \equiv \mathrm{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0)$ with $\lambda_i \not\equiv 0$. Then r is the rank of Δ_G over $\mathbb{F}_p$, the class of $D = \lambda_1 \cdots \lambda_r$ modulo

squares does not depend on P, and

$$\Gamma_G(p) = p^{\,k-r}\left(\frac{D}{p}\right) g_p^r,$$

with the convention that the right side is p^k when $r = 0$.

Proof. Energy is the integer quadratic form $a^\top \Delta_G a$, so $\omega^{E_G(a)}$ depends only on a mod p. Every symmetric matrix over a field of odd characteristic is congruent to a diagonal one: if some diagonal entry is nonzero, move it to the pivot and clear its row and column by the substitutions $e_j \mapsto e_j - (m_{kj}/m_{kk})e_k$; if all remaining diagonal entries vanish but $m_{ij} \neq 0$, replace e_i by $e_i + e_j$, whose new diagonal entry is $2m_{ij} \neq 0$ because p is odd; if the remaining block is zero, stop. Since P is invertible, $a = Pb$ is a bijection of $\mathbb{F}_p^k$, so

$$\Gamma_G(p) = \sum_b \omega^{\sum_i \lambda_i b_i^2} = \prod_{i \le r} \Big(\sum_u \omega^{\lambda_i u^2}\Big) \cdot p^{\,k-r}.$$

Each zero diagonal entry is a singular direction and contributes a full factor p; these include the constant vector of every connected component, and further directions at primes dividing invariant factors of the Laplacian. For $\lambda \neq 0$, substituting $u \mapsto \lambda^{-1}u$ when λ is a square, and counting the solutions of $u^2 = v$ when it is not, gives $\sum_u \omega^{\lambda u^2} = \left(\frac{\lambda}{p}\right) g_p$. Multiplicativity of the Legendre symbol gives the formula. Rank is a congruence invariant. The radical $R = \ker \Delta_G$ is intrinsic, and the induced nondegenerate form on $\mathbb{F}_p^k/R$ has Gram determinant D in the basis given by the images of the first r columns of P. A change of basis multiplies it by a nonzero square, so the square class is well defined. □

For a connected graph with $p \nmid \tau(G)$, the rank is $k-1$ and D has the square class of $\tau(G)$, which recovers SNG-I022. The triangle at $p = 3$ is an exceptional prime: $\tau = 3$, and $\Delta_{K_3} \equiv 2J \pmod 3$ has rank one, with $D = 2$ a non-square. Hence $\Gamma_{K_3}(3) = -9g_3 = -9 - 18\omega$, whereas the nondegenerate formula would predict a multiple of g_3^2. The implementation `src/singularity/gauss.py` returns the diagonalizing matrix as a witness and rechecks $P^\top \Delta_G P$ and $\det P \neq 0$ exactly. It compares $p^{k-r}(D/p)g_p^r$ with the direct residue-cube sum in $\mathbb{Z}[t]/\Phi_p(t)$, with no square root of p, on all labeled graphs with at most four vertices for $p = 3, 5, 7$ and all five-vertex graphs for $p = 3$. It rejects $p = 2$, composite moduli and prime powers, which this theorem does not cover. This provides a defined arithmetic role for discriminant primes: they signal additional modular degeneracy. It does not make every finite-conductor phase statistic complete for graph reconstruction.

11.6 What the new direction actually adds

The archival programs often sought an invariant that survived a passage from a large object to a smaller representation. That ambition remains useful, but the present project asks a more relational inverse question: what is preserved when one counts every nested height pattern while forgetting which vertex carried it? A binary subset census counts one boundary at a time. A two-level height configuration simultaneously places two nested boundaries and records edges that cross both. The exact co-binary example proves that this added relation is not determined by the entire one-boundary census.

The difference should not be called a higher-body interaction in the energy. The energy remains a sum of pairwise edge terms. Higher relational information appears because the observation organizes those same pairwise terms across several nested threshold sets. This distinction matters both mathematically and historically. The project has changed what is retained about a population, rather than positing a new force or claiming a new physical realization.

The resulting research program is therefore internal to mathematics at this stage. It studies a hierarchy of marked populations, their inverse problems, exact computation, and arithmetic singularities after a precisely defined Euler lift. Older physical, commercial, or external ambitions neither prove its significance nor limit the objects it can now examine. The useful inheritance is the ability to calculate exact populations, distinguish insufficient representations, and move consistently between valuation, lattice and conductor descriptions. The new work must earn its further claims through theorems or explicit counterexamples about the height hierarchy itself.

One further lesson from the archive is that a change of coefficient domain and a change of question are separate moves. Exact cyclotomic arithmetic can certify a finite phase sum that floating-point recognition only suggests. Exact integer arithmetic can certify the histogram of a large finite population. Neither act automatically changes what the histogram measures. A perfectly certified binary census remains a binary census. The co-binary example makes that point without relying on measurement noise, numerical instability, unverified physical premises or imperfect training data: even exact binary information can be insufficient for an exact higher-height question.

Conversely, enlarging the question does not require abandoning the old exact arithmetic. The same graph can be studied through an integer coefficient table, a real parameter $0 < q < 1$, a cyclotomic specialization, a cut-lattice theta series, or a Dirichlet series. Each view should carry an explicit account of the markers it retains. A statement about the full marked hierarchy cannot be replaced by a statement about one root-of-unity specialization. A statement about an Euler singularity cannot substitute for equality of every local coefficient. The creative move is the selection of the relational population, while the archival discipline supplies the maps that let its different views be compared honestly.

Chapter 12

Exact computation, certificates, and reproducible experiments

12.1 What a computation is intended to establish

The computational object is a population of labeled integer assignments. For a simple graph $G = (V, E)$, with $V = \{0, \dots, n-1\}$, an assignment $a \in \mathbb{N}^n$ has volume $v(a) = \sum_i a_i$, energy $E_G(a) = \sum_{\{i,j\} \in E} (a_i - a_j)^2$, and span $\max_i a_i - \min_i a_i$. The local series is $L_G(x, q) = \sum_a x^{v(a)} q^{E_G(a)}$. Computing through volume R means computing every coefficient of every polynomial $[x^v]L_G$ for $0 \le v \le R$. It does not mean evaluating the series at a few floating-point points or recording only energies below a second cutoff. An additional energy cutoff is legitimate only when it is part of the declared output and its omitted region is explicit.

The distinction between labeled profiles and isomorphism classes is essential. Graph relabeling does not change the resulting polynomial, but assignments related by a graph automorphism remain distinct members of the population. Dividing counts by an automorphism-group order generally gives incorrect coefficients because stabilizers vary. Orbit enumeration can accelerate computation, provided each representative carries its exact orbit size. The first implementation should retain labeled assignments; a later orbit implementation should reproduce those counts before becoming a source of larger examples.

Finite coefficient equality is a finite claim. Equality through volume R is evidence about a truncation and proves equality of that truncation. It does not prove equality of the full local series. Conversely, a single differing coefficient is a complete witness that two full series differ. The software should make this asymmetry visible in its result types: a discrepancy record is conclusive, whereas an unresolved collision retains its tested bound. A graph pair may be cospectral, binary-census equivalent, or equal through a stated volume without being equivalent for the relational height population.

12.2 A reference enumerator whose meaning is transparent

At a fixed volume v, enumerate all weak compositions of v into n parts. There are exactly $\binom{v+n-1}{n-1}$ such assignments when $n > 0$. For each tuple, calculate energy directly from the normalized edge list, then increment the corresponding integer histogram. Python integers suffice: coefficients are arbitrary-precision integers and no

rounding is involved. A sparse dictionary indexed by energy is preferable to a dense array when the attainable energies occupy a small subset of their possible range. The mathematical definition, rather than an optimization, remains the specification of this enumerator.

An exact and useful bound is $E_G(a) \le \Delta(G)v^2$ for nonnegative assignments of volume v. Indeed, $(a_i - a_j)^2 \le a_i^2 + a_j^2$, so the energy is at most $\sum_i d_i a_i^2 \le \Delta \sum_i a_i^2 \le \Delta v^2$. Concentrating all volume at a vertex of maximum degree attains the bound. The bound therefore checks both data validity and the maximum polynomial degree. It is not a reason to allocate every intermediate degree. At zero volume, the unique assignment is zero and contributes one at energy zero.

The reference cost through R is proportional to $|E|\binom{R+n}{n}$, up to tuple-generation and integer-arithmetic costs. This follows from summing the weak-composition counts, not from an assumption that all energies are distinct. Memory is controlled by the number of nonzero histogram entries retained. Timings must report n, $|E|$, R, the number of assignments visited, histogram support sizes, interpreter version, and hardware. An isolated runtime without these quantities cannot distinguish a better algorithm from an easier instance.

12.3 Three independent representations of the same coefficients

Translation normalization supplies the first independent check. Write $a = b + c\mathbf{1}$, where $c = \min_i a_i$ and $\min_i b_i = 0$. Energy is unchanged and volume increases by nc. Consequently $B_G = (1 - x^n)L_G$, where B_G counts minimum-zero profiles. A direct minimum-zero enumerator should agree with the coefficient subtraction $B_v = L_v - L_{v-n}$, taking negative indices as zero. This tests a bijection and a polynomial operation against an independent predicate. It also catches the mistake of normalizing each connected component separately when the declared convention uses one global minimum.

The second representation uses nested level sets. For a normalized profile of span h, put $S_t = \{i : a_i \ge t\}$. Then $\emptyset \ne S_h \subseteq \cdots \subseteq S_1 \subsetneq V$, and repeated sets are permitted. Volume is $\sum_t |S_t|$. Define $Q(S,T) = \sum_{\{i,j\}\in E}(1_S(i) - 1_S(j))(1_T(i) - 1_T(j))$. For $T \subseteq S$, this is the number of edges between T and $V \setminus S$. Expanding a square gives

$$E_G(a) = \sum_{t=1}^{h} |\delta S_t| + 2 \sum_{1\le u<t\le h} Q(S_u, S_t).$$

A level-chain enumerator therefore has a genuinely different failure surface from a weak-composition enumerator. It should reconstruct the tuple as well as its three statistics and compare complete histograms. Omitting repetitions silently loses profiles with height gaps. Omitting the mixed terms silently replaces squared differences by absolute differences. These are mathematically different models, even when a few tiny examples happen to agree.

The fully marked series $B_G(x, q, z)$ retains span. Its span-one sector is exactly the nonconstant binary population: $[z]B_G = \sum_{\emptyset\ne S\subsetneq V} x^{|S|}q^{|\delta S|}$. This identifies a direct check against a bit-mask cut enumerator. The empty and full subsets require separate treatment in an unnormalized binary census. The marked identity proves that the full

marked series contains the binary sector; setting $z = 1$ mixes sectors and does not establish a method for recovering them.

12.4 Exact polynomials before numerical evaluation

Each volume coefficient is stored as an integer polynomial in q. Addition, multiplication, subtraction, and comparison use degree-indexed sparse maps. Rational coefficients are needed for logarithms, division of formal series, and Euler-factor extraction, so those operations use reduced fractions. A numerical value such as $q = 1/2$ is substituted only after the polynomial identity has been established. Equality at one rational point is weaker than polynomial equality; a discrepancy can also disappear at an exceptional specialization.

The two supplied graph pairs illustrate this discipline. The spectral pair has its first local discrepancy at volume two, with polynomial $2q^2(1-q)^3(1+q)$. The binary-census pair agrees in its complete binary population but first differs at volume three, with polynomial $q^3(1-q)(1-q^3)^2(1-q^4)^2$. The corresponding values at one half are $3/32$ and $11025/262144$. A release should retain expanded coefficient maps, factorizations, and exact substitutions. The factorization explains signs and exceptional endpoints; the expanded maps make recomputation straightforward. The factored expressions alone do not certify that every lower-volume coefficient was checked.

For connected graphs, the endpoint checks are particularly strong. At $q = 1$, the local series is $(1 - x)^{-n}$; at $q = 0$, only constant profiles survive, giving $(1 - x^n)^{-1}$. Thus the normalized series becomes $(1 - x^n)(1 - x)^{-n}$ and 1, respectively. A disconnected graph has additional zero-energy profiles, so the connected endpoint formula must not be used as a universal assertion. Empty-edge graphs and isolated vertices are useful tests precisely because they expose hidden connectedness assumptions.

12.5 Ranking, unranking, and exact sampling

The finite population machinery can provide reproducible witnesses as well as totals. For fixed volume, order weak compositions lexicographically by labeled coordinates. Once a prefix is fixed, its unconstrained completion count is a binomial coefficient. Ranking sums counts of earlier possible next entries; unranking subtracts those counts until the selected branch is found. These operations establish an explicit bijection between assignments and integers in a known interval. A uniformly selected integer therefore gives an exactly uniform assignment, without floating-point probabilities.

Conditioning on energy or span requires completion counts for the conditioned population, not reuse of the unconstrained formula. A memoized state can retain the assigned boundary heights, remaining volume, accumulated energy, and any extrema required by the marking. Whenever an edge has both endpoints assigned, charge its energy once. The remaining state must retain every quantity on which future contributions depend. A compressed state justified only by empirical agreement is a heuristic until its completion-count equivalence is proved. Tree decompositions offer a principled route to smaller boundaries, but the correctness argument is still the same conditional partition of the original labeled population.

The released `HeightPopulation` implementation supports ranking for finite bounded classes whose completion counts are exact and independently checked. The round-trip identities $\mathrm{unrank}(\mathrm{rank}(a)) = a$ and $\mathrm{rank}(\mathrm{unrank}(j)) = j$ must hold over the com-

plete small population. Sampling is useful for exploring typical profiles, but sampling cannot replace complete enumeration when the reported result is a full coefficient or an absence claim. Each sampled witness should include its rank, population size, seed provenance when applicable, and the exact conditioning predicate.

12.6 Certificates and the trusted boundary

The implemented certificate format is `singularity.certificate/1`. It records the finite graph source and its digest, the maximum retained volume, local coefficients, minimum-zero coefficients, fully marked coefficients, total enumerated profile count, and formal Euler exponents. Local integer counts are JSON integers. Exact rational polynomial data use triples `[degree, numerator, denominator]` with integer entries and a positive denominator. The released Python reader preserves these integers exactly. A JavaScript transport should convert large counts to decimal strings or BigInt before arithmetic, because ordinary JavaScript numbers do not preserve integers beyond $2^{53} - 1$; that transport change is a future schema extension rather than a property of the current format.

A pair certificate adds the second graph and digest, its retained local coefficients, the first discrepancy within the common truncation, and an optional supplied factorization. Replay independently enumerates original labeled profiles, checks their energies against the nested-threshold identity, reconstructs all recorded sectors, and replays the formal Euler recomposition. The receipt records a total profile count rather than a separate count for every volume. It does not encode a q-specialization or assert an analytic continuation. Those conclusions belong to the separately stated mathematical theorem. A source digest identifies the declared graph bytes; correctness comes from the independent calculation and mathematical identities.

Replay first runs a structural parser. It rejects unknown or missing fields, non-integer or Boolean counts, negative degrees, duplicate or unsorted terms in any polynomial or marked row, rational coefficients not in lowest terms, invalid or non-canonical edges in either graph, and the superseded roadmap name `relational-height/1`. Only a well-formed packet reaches arithmetic. Recomputation then uses two routes, neither of which is the builder's weak-composition engine: a replay-local stars-and-bars generator, and the level-chain search of Proposition SNG-F21. Both must reproduce the marked cells, and the work budget counts both. A packet whose counts are altered is rejected even when its digest, identity flags and every other field are left untouched. The format name, field meanings and this decision are fixed in `docs/specifications/COEFFICIENTS-v1.md`.

There are several different assurance levels. Direct Python enumeration is an exact finite computation relative to the interpreter and implementation. Independent enumerators reduce common implementation risk. A proof-assistant theorem about the enumeration algorithm is stronger, but does not automatically verify a particular exported table. A kernel-checked replay of the concrete table crosses that remaining boundary. Documentation should state which level each claim has reached. A schema and a future formalization plan are not present proof-assistant verification.

For the two decisive pair certificates, that last level has now been reached for the unmarked tables. `formal/` is a core Lean 4.12.0 package with no Mathlib. In it, `Singularity/Normalization.lean` and `Singularity/Thresholds.lean` (P07, P08) define profiles, volume and squared-difference energy, and prove the translation normalization and the nested threshold identity. `Singularity/Certificates.lean` (P33) proves that a weak-composition enumerator is exactly the duplicate-free list of profiles of

each volume, links its counts to the semantic coefficient definition, and then has the kernel evaluate the transcribed tables. `Singularity/Visibility.lean` proves the finite first-discrepancy cancellation for arbitrary series over any commutative ring and instantiates it on both certificates. Its original abstract positive-real interface is superseded by the actual Mathlib development in `formal-analysis/`: `SingularityAnalysis.graph_visibility` proves the convergent ratio and regularized prime products, zeta factorization, positive regular factor and right-hand normalized limit from the numerical q-range and certified finite coefficient agreement. The local remainder, prime summability and zeta-limit ingredients are proved, rather than retained in an analytic premise structure. For the spectral pair and the binary-census pair, the kernel checks:

- every local coefficient of both graphs through volume four, and the minimum-zero table of the certificate graph;
- equality of the two graphs at every lower volume;
- the first-difference polynomial and the expansion of its stated factorization.

Theorems also show that changing one edge, or one decisive coefficient, is rejected. The trusted boundary is the transcription: `formal/tools/gen_certificates.py` copies the JSON numbers into Lean literals, records the input SHA-256 digests, and has a `--check` mode. The span-marked rows and Euler exponents are not covered. The actual positive-real specializations `SingularityAnalysis.spectral_visibility_half` and `SingularityAnalysis.binary_visibility_half` are covered, with exact exponents 3/32 and 11025/262144; local-complex continuation, certified numerical intervals and synchronization limits are not formalized. `scripts/check_formal.py` builds the package and fails on any axiom outside `propext`, `Quot.sound` and `Classical.choice`; CI runs it on a clean download of the pinned release.

12.7 Analytic experiments and graph boundaries

For a first discrepancy at volume r, a numerical prime product should evaluate the normalized ratio after extracting $\zeta(rs)^{\delta}$, not multiply divergent raw products and interpret numerical overflow as evidence. The remainder product converges on the positive real region $s > 1/(r+1)$ under the stated local cancellation argument. A plot near $1/r$ should therefore compare the normalized remainder with a certified tail bound. Finite prime cutoffs need their own error estimates. Exact local coefficients establish the exponent; a fitted slope does not establish it independently.

Graph isomorphism is a separate task. Degree sequences, Laplacian spectra, binary census, and truncated height populations are filters, not general isomorphism decision procedures. Small examples can be checked by exhaustive permutation with an explicit witness or exhaustive exclusion. Larger censuses should use a documented canonical-labeling implementation and preserve its version and generation parameters. A claim that an invariant separates every graph in a census requires an independently specified complete graph population. A claim that it reconstructs all finite graphs requires a theorem. Neither follows from many successful pair comparisons.

The reproducible unit is a command, its declared finite input population, a complete output artifact, and an explanation of what that artifact proves. Small deterministic fixtures belong in routine checks. Larger censuses belong in explicit experiment scripts with resumable checkpoints and resource limits. The reference enumerator should remain available even after faster methods are introduced: its value is interpretability and independent agreement. The project becomes stronger when a reader can reproduce a modest decisive computation in minutes and then understand exactly which larger statements remain mathematical questions.

Chapter 13

A mathematical research program with distinct proof routes

13.1 The question that organizes the program

The program asks how much relational information survives when labeled integer height profiles are counted by volume and quadratic edge energy, and how a first finite discrepancy becomes visible in an arithmetic Euler product. It is a question about explicit combinatorial and analytic objects. It does not require a universal theory of every obstruction, a claim that all graph information has been recovered, or an interpretation of the Euler product as a source of additional graph data. The local series supplies the data; the arithmetic product changes where and how a discrepancy is expressed.

Three levels of progress should remain distinct. A structural theorem explains an entire graph class or a transformation law. An exact counterexample separates two stated invariants. A complete finite census establishes a bounded classification relative to a documented input population. Each can support a paper, but they answer different questions. The 36 pushes in `docs/development/PUSHES.md` are a dependency plan for obtaining these results, not a list of results already achieved. Negative outcomes are legitimate endpoints when they are recorded with the same precision as successful separations.

13.2 Tree messages and separator geometry

Trees provide the first substantial algorithmic theorem target because conditioning on a vertex height separates the descendant subtrees. Root each component of a forest at its least original label and orient every edge from parent to child. For a vertex v, write T_v for the set of descendants of v, including v, and $E(T_v)$ for the edges with both endpoints in T_v. Fix a height bound $H \geq 0$. For $0 \leq t \leq H$ define the **message**

$$F_v(t; x, q) = \sum_{\substack{b \in \{0,\dots,H\}^{T_v} \\ b_v = t}} x^{\sum_{w \in T_v} b_w} \, q^{\sum_{\{u,w\} \in E(T_v)} (b_u - b_w)^2},$$

a polynomial whose $x^m q^e$ coefficient counts labeled assignments of the subtree with root height t, volume m and internal energy e.

Proposition SNG-F23 (forest message recursion). For every vertex v with children $u_1, \dots, u_c$,

$$F_v(t) = x^t \prod_{i=1}^{c} \Big(\sum_{j=0}^{H} q^{(t-j)^2} F_{u_i}(j)\Big),$$

and the bounded-box series of the forest is $C_H = \prod_r \sum_{t=0}^{H} F_r(t)$, the product over component roots r.

Proof. Induct on $|T_v|$. The vertex sets $\{v\}, T_{u_1}, \dots, T_{u_c}$ partition T_v, and the edges of $E(T_v)$ are partitioned into the c connecting edges $\{v, u_i\}$ and the sets $E(T_{u_i})$; no other edge exists, because a forest has no edge between different child subtrees. An assignment b on T_v with $b_v = t$ is therefore the same as a choice of heights $j_i = b_{u_i}$ and of independent assignments on each T_{u_i} with root height j_i. Its volume is t, charged once at v by the factor x^t, plus the subtree volumes; its energy is the sum of the internal subtree energies and the connecting terms $(t - j_i)^2$, each charged once in the factor for u_i. Multiplication of generating polynomials realizes exactly this product of independent choices, so coefficients count labeled assignments without repetition or loss. The base case $T_v = \{v\}$ gives $F_v(t) = x^t$. Components are disjoint and joined by no edge, so the forest series is the product over roots of the summed root messages. □

The implementation computes messages in reverse depth-first order, so each child is complete before its parent. After F_u is folded into its parent it is deleted. This preserves the result because each non-root vertex has exactly one parent: no later step reads F_u. The roots' messages are retained and exposed as `root_messages`, keyed by original labels. Three cost measures are recorded separately by `transfer_statistics`: height states $(H+1)n$, the total and maximum number of sparse (volume, energy) cells across messages, and the largest integer bit length. None is by itself a running-time bound, since sparse convolution cost depends on support products.

Proposition SNG-F24 (safe volume truncation). Let $R \geq 0$. If every intermediate sum and product in SNG-F23, the convolution over components and the minimum-zero subtraction below are reduced by deleting monomials $x^m q^e$ with $m > R$, the result agrees with the unreduced computation in every coefficient with $m \leq R$. If $H \geq R$, these coefficients equal $[x^m]L_G$ for $m \leq R$.

Proof. Deleting monomials of x-degree above R is the quotient map $\pi : \mathbb{Z}[q][x] \to \mathbb{Z}[q][x]/(x^{R+1})$, a ring homomorphism, because the monomials of degree above R form an ideal when all x-exponents are nonnegative: a product of a monomial of degree $m > R$ with any polynomial of nonnegative degrees has every degree at least m. Hence $\pi(fg) = \pi(\pi f\, \pi g)$ and $\pi(f + g) = \pi f + \pi g$, so reducing after every operation equals reducing the final answer. Multiplication by $q^{(t-j)^2}$ does not change x-degree. Every message, the component product and C_{H-1} have nonnegative x-exponents, which is the only property used. For the subtraction $D_H = C_H - x^n C_{H-1}$, the term $x^n C_{H-1}$ is needed only through degree R, so C_{H-1} is needed only through degree $R - n$; the implementation computes it with that shifted budget. Finally, a nonnegative profile of volume $m \leq R$ has every height at most $m \leq R \leq H$, so it lies in the box $\{0, \dots, H\}^n$, and the box and unrestricted counts agree at that volume. When no budget is given the implementation uses $R = nH$, which deletes nothing because every box profile has volume at most nH. □

The argument would fail for a representation with negative volume exponents, such as a series divided by a power of x. The retained benchmark therefore records two

distinct populations: the low-volume part of an $H = 2$ box on paths of up to ninety-six vertices, which is a bounded-height population and not the unrestricted series, and $H = R$ runs on a forty-vertex ternary tree, a sixty-three-vertex binary tree and a hundred-vertex path, which compute complete unrestricted local coefficients through $R = 8$ or 10. The latter are checked against two tree-independent identities: the profile count $\binom{m+n-1}{n-1}$ and the energy sum $2|E|\binom{m+n}{n+1}$ at each volume, the second because $\sum_a (a_u - a_w)^2 = 2\binom{m+n}{n+1}$ over weak compositions of m.

Proposition SNG-F25 (span by bounded-box subtraction). Let C_H count all profiles in $\{0, \dots, H\}^n$ and $C_{-1} = 0$. Then $D_H = C_H - x^n C_{H-1}$ counts profiles with minimum zero and maximum at most H, and $D_H - D_{H-1}$ (with $D_{-1} = 0$) counts minimum-zero profiles of span exactly H. All counts are nonnegative.

Proof. The profiles in the box with minimum at least one are exactly $b + \mathbf{1}$ for $b \in \{0, \dots, H-1\}^n$. The map $b \mapsto b + \mathbf{1}$ is a bijection, adds n to volume and leaves energy unchanged, because every edge difference is unchanged; this is one global translation, also for disconnected graphs. Subtracting $x^n C_{H-1}$ removes these and leaves the box profiles with minimum zero, a subset of the first box. A minimum-zero profile has span equal to its maximum. Minimum-zero profiles with maximum at most $H - 1$ form a subset of those with maximum at most H, and the set difference is exactly those of span H. Both subtractions remove the counts of a subset, so no final count is negative. For $H = 0$, $D_0 = C_0 = 1$ is the zero profile. □

`marked_forest_coefficients` implements this with the shifted budget of SNG-F24 and raises, carrying its input, if a negative difference ever appears. For every labeled forest on at most five vertices through volume five, and for named paths, stars, balanced and branched trees and a disconnected forest through volume six, its $(\text{volume}, \text{energy}, \text{span})$ cells equal both the weak-composition and level-chain enumerators.

For general graphs, a tree decomposition replaces a single parent height by a tuple of heights on a separator. A **tree decomposition** of G is a tree $\mathcal{T}$ whose nodes carry bags $X_s \subseteq V$ such that every vertex lies in some bag, both endpoints of every edge lie in a common bag, and for each vertex the bags containing it form a connected subtree. Choose a root bag. For a vertex v the bags containing it form a subtree with a unique node nearest the root; call it the **owner** of v. For an edge uv the bags containing both endpoints form the intersection of two subtrees, again a subtree with a unique top node, its owner. For a node s let $\mathcal{T}_s$ be its subtree, $V_s = \bigcup_{r \in \mathcal{T}_s} X_r$, and W_s and E_s the vertices and edges owned by nodes of $\mathcal{T}_s$. For $\sigma \in \{0, \dots, H\}^{X_s}$ define

$$M_s(\sigma) = \sum_{\substack{b \in \{0,\dots,H\}^{V_s} \\ b|_{X_s} = \sigma}} x^{\sum_{v \in W_s} b_v} q^{\sum_{uv \in E_s} (b_u - b_v)^2}.$$

Proposition SNG-F26 (bag-conditioned recursion). Define the local charge by

$$w_s(\sigma) = x^{\sum_{v \text{ owned by } s} \sigma_v} q^{\sum_{uv \text{ owned by } s} (\sigma_u - \sigma_v)^2}.$$

Then

$$M_s(\sigma) = w_s(\sigma) \prod_{r \text{ child of } s} \sum_{\substack{\tau \in \{0,\dots,H\}^{X_r} \\ \tau|_{X_r \cap X_s} = \sigma|_{X_r \cap X_s}}} M_r(\tau),$$

and the bounded-box series is $C_H = \sum_\sigma M_{\text{root}}(\sigma)$. Every vertex volume and every edge energy is charged exactly once.

Proof. Every vertex lies in some bag and every edge in some common bag, so each has exactly one owner, and the owners partition the charges; at the root $W = V$ and $E_{\text{root}} = E(G)$. Induct on $|\mathcal{T}_s|$. The owned sets W_s, E_s are the disjoint union of the charges owned at s and the sets W_r, E_r of the children. A vertex $v \in V_r \setminus X_s$ lies in no bag outside $\mathcal{T}_r$: otherwise the connected subtree of bags containing v would pass through s. Hence the sets $V_r \setminus X_s$ for distinct children are disjoint and disjoint from X_s, and an assignment on V_s extending σ is exactly a choice, for each child independently, of an assignment on V_r agreeing with σ on $X_r \cap X_s$. Every charge owned in $\mathcal{T}_r$ involves only vertices of V_r, so its contribution depends only on that child's choice. Grouping those choices by $\tau = b|_{X_r}$ and applying the induction hypothesis gives the displayed sum for each child; the product realizes independent choices, and w_s adds the charges owned at s, each once. A vertex shared by a parent and child bag carries one height because it is conditioned on both, and its volume is added only at its owner, so repeated bags and joins never double-count it. □

Safe truncation (SNG-F24) applies verbatim because every message has nonnegative volume exponents, and SNG-F25 gives the minimum-zero and exact-span sectors with one global translation. The implementation `src/singularity/decomposition.py` checks the decomposition axioms and the tree structure before any arithmetic and rejects a decomposition that misses a vertex or an edge or splits the bags of a vertex. Its costs are reported separately: $\sum_s (H+1)^{|X_s|}$ bag assignments, sparse message support and integer bit length. On several distinct valid decompositions each of the five- and six-cycles, a ladder, K_4 (including redundant repeated bags), two disjoint triangles (including an empty connecting bag) and K_5, the bounded, minimum-zero and marked outputs equal Cartesian enumeration, weak compositions and level chains; width-one decompositions of forests, including disconnected ones, reproduce the forest engine.

13.3 What nested boundaries remember

The span-one sector records a cut-size distribution indexed by the size of a side. Span two introduces ordered nested pairs $T \subseteq S$, with volume $|S| + |T|$ and energy $|\delta S| + |\delta T| + 2e(T, V \setminus S)$. Thus two cuts with the same individual sizes and boundary sizes can contribute differently because the graph places their boundaries differently. This is an explicit refinement mechanism. It should not be described as a higher-body interaction: the original energy remains a sum over graph edges.

A useful new intermediate object is the finite nested-pair census indexed by $(|S|, |T|, |\delta S|, |\delta T|, e(T, V \setminus S))$. Its purpose is diagnostic. If binary-equivalent graphs first separate at span two, the census can identify which mixed boundary statistic causes that separation. Higher nested censuses are defined similarly, but their dimension grows quickly. The right theorem is not that more marks are always better; it is an explicit projection identity and, when available, a pair of graphs demonstrating that one projection loses information retained by another.

Proposition SNG-F40 (nested-pair census and its span-two projection). For a graph on V, $|V| = n$, let $\mathcal{N}_G$ count all pairs $T \subseteq S \subseteq V$, with empty, full and equal sets allowed, by $(|S|, |T|, c(S), c(T), e(T, V \setminus S))$. Then: (i) $\mathcal{N}_G$ has total mass 3^n; (ii) it is invariant under $(s, t, c_S, c_T, x) \mapsto (n-t, n-s, c_T, c_S, x)$; (iii) the cells with $t \geq 1$, $s < n$ project onto the span-two sector,

$$[z^2]\mathcal{B}_G(x, q, z) = \sum_{t \geq 1,\ s < n} \mathcal{N}_G(s, t, c_S, c_T, x')\, x^{s+t} q^{c_S + c_T + 2x'};$$

(iv) the cells with $t = 0$ form the complete binary cut-volume census.

Proof. A pair is a choice, for each vertex, of one of the three options $V \setminus S$, $S \setminus T$, T, which gives (i). The map $(T, S) \mapsto (V \setminus S, V \setminus T)$ is an involution on nested pairs. It exchanges the two boundaries, and it preserves the crossing set: edges joining T to $V \setminus S$ are the edges joining $V \setminus S$ to $V \setminus (V \setminus T)$. This gives (ii); on profiles it is $a \mapsto 2 - a$. For (iii), the pairs with $\emptyset \neq T \subseteq S \subsetneq V$ are the threshold chains of minimum-zero profiles of span two (SNG-F09 with $h = 2$), including $T = S$ for the gap profiles $2 \cdot \mathbf{1}_S$. Volume and energy then follow from SNG-F09 and SNG-F10. Part (iv) is immediate. □

The census is strictly finer than the span-two coefficients, which see only $c_S + c_T + 2x$. It explains the binary-census pair cell by cell. The two graphs have identical $t = 0$ cells, by (iv) and SNG-E003, yet they already differ in 591 cells of $\mathcal{N}$. At volume three the differing cells all have $S = \{i, j\}$, $T = \{i\}$, with statistics $(d_i + d_j - 2A_{ij},\ d_i,\ d_i - A_{ij})$. These are exactly the ordered adjacent-degree triples. The binary census records only the unordered sum $d_i + d_j - 2A_{ij}$ of a two-element set, while a nested pair also orders the two vertices and records which of them is adjacent to the outside. Projecting the differing cells to energy reproduces the discrepancy polynomial $q^3(1-q)(1-q^3)^2(1-q^4)^2$ of SNG-E006. The census is a finite invariant with no claimed completeness. `src/singularity/nested_census.py` computes it. `tests/test_nested_census.py` checks (i)–(iii) on all 75 labeled graphs with at most four vertices and on the reference graphs against weak compositions and level chains. `results/nested_census.json` retains the complete tables of the reference pairs with graph digests.

Minimum cuts provide a complementary asymptotic direction. On a connected graph, a nonconstant binary profile has positive energy, and the least such energy is the edge-connectivity. Its side-size multiplicities are visible in the marked binary sector. Nonbinary profiles can have energy tied to several nested boundaries, so identifying the first low-energy terms of the complete normalized series requires an argument about all heights. A promising target is a characterization of the minimum positive energy and its minimizing profiles, followed by a classification of the next energy layer. Each statement should distinguish energy ordering from volume ordering, since the Euler visibility theorem uses the latter.

13.4 Matroids, cut lattices, and the role of a positive chamber

The quadratic form $a \mapsto a^\top L_G a$ factors through integer height vectors modulo constants. Sending a height vector to its oriented edge differences identifies this quotient, in the connected case, with the integral cut lattice equipped with its Euclidean energy. This is established territory. The additional marks in this project are not automatically invariants of an abstract lattice: choosing the minimum-zero representative and taking its coordinate sum uses the vertex-height realization of that lattice. Span also uses that realization.

The corresponding research question is precise: which parts of $B_G(x, q, z)$ are determined by the metric cut lattice, which are determined by the graphic matroid, and which require additional vertex information? Whitney switches are a natural test population because they preserve cycle-matroid structure under the appropriate hypotheses. A complete search can either produce a separating pair, thereby disproving a proposed factorization through the matroid, or identify a transformation law worth proving. Failure to find a small separating pair is not proof of matroid invariance.

Theorem SNG-F46 (marks versus the cut lattice). Let G, G' be connected graphs on n vertices and $\Lambda_G = \mathbb{Z}^n/\mathbb{Z}\mathbf{1}$ with energy E_G. For $[a] \in \Lambda_G$ let $\nu([a])$ and $\sigma([a])$ be the volume and span of the minimum-zero representative. (i) If a bijection $\Phi : \Lambda_G \to \Lambda_{G'}$ satisfies $E_{G'} \circ \Phi = E_G$, $\nu \circ \Phi = \nu$ and $\sigma \circ \Phi = \sigma$, then $\mathcal{B}_G = \mathcal{B}_{G'}$. (ii) Energy preservation alone does not suffice, even for an affine unimodular isometry induced by an edge bijection that preserves the cycle matroid. Let $G = B_1 \cup B_2$ and $G' = B_1 \cup_{\mathrm{tw}} B_2$ be the straight and twisted gluings of blocks with terminals s, t. The map φ, equal to the identity on B_1 and to $a_v \mapsto a_s + a_t - a_v$ on the interior of B_2, is an involutive affine bijection of $\mathbb{Z}^n$ commuting with constant shifts and satisfying $E_{G'}(\varphi(a)) = E_G(a)$. For the path blocks of SNG-F42, G, G' are $G_{2m,2k}$ and $G_{2m,2m}$, whose marked series differ at volume $2k+1$ by SNG-F36.

Proof. (i) The minimum-zero representatives are in bijection with Λ_G, and each marked coefficient counts the classes with prescribed (ν, E, σ). The coefficients are finite by SNG-F01, and Φ carries each such finite set bijectively onto the corresponding set for G'. (ii) Edges of B_1 keep both endpoint heights. On an edge uv of B_2 the difference $a_u - a_v$ becomes $-(a_u - a_v)$, including at the terminals, which the twist exchanges. So every squared difference is preserved. Adding $c\mathbf{1}$ changes $a_s + a_t - a_v$ by c, so φ descends to the quotient, and applying it twice is the identity. The edge correspondence carries cycles to cycles: a cycle meeting both blocks passes through both terminals, and the twist reverses only its B_2 segment. Hence the cycle matroid, and with it the cut lattice and its energy theta series, is preserved. The marks are not. For an interior vertex v of B_2, the unit vector e_v has normalized volume 1, while $\varphi(e_v)$ equals -1 at v and 0 elsewhere, whose minimum-zero representative $\mathbf{1} - e_v$ has volume $n - 1$. □

So the extra information in $\mathcal{B}_G$ beyond the cut lattice is exactly the dependence of ν and σ on the vertex realization. A matroid-preserving switch moves it, and the twist pairing of SNG-F41 measures by how much. `twist_lattice_map` in `src/singularity/terminals.py` implements φ, and `tests/test_lattice_marks.py` checks energy preservation, shift equivariance, involution and the volume change exactly.

This path is especially useful because it can explain a collision rather than merely record one. If an equality comes from an integral lattice isometry that also preserves the chosen height chamber and marks, then a structural bijection may exist. If an isometry preserves energy but moves the chamber or the volume functional, the resulting failure identifies exactly which extra structure the series observes. The intended publication contribution would be a theorem describing such preservation or failure, with an explicit worked graph pair, rather than a broad claim that graph polynomials and lattices are connected.

13.5 Root-of-unity specializations as finite exact laboratories

Let ω have order m. Write every nonnegative height uniquely as $a_i = r_i + mb_i$, with $0 \le r_i < m$. Since energy is an integer quadratic form, $E_G(a) \equiv E_G(r) \pmod m$. Summing the independent nonnegative b_i gives the exact identity

$$L_G(x, \omega) = \frac{\sum_{r \in \{0,\dots,m-1\}^n} x^{\sum_i r_i} \omega^{E_G(r)}}{(1 - x^m)^n}.$$

Proposition SNG-F28 (energy roots of unity give rational series). Let $m \ge 1$

and let ω be a primitive m-th root of unity. Then $L_G(x,\omega) = N_{G,m}(x)/(1-x^m)^n$, where

$$N_{G,m}(x) = \sum_{r\in\{0,\dots,m-1\}^n} x^{|r|}\omega^{E_G(r)}$$

is a polynomial of degree at most $n(m-1)$ with coefficients in $\mathbb{Z}[\omega] \cong \mathbb{Z}[t]/\Phi_m(t)$. Consequently, for every integer $N \geq 0$,

$$[x^N]L_G(x,\omega) = \sum_{\substack{0\leq j\leq n(m-1)\\ j\equiv N \pmod m}} [x^j]N_{G,m}\cdot\binom{(N-j)/m+n-1}{n-1}.$$

Proof. Division with remainder writes each $a_i \geq 0$ uniquely as $r_i + mb_i$ with $0 \leq r_i < m$ and $b_i \geq 0$, so $a \mapsto (r,b)$ is a bijection from $\mathbb{Z}_{\geq 0}^n$ to $\{0,\dots,m-1\}^n \times \mathbb{Z}_{\geq 0}^n$ with $|a| = |r| + m|b|$. For each edge, $(a_u - a_v)^2 = (r_u - r_v)^2 + 2m(r_u - r_v)(b_u - b_v) + m^2(b_u - b_v)^2$, so $E_G(a) \equiv E_G(r) \pmod m$ and $\omega^{E_G(a)} = \omega^{E_G(r)}$. Summing over b separately gives the factor $\sum_{b\geq 0} x^{m|b|} = (1-x^m)^{-n}$, whose x^{mk} coefficient is the number $\binom{k+n-1}{n-1}$ of weak compositions of k into n parts. Multiplying by $N_{G,m}$ and reading the coefficient of x^N gives the remote formula, which involves only the finitely many numerator coefficients. Each ω^E is represented by the reduction of $t^{E \bmod m}$ modulo the monic integer polynomial Φ_m, so all arithmetic is exact in $\mathbb{Z}[t]/\Phi_m$; for $m = 1$, $\Phi_1 = t - 1$, the ring is $\mathbb{Z}$ and the formula counts all profiles. □

The identity concerns one specialization. Equality of two graphs' specialized series at one root of unity does not imply equality of the unspecialized series: at $m = 2$ both spectral graphs have the same numerator because they have the same degree parities, although their volume-two coefficients differ, and they already differ at $m = 3$. No positivity or real-logarithm argument for $0 < q < 1$ transfers to complex ω. `src/singularity/conductor.py` implements $N_{G,m}$ under an explicit residue-cube budget m^n and evaluates the remote formula without enumerating any height box; `tests/test_conductor_weighted.py` checks Φ_m against the independent product $\prod_{d|m}(t^d-1)^{\mu(m/d)}$ for $m \leq 24$, the numerator expansion against direct specialization of enumerated coefficients for composite orders and graphs with isolated vertices, and the retained fixtures in `results/conductor_fixtures.json`.

An infinite height population therefore becomes a finite residue-cube numerator at each root-of-unity specialization. This identity is a direct consequence of residue decomposition; it is not a claim of new arithmetic theory. It offers a particularly clean computational bridge because all coefficients can be represented exactly in a cyclotomic ring, and equality can be decided by reduction modulo its defining polynomial.

At $q = -1$, the result collapses further. Squares agree with their bases modulo two, so $E_G(a) \equiv \sum_i d_i a_i \pmod 2$, and

$$L_G(x,-1) = \prod_i (1-(-1)^{d_i}x)^{-1}.$$

This specialization sees only degree parity counts. It supplies both a theorem and a warning: an elegant arithmetic specialization can discard most graph information. At odd primes, evaluating the finite numerator at $x = 1$ produces a quadratic Gauss sum for the Laplacian over a finite field. Rank and the discriminant of its nonsingular part become relevant. The singular constant direction must be accounted for, and prime

powers need their own lifting argument; an odd-prime formula must not be reused at two.

There is a separate root-of-unity filter in the volume variable. Averaging $L_G(\omega^j x, q)$ with character weights isolates volumes in a chosen congruence class.

Proposition SNG-F27 (volume congruence filter). Let $m \geq 1$, let ω be a primitive m-th root of unity, let $a \in \mathbb{Z}$, and let $F = \sum_{v \geq 0} F_v(q) x^v$ be a formal series, or a truncation through any degree, with $F_v \in \mathbb{Q}[q]$. Then

$$\frac{1}{m} \sum_{j=0}^{m-1} \omega^{-aj} F(\omega^j x, q) = \sum_{v \equiv a \pmod{m}} F_v(q) x^v.$$

Proof. The coefficient of x^v on the left is $F_v(q) \cdot \frac{1}{m} \sum_{j=0}^{m-1} \zeta^j$ with $\zeta = \omega^{v-a}$. If $v \equiv a \pmod{m}$, then $\zeta = 1$ and the average is 1. Otherwise $\zeta \neq 1$ and $\zeta^m = 1$, so $(\zeta - 1) \sum_j \zeta^j = \zeta^m - 1 = 0$ forces $\sum_j \zeta^j = 0$; this holds in the field $\mathbb{Q}(\omega) \cong \mathbb{Q}[t]/\Phi_m(t)$, hence in the integral representation $\mathbb{Z}[t]/\Phi_m(t)$. The identity is coefficientwise, so it holds for every truncation and for any residue representative a; for $m = 1$ it is the identity map. □

The implementation in `src/singularity/congruence.py` evaluates the left side literally in $\mathbb{Z}[t]/\Phi_m(t)$, including the nonprimitive powers ω^j, divides only the completed sum by m, and raises if the sum is not an integer multiple of the identity. Tests compare it with direct degree selection and with tuple enumeration filtered by each profile's own volume on all labeled graphs with at most four vertices, for several moduli including composite ones and truncations shorter than the modulus. The filter acts on the volume variable; the energy specialization $q = \omega$ above is a different operation. It does not isolate binary profiles or recover span after setting $z = 1$: volume classes merge different spans with equal volume and energy, as $2e_i$ and $e_i + e_j$ on an edgeless graph already show. Both kinds of specialization are worthwhile because they provide finite exact comparison problems with different information loss. The positive-real visibility theorem for $0 < q < 1$ does not automatically extend to complex q, where local factors may vanish and a logarithm needs an explicitly chosen domain.

13.6 Order chambers and weighted lattice-point geometry

Every height profile has a unique order obtained by sorting vertices first by height and then by label. For a permutation π in that order, introduce $d_0 = a_{\pi_1}$ and $d_j = a_{\pi_{j+1}} - a_{\pi_j}$ for $1 \leq j < n$. These gaps are nonnegative, with $d_j > 0$ whenever $\pi_j > \pi_{j+1}$, because an adjacent descent cannot occur inside a tied block. The resulting chambers are disjoint, including their prescribed boundaries. Volume becomes $nd_0 + \sum_{j=1}^{n-1} (n-j) d_j$, and energy becomes an explicit quadratic form in the remaining gaps.

Proposition SNG-F29 (disjoint order chambers). Index positions from 0. For a permutation $\pi = (\pi_0, \ldots, \pi_{n-1})$ let C_π be the set of gap vectors $d \in \mathbb{Z}^n_{\geq 0}$ with $d_j \geq 1$ whenever $j \geq 1$ and $\pi_{j-1} > \pi_j$. The map $(\pi, d) \mapsto a$, $a_{\pi_k} = d_0 + \cdots + d_k$, is a bijection

from $\bigsqcup_\pi C_\pi$ onto $\mathbb{Z}^n_{\geq 0}$. Writing $T_j = \{\pi_j, \dots, \pi_{n-1}\}$,

$$
\begin{aligned}
|a| &= nd_0 + \sum_{j=1}^{n-1} (n-j)d_j, \\
E_G(a) &= \sum_{i,j=1}^{n-1} Q_{ij} d_i d_j, \\
Q_{jj} &= c_G(T_j), \\
Q_{ij} &= e_G\big(T_{\max(i,j)}, V \setminus T_{\min(i,j)}\big).
\end{aligned}
$$

and on $d_0 = 0$ (the normalized profiles) the span is $\sum_{j\geq 1} d_j$.

Proof. Given a, let π list the vertices by increasing (a_v, v) and set $d_0 = a_{\pi_0}$, $d_j = a_{\pi_j} - a_{\pi_{j-1}}$. The gaps are nonnegative, and if $\pi_{j-1} > \pi_j$ the two heights cannot be equal, because equal heights are sorted by increasing label; so $d \in C_\pi$ and the map hits a. Conversely, for $d \in C_\pi$ the heights a_{π_k} are nondecreasing in k, and two consecutive positions tie only when $d_j = 0$, which forces $\pi_{j-1} < \pi_j$. Sorting a by (a_v, v) therefore returns π, and the gaps are recovered by differences, so (π, d) is determined by a: the chambers are disjoint. Vertex π_k has height $d_0 + \cdots + d_k$, so gap d_j is counted at the $n - j$ positions $k \geq j$, giving the volume. An edge with endpoints at positions $p < q$ has difference $d_{p+1} + \cdots + d_q$; squaring and summing over edges gives $\sum Q_{ij} d_i d_j$, where Q_{ij} counts edges whose position interval contains both i and j. Such an edge has its upper endpoint in $T_{\max(i,j)}$ and its lower endpoint outside $T_{\min(i,j)}$, and conversely; for $i = j$ this is the cut of T_j. The minimum height is d_0 and the maximum $d_0 + \sum_{j\geq 1} d_j$. □

The matrix Q is the nested-boundary matrix of SNG-F10 for the chain $T_1 \supseteq \cdots \supseteq T_{n-1}$, with gap d_j recording how many threshold levels coincide with T_j. Closed weak-order cones would place a profile with a tied block of size s in $s!$ cones; the strict descent condition removes exactly that overcount. `src/singularity/chambers.py` enumerates the $n!$ chambers and their gap vectors within a volume budget, evaluates volume and energy from Q, and never constructs height tuples; its factorial cost restricts it to small graphs. Its local and marked histograms equal the weak-composition and level-chain enumerators on all 75 labeled graphs with at most four vertices through volume six, and the chamber of every profile in $\{0, \dots, 3\}^4$ is recovered uniquely.

This description links the population to familiar order and lattice-point methods while retaining a concrete accounting benefit. It can expose which cross terms depend on the relative placement of cuts, and it gives another independent enumerator. The marks are not automatically Ehrhart polynomials. Unweighted integer points in a dilated fixed rational polytope admit established Ehrhart theory; weighting them by q to a quadratic energy introduces an additional structure. Any reciprocity statement must specify whether it concerns unweighted counts, a formal weighted series, or a transformed analytic function, and must prove the claimed boundary terms.

Proposition SNG-F32 (poset-constrained heights and order polytopes). Let P be a finite poset on V, $|V| = n$, and $\mathcal{O}(P) = \{x \in [0,1]^n : x_i \leq x_j \text{ whenever } i <_P j\}$, with the inequalities oriented as in the height constraint. For every integer $H \geq 0$:

1. The weak assignments $a \in \{0, \dots, H\}^n$ with $a_i \leq a_j$ for $i <_P j$ are exactly the lattice points of $H\mathcal{O}(P)$. Their number is $\Omega(P, H+1)$, where $\Omega(P, m)$ is Stanley's

order polynomial, which counts order-preserving maps $P \to \{1, \dots, m\}$. For any natural labeling, $\Omega(P, H+1) = \sum_{\sigma \in \mathcal{L}(P)} \binom{H+n-\operatorname{des}\sigma}{n}$.

2. In the chamber decomposition of SNG-F29, a relation $i <_P j$ holds automatically when i precedes j in the chamber order. When i follows j, it forces every gap between them to be zero, which is impossible across a label descent. Each surviving chamber contributes $\binom{H-s+f}{f}$ points, where f counts free gaps and s is the forced minimum.
3. Strict assignments, with $a_i < a_j$, form a different population. For a chain they number $\binom{H+1}{n}$ instead of $\binom{H+n}{n}$.
4. The weighted count $\sum_{a \in H\mathcal{O}(P) \cap \mathbb{Z}^n} x^{|a|} q^{E_G(a)}$ is a finite weighted sum over the same points. For fixed $0 < q < 1$ it need not be a polynomial in H.

Proof. (1) Scaling $\mathcal{O}(P)$ by H gives $0 \le x_i \le H$ and the same order inequalities, so its integer points are the weak assignments. The map $a \mapsto a + \mathbf{1}$ identifies them with order-preserving maps to $\{1, \dots, H+1\}$, which is the argument shift. For a natural labeling, each weak map lies in exactly one chamber of SNG-F29 (sorting by height and then label), and the chambers it can occupy are exactly the linear extensions, since ties are broken by increasing label and a natural labeling increases along relations. In the chamber of σ, the gaps $d_1, \dots, d_{n-1}$ are at least one at the $\operatorname{des}\sigma$ descents. The points with maximum at most H correspond to nonnegative solutions of $\sum_{j \ge 0} d'_j \le H - \operatorname{des}\sigma$, which number $\binom{H-\operatorname{des}\sigma+n}{n}$. (2) is the same count without assuming a natural labeling. If i precedes j the heights are nondecreasing along the order. If i follows j, then $a_i \ge a_j$, so the relation requires equality and every intermediate gap must vanish. (3) A strict chain assignment chooses n distinct values in $\{0, \dots, H\}$. (4) For the two-element chain with the edge $\{0, 1\}$, the weighted count is $W_H(q) = \sum_{d=0}^{H} (H+1-d) q^{d^2}$. Its second difference in H is $q^{(H+2)^2}$, which is positive and tends to 0. The second difference of a polynomial is a polynomial, and one that tends to 0 vanishes identically, so W_H is not a polynomial in H. The unweighted count on the same points is the polynomial $\binom{H+2}{2}$. □

Statement (1) is classical [Stanley1986; Stanley2012]. The project adds two things: the chamber form (2), which applies to arbitrary labels and is checked against direct enumeration, and the explicit warning (4). Quadratic weighting keeps the lattice points of the order polytope but leaves the polynomial world of ordinary Ehrhart theory, so a reciprocity statement for weighted counts needs its own proof (P22). `src/singularity/orders.py` validates acyclicity and takes the transitive closure. Its direct, chamber and linear-extension counts agree on chains, a reversed chain, antichains, the diamond, a poset with unnatural labels, a disconnected poset and a redundant relation for several complete height ranges.

Two reciprocity statements hold for these counts, and both are established background rather than project results [Stanley2012]. For strict assignments $a_i < a_j$ with values in $\{0, \dots, H\}$, the count is $(-1)^n \Omega(P, -(H+1))$, where $\Omega(P, m)$ is the order polynomial of SNG-F32 evaluated as a polynomial at a negative integer. For unbounded heights, let $W_P(x) = \sum x^{|a|}$ over weak maps $a \ge 0$ and $S_P(x) = \sum x^{|a|}$ over strict maps $a \ge 1$. Both are rational. In chamber form, with ties broken by increasing natural label for W and by decreasing natural label for S,

$$W_P = \frac{\sum_{\sigma \in \mathcal{L}(P)} x^{\sum_{j \in \operatorname{Des}\sigma} (n-j)}}{\prod_{i=1}^{n} (1 - x^i)}, \qquad S_P = \frac{\sum_{\sigma \in \mathcal{L}(P)} x^{\,n + \sum_{j \in \operatorname{Asc}\sigma} (n-j)}}{\prod_{i=1}^{n} (1 - x^i)},$$

and $S_P(x) = (-1)^n W_P(1/x)$. Breaking ties by decreasing label makes every tied block a decreasing run of a linear extension, hence an antichain. This is exactly what strictness permits. The two exponents add to $n(n+1)/2$, which gives the identity.

Theorem SNG-F34 (energy weights obstruct P-partition reciprocity). Let P be the two-element chain $0 < 1$ and $G = K_2$ on the same vertices, and put $W(x,q) = \sum_{0\le a_0\le a_1} x^{a_0+a_1} q^{(a_1-a_0)^2}$ and $S(x,q) = \sum_{1\le a_0<a_1} x^{a_0+a_1} q^{(a_1-a_0)^2}$. Then

$$W = \frac{\theta(x,q)}{1-x^2}, \qquad S = \frac{x^2(\theta(x,q)-1)}{1-x^2}, \qquad \theta(x,q) = \sum_{d\ge 0} x^d q^{d^2},$$

and neither W nor S is a rational function of x over $\mathbb{Q}(q)$. Consequently $W(1/x,\cdot)$ has no meaning as a rational function, and no identity of the form $S(x,q) = \pm W(1/x, \psi(q))$ extends the classical reciprocity, which holds at $q = 1$.

Proof. Writing $a = (t, t+d)$ gives the closed forms. If W were rational in x over $\mathbb{Q}(q)$, so would be $\theta = (1-x^2)W$. Its coefficients q^{d^2} would then satisfy a linear recurrence $\sum_{i=0}^{k} c_i q^{(d+i)^2} = 0$ for all $d \ge d_0$, with $c_i \in \mathbb{Q}(q)$ not all zero. Dividing by q^{d^2} gives $\sum_i c_i q^{i^2} y^i = 0$ with $y = q^{2d}$. This polynomial in y would vanish at the infinitely many distinct elements q^{2d} of $\mathbb{Q}(q)$, hence identically, forcing every $c_i = 0$. The same argument applies to S. At $q = 1$, $\theta = 1/(1-x)$, $W = 1/((1-x)(1-x^2))$ and $S = x^3/((1-x)(1-x^2)) = W(1/x)$. □

The obstruction concerns rationality in x, which the classical proof needs. It does not rule out other weighted identities with additional boundary terms or a transformed analytic function; any such claim would need its own definition of the evaluation at $1/x$ or at negative bounds. The bounded weighted count is not polynomial in H either (SNG-F32), so evaluation at negative H is also undefined. `tests/test_reciprocity.py` checks both background reciprocities on six complete small posets, including singleton, chain, antichain, diamond, unnatural labeling and a disconnected poset. It checks the closed forms of W and S through x^{12} by direct enumeration and confirms that the Hankel determinants $\det[q^{(i+j)^2}]_{0\le i,j\le k}$ for $k \le 3$ are nonzero polynomials. These determinants are the finite certificates excluding recurrences of order k that start at $d = 0$.

The intended order-theoretic theorem target is a chamber formula with a correct treatment of ties, followed by reductions for graph families with a controlled set of chamber types. An algorithm summing all $n!$ orders may be a useful check without being a useful large-scale algorithm. A later symmetry reduction must preserve the original labeled multiplicities. Successful chamber identities would provide exact explanations for some coefficient factorizations, rather than relying on numerical pattern recognition.

13.7 Visibility strata and formal proof priorities

For fixed graph pairs, coefficient differences are polynomials in q. Their common vanishing defines exceptional parameter values where the first discrepancy moves to a later volume. The generic first discrepancy and its exceptional set should be computed separately. At a rational parameter, direct substitution determines whether a proposed exponent survives. At an algebraic parameter, reduction by a minimal polynomial gives an exact test. Sampling several real values cannot prove that a discrepancy has no interior zero.

Proposition SNG-F31 (exceptional strata are finite and exactly computable). Let $\Delta_v = [x^v](L_G - L_H) \in \mathbb{Z}[q]$ for $0 \le v \le R$, let r be the least v with $\Delta_v \neq 0$, and for $q_0 \in (0,1)$ let $r(q_0)$ be the least $v \le R$ with $\Delta_v(q_0) \neq 0$, if any. Then $r(q_0) = r$ except at the finitely many roots of Δ_r in $(0,1)$; at such a root α, $r(q_0)$ is the least $v > r$ with $\Delta_v(\alpha) \neq 0$, or undefined through R. Every such root is found by Sturm isolation of the square-free part of Δ_r, and whether $\Delta_v(\alpha) = 0$ is decided exactly.

Proof. For $v < r$, $\Delta_v = 0$ identically, so $r(q_0) \ge r$ for every q_0, with equality exactly when $\Delta_r(q_0) \neq 0$. A nonzero polynomial has finitely many roots. Let $f = \Delta_r / \gcd(\Delta_r, \Delta_r')$, which has the same roots, all simple. Sturm's theorem states that the number of distinct roots of f in $(a,b]$ equals the drop in sign changes of its Sturm sequence from a to b; subtracting a root at b, found by exact evaluation, gives the number strictly inside. Bisection at rational midpoints, recording any midpoint that is a root, splits every interval with two or more roots. It terminates because distinct roots are separated by a positive distance. Each surviving interval then contains exactly one root, so the list is complete. For an isolated root α of f in (a,b), $\Delta_v(\alpha) = 0$ if and only if α is a common root of f and Δ_v, that is, a root of $\gcd(f, \Delta_v)$. That gcd divides f, so its roots in (a,b) are among those of f, namely α alone, and a Sturm count of the gcd on the interval decides the question. Rational arithmetic is exact throughout. □

`src/singularity/parameters.py` implements this. It reports rational roots exactly, also through the rational-root theorem when the coefficients are small. It represents the others by a square-free defining polynomial and an isolating interval. The defining polynomial divides a discrepancy and need not be the minimal polynomial. It also computes multiplicities by repeated derivatives and evaluates the endpoints $q = 0, 1$ separately. For the spectral, binary-census and cube–Möbius pairs it finds no interior root of the leading discrepancy, so the first visible volume is 2, 3 and 4 at every $q \in (0,1)$ (`results/parameter_strata.json`). At both endpoints the binary pair agrees through the tested bound. Artificial discrepancies with interior irrational and rational roots, repeated roots and identically zero differences are tested as well.

The analytic path starts with a finite certified coefficient discrepancy, then proves a local factor expansion and a uniformly convergent normalized prime product in the stated real domain. More cancellations can move the remainder convergence boundary; they do not justify a global meromorphic continuation of the original product. Nonintegral exponents require positive-real branches or explicitly described local complex branches. The project should seek quantitative remainder bounds uniform on compact parameter intervals away from exceptional points, so numerical demonstrations become reproducible approximations to an already stated theorem.

Formalization should follow the dependency order of the mathematics. First prove finite graph energy definitions, translation normalization, binary extraction, and the nested-level identity. Next verify finite coefficient tables and polynomial discrepancy certificates. Only then formalize the analytic comparison theorem against whatever zeta and infinite-product library is actually available. If a library theorem is assumed or imported externally, state that boundary. The central objective is a short chain from an explicit labeled population to a specific real singularity, with every link visible to a reader and no broader reconstruction or formal-verification claim added by implication.

Chapter 14

Literature, attribution, and publication claims

14.1 The contribution must be stated at the right scale

The ingredients of this project have substantial precedents. Graph Laplacians, integer cut lattices, Ising partition functions, Gaussian fields, order polytopes, Euler transforms, and zeta-factor extraction are established mathematics. Combining their names is not a novelty claim. A defensible contribution must identify a specific population, a theorem about information retained by its particular marks, a verified separation from a stated invariant, a new algorithm with a correctness argument, or a classification result that was not already supplied by the background literature.

The present formulation counts nonnegative labeled vertex heights by total height and squared edge differences, with an optional minimum-zero normalization and span mark. Its immediate value is that this object supports a short exact route from finite profiles to an arithmetic comparison. That route is worth explaining even when individual steps are elementary. Whether the resulting invariant, a specific identity, or a particular example has historical priority remains a literature question. The repository should claim what it establishes, not use the absence of a familiar name as evidence of originality.

There is also a distinction between mathematical novelty and a useful research artifact. A reproducible pair of examples with explicit edges, exact coefficient maps, an independent implementation, and a readable proof can be valuable without being advertised as a new general theory. A literature-ready manuscript should explain the closest known objects first, then state the narrower result. It should give readers enough detail to decide whether the distinction is substantial. Publication merit is strengthened by a theorem that explains a family of examples, rather than by a larger vocabulary around one computation.

14.2 Binary heights and the established Ising polynomial

Define the complete binary census by $C_G(u,v) = \sum_{S \subseteq V} u^{|S|} v^{|\delta S|}$. This is a conventional form of a bivariate Ising polynomial. With spins $\sigma_i \in \{-1, 1\}$, take S to be the negative-spin set. Then $\sum_i \sigma_i = n - 2|S|$ and $\sum_{\{i,j\} \in E} \sigma_i \sigma_j = |E| - 2|\delta S|$. The standard Ising partition sum with uniform coupling and field is therefore $e^{\beta|E| + hn} C_G(e^{-2h}, e^{-2\beta})$, under this sign convention. This change of variables should

accompany comparisons with the literature; polynomial names alone can conceal differing normalizations.

Andrén and Markström studied the bivariate Ising polynomial explicitly. Sokal's account of the multivariate Tutte polynomial explains the larger Potts and matroid setting. Neither reference justifies claiming that every multilevel integer-height population is determined by the binary census. The supplied binary-census pair provides an exact finite separation in the opposite direction for this particular series. The appropriate statement is that these two specified graphs have equal binary census and unequal height coefficients, with the precise certificate supplied. It is not that the project first discovered that distinct graphs can share an Ising polynomial.

The span mark clarifies the information relation. The full marked normalized series contains the nonconstant binary census at span one. The unmarked specialization sums all spans together. No recovery theorem for the binary sector from that specialization has been established here. This boundary belongs in a paper because it distinguishes a proved inclusion of information from a plausible but unproved relationship between two projections.

14.3 Cut lattices, spectra, and Gaussian comparisons

Bacher, de la Harpe, and Nagnibeda develop the integral flow and cut lattices associated with graphs. Their work is a direct antecedent for interpreting squared edge differences as a lattice energy. In a connected graph, integer heights modulo constants map to integral edge gradients; changing an edge orientation changes signs of coordinates, not the energy. A manuscript should cite this framework before discussing its additional chamber and volume marks. The existence of a cut-lattice description is background, while any theorem describing which of these additional marks survive a lattice isometry is a separate contribution requiring proof.

The usual continuous Gaussian integral on the zero-mean subspace depends only on the nonzero Laplacian eigenvalues. For example, integrating $\exp(-tx^{\top}L_G x)$ with Euclidean measure on that subspace gives $\pi^{(n-1)/2}t^{-(n-1)/2}/\sqrt{\det{}' L_G}$. This makes cospectrality a natural comparison, but does not imply that every discrete Gaussian or lattice theta series is spectral. Integrality, a positive-height chamber, and a volume mark retain additional structure. The distinction is classical in spirit and should not be presented as a discovery that continuous spectra can lose combinatorial information.

Rue and Held provide an authoritative account of Gaussian Markov random fields and precision-matrix methods. Their subject helps explain why Laplacians appear, not why this specific height series must be novel. The spectral example in this repository establishes a concrete strict comparison with the Laplacian characteristic polynomial on the given graph pair. A statement about all graph spectra, all lattice theta functions, or complete reconstruction would go beyond that evidence.

14.4 Orders, matroids, and Euler products

Stanley's order-polytope and chain-polytope theory is relevant when height inequalities are organized by partial orders. His work on acyclic orientations is relevant when edge directions are induced by vertex orders. The project should use these theories where their hypotheses actually hold. A finite height box can be decomposed into ordered chambers; this does not make a quadratic-energy weighted sum an ordinary

Ehrhart polynomial. A claimed reciprocity theorem must specify its weights, strictness convention, and boundary strata. An analogy with a familiar theorem is a guide to a proof, not a substitute for one.

The matroid question likewise needs a precise target. Sokal treats graph and matroid partition functions, while cut-lattice theory supplies another route to graph equivalences. A relational height series may retain information about the vertex realization that a matroid forgets. To establish this, one should either produce a certified pair related by an appropriate matroid-preserving operation and separated by the series, or prove a transformation theorem. Neither the phrase 'matroid refinement' nor a generic graph collision is sufficient.

Euler transforms and Möbius inversion explain how local formal coefficients can be reorganized into cyclotomic factors. For a scalar formal series $A(x) \in 1 + xK[[x]]$, with K of characteristic zero, write $\log A(x) = \sum_{n\geq 1} c_n x^n$. There are unique formal exponents $b_n \in K$ with $A(x) = \prod_{n\geq 1}(1-x^n)^{-b_n}$, and $nc_n = \sum_{d|n} db_d$. Moree's work on the formal series Witt transform is a relevant account of the associated cyclotomic identities and Möbius transforms. When q is a parameter, scalar exponents need not be nonnegative integers or even integer-coefficient polynomials in q.

The analytic Euler product requires a separate argument. Extracting finitely many zeta factors can improve the convergence of a remainder, but a formal infinite factorization is not automatically a global analytic continuation. In this project, the first-discrepancy statement uses a finite local expansion and a normalized remainder converging in a specified region. With a nonintegral discrepancy exponent, a real positive branch is available; a blanket meromorphic-continuation claim is not. The arithmetic lift reexpresses local information and supplies a singularity comparison. It does not increase the invariant's graph-reconstruction power. The released `formal-analysis/` package now verifies the positive-real comparison using actual Mathlib real numbers, real powers, infinite products and the Riemann zeta function; the generic graph theorem and two half-parameter pair theorems discharge all analytic premises. This formal scope does not include local-complex continuation, numerical interval endpoints or the synchronization limit. A kernel check does not establish historical novelty, and the bibliography remains the basis for that separate comparison.

14.5 A claim register for papers and releases

An established-background claim should name its source and retain its hypotheses. A proved proposition should have a complete argument in the manuscript, with graph order, connectedness, and parameter restrictions visible. An exact finite claim should give the finite input population and a reproducible coefficient or graph certificate. A conjecture should be phrased so that a counterexample or a proof can settle it. A proposed algorithm should not be described as implemented until its code and validation are present. These categories are editorial tools for keeping a mathematical narrative honest, rather than an additional software framework.

For the current examples, acceptable language is: 'For the two graphs with the displayed edge lists, the Laplacian characteristic polynomials agree, while the local height series first differ at volume two by the displayed polynomial.' The binary example has the analogous finite census statement and a first difference at volume three. If independent enumeration has been run, that fact can be added with its method. 'The invariant reconstructs graphs,' 'all graphs were checked,' 'the proof is kernel verified,' and 'this is the first such invariant' require evidence of a different kind and

should be absent unless that evidence is supplied.

The strongest candidate conjectures concern explicit comparison boundaries: whether a marked series factors through a given lattice or matroid structure, whether a specified family has a bounded first-separation volume, or whether a nested-cut hierarchy is strict at arbitrarily many levels. An exhaustive finite experiment can guide these conjectures and may disprove them. It cannot establish an unrestricted theorem. If a proposed direction reproduces an already known object after a change of variables, the correct outcome is an attribution and a revised question.

14.6 Candidate paper abstracts with stated prerequisites

Candidate A: Relational height populations and arithmetic visibility. The following abstract is suitable only when the finite examples and analytic comparison have their full proofs and released certificates: 'We study labeled nonnegative integer heights on a finite graph, weighted by total height and quadratic edge energy. Translation normalization gives minimum-zero representatives, and a span marking resolves nested boundary contributions. Explicit graph pairs separate this population from the Laplacian spectrum and from a complete binary cut census. For a fixed real energy parameter, we show that the first differing local coefficient determines a zeta-factor exponent in the ratio of the associated prime products. The remainder is controlled in a specified real convergence region. The examples and coefficient identities are supplied with exact reproducible data.' This abstract claims no general reconstruction theorem or historical priority.

Candidate B: Exact height enumeration on trees and bounded separators. This is a future methods paper, conditional on the dynamic programs, correctness proofs, and measured experiments being completed: 'We develop exact volume-truncated algorithms for graph height populations using separator-conditioned generating functions. A bounded-height formulation recovers minimum-zero and span-marked coefficients by translation subtraction. Independent composition and level-chain enumerators establish finite reference data, and exact ranking provides reproducible conditioned witnesses. We analyze state growth and polynomial-convolution costs separately and compare complete coefficient ranges on tree and bounded-width families.' Speedups and maximum instance sizes must be inserted from actual measurements, not anticipated results.

Candidate C: What graph structure survives a height population? This is a structural question rather than an available result: 'We compare marked integer-height populations with metric cut lattices and graph matroids, identifying transformation laws and explicit obstructions to preservation of the volume chamber.' A paper with this abstract needs an actual theorem or a decisive classification. The current program supplies the question and test apparatus. It does not yet supply the answer. A negative result showing that a proposed refinement collapses to known data would also be publishable if its statement and proof are substantive.

14.7 Verified references and their intended use

Roland Bacher, Pierre de la Harpe, and Tatiana Nagnibeda, 'The lattice of integral flows and the lattice of integral cuts on a finite graph,' *Bulletin de la Société Mathématique de France* **125** (1997), 167–198. DOI: 10.24033/bsmf.2303. The Numdam

record and article provide the primary source. Use this for the established lattice setting, not as an attribution for the particular marked series defined here.

Daniel Andrén and Klas Markström, 'The bivariate Ising polynomial of a graph,' *Discrete Applied Mathematics* **157** (2009), 2515–2524. DOI: 10.1016/j.dam.2009.02.021. Use this for the binary comparison and for checking normalization conventions. Equality of a binary census is an established equivalence relation; its failure to determine this particular multilevel series is the narrower issue tested here.

Alan D. Sokal, 'The multivariate Tutte polynomial (alias Potts model) for graphs and matroids,' in *Surveys in Combinatorics 2005*, 173–226. Primary author manuscript: arXiv:math/0503607. Use this for partition-function and matroid background. The document explicitly addresses the flexibility of multivariate weights; any claim of improvement must compare actual specializations and information, rather than compare names.

Håvard Rue and Leonhard Held, *Gaussian Markov Random Fields: Theory and Applications*, Chapman & Hall/CRC, 2005. DOI: 10.1201/9780203492024. The publisher record verifies authorship and edition. Use this for Gaussian precision-matrix context; the elementary restricted Gaussian integral should still be derived with the measure convention used in the manuscript.

Richard P. Stanley, 'Two poset polytopes,' *Discrete & Computational Geometry* **1** (1986), 9–23. DOI: 10.1007/BF02187680. The author-hosted article is the primary accessible text. Use this for order and chain polytopes and their lattice-point interpretation, with a separate proof for any quadratic-energy extension.

Richard P. Stanley, 'Acyclic orientations of graphs,' *Discrete Mathematics* **5** (1973), 171–178. DOI: 10.1016/0012-365X(73)90108-8. The 2006 reprint has a different DOI; cite the original year when discussing the original theorem. Use this for the classical orientation and reciprocity background, not for a weighted identity that has not been established.

Roland Bauerschmidt, Jiwoon Park and Pierre-François Rodriguez, 'The Discrete Gaussian model, I. Renormalisation group flow at high temperature,' *Annals of Probability* **52** (2024), no. 4, 1253–1359. DOI: 10.1214/23-AOP1658. The Project Euclid record and arXiv:2202.02286 were checked on 18 October 2026.

Pieter Moree, 'The formal series Witt transform,' *Discrete Mathematics* **295** (2005), 143–160. DOI: 10.1016/j.disc.2005.03.004. The author's institutional publication record supplies an accessible description and published version. Use this for formal cyclotomic and Möbius-transform context. The real analytic remainder argument in this project must be stated and proved independently.

Brendan D. McKay and Adolfo Piperno, 'Practical graph isomorphism, II,' *Journal of Symbolic Computation* **60** (2014), 94–112. DOI: 10.1016/j.jsc.2013.09.003. Primary author manuscript: arXiv:1301.1493. Use this when a census relies on canonical labeling or on nauty/Traces. A citation does not substitute for recording the actual executable version, graph-generation command, and completeness boundary of a particular experiment.

These references were checked against primary author, publisher, or journal records for this release. They are a starting bibliography, not a completed priority search. Before journal submission, the author should search directly for integer-valued graph Gaussian models, graph height partition functions, multivariate theta series, and refinements of Ising-equivalent graphs, then compare definitions term by term. A responsible priority statement belongs at the end of that process.

Chapter 15

Context operators, cheapest experiments and periodic sectors

This chapter adds an experimental layer to the boundary machinery of Chapter 10. A two-terminal graph state is changed by attaching declared contexts and is read through its local coefficients. The questions are which states can ever be told apart, by which experiment, at what cost, and how much state an observer must keep. Three ingredients are standard: linear observability, minimal realization of weighted automata, and the invariant-subspace computation behind them. The graph-specific parts are new to this project. They are the context compiler and its correctness conditions, the experiment language with costs, independently replayable certificates that tie the retained rows to actual experiment words, the cost-filtered pruning theorem, the top-volume corank bound and the periodic cutoff. The last section integrates the October 2026 edge-factor supplement.

Throughout, $\mathbb{Q}$ is the field of rationals and every computation is exact. "Proved" refers to the proofs written here. "Checked" refers to an exact finite computation with a declared population and budget, retained in `results/operator_experiments.json` and replayed by `scripts/operator_experiments.py --check`. None of the new statements is formalized in Lean unless Section 15.10 says so.

15.1 Background and what is new

For a linear system with state $v \in K^n$, generators M_i and output rows L_j, the set of states with identical output under every input word is an invariant subspace, and the quotient by it is the minimal linear realization. Kalman (1963) developed this for linear dynamical systems. Schützenberger (1961) introduced the weighted-automaton form of rational series, and Carlyle and Paz (1971) proved the Hankel-rank minimal realization for such automata. Berstel and Reutenauer's monograph gives a modern account of noncommutative rational series and their minimization. These are the established background of Sections 15.2 and 15.3. We do not claim them.

The bibliographic data of these four sources agree across independent library and index records. These include the MIT *Information and Control* author index, the Cambridge copyright record of the Berstel–Reutenauer volume, and reference lists citing Kalman and Carlyle–Paz. In this session the publisher pages, Crossref and author-hosted copies were unreachable through the network policy. Their primary-record check, and the DOIs other than Berstel–Reutenauer's catalogue DOI, remain open.

The project-specific content is as follows:

- a compiler from symmetric two-terminal graph contexts to exact rational operators on truncated boundary kernels, with the conditions under which it is correct for every word length (SNG-O05);
- certificates whose retained rows are actual experiments with their words and costs, replayable without rediscovery, with an exactly typed in-memory verifier and a strict portable format (Section 15.4);
- the cost-filtered pruning theorem, which makes the first retained row nonzero on a difference a globally cheapest distinguishing experiment (SNG-O03), together with the counterexample showing that it does not maximize signal (SNG-O04);
- the top-volume corank bound for graph contexts (SNG-O06) and the exact evidence at budgets four and seven;
- the exact robustness statement for finite candidate families (SNG-O07);
- the periodic residue sector with a palindromic numerator and a sufficient cutoff (SNG-O08, SNG-O09).

15.2 Operator systems, experiments and word order

An **operator system** over $\mathbb{Q}$ consists of the following data:

- the dimension $n \geq 0$ of the formal state space $\mathbb{Q}^n$, whose states are columns;
- generators $M_1, \dots, M_g \in \mathbb{Q}^{n\times n}$, with $g \geq 0$;
- observation rows $L_1, \dots, L_o \in \mathbb{Q}^{1\times n}$, with $o \geq 0$;
- attachment costs $c_i > 0$ and observation costs $\gamma_j \geq 0$.

A **word** is a finite sequence $w = (i_1, \dots, i_m)$ of generator indices, read chronologically: i_1 is applied first. It acts on states by

$$M_w = M_{i_m} \cdots M_{i_1}, \qquad v \mapsto M_w v.$$

The **experiment** (j, w) is the row $L_j M_w$. Its cost is $\gamma_j + \sum_k c_{i_k}$, and its reading on v is $L_j M_w v$. Right multiplication of an experiment row by M_i gives $L_j M_w M_i = L_j M_{(i,w)}$, the experiment whose word has i **prepended**: the new attachment happens first. This convention matters only for noncommuting generators. The test suite checks it on

$$M_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad M_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad L = (1, 0).$$

Here $M_1 M_0 (0,1)^T = (1,2)^T \neq (1,1)^T = M_0 M_1 (0,1)^T$, and the closure retains the words $()$ and (0). Parallel graph attachments commute, so graph data alone cannot detect an order error.

The **observation space** $\mathcal{O}$ is the span of all experiment rows. Two states are **future-equivalent** if every experiment reads the same value on them.

15.3 Closure, quotient and cheapest experiments

Theorem SNG-O01 (closure certificate). Let $B \in \mathbb{Q}^{r\times n}$ have rows $b_1, \dots, b_r$, each equal to an actual experiment $L_{j_k} M_{w_k}$. Suppose there are matrices $A \in \mathbb{Q}^{o\times r}$ and $T_i \in \mathbb{Q}^{r\times r}$ with

$$L = AB, \qquad BM_i = T_i B \quad (1 \leq i \leq g),$$

where L stacks the observation rows. Then row $B = \mathcal{O}$. For every word w, the experiment rows satisfy $LM_w = AT_w B$, where $T_w = T_{i_1} T_{i_2} \cdots T_{i_m}$.

Proof. Induct on $m = |w|$. The case $m = 0$ is $L = AB$. If $LM_w = AT_wB$, then

$$LM_{(i,w)} = LM_wM_i = AT_wBM_i = AT_wT_iB = AT_{(i,w)}B,$$

so every experiment row lies in row B. Conversely, each b_k is itself an experiment row, so row $B \subseteq \mathcal{O}$. □

The identities say that the retained space contains every future observation. The words say that it contains nothing else. Neither half needs a horizon or a bound on word length. Gaussian-elimination residuals are useful internally, but they are generally not experiment rows, so a certificate built from them would not prove reachability. The implementation therefore keeps original experiment rows as B.

Theorem SNG-O02 (invariant kernel, quotient and linear minimality). Under SNG-O01, put $N = \ker B$. Then:

1. $M_iN \subseteq N$ for every i;
2. every experiment vanishes on N, and two states are future-equivalent if and only if their difference lies in N;
3. the quotient state $y = Bv$ evolves by $y \mapsto T_iy$ and is read by $Lv = Ay$;
4. if $\phi : \mathbb{Q}^n \to W$ is linear and every experiment is constant on the fibres of ϕ, then $\ker \phi \subseteq N$. Consequently $\operatorname{rank} \phi \geq \operatorname{rank} B$.

Proof. (1) If $Bv = 0$, then $BM_iv = T_iBv = 0$. (2) By SNG-O01 every experiment row is a combination of rows of B, so it vanishes on N. Conversely, if $d \notin N$, some row b_k of B is nonzero on d, and b_k is an experiment. (3) $BM_iv = T_iBv$ and $Lv = ABv$. (4) If $\phi(d) = 0$, then d and 0 lie in one fibre, so every experiment, and in particular every b_k, vanishes on d; hence $d \in N$. Then $\operatorname{rank} \phi = n - \dim \ker \phi \geq n - \dim N = \operatorname{rank} B$. □

Part (4) is minimality among **linear** compressions of the **full declared formal state space**. It says nothing about nonlinear encodings. It also says nothing about the smaller set of states that graphs actually realize: a family occupying an affine subspace or a semialgebraic subset can need fewer coordinates. Section 15.11 lists this as open.

Theorem SNG-O03 (cost-filtered pruning and cheapest witnesses). Assume the following:

- the rows of B are actual experiments with costs $\beta_1 \leq \cdots \leq \beta_r$;
- $L = AB$ and $BM_i = T_iB$;
- $A_{jk} \neq 0$ implies $\beta_k \leq \gamma_j$;
- $(T_i)_{ak} \neq 0$ implies $\beta_k \leq \beta_a + c_i$.

Then for every real C, every experiment of cost at most C lies in the span of the rows b_k with $\beta_k \leq C$. Hence, for any difference $d \in \mathbb{Q}^n$:

- if some experiment is nonzero on d, then the first row b_k with $b_kd \neq 0$ is an experiment of globally minimum cost among all experiments nonzero on d, within the declared generator language;
- if no row of B is nonzero on d, no experiment distinguishes the two states.

Proof. Induct on word length. For the empty word, $L_j = \sum_k A_{jk}b_k$, and every k used has $\beta_k \leq \gamma_j$. Suppose $L_jM_w = \sum_a \lambda_a b_a$ with $\beta_a \leq \gamma_j + c(w)$ whenever $\lambda_a \neq 0$. Then

$$L_jM_{(i,w)} = \sum_a \lambda_a b_a M_i = \sum_a \sum_k \lambda_a (T_i)_{ak} b_k,$$

and every k with a nonzero contribution has $\beta_k \le \beta_a + c_i \le \gamma_j + c(w) + c_i$, the cost of $(j,(i,w))$. Now let e be an experiment of cost C with $ed \neq 0$. Since e lies in the span of the rows of cost at most C, one of those rows is nonzero on d. The first row nonzero on d therefore costs at most C, and it is itself an experiment. □

The algorithm produces such a certificate. It processes candidate experiments in nondecreasing cost from a priority queue, keeps a candidate only if it is independent of the rows already kept, and extends each kept row by every generator. A discarded candidate lies in the span of kept rows of cost at most its own. Right multiplication preserves that relation and adds the same positive cost c_i to every term, so the extensions of discarded rows are never needed. This is where strict positivity of attachment costs is used. Observation costs may be zero. The verifier checks the cost conditions on every nonzero coefficient of A and T_i.

Proposition SNG-O04 (cheapest is not strongest). Take $n = 1$, a single observation $L = (1)$ and a single generator $M = (2)$. Then:

- the closure has rank one and the cheapest witness for $d = 1$ is the direct reading, with cost 0 and signal 1;
- the experiment with word $(0,0,0)$ reads 8;
- dependence pruning discards every extension, although the extensions amplify the discrepancy without bound.

Minimum cost, maximum margin under a cost budget, and minimum-cost shared panels for several candidate pairs are therefore three different optimization problems. SNG-O03 solves only the first.

15.4 Certificates and their contracts

A closure certificate contains the experiments of B (observation index, word, row and cost), the matrices A and T_i, the reduced-row-echelon pivots of B and the canonical kernel basis of N.

The verifier recomputes every listed experiment from its word, rechecks independence, the pivots and the kernel, both identity families, the invariance of N and the cost filtration. It never reruns the closure search. It is exact in two senses. Arithmetic is rational, and types are checked: every stored number must be a `Fraction`, every index and pivot an `int` that is not a `bool`, and every container a tuple of the stated size. Python equality identifies `Fraction(0)`, `0`, `0.0` and `False`, so a value check alone would accept a forged cost of `0.0`. The portable JSON format `singularity.operator-closure/1` encodes rationals as reduced pairs `[numerator, denominator]` and replays with resource limits. In memory and on disk the two contracts agree.

A rank-zero closure is legitimate: the zero observation, or no observations at all. Its certificate has no rows, and reconstruction must return the zero vector of the ambient dimension. Earlier checkpoint code lost that dimension when reconstructing from an empty matrix and rejected such certificates. The reconstruction now takes the dimension explicitly. The test suite covers dimension zero, rank zero, and empty observation and generator families.

Exception contract. Non-exact public numbers (`bool`, `float`, `str`) raise `TypeError`. Valid types with invalid values or shapes raise `ValueError`. Portable replay converts every defect to `ValueError`.

15.5 The graph context compiler

A **context** is a simple graph C with ordered terminals (a, b), subject to three conditions:

- $ab \notin E(C)$;
- there is a graph automorphism σ of C with $\sigma(a) = b$ and $\sigma(b) = a$, and the implementation checks the actual permutation;
- on attachment the interior of C is copied freshly, disjoint from everything else, and the terminals of C are identified with those of the current graph.

A **base block** is any two-terminal simple graph, symmetric or not, and it may contain its terminal edge. Write $K_G(s,t) \in \mathbb{Z}[q][x]$ for the interior kernel of G at terminal heights (s,t). It sums $x^{\text{interior volume}} q^{\text{energy}}$ over nonnegative interior heights. Fix a budget R and $q \in \mathbb{Q} \cap (0,1)$. The state coordinates are indexed by orbits $o = \{s \le t\}$ with $s + t \le R$ and volumes $s + t \le v \le R$:

$$Y_o[v] = [x^v]\, x^{s+t} K_G^+(o), \qquad K_G^+(o) = \begin{cases} K_G(s,t) + K_G(t,s) & s < t, \\ K_G(s,s) & s = t. \end{cases}$$

These are raw labeled population counts. Nothing is averaged, symmetric coordinates are not identified, and no normalization changes as the graph grows.

Theorem SNG-O05 (compiler correctness). Under the conditions above, attaching C maps the state of G to the state of the glued graph by

$$Y'_o[v] = \sum_{j=0}^{v-s-t} [x^j] K_C(o)\, Y_o[v-j],$$

and the local coefficient at volume $v \le R$ is $\sum_o Y_o[v]$. The glued graph is again a simple two-terminal graph with the same terminals. Hence the formula holds for every word of attachments of every length, with no truncation error.

Proof. Every edge of C has an interior endpoint, because $ab \notin E(C)$. The fresh copy makes the interiors of G and C disjoint. So for fixed terminal heights the energy splits as $E_G + E_C$, the volume as $s + t$ plus the two interior volumes, and the interior height choices are independent. The local series of the glued graph is therefore $\sum_{(s,t)} x^{s+t} K_G(s,t) K_C(s,t)$ over ordered terminal heights. Since σ swaps the terminals, $K_C(s,t) = K_C(t,s)$. Grouping (s,t) and (t,s) gives $\sum_o x^{s+t} K_G^+(o) K_C(o)$. Extracting $[x^v]$ gives the update and the readout. Interior volumes are nonnegative, so $Y'_o[v]$ uses only $Y_o[v']$ with $v' \le v$. The truncated coordinates are therefore closed under the update, and the budget never has to grow with word length. No terminal edge is added and the interior is fresh, so simplicity and the terminal labels persist. □

The implementation `BoundaryExperimentSystem` compiles a declared list of contexts at given R and q. The checked evidence is listed here; all of it is in the retained receipt.

- **Standard contexts.** The pendant-leaf pair attaches one new leaf at each terminal. The shared neighbour attaches one new vertex adjacent to both.
- **Budget four (SNG-E009).** At $R = 4$, $q = 1/2$ the state dimension is 22 and the closure has full rank 22. The number of retained rows of word length at most k is

  $$5,\ 11,\ 15,\ 17,\ 18,\ 19,\ 20,\ 21,\ 22 \qquad (k = 0, \dots, 8).$$

  By SNG-O03 with unit attachment costs, this is the rank of all experiments of length at most k. The first four values are confirmed by an independent rank

computation over all words. The portable certificate is retained, replays without discovery, and equals a fresh discovery byte for byte as JSON.

- **Budget seven (SNG-E010).** At $R = 7$ the dimension is 70 and the rank is 69. The kernel is spanned by
$$e_{((3,4),7)} - e_{((0,5),7)}.$$
Both generators multiply it by $q^{25} = 2^{-25}$, and every output annihilates it. This is a **formal** interface ambiguity in the declared coordinate space. No pair of graphs realizing this difference is asserted.
- **Path fixture (SNG-E011).** Take the path $0 - 1 - 2$ at $R = 1$, $q = 1/2$, with terminals at the two endpoints or at an endpoint and the middle vertex. Both direct observation vectors are $(1, 5/4)$. At volume one, by direct height enumeration on the glued graphs, the pendant-leaf pair gives $7/4$ and $15/8$ and the shared neighbour gives 1 and $9/8$. With costs 2 and 1, the certified cheapest witness is the shared neighbour, with cost 1 and discrepancy $-1/8$. All fifteen words of length at most three were checked against direct enumeration at $R = 3$ for both placements.

Proposition SNG-O06 (top-volume corank bound). Let every context in the language satisfy the conditions above, and let d_C be the number of interior neighbours of a terminal of C, which is the same for both terminals. For each orbit $o = (s, t)$ with $s + t \le R$, the coordinate vector $e_{(o,R)}$ is an eigenvector of every compiled operator, with eigenvalue $q^{d_C(s^2+t^2)}$, and its readout is the volume-R unit. Consequently, if orbits $o \neq o'$ with $s + t, s' + t' \le R$ have $s^2 + t^2 = s'^2 + t'^2$, then $e_{(o,R)} - e_{(o',R)}$ is invisible, and

$$\text{corank} \;\ge\; \sum_{\text{classes}} \left(\#\{o :\; s + t \le R,\; s^2 + t^2 = c\} - 1\right).$$

Proof. By SNG-O05, the column (o, R) of the operator has nonzero entries only at (o, v) with $v \ge R$, so only at (o, R). The entry is the interior-volume-zero kernel coefficient $[x^0]K_C(o)$. With every interior height zero, only the terminal–interior edges carry energy, which totals $d_C s^2 + d_C t^2$. The readout of the coordinate (o, R) is the coefficient at volume R. Differences within one class are therefore common eigenvectors with zero readout. Differences from distinct classes have disjoint supports, so they are independent. □

The bound is zero for $R \le 6$. It is one for $R = 7, 8, 9$, from $0^2 + 5^2 = 3^2 + 4^2$, and two at $R = 10$, adding $1^2 + 7^2 = 5^2 + 5^2$. It equals the computed corank at $R = 4$ and $R = 7$. Whether it is sharp for larger budgets is open. Fixed-budget closure is a statement about a finite truncation: it does not prove that an untruncated generating function is rational or that the visible rank stabilizes as R grows.

15.6 Exact robustness for finite candidate families

Experiments that return polynomial curves in q are compared in the integral metric of $L^2([a, b])$. Its Gram matrix on monomials is $G_{ij} = (b^{i+j+1} - a^{i+j+1})/(i + j + 1)$. The coefficient dot product is a different quantity. For the path discrepancy $\Delta(q) = -q(1 - q)^2$, with coefficients $(0, -1, 2, -1)$,

$$\int_0^1 \Delta(q)^2 \, dq = \frac{1}{105}, \qquad (0, -1, 2, -1) \cdot (0, -1, 2, -1) = 6.$$

In the monic orthogonal basis on $[0,1]$, the coordinates of Δ are $(-1/12, 1/10, 1/2, -1)$ with squared norms $(1, 1/12, 1/180, 1/2800)$, and Parseval returns $1/105$. All Gram matrices are checked positive definite by exact LDL^{T}.

Proposition SNG-O07 (certified nearest-candidate identification). Suppose:

- the candidate family is a finite set $\mathcal{F}$ of observation vectors with minimum pairwise distance δ in a declared positive definite metric;
- the true vector m lies in $\mathcal{F}$;
- the observation is $y = m + e$ with $\|e\| \le \varepsilon$.

If $2\varepsilon < \delta$, then m is the unique nearest candidate to y.

Proof. For $c \in \mathcal{F} \setminus \{m\}$,

$$\|y - c\| \;\ge\; \|m - c\| - \|e\| \;\ge\; \delta - \varepsilon \;>\; \varepsilon \;\ge\; \|y - m\|. \qquad \square$$

At $2\varepsilon = \delta$ the observation can be equidistant from two candidates, so equality certifies nothing. The noise model is deterministic and bounded. Nothing outside the declared family is identified. Whole-curve separation is not pointwise separation: $(-1/2) + q$ has positive L^2 norm on $[0,1]$ but vanishes at $q = 1/2$. A single specialized interface value cannot reconstruct a parameter polynomial. The finite-pool ranking scores $\delta^2/(\text{cost} \cdot \text{noise}^2)$ over an explicit pool. It is not a search over attachment words.

15.7 The periodic sector

Fix a graph on n labeled vertices (isolated vertices and disconnected graphs allowed), a modulus $M \ge 1$, and integer exponents a_e, where zero and negative values are allowed. Put $t_e = \zeta_M^{a_e}$ in $\mathbb{Z}[t]/\Phi_M(t)$ and

$$F(x) = \sum_{h \in \mathbb{N}^n} x^{|h|} \prod_{uv \in E} t_{uv}^{(h_u - h_v)^2}.$$

Theorem SNG-O08 (residue numerator). With $K = n(M-1)$,

$$F(x) = \frac{P(x)}{(1 - x^M)^n}, \qquad P(x) = \sum_{\rho \in \{0,\dots,M-1\}^n} x^{|\rho|} \prod_{uv} t_{uv}^{(\rho_u - \rho_v)^2}, \qquad \deg P \le K.$$

Proof. Write $h = \rho + Mb$ uniquely with $0 \le \rho_v < M$ and $b \in \mathbb{N}^n$. Then $(h_u - h_v)^2 \equiv (\rho_u - \rho_v)^2 \pmod{M}$, and $t_e^M = 1$, so each weight depends only on ρ. Summing $x^{M|b|}$ over b gives $(1 - x^M)^{-n}$. $\square$

Theorem SNG-O09 (palindrome and sufficient cutoff). Then $P_k = P_{K-k}$ for every k. Let two models have the same n and M. If their series agree at every volume up to $\lfloor K/2 \rfloor$, they agree at every volume.

Proof. The complement $\rho \mapsto (M-1)\mathbf{1} - \rho$ negates every difference $\rho_u - \rho_v$, so it preserves every squared difference. It sends $|\rho|$ to $K - |\rho|$. It is a bijection of the residue box, so $P_k = P_{K-k}$. Since $P = F \cdot (1 - x^M)^n$ and the factor is $1 + O(x^M)$, P_k is determined by $F_0, \dots, F_k$. Agreement through $\lfloor K/2 \rfloor$ fixes $P_0, \dots, P_{\lfloor K/2 \rfloor}$. The palindrome fixes the remaining coefficients, and P determines F. $\square$

This is a sufficient cutoff, not a sharpness claim. The argument uses squared edge differences only. A unary or vertex weight is not preserved by complementation, and no such extension is claimed. The following were checked:

- **A certified equal pair (SNG-E012).** At $n = 3$, $M = 2$, with every exponent one, the path $0 - 1 - 2$ and the edge 01 with isolated vertex 2 have the same numerator $(1, -1, -1, 1)$. The cutoff is 1, so their complete series are equal. Direct enumeration confirms agreement through volume 14. At $M = 3$ the same pair separates at volume one, with $F_1 = 2\zeta + \zeta^2$ versus $2\zeta + 1$.
- **Exhaustive population.** Every graph on $n \leq 3$ labeled vertices, every $M \in \{1, 2, 3, 4\}$ and every exponent assignment from $\{-1, 0, 1, 2\}$: 524 models. The residue identity was checked against direct height enumeration through volume $K+2M+1$, the palindrome on every numerator, and the cutoff theorem on every pair with equal (n, M).

15.8 Edge-factor families and compressed interfaces

The October 2026 supplement `docs/research/VAKIL_BRIDGES.md` gives full proofs of three results, recorded here with identifiers. It applies standard tools from Vakil's draft of *Foundations of Algebraic Geometry* to the height series.

Proposition SNG-O10 (finite determination for edge-factor families). Place an independent factor t_e on every edge of a fixed template. For two fixed templates there is an R_* such that, over every commutative $\mathbb{Q}$-algebra, agreement through volume R_* is equivalent to equality of the complete series. The proof is the Hilbert basis theorem applied to the chain of coefficient-difference ideals. No effective R_* is given. An ideal that stops growing once can grow later: the coefficients $0, \ldots, 0, t$ are an example. Noetherian finite determination is not an algorithm.

Proposition SNG-O11 (triangle tangent ranks). On the positive octant, the three top-moment probes of the edge-factor triangle have Jacobian rank 3, 2 or 1 according as the edge factors take three, two or one distinct values. No further symmetric coefficient raises the rank.

Proposition SNG-O12 (parallel composition and terminal forgetting). Compressed charge interfaces multiply under parallel composition of blocks with disjoint interiors and edge-disjoint terminal graphs. Forgetting a terminal t at interior height h sends a cell by

$$(j, Q, E) \mapsto (j + h,\ Q_{-t} + h\zeta,\ E + (d_t + |\zeta|)h^2 - 2hQ_t),$$

and a source interface complete through R suffices for output budget R.

These are applications of standard principles, not new algebraic geometry. They are implemented in `edge_factors.py` and `charges.py` and checked by `scripts/vakil_bridge.py --check`.

15.9 Interfaces and runnable examples

The public contracts are specified in `docs/specifications/OPERATORS-v1.md`. A minimal session:

```
from fractions import Fraction
from singularity.graph import Graph
from singularity.graph_experiments import BoundaryExperimentSystem
from singularity.observability import observable_closure,
 ↪  minimum_distinguishing_experiment
```

```
system = BoundaryExperimentSystem(1, Fraction(1, 2))
operators = system.operator_system(attachment_costs=(2, 1))
certificate = observable_closure(operators)
path = Graph(3, ((0, 1), (1, 2)))
d = tuple(a - b for a, b in zip(system.compile_graph(path, (0, 2)),
 ↪  system.compile_graph(path, (0, 1))))
witness = minimum_distinguishing_experiment(operators, d, certificate)
assert (witness.word, witness.cost) == ((1,), 1)
```

```
from singularity.graph import Graph
from singularity.periodic import PeriodicModel, compare_models
result = compare_models(PeriodicModel(Graph(3, ((0, 1), (1, 2))), 2),
 ↪  PeriodicModel(Graph(3, ((0, 1),)), 2))
assert result.equal and result.cutoff == 1
```

15.10 Formal status

The Lean packages `formal/` and `formal-analysis/` predate this chapter. Their declarations do not formalize SNG-O01 to SNG-O12. A separate formalization of the closure, quotient, cost-filtration and periodic-complement arguments is reported in `formal-analysis/README.md` only to the extent that it compiles without admitted results. Until then, all statements in this chapter are mathematical exposition with exact finite checks.

15.11 Open questions

The following are open:

- **Restricted state families.** Find the minimal dimension of a compression that is exact only on the states graphs actually realize, and decide whether such a family can be closed under attachment with typed transitions.
- **Sharpness of the corank bound.** Is SNG-O06 sharp for every budget, and does the visible rank have a closed form in R?
- **Margins.** Develop bounded maximum-margin experiment search under a cost budget. An unrestricted optimum need not exist, because signals can grow without bound (SNG-O04).
- **Shared panels.** Find minimum-cost shared panels that separate every pair in a candidate family, ideally with joint robustness. Exact methods face set-cover hardness in general.
- **Parameter dependence.** Determine where the rank depends on q: generic rank over $\mathbb{Q}(q)$ versus rank at every specialization.
- **Sharper periodic cutoffs and noncommuting operations.** Find periodic cutoffs below $\lfloor K/2 \rfloor$ for structured families, and genuinely noncommuting graph operations with closed state semantics.

Chapter 16

Implementation and retained computational results

The mathematical manuscripts describe a research program, while this chapter records the executable boundary of version 0.3.0. The core package requires Python 3.10 or newer and uses only the standard library. Documentation generation additionally uses Pandoc and a LaTeX installation. The rendered PDF and HTML are distributed so that a reader does not need those authoring tools merely to read the work.

16.1 Installing and running the project

From the repository root, `python -m pip install -e .` installs the command `singularity`. An installation-free alternative is to set `PYTHONPATH=src` and invoke `python -m singularity`. The commands emit structured JSON so a retained calculation can be consumed by another experiment. Human explanations and proofs live in the monograph; JSON field names are not intended to replace the mathematical exposition.

`python -m singularity examples` prints the four original seven-vertex sources. `python -m singularity compare binary_g binary_h --max-volume 3` recovers their binary-census equality and first local difference. The parameter `--q 1/2` uses an exact rational, and the reported value is an exact fraction. A vanishing specialization is reported without pretending that a symbolic discrepancy remained visible at that parameter.

`python -m singularity profile --example binary_g --max-volume 4` returns the degree data, exact Laplacian polynomial, binary census, local coefficients, normalized coefficients and marked height sectors. The input can instead be a JSON object with an integer vertex count and an explicit edge list. Original vertex labels are retained throughout all finite populations. Isomorphism is a separate mathematical quotient, never an implicit division of multiplicities.

16.2 Implemented mathematical modules

The meaning of every public coefficient, population and certificate interface is fixed in the coefficient specification. The finite graph population used by census comparisons is fixed in the census specification.

Module	Exact object	Current capability
`graph`	Finite simple graph and integer quadratic energy	Validation, components, relabeling, cuts, Laplacian, signed integer height energy
`heights`	Total-height and span-marked populations	Weak compositions, local coefficients, minimum-zero normalization, marked sectors
`invariants`	Binary and linear-algebraic comparisons	Complete binary cut census, minimum-cut side sizes, integer characteristic polynomial, first finite difference
`population`	Constrained original-labeled height tuples	Exact counts, lexicographic rank/unrank, feasibility witnesses and finite bound certificates
`polynomials`	Sparse rational q-polynomials and formal x-series	Exact arithmetic, logarithms, specialization, JSON conversion and factor replay
`euler`	Formal zeta exponents	Recursive and Mobius inversion, generalized-binomial recomposition, parameter-specific first discrepancy
`transfer`	Bounded-height forest populations	Separator messages on trees/forests (SNG-F23), safe volume pruning (SNG-F24), root messages, separate cost measures, minimum-zero subtraction and exact-span marked histograms (SNG-F25)
`decomposition`	Supplied tree decompositions of arbitrary graphs	Axiom checker, owned-charge bag messages (SNG-F26), truncation, minimum-zero and exact-span sectors, elimination-order constructor
`visibility`	Finite visibility filtrations	Prefix-equivalence classes in symbolic, marked, rational and algebraic domains, split witnesses, unresolved classes (SNG-F33)
`parameters`	Exceptional energy parameters	Square-free parts, Sturm sequences, exact real-root isolation, algebraic specialization, parameter strata (SNG-F31)
`gauss`	Odd-prime Laplacian Gauss sums	Congruence diagonalization with witness, rank, discriminant character, rank-sensitive formula versus direct residue sum (SNG-F30)
`congruence`	Volume residue classes	Direct degree selection, exact character averaging in Z[t]/Phi_m (SNG-F27), direct tuple filtering
`certified`	Certified prime-product evaluation	Normalized local factors on a decomposition with outward rounding, quadratic height tails, interval logarithms, coefficient-aware omitted-prime bound (SNG-F38, SNG-F39)

Module	Exact object	Current capability
`nested_census`	Nested-pair boundary census	Five-statistic tensor over all T within S, reflection, span-two projection, binary marginal (SNG-F40)
`terminals`	Terminal kernels and gluing	Kernels conditioned on terminal heights, gluing for any number of terminals, Whitney-twist antisymmetric pairing (SNG-F41, SNG-F42)
`contexts`	Restricted-context quotients	Symmetrized kernels, recovery from swap-symmetric probes and exact interface ranks under a volume budget (SNG-F47, SNG-F48)
`charges`	Boundary charges and context transport	Charge tables rebuilding terminal kernels, the dominating-terminal sequence, closure operators, single-closure recovery and transport (SNG-F53, SNG-F54)
`locality`	Connected-support compiler	Exact-support logarithms over connected sets, pinned series for any degree shift (L_G and A_H), the 2-regular reconstruction (SNG-F55 to SNG-F57); regular-graph loop decomposition and first-overlap correction (SNG-F60, SNG-F61)
`families`	Pendant-cycle family	Generators, closed binary discrepancy, decorated decks, optimal probe parameter (SNG-F36, SNG-F37)
`conductor`	Root-of-unity local series	Integer cyclotomic rings, finite residue numerators, arbitrary-index exact coefficients with explicit residue-box budget
`weighted`	Positive integral vertex masses and edge weights	Parallel-weight aggregation and hard equality contraction preserving original energy and mass-weighted height
`arithmetic`	Prime-valuation populations	Positive integer factorization, finite Dirichlet coefficients, conservative exact local remainder and tail constants
`primitive`	Gcd-one normalized height profiles	Mobius extraction, independent gcd enumeration and exact scaling recomposition
`chains`	Weakly nested level chains	Direct chain generation with repeated levels, nested-boundary energy, marked/normalized/local coefficients independent of weak compositions
`chambers`	Half-open order chambers	Gap coordinates, nested-boundary gap form, unique chamber reconstruction, factorial-cost third enumerator (SNG-F29)

Module	Exact object	Current capability
`orders`	Poset-constrained heights	Validated posets, weak and strict counts, chamber and linear-extension order-polynomial formulas, weighted histograms on order-polytope points (SNG-F32)
`lowvolume`	Closed coefficient formulas through volume three	Degree, ordered adjacent-pair and induced-triple statistics; pattern sectors; regular closed form
`certificates`	Source-bound finite mathematical claims	Structural parser, two independent replay routes (stars-and-bars and level chains), nested-cut checks, normalization and factorization checks
`observability`	Finite operator systems and experiments	Exact closure with actual experiment words, certificates of L=AB and BM_i=T_iB, invisible subspace, quotient dynamics, globally cheapest distinguishing experiments under positive costs, strict in-memory and portable replay (SNG-O01–SNG-O04)
`graph_experiments`	Symmetric two-terminal context operators	Checked contexts with terminal-swap automorphisms, compiled truncated kernel operators, raw state compilation for asymmetric base blocks, direct-enumeration cross-check (SNG-O05, SNG-O06)
`experiment_design`	Exact polynomial-curve metrics	Gram matrices, LDL positivity, orthogonal coordinates, separations, certified nearest-candidate identification and finite-pool scores (SNG-O07)
`periodic`	Root-of-unity edge phases	Residue numerators in Z[t]/Phi_M, palindromes, independent height enumeration and the sufficient cutoff comparison (SNG-O08, SNG-O09)

The primitive module implements a useful extension of the foundational scaling identity. For every nonzero normalized profile a, its positive coordinate gcd d is unique; dividing by d produces a primitive profile. Total height and span scale by d, while energy scales by d squared. Consequently full marked populations are recovered from primitive populations by a finite divisor sum in every retained volume. The inverse operation uses the ordinary Mobius function on the scaling factor. The released test checks both directions on every simple graph through four vertices at a declared finite depth.

16.3 Population navigation and exact bounds

The command `population --example binary_g --volume 3 --span 2 --sampl e 5` compiles the 42 ordered profiles of type 2e_i+e_j. Additional exact or upper-bound constraints may restrict energy, total height, coordinate height and span. The compiler records why its search domain is finite. A bound supplied by the total height

is different from a bound inferred from connectedness and energy; that distinction remains visible in the report.

The ranking order is lexicographic in the original coordinate tuple. `--rank` checks a proposed tuple and reports its membership and index; `--unrank` reconstructs an original tuple from an index. Prefix completion counts make these operations possible without storing every completed tuple. Their computational cost still depends on the graph's frontier and on the active constraints. The package does not advertise a uniform polynomial-time procedure for arbitrary graphs or arbitrary energy budgets.

16.4 Forest transfer experiment

The forest engine conditions each subtree on the height at its root. Combining a child charges the connecting edge exactly once and convolves the child's volume-energy histogram. A final convolution joins disconnected components. The implementation rejects cycles instead of applying a tree formula to an unsupported graph. A Cartesian enumerator provides an independent oracle on small graphs.

For the path on 96 vertices, heights in 0,1,2, and total height at most 12, the retained population has exactly 2,506,262,962,660,245 original labeled profiles. This is a restricted low-volume population, rather than the entire height cube of size 3^96. Every volume marginal is independently checked against

$$[x^v](1+x+x^2)^{96} = \sum_{j\geq 0}(-1)^j \binom{96}{j}\binom{v-3j+95}{95},$$

with inadmissible binomial coefficients treated as zero. The report in `results/transfer_benchmark.json` records both counts and measured execution times. Timing is an environmental observation. The arithmetic identities and retained population definition are the reproducible mathematical conclusions.

16.5 Complete small-graph census

The script `scripts/census.py` enumerates every labeled simple graph through six vertices, filters connected graphs, and takes an exact isomorphism quotient using degree-respecting permutations. This is an intentionally modest exhaustive baseline. It avoids a hidden external graph database or an unrecorded canonical-labeling dependency. Its work limit is explicit: graph orders one through six.

Vertices	Connected labeled graphs	Connected isomorphism classes	Degree-layer classes	Through volume 2	Through volume 3
1	1	1	1	1	1
2	1	1	1	1	1
3	4	2	2	2	2
4	38	6	6	6	6
5	728	21	19	21	21
6	26704	112	68	100	112

Here the degree layer is the complete coefficient polynomial at volume one, not a single numeric evaluation. Local and fully marked prefixes produce the same class

counts on this retained census. All 112 connected six-vertex classes separate by volume three, and the complete binary census also separates them. The seven-vertex binary collision demonstrates why this finite observation cannot be promoted to a reconstruction theorem. The census gives a reproducible starting point for measuring how visibility changes as the graph population expands.

16.6 Reproduction and document generation

`make test` runs the standard-library suite. `make examples` regenerates the three reference comparisons, graph inputs, finite certificates and a remote conductor coefficient. `python scripts/reproduce.py --check` recomputes these results and compares them with their retained JSON. `python scripts/benchmark_transfer.py` reruns the path experiments. The exhaustive census is separate because its search scope is broader than the routine example replay.

`make book` assembles the chapter sources into a navigable monograph, generates LaTeX and HTML, and compiles the mathematical PDF. The shorter article in `docs/papers/visibility.md` provides a paper-shaped entry to the central construction and visibility theorem. `make release` performs the checks and creates a complete archive with source, documents, tests, inputs and results. Generated archives exclude earlier archives, temporary files and the local git object directory, preventing recursive packaging.

The released finite calculations are computational evidence for their declared inputs. The proofs in the manuscript establish the mathematical identities and analytic transfer statements with their stated hypotheses. Neither the software tests nor a source digest imply a proof-assistant verification of the implementation. This separation lets later pushes strengthen formal coverage without rewriting the meaning of the existing results.

Formal development: `formal/` is a core Lean 4.12.0 package (no Mathlib) with kernel-checked theorems for translation normalization (P07) and the nested threshold energy identity (P08); `scripts/check_formal.py` builds it and audits the axioms of the main theorems (see `formal/README.md`).

Chapter 17

Thirty-six concrete research pushes

This is a proposed sequence after the initial release. No entry is marked complete by its presence here. A push closes when its mathematical output and acceptance criterion are both met; a negative theorem or certified counterexample can be a successful outcome. Pushes are ordered for review and integration, while the explicit dependencies identify which preparatory work can proceed independently. Every completed push should update the manuscript, release its finite inputs and outputs, and record unresolved questions in ordinary prose. A larger computation is justified by the question it settles, not by its runtime.

The sequence follows six linked paths: exact populations, separator algorithms, nested boundaries and matroids, arithmetic specializations, order geometry, and visibility. Formal proof work follows concrete definitions and certificates. General graph reconstruction is a question to be investigated, never an acceptance criterion satisfied by a finite census. The two supplied graph pairs remain fixed regression examples throughout, so an expansion cannot silently change their labeling, normalization, or first-discrepancy statements.

17.1 P01 — Freeze the coefficient semantics

Inputs: the graph and height definitions of the initial release. **Outcome:** a versioned mathematical data specification covering labels, minimum-zero normalization, span, and truncation. **Acceptance:** every public function has an explicit finite meaning, and disconnected examples distinguish global from componentwise translation. **Dependencies:** the initial release. This push settles conventions before optimization introduces alternative representations.

17.2 P02 — Add independent level-chain enumeration

Inputs: P01 and the nested-boundary identity. **Outcome:** a second coefficient enumerator using nonempty nested level sets with repetitions. **Acceptance:** complete marked histograms agree with weak compositions on a declared small graph population, including profiles with height gaps. **Dependencies:** P01. The implementation must charge mixed boundary terms rather than substitute absolute-difference energy.

17.3 P03 — Publish portable coefficient certificates

Inputs: both enumerators and canonical graph serialization. **Outcome:** `relational-height/1` records with exact integer or rational coefficients and source digests. **Acceptance:** an independent reader rejects malformed degrees, duplicate terms, invalid edges, and altered coefficients, while reproducing all released small tables. **Dependencies:** P01–P02. Hash agreement alone does not close this push.

17.4 P04 — Give the spectral pair a short coefficient proof

Inputs: the fixed seven-vertex spectral pair. **Outcome:** a human-readable volume-two classification yielding $2q^2(1-q)^3(1+q)$. **Acceptance:** classify every profile type at volumes zero, one, and two; compare exact Laplacian characteristic polynomials; match both enumerators. **Dependencies:** P03. The proof should explain the difference through explicit degree and adjacency contributions.

17.5 P05 — Explain the binary-census separation

Inputs: the second fixed graph pair and its 128 binary assignments. **Outcome:** a full binary equality certificate and a volume-three proof of the displayed discrepancy. **Acceptance:** identify the span-two profile types responsible, account for every lower coefficient, and recover the exact value at $q = 1/2$. **Dependencies:** P02–P03. Binary sampling is insufficient.

17.6 P06 — Derive universal low-volume formulas

Inputs: arbitrary simple graphs and profile partitions through volume three. **Outcome:** formulas using degree statistics and explicitly defined adjacent-vertex statistics. **Acceptance:** prove each formula by a disjoint profile classification and compare it against complete small-graph enumeration. **Dependencies:** P04–P05. Any proposed graph statistic must have a definition independent of the formula it explains.

17.7 P07 — Formalize translation normalization

Inputs: the finite graph energy definition and minimum-zero bijection. **Outcome:** a proof-assistant theorem for $B = (1 - x^n)L$ at coefficient level. **Acceptance:** prove injectivity, surjectivity, volume shift, and energy preservation without admitted lemmas; publish the actual build command and dependencies. **Dependencies:** P01. No analytic infinite-product library is needed for this push.

17.8 P08 — Formalize the nested-level energy identity

Inputs: finite height tuples and threshold sets. **Outcome:** kernel-checked volume, span, and quadratic energy identities. **Acceptance:** allow repeated levels, prove the mixed term equals the stated edge count for nested sets, and derive binary extraction. **Dependencies:** P07. The theorem should expose the pairwise edge model rather than introduce a different interaction law.

17.9 P09 — Implement bounded-height tree messages

Inputs: rooted trees, height bound H, and volume bound R. **Outcome:** exact separator-conditioned message polynomials. **Acceptance:** prove the child-product recursion, count every vertex and edge once, and match reference coefficients on paths, stars, and branched trees. **Dependencies:** P02. Report polynomial support sizes separately from height-state counts.

17.10 P10 — Prove safe volume truncation

Inputs: P09's recursion and nonnegative volume exponents. **Outcome:** a theorem that intermediate reduction modulo x^{R+1} preserves the requested output. **Acceptance:** show that $H = R$ suffices for unrestricted coefficients through R, then demonstrate larger tree computations with identical semantics. **Dependencies:** P09. Benchmarks must compare complete coefficient ranges.

17.11 P11 — Recover span by bounded-box subtraction

Inputs: the bounded-height polynomials C_H. **Outcome:** a marked tree algorithm using $D_H = C_H - x^n C_{H-1}$ and exact-span differences. **Acceptance:** prove the shift subtraction and compare every (v, e, h) count with direct marked enumeration. **Dependencies:** P09–P10. Negative intermediate coefficients must be investigated rather than clipped to zero.

17.12 P12 — Extend messages to bounded separators

Inputs: a graph with an explicitly supplied tree decomposition. **Outcome:** a bag-conditioned exact algorithm and decomposition checker. **Acceptance:** verify the decomposition axioms, prove local charge bookkeeping, and compare cycles and bounded-width examples with reference data. **Dependencies:** P10–P11. State-count and sparse polynomial-convolution costs must remain separate in the complexity discussion.

17.13 P13 — Classify the first positive-energy layer

Inputs: connected graphs and marked coefficients. **Outcome:** a theorem describing minimum-energy nonconstant profiles and their relation to minimum cuts. **Acceptance:** handle possible nonbinary competitors explicitly, prove the stated multiplicities, and compare with exhaustive profiles. **Dependencies:** P06 and P08. Energy order must not be confused with volume order.

17.14 P14 — Compute the nested-pair census

Inputs: all $T \subseteq S \subseteq V$ in a declared graph population. **Outcome:** the five-statistic nested-boundary tensor described in Chapter 13. **Acceptance:** recover the span-two generating function by an explicit projection and explain the binary-census example's separation. **Dependencies:** P05 and P08. A new tensor name is not itself a structural result.

17.15 P15 — Separate lattice information from chamber information

Inputs: the cut-lattice realization and minimum-zero volume functional. **Outcome:** a precise preservation question with proved sufficient conditions for marked-series equality. **Acceptance:** specify which lattice isometries preserve height volume and span, and exhibit a nontrivial worked instance or a failure of a proposed condition. **Dependencies:** P08 and P14. Abstract energy preservation alone is insufficient.

17.16 P16 — Test Whitney switches systematically

Inputs: a documented family of admissible two-terminal gluings and switches. **Outcome:** exact comparison tables and either a separating pair or a transformation conjecture. **Acceptance:** verify the matroid-preserving hypotheses independently and retain tested volume bounds for unresolved pairs. **Dependencies:** P15. No finite failure to separate may be reported as universal matroid invariance.

17.17 P17 — Add exact volume congruence filters

Inputs: local polynomials and a positive modulus. **Outcome:** character-filtered populations for volume residue classes. **Acceptance:** prove the root-of-unity averaging identity and compare it with direct modular filtering of assignments. **Dependencies:** P03. This filter must not be described as extracting span or the binary sector after those marks have been summed away.

17.18 P18 — Prove root-of-unity rationality in energy

Inputs: an energy root of unity of order m. **Outcome:** the residue-cube numerator over $(1 - x^m)^n$, represented in an exact cyclotomic ring. **Acceptance:** prove the coordinate residue decomposition and reproduce local coefficients through an independent bound. **Dependencies:** P03 and P17. Complex specialization does not inherit positive-real branch assertions.

17.19 P19 — Analyze finite quadratic Gauss sums

Inputs: P18 numerators at $x = 1$ and odd-prime moduli. **Outcome:** formulas in terms of Laplacian rank and nonsingular discriminant, with singular directions accounted for. **Acceptance:** compare direct residue enumeration with diagonalized forms and give explicit exceptional-prime examples. **Dependencies:** P18. Prime powers and characteristic two remain separate until proved.

17.20 P20 — Build the disjoint order-chamber formula

Inputs: vertex permutations with label-based tie breaking. **Outcome:** a gap-coordinate expression for volume and quadratic energy. **Acceptance:** prove that every profile belongs to exactly one chamber, including tied heights, and match complete histograms. **Dependencies:** P02 and P06. Summing closed chambers without boundary corrections fails this criterion.

17.21 P21 — Relate constrained heights to order polytopes

Inputs: an explicit finite poset of height inequalities. **Outcome:** a comparison between constrained populations and established order-polytope lattice counts. **Acceptance:** derive the unweighted bounded-height specialization with correct weak and strict inequalities, then identify what quadratic weighting adds. **Dependencies:** P20. Ordinary Ehrhart conclusions require an ordinary fixed-polytope counting problem.

17.22 P22 — Establish one genuine reciprocity statement

Inputs: a precisely chosen bounded order or chamber population. **Outcome:** a proved reciprocity identity with explicit boundary terms, or a counterexample to the proposed weighted extension. **Acceptance:** verify complete small instances and give a full argument for the stated scope. **Dependencies:** P21. An analogy to Ehrhart or chromatic reciprocity does not close the push.

17.23 P23 — Implement formal logarithmic extraction

Inputs: exact local coefficient polynomials through volume R. **Outcome:** formal logarithm, inverse series, and finite Euler exponents using rational arithmetic. **Acceptance:** reconstruct the input truncation by multiplication and verify the Möbius coefficient identities. **Dependencies:** P03. Distinguish scalar exponents depending on q from integer combinatorial multiplicities.

17.24 P24 — Determine exceptional energy parameters

Inputs: discrepancy polynomials for a graph pair. **Outcome:** exact rational or algebraic parameter strata where the first differing volume changes. **Acceptance:** isolate every real root in $(0, 1)$ for the tested leading polynomials and recompute at each exceptional value. **Dependencies:** P23. A plot or a finite parameter grid cannot establish absence of roots.

17.25 P25 — Organize successive visibility levels

Inputs: coefficient differences through a declared volume bound. **Outcome:** a filtration by first discrepancy, including parameter-dependent refinements. **Acceptance:** prove every reported equality below the first differing coefficient and label unresolved collisions with their tested bound. **Dependencies:** P23–P24. Claims of an unrestricted separation depth require an additional theorem.

17.26 P26 — Make analytic remainder bounds explicit

Inputs: local coefficient growth bounds and a first discrepancy at r. **Outcome:** a quantified normalized-factor estimate sufficient for convergence when $s > 1/(r + 1)$. **Acceptance:** display constants, parameter restrictions, and treatment of finitely many small primes. **Dependencies:** P25. A formal expansion without a controlled remainder is not the analytic theorem.

17.27 P27 — Produce controlled real singularity experiments

Inputs: P26 bounds and exact discrepancy exponents. **Outcome:** prime-product evaluations of the normalized positive-real remainder near $s = 1/r$. **Acceptance:** attach finite-cutoff error intervals and recover the predicted asymptotic within them. **Dependencies:** P26. Numerical slope fitting illustrates an established exponent; it does not determine the exponent's proof.

17.28 P28 — State the available complex domain exactly

Inputs: fixed admissible q, local factor zeros, and normalized products. **Outcome:** a local complex branch statement on an explicitly described nonvanishing domain, or a documented obstruction. **Acceptance:** specify logarithm choices and convergence uniformly on compact subsets. **Dependencies:** P26. Do not infer global meromorphic continuation from real-axis control.

17.29 P29 — Specify a complete finite graph census

Inputs: a chosen graph class, order bound, and graph-generation method. **Outcome:** a canonical input manifest with isomorphism handling documented. **Acceptance:** independently reconcile population counts, generation options, and graph hashes. **Dependencies:** P03. Degree and spectrum filters are not a substitute for a complete census or a graph-isomorphism decision.

17.30 P30 — Measure collision classes across invariants

Inputs: P29's fixed graph population and exact coefficient algorithms. **Outcome:** collision partitions for spectra, binary census, and specified height truncations. **Acceptance:** release full class memberships, first-discrepancy witnesses, resource limits, and unresolved classes. **Dependencies:** P12, P14, and P29. Report bounded separation percentages with their exact denominator.

17.31 P31 — Resolve targeted collisions structurally

Inputs: the smallest unresolved classes from P30. **Outcome:** an explicit bijection, a higher-volume separation, or a precise remaining conjecture for each selected pair. **Acceptance:** explain the mechanism rather than only increase a cutoff; retain negative results. **Dependencies:** P16 and P30. A full-series equality claim needs an all-volume argument.

17.32 P32 — Seek a separation bound for one graph family

Inputs: low-volume formulas, tree messages, and collision evidence. **Outcome:** a proved first-separation bound for a narrowly stated family, or an explicit family defeating the proposed bound. **Acceptance:** parameterize the family and prove the statement uniformly. **Dependencies:** P06, P12, and P31. Complete graph reconstruction remains outside the claim unless proved separately.

17.33 P33 — Verify concrete coefficient certificates in a kernel

Inputs: P03 certificates and the formal finite identities. **Outcome:** a checker connecting released tables to the mathematical coefficient definition. **Acceptance:** kernel-check both decisive pair certificates and reject a changed coefficient or graph edge. **Dependencies:** P07–P08 and P23. A verified parser alone does not verify enumerated multiplicities.

17.34 P34 — Formalize the visibility comparison boundary

Inputs: the finite first-discrepancy theorem and available analytic libraries. **Outcome:** a formal statement connecting the normalized local expansion to the real zeta-factor asymptotic. **Acceptance:** expose every imported theorem and any unformalized premise; claim complete verification only when the full chain compiles. **Dependencies:** P26 and P33. Missing library infrastructure is recorded explicitly.

17.35 P35 — Release the first focused manuscript

Inputs: the exact examples, finite proofs, and real comparison theorem. **Outcome:** a literature-positioned paper matching Candidate A in Chapter 14, with source and reproducible artifacts. **Acceptance:** independently replay decisive coefficients, reconcile every claim with its proof or certificate, and remove unsupported priority language. **Dependencies:** P04–P05, P26–P27, and P33; formal analytics are included only to their achieved scope.

17.36 P36 — Choose the second paper from actual results

Inputs: the tree, matroid, arithmetic, and order outcomes. **Outcome:** one coherent methods or structural paper, plus an updated open-problem ledger. **Acceptance:** select a theorem or decisive negative classification as its center, complete the relevant literature comparison, and package reproducible data. **Dependencies:** P19, P22, P31–P32, and P35. Do not combine unfinished paths merely to make the manuscript larger.

The final decision is mathematical. If the work yields only a useful enumerator and two small examples, the release should say so clearly. If it yields a family theorem, a genuine comparison boundary, or a sharp analytic statement with quantitative control, the paper should lead with that result. The plan is designed to make these outcomes distinguishable early enough that the next few dozen pushes accumulate knowledge rather than accumulate terminology.